Dynamics of Particles

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Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

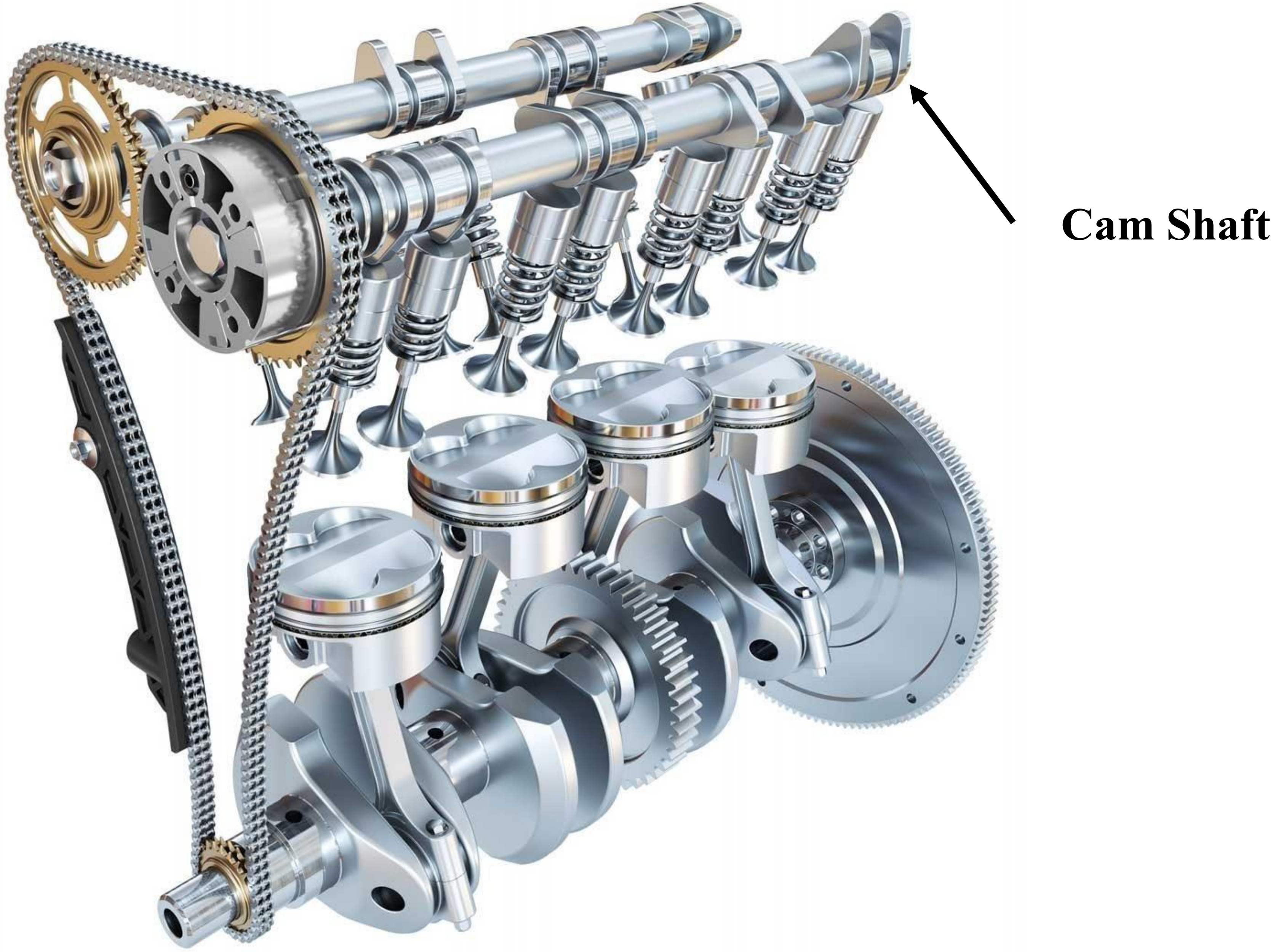
Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Kinematics of Particles

- ## What is Kinematics?

- Kinematics is the **branch of dynamics** which describes the **motion of bodies without reference to the forces** which either cause the motion or are generated as a result of the motion
- **Geometry of Motion**
- **Engineering applications of Kinematics**
  - Design of cams, gears, linkages, and other machine elements to control or produce certain desired motions
  - Calculation of flight trajectories for aircraft, rockets, and spacecraft
- **Prerequisite for kinetics**


Cam Shaft

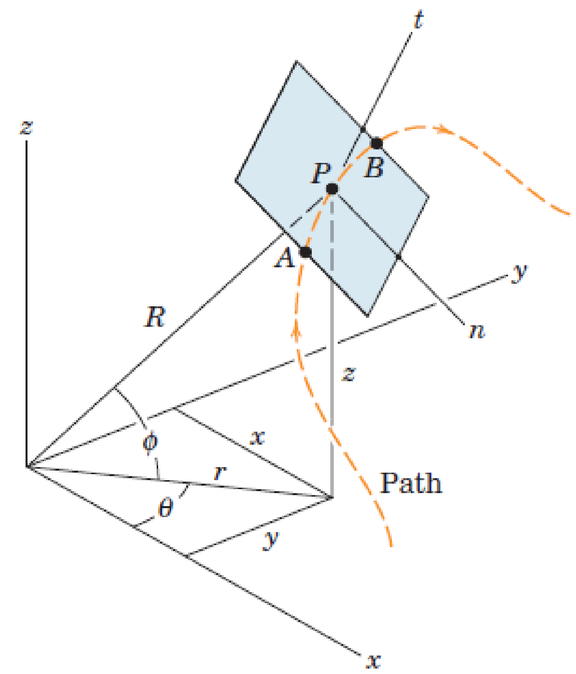
Kinematics of Particles

- **Particle?**

- A particle is a body whose physical dimensions are so small compared with the radius of curvature of its path that we may treat the motion of the particle as that of a point
- **Example:** Wingspan of a jet transport flying between Jodhpur and Kerala

Coordinate Systems

- **Coordinate systems**
- Rectangular Coordinate (x-y-z)
- Cylindrical Coordinate (r-θ-z)
- Spherical Coordinate (R-θ-ϕ)
- Normal-tangential coordinates:
  - measurements along the tangent t and normal n to the curve
  - The direction of n lies in the **local plane of the curve**
  - These measurements are called **PATH VARIABLES**.

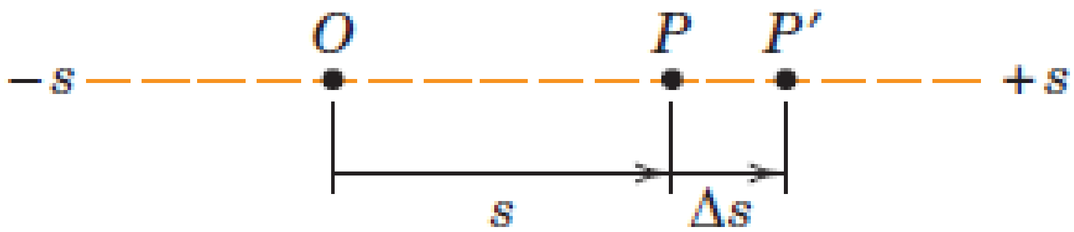


Plane motion

- **Plane motion**
- All movements occur in or can be represented as occurring in a single plane
- In many cases, motions of machines and structures in engineering can be represented as plane motion

Rectilinear motion

- **Motion along a line**

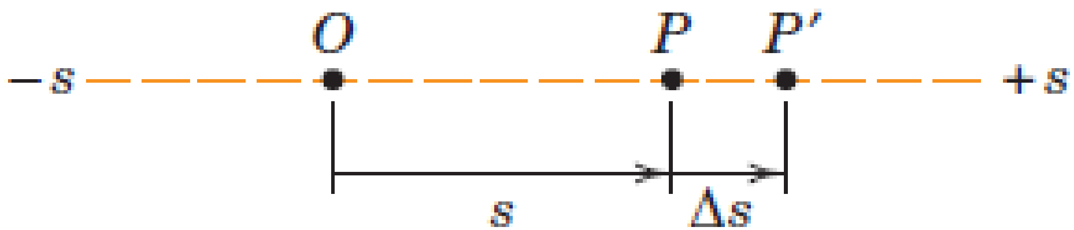


- Average velocity $\boxed{v_{av} = \dfrac{\Delta s}{\Delta t}}$

- Instantaneous velocity $\boxed{v = \lim_{\Delta t \to 0} \dfrac{\Delta s}{\Delta t}}$

$\boxed{v = \dfrac{ds}{dt}}$ Time rate of change of position coordinate, s

Rectilinear motion



- Average acceleration

$$\boxed{a_{av} = \frac{\Delta v}{\Delta t}}$$

- Instantaneous acceleration

$$\boxed{a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}}$$

$$\boxed{a = \frac{dv}{dt}}$$

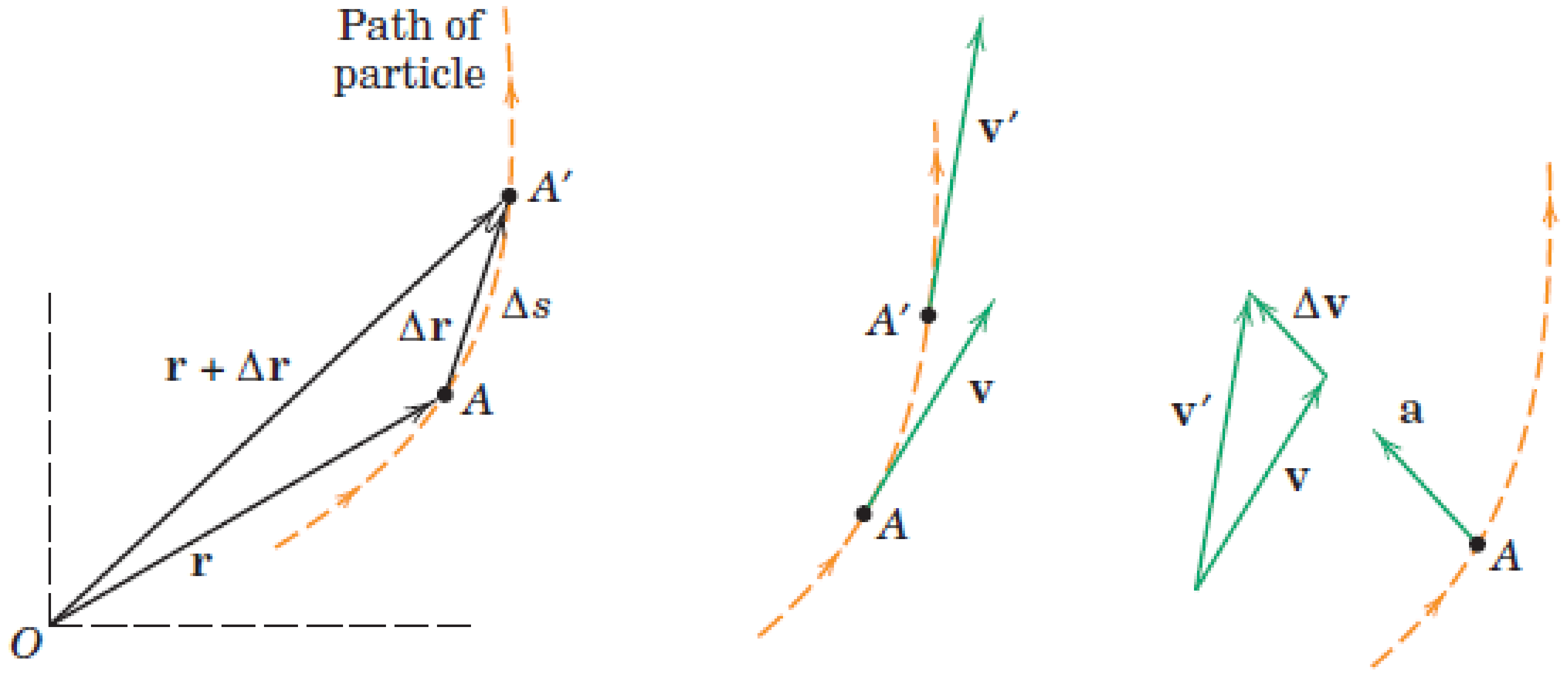
Time rate of change of velocity, *v*

$$\boxed{a = \frac{d^2S}{dt^2}}$$

$$\boxed{v dv = a ds} \qquad \boxed{\dot{s} d\dot{s} = \ddot{s} ds}$$

Plane Curvilinear motion

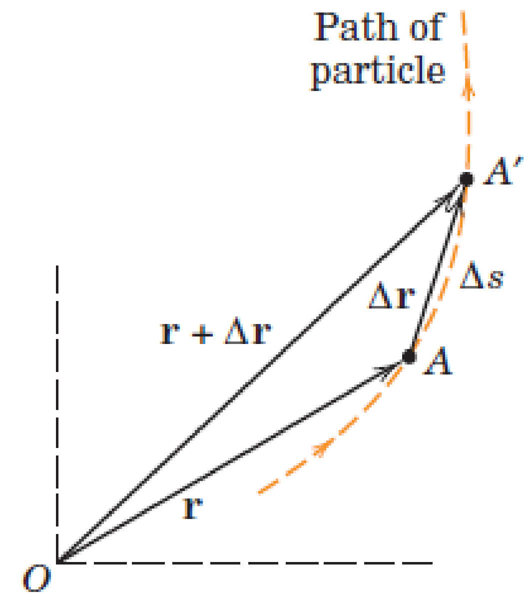
- **Motion along a plane curve**



- Average velocity $\boldsymbol{v}_{av} = \dfrac{\Delta \boldsymbol{r}}{\Delta t}$

Plane Curvilinear motion

- **Motion along a plane curve**



- Instantaneous velocity

$$\boldsymbol{v} = \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{r}}{\Delta t}$$

$$\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = \dot{\boldsymbol{r}}$$

- Average speed

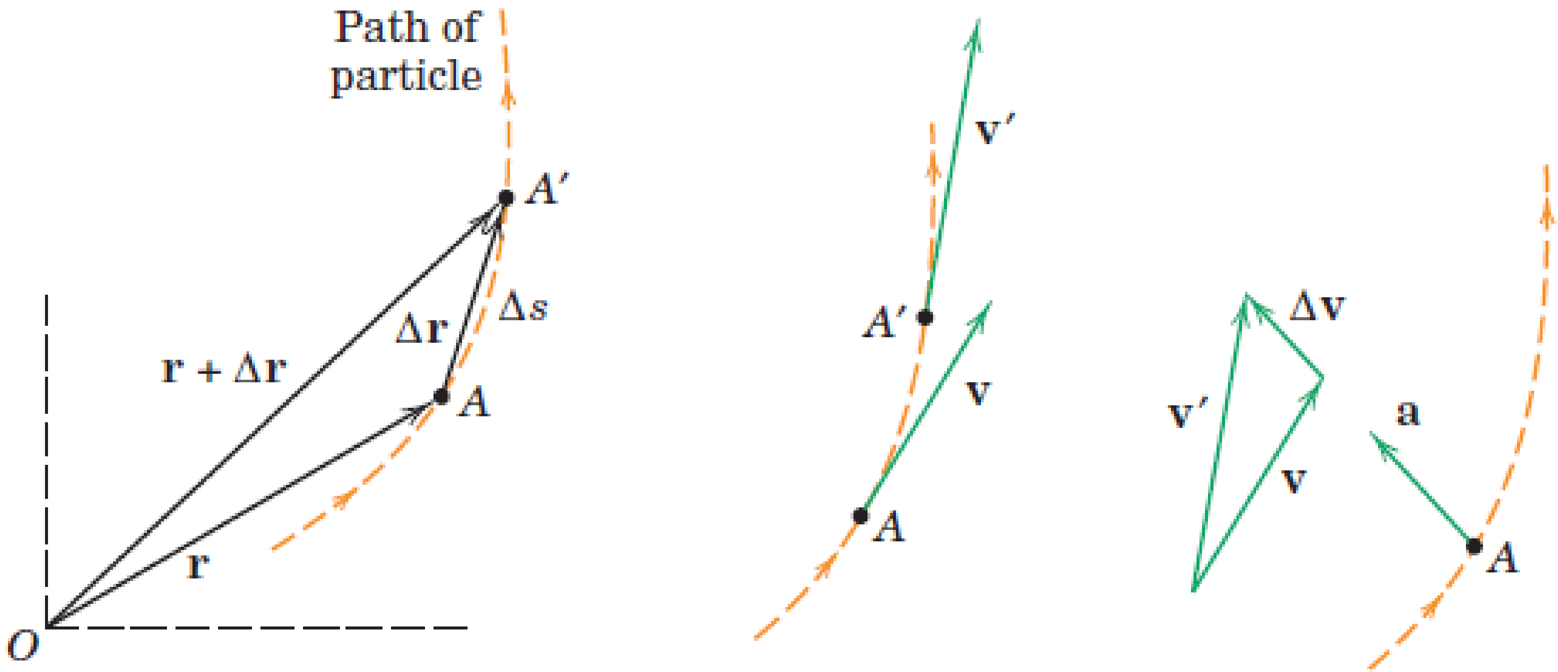
$$\text{speed}_{av} = \frac{\Delta s}{\Delta t}$$

- Speed

$$v = |\boldsymbol{v}| = \frac{ds}{dt} = \dot{s}$$

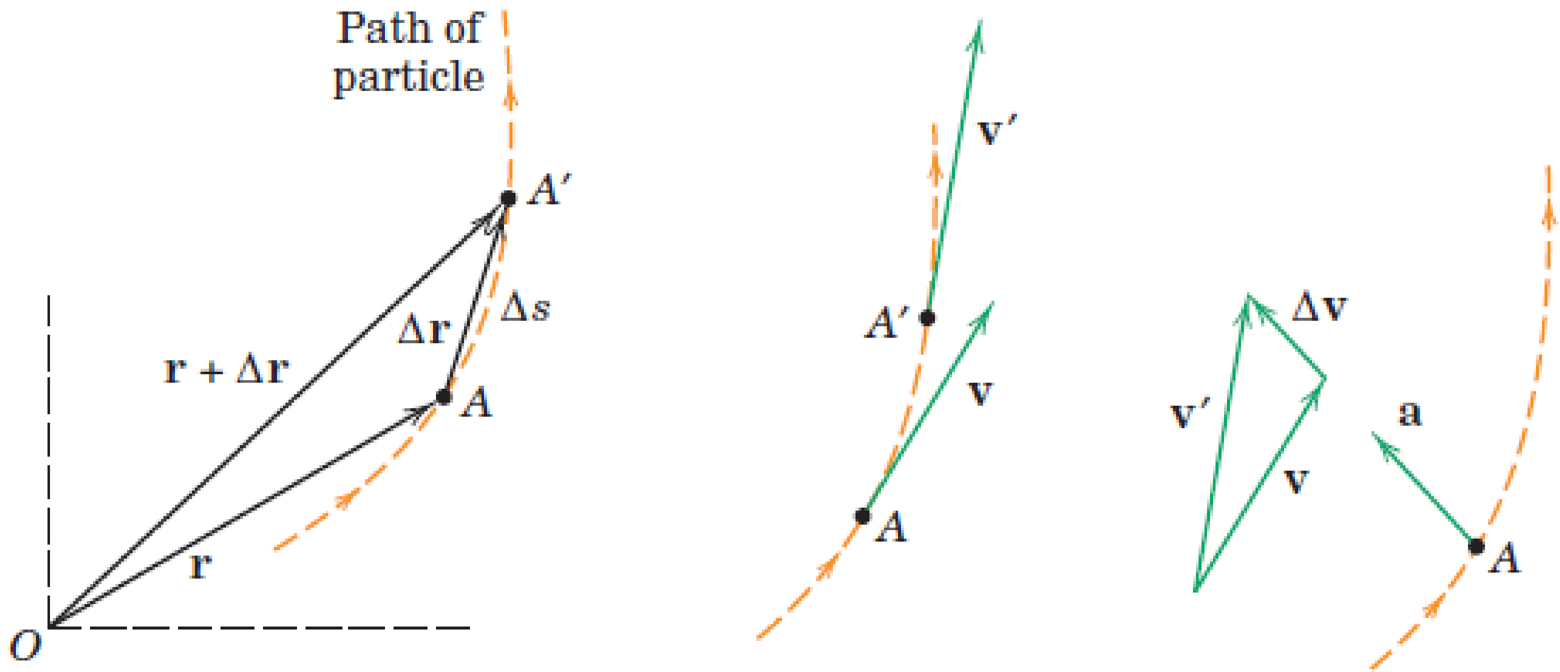
Plane Curvilinear motion

- **Motion along a plane curve**



- Average acceleration $$\boldsymbol{a}_{av} = \frac{\Delta \boldsymbol{v}}{\Delta t}$$

Plane Curvilinear motion

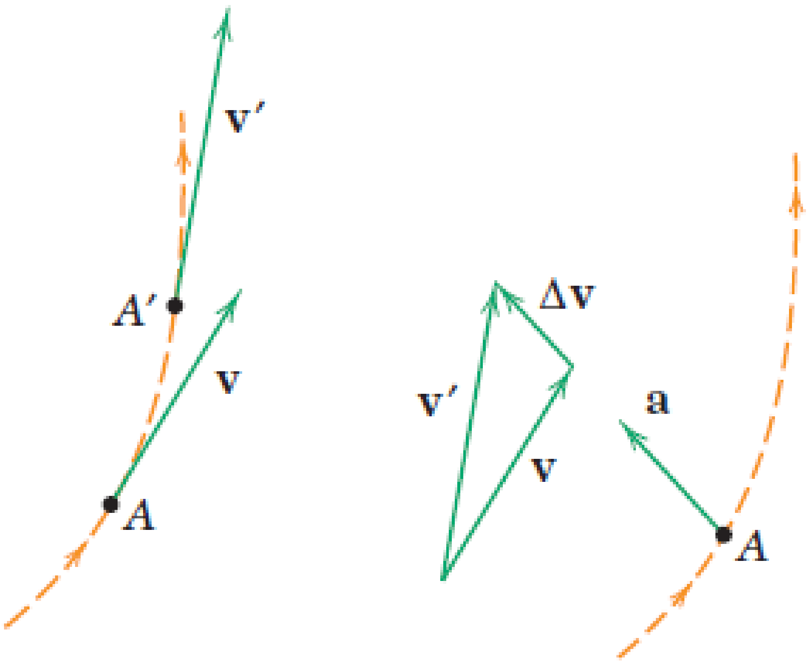


- Instantaneous acceleration

$$\boldsymbol{a} = \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{v}}{\Delta t} \qquad \boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}}$$

Plane Curvilinear motion

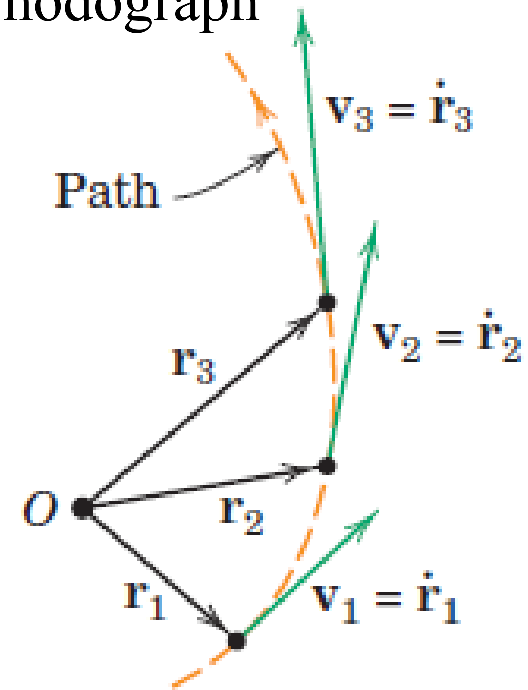
- **Motion along a plane curve**
- Average acceleration
- Instantaneous acceleration
  - In general, the direction of the acceleration of a particle in curvilinear motion is neither tangent to the path nor normal to the path.

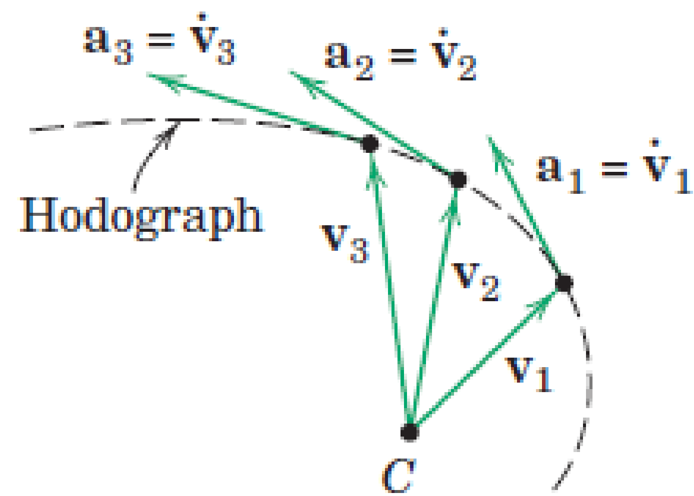


  - Acceleration ***component which is normal to the path points toward the center of curvature of the path***.

Plane Curvilinear motion

- **Hodograph**
- Velocity vectors plotted from some arbitrary point C, a curve, called the hodograph, is formed. The derivatives of these velocity vectors will be the acceleration vectors which are tangent to the hodograph





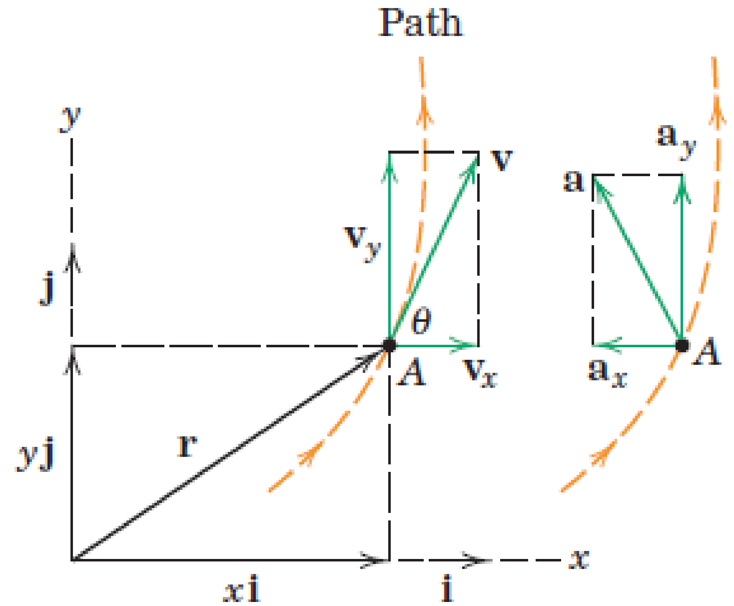
Rectangular Coordinates

- **Plane Motion**
- Position, Velocity and Acceleration vectors

$$\boldsymbol{r} = xi + yj$$

$$\boldsymbol{v} = \dot{\boldsymbol{r}} = \dot{x}i + \dot{y}j$$

$$\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{\boldsymbol{r}} = \ddot{x}i + \ddot{y}j$$



$$v_x = \dot{x} \qquad a_x = \dot{v}_x$$

$$v_y = \dot{y} \qquad a_y = \dot{v}_y$$

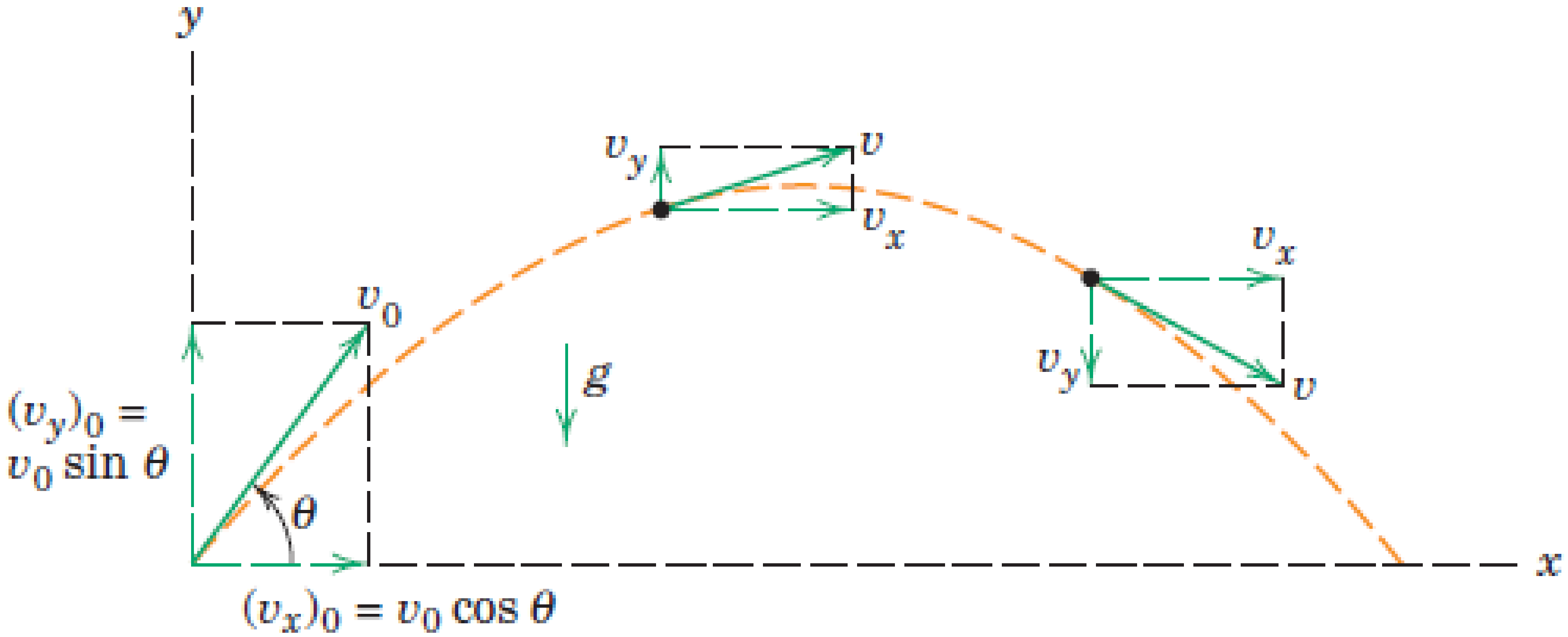
$$v^2 = {v_x}^2 + {v_y}^2$$

$$a^2 = {a_x}^2 + {a_y}^2$$

$$\tan\theta = \frac{v_y}{v_x} = \frac{dy}{dx}$$

Projectile Motion

- **Plane Motion**



$v_x = ?$ $\quad a_x = ?$

$v_y = ?$ $\quad a_y = ?$

Projectile Motion

- **Plane Motion (study Curvilinear motion using rectilinear motion)**
- Position, Velocity and Acceleration vectors

$$v = v_0 + at \qquad v^2 = {v_0}^2 + 2a(s - s_0) \qquad s = s_0 + v_0 t + \frac{1}{2}at^2$$

$$a_x = 0$$

$$v_x = (v_x)_0$$

$$x = x_0 + (v_x)_0\, t$$

$$a_y = -g$$

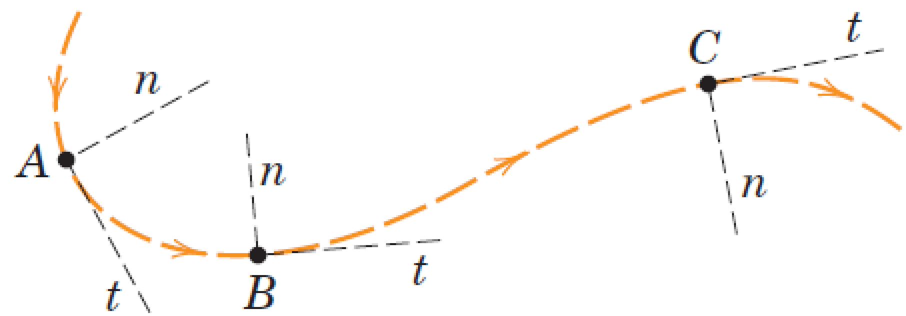
$$v_y = (v_y)_0 - gt$$

$$y = y_0 + (v_y)_0\, t - \frac{1}{2}gt^2$$

$${v_y}^2 = {(v_y)_0}^2 - 2g(y - y_0)$$

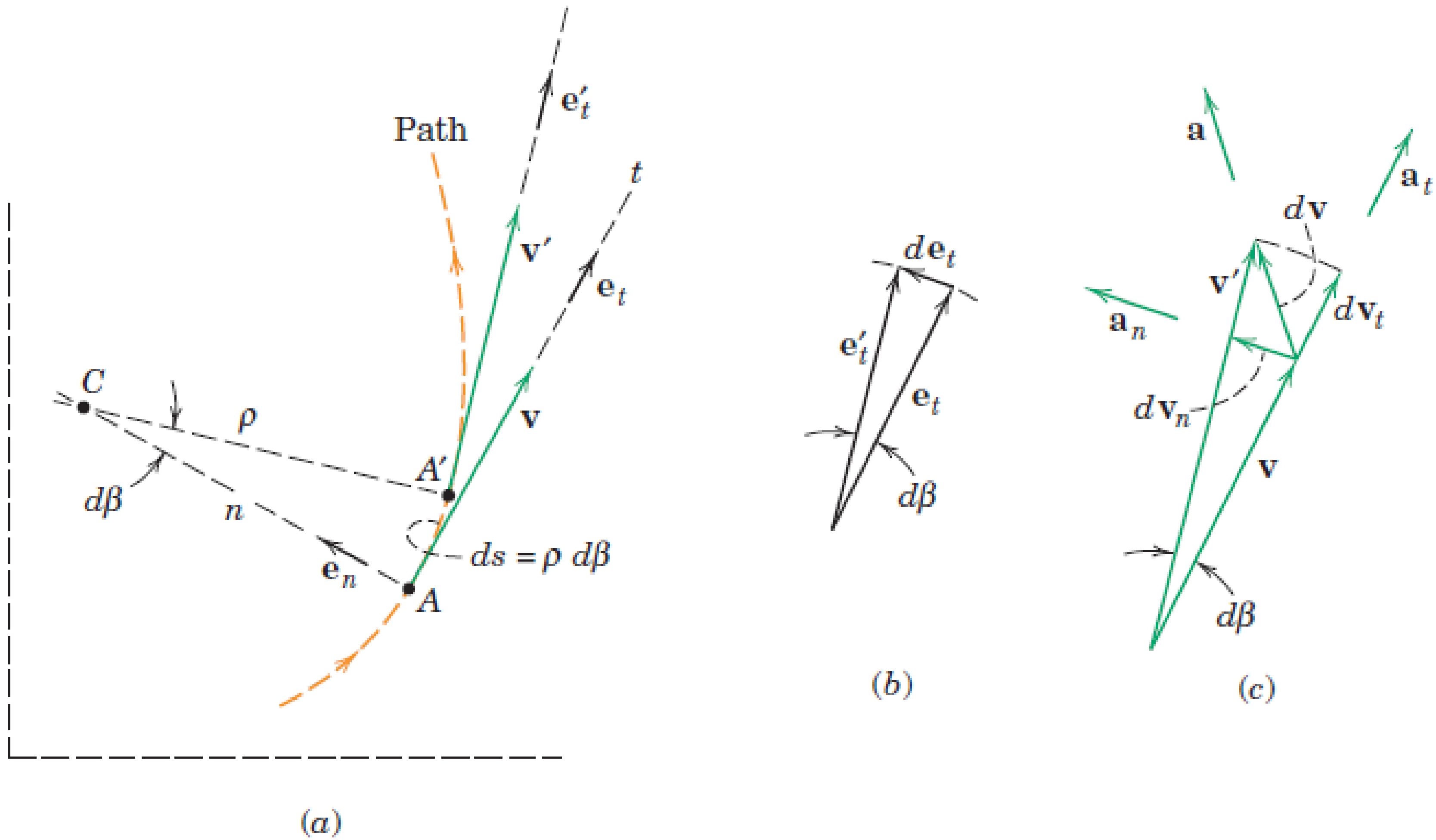
Normal and Tangential Coordinates (n-t)

- Path Variables: Measurements along normal and tangential directions

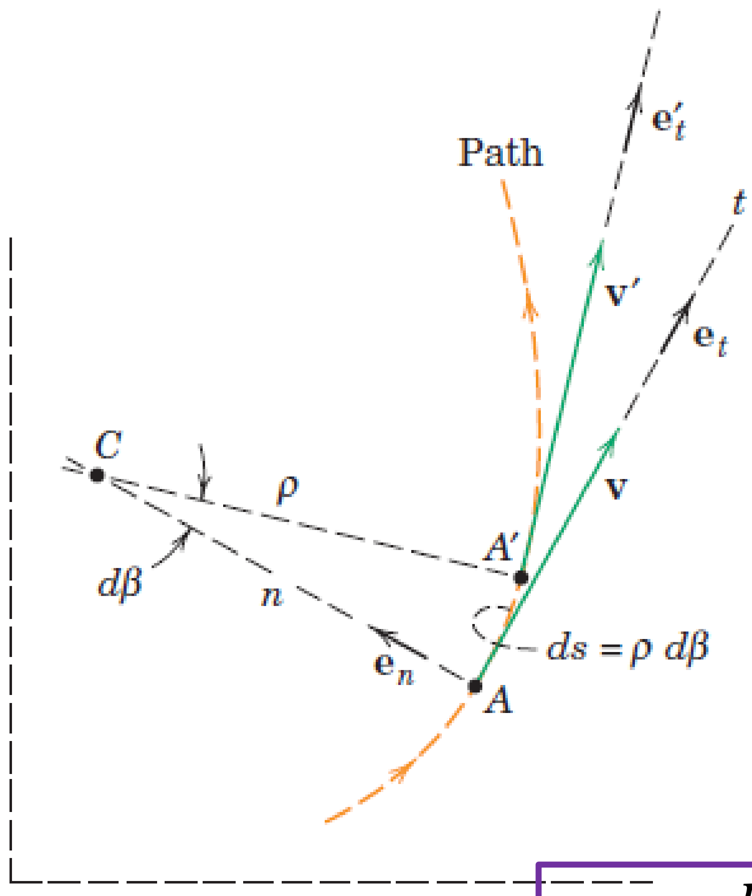


- Positive direction for *n* at any position is always taken toward the center of curvature of the path
- Positive *n* - direction will shift from one side of the curve to the other side if the curvature changes direction.

Normal and Tangential Coordinates (n-t)



Normal and Tangential Coordinates (n-t)



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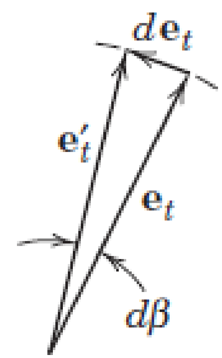
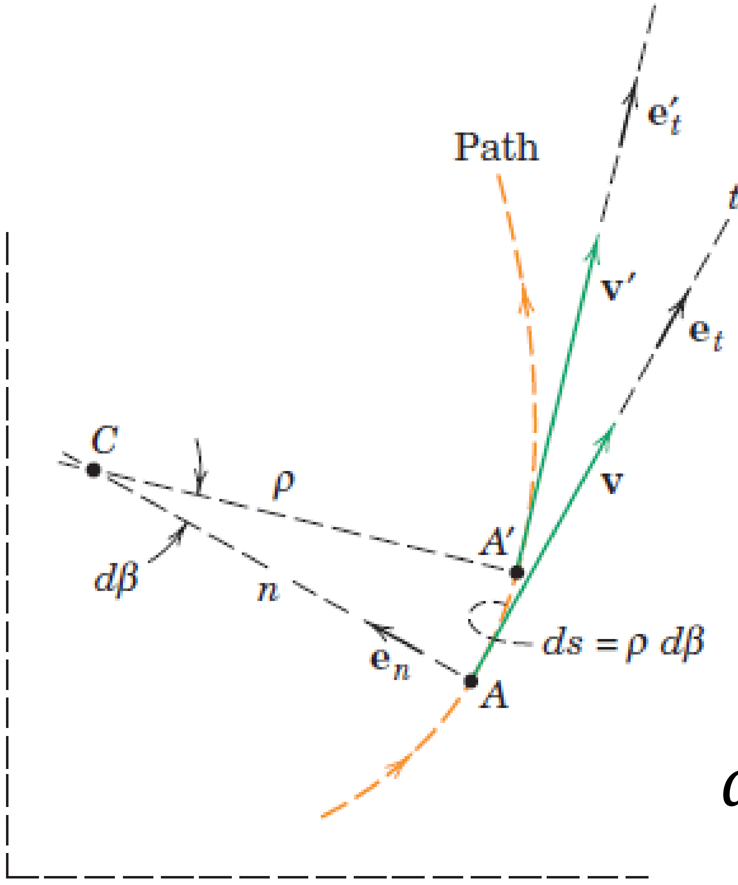


$$|de_t| = d\beta\; e_t \qquad de_t = e_n d\beta$$

$$\frac{de_t}{dt} = e_n \frac{d\beta}{dt} \qquad \boxed{\dot{e}_t = e_n \dot{\beta}}$$

$$\boxed{v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho\dot{\beta}} \qquad \boxed{\dot{e}_t = e_n \frac{v}{\rho}}$$

Normal and Tangential Coordinates (n-t)



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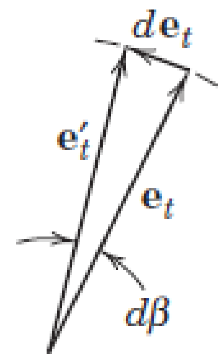


$$\boxed{v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho\dot{\beta}}$$

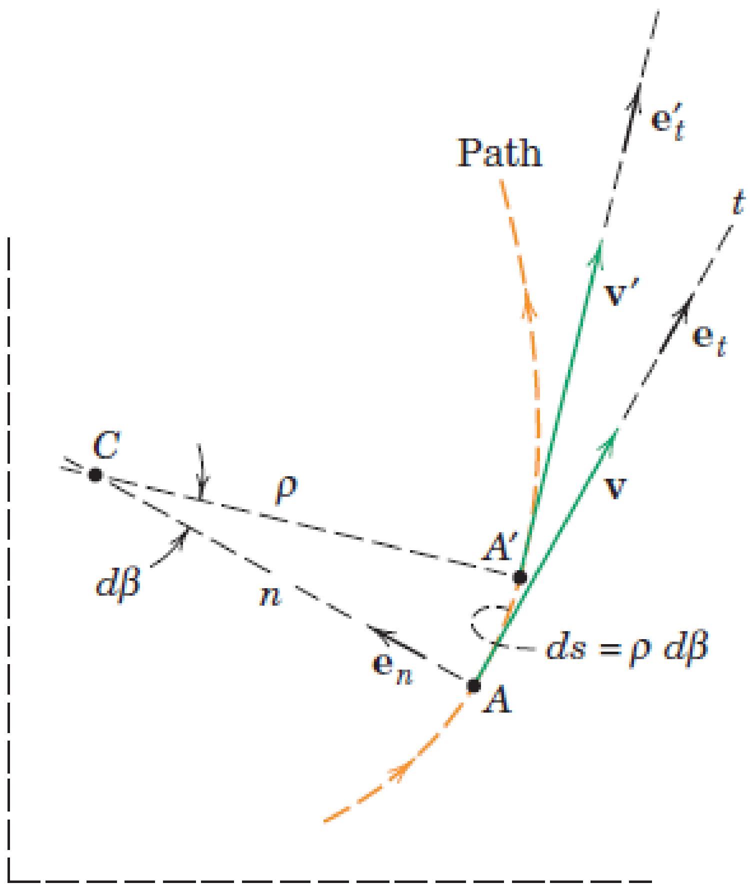
$$v = ve_t = \rho\dot{\beta}e_t$$

$$a = \frac{dv}{dt} = \frac{d(ve_t)}{dt} = \dot{v}e_t + v\dot{e}_t$$

$$\boxed{\dot{e}_t = e_n\dot{\beta}}$$

$$\boxed{\dot{e}_t = e_n\frac{v}{\rho}}$$

Normal and Tangential Coordinates (n-t)



$$\boxed{v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho\dot{\beta}}$$

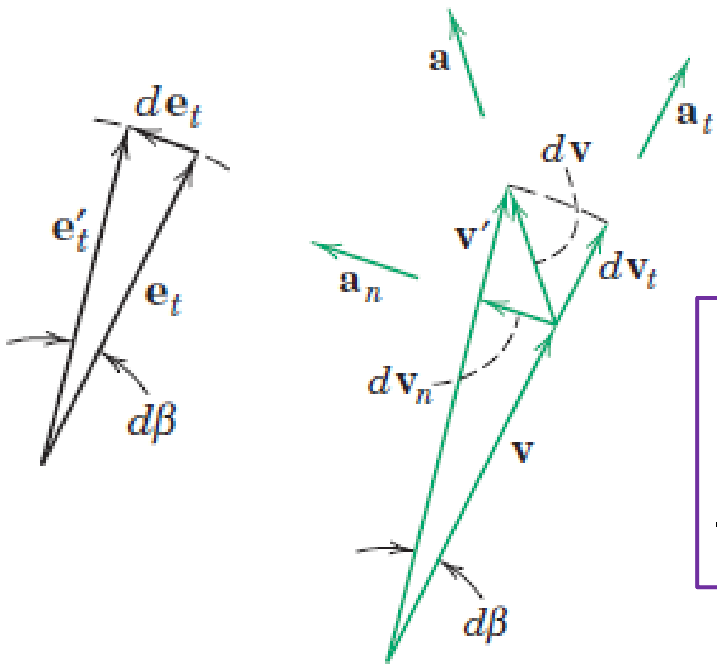
$$v = ve_t = \rho\dot{\beta}e_t$$

$$a = \frac{dv}{dt} = \frac{d(ve_t)}{dt} = \dot{v}e_t + v\dot{\beta}e_n$$

$$a_t = \dot{v} = \ddot{s}$$

$$a_n = v\dot{\beta} = \rho\dot{\beta}^2 = \frac{v^2}{\rho}$$

Normal and Tangential Coordinates (n-t)



$$\boxed{v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho\dot{\beta}}$$

$$v = ve_t = \rho\dot{\beta}e_t$$

$$\boldsymbol{dv_n} = |\boldsymbol{v}|d\beta$$

Acceleration contribution:

$$\frac{\boldsymbol{dv_n}}{dt} = \frac{|\boldsymbol{v}|d\beta}{dt} = |\boldsymbol{v}|\dot{\beta} = \rho\dot{\beta}^2 \text{ in } e_n \text{ direction}$$

$$a = \dot{v}e_t + v\dot{\beta}e_n$$

dv_t: **change in magnitude of *v***

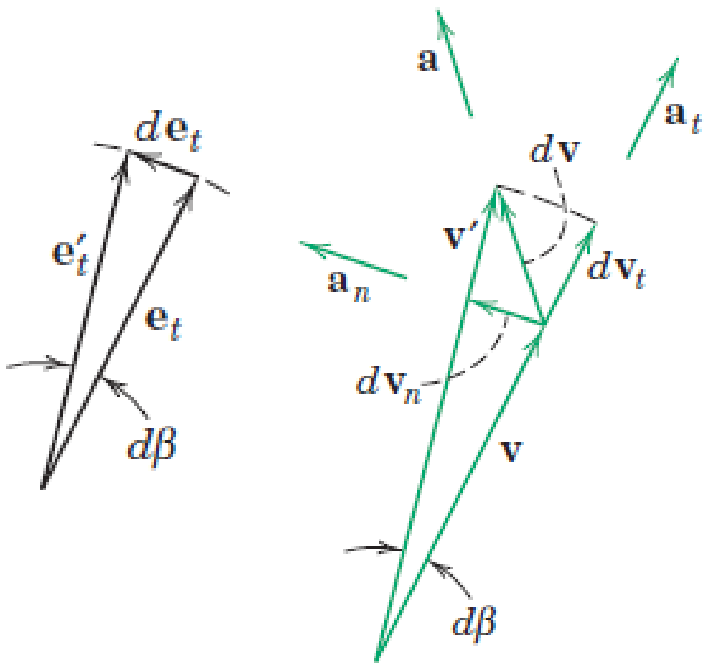
Acceleration contribution:

$$\frac{dv_t}{dt} = \frac{d(\rho\dot{\beta})}{dt} \quad \text{in } e_t \text{ direction}$$

$$\frac{dv_t}{dt} = \ddot{s} \quad \text{in } e_t \text{ direction}$$

Normal and Tangential Coordinates (n-t)

$$a = \dot{v}e_t + v\dot{\beta}e_n$$



$$\boldsymbol{dv_n} = |\boldsymbol{v}|d\beta$$

Acceleration contribution:

$$\frac{\boldsymbol{dv_n}}{dt} = \frac{|\boldsymbol{v}|d\beta}{dt} = |\boldsymbol{v}|\dot{\beta} = \rho\dot{\beta}^2 \text{ in } e_n \text{ direction}$$

dv_t: **change in magnitude of *v***

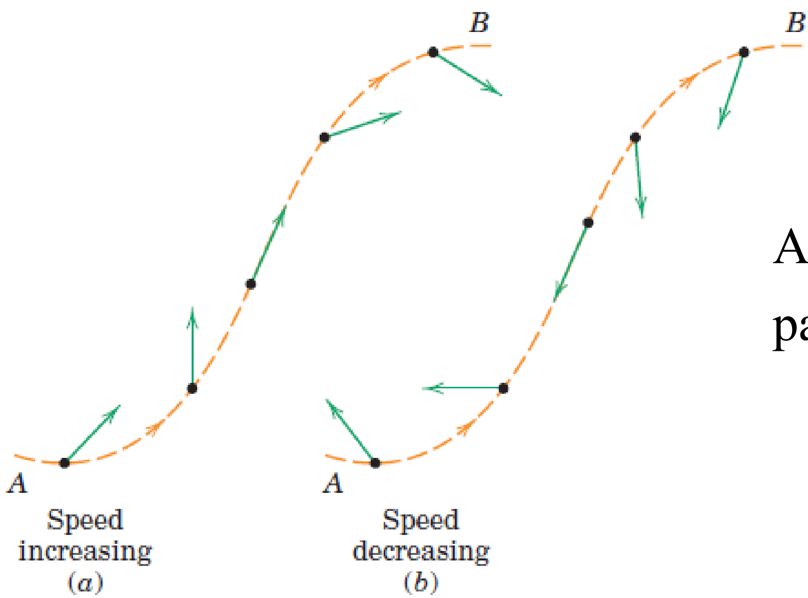
Acceleration contribution:

$$\frac{dv_t}{dt} = \frac{d(\rho\dot{\beta})}{dt} \quad \text{in } e_t \text{ direction}$$

$$\frac{dv_t}{dt} = \ddot{s} \quad \text{in } e_t \text{ direction}$$

- a_n is *always directed toward the center of curvature C*
- a_t positive *t*-direction of motion if the speed *v* is increasing and negative *t*-direction if the speed is decreasing.

Normal and Tangential Coordinates (n-t)



Acceleration vectors for particle moving from *A* to *B*

(a) increasing speed and (b) decreasing speed

- For (a), tangential acceleration is in positive *t- direction* and for (b), tangential acceleration is in negative *t*-direction
- At an inflection point on the curve, the normal acceleration $\frac{v^2}{\rho}$ goes to zero because ρ becomes infinite

Normal and Tangential Coordinates (n-t)

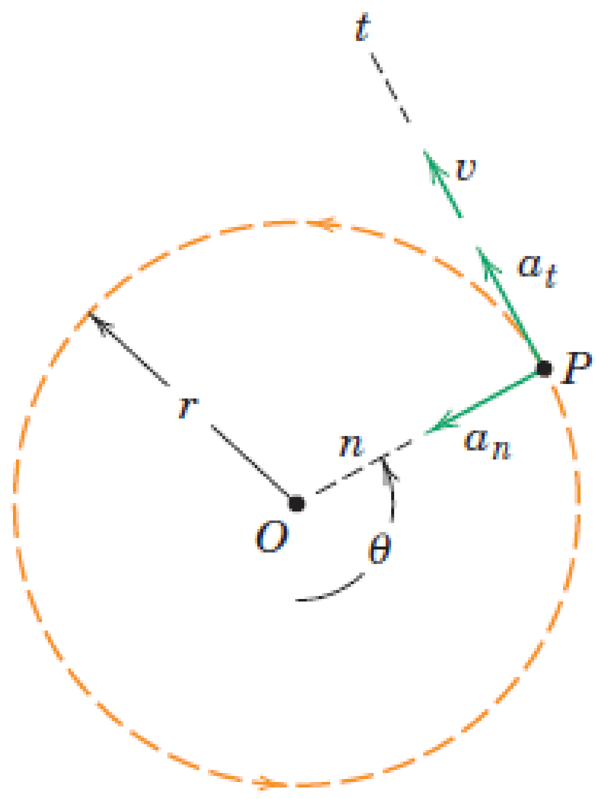
- **Circular Motion**

$$v = r\dot{\theta} \qquad\qquad \rho = r$$

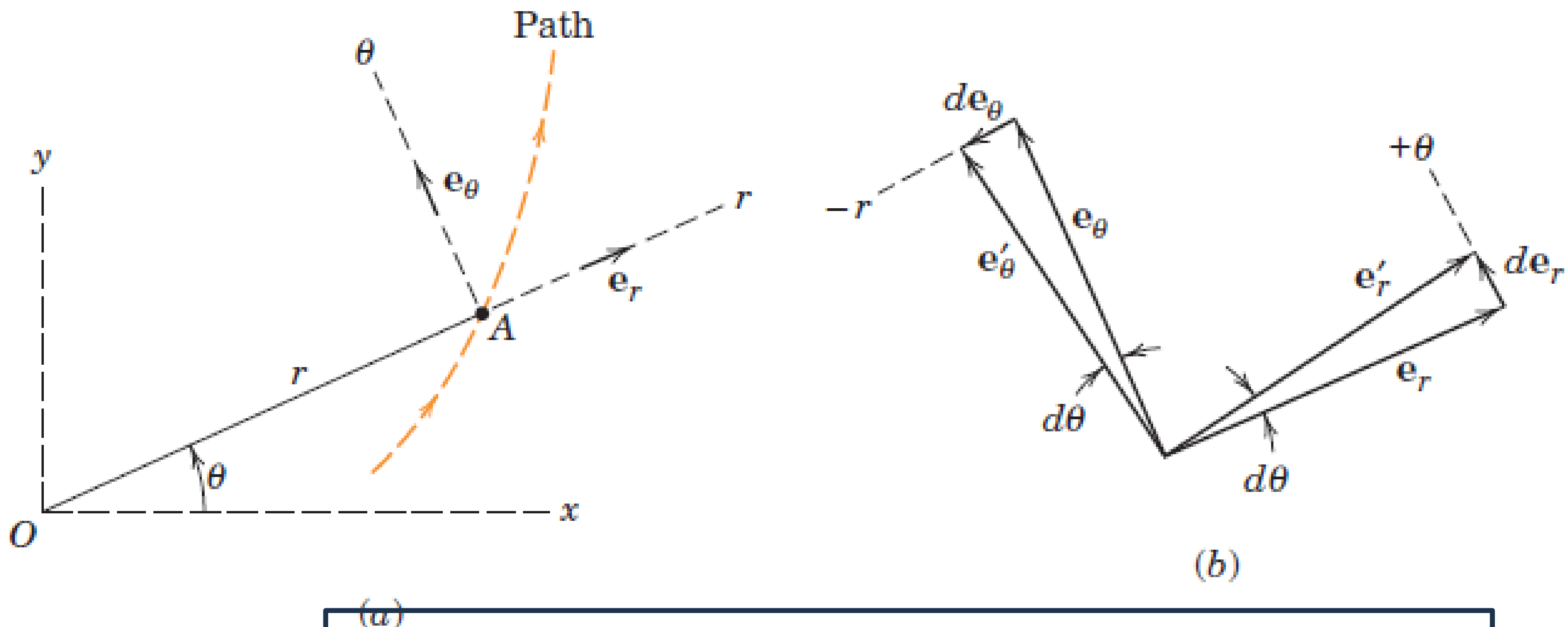
$$a = \dot{v}e_t + v\dot{\beta}e_n$$

$$a_t = \dot{v} = r\ddot{\theta}$$

$$a_n = \frac{v^2}{r} = r\dot{\theta}^2 = v\dot{\theta}$$



Polar Coordinates (r-θ)



$$\boldsymbol{r} = r\boldsymbol{e_r}$$

$$\boxed{\frac{d\boldsymbol{e_r}}{d\theta} = \boldsymbol{e_\theta} \qquad \frac{d\boldsymbol{e_r}}{dt} = \frac{d\boldsymbol{\theta}}{dt}\boldsymbol{e_\theta} \qquad \dot{\boldsymbol{e_r}} = \dot{\theta}\boldsymbol{e_\theta}}$$

$$\boxed{\frac{d\boldsymbol{e_\theta}}{d\theta} = -\boldsymbol{e_r} \qquad \frac{d\boldsymbol{e_\theta}}{dt} = -\frac{d\boldsymbol{\theta}}{dt}\boldsymbol{e_r} \qquad \dot{\boldsymbol{e_\theta}} = -\dot{\theta}\boldsymbol{e_r}}$$

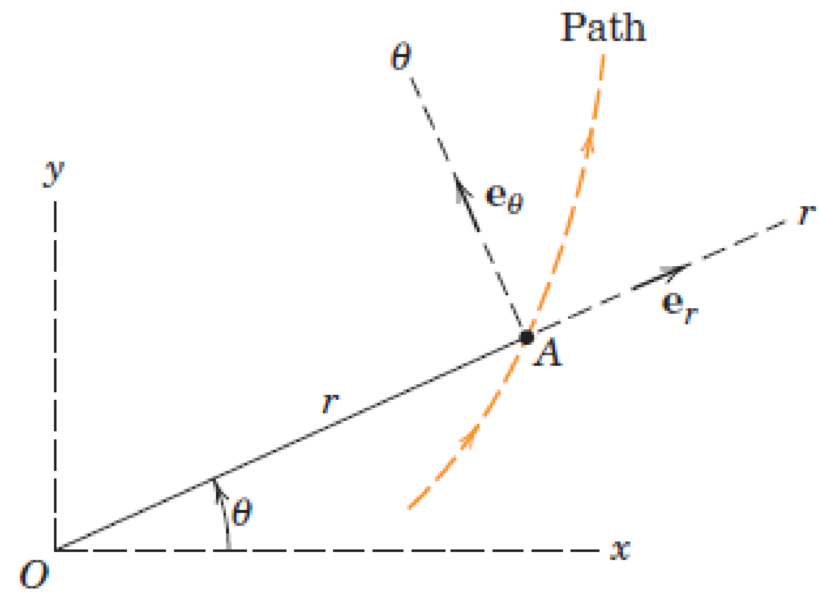
Polar Coordinates (r-θ)

$$\boldsymbol{r} = r\boldsymbol{e_r}$$

$$\boldsymbol{v} = \dot{\boldsymbol{r}} = \dot{r}\boldsymbol{e_r} + r\dot{\boldsymbol{e}}_{\boldsymbol{r}}$$

$$\dot{\boldsymbol{e}}_{\boldsymbol{r}} = \dot{\theta}\boldsymbol{e_\theta}$$

$$\boldsymbol{v} = \dot{r}\boldsymbol{e_r} + r\dot{\theta}\boldsymbol{e_\theta}$$



(a)

- **r**- component of **v:** rate at which the vector **r** stretches
- θ- component of **v:** due to the rotation of **r**

Polar Coordinates (r-θ)

$$\boxed{\boldsymbol{v} = \dot{r}\boldsymbol{e_r} + r\dot{\theta}\boldsymbol{e_\theta} \qquad \dot{\boldsymbol{e}}_{\boldsymbol{r}} = \dot{\theta}\boldsymbol{e_\theta} \qquad \dot{\boldsymbol{e}}_{\boldsymbol{\theta}} = -\dot{\theta}\boldsymbol{e_r}}$$

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\boldsymbol{e}}_{\boldsymbol{r}} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} + r\dot{\theta}\dot{\boldsymbol{e}}_{\boldsymbol{\theta}}}$$

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} - r\dot{\theta}\dot{\theta}\boldsymbol{e_r}}$$

$$\boxed{\boldsymbol{a} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\boldsymbol{e_\theta}}$$

Polar Coordinates (r-θ)

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\boldsymbol{e}}_{\boldsymbol{r}} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} + r\dot{\theta}\dot{\boldsymbol{e}}_{\boldsymbol{\theta}}}$$

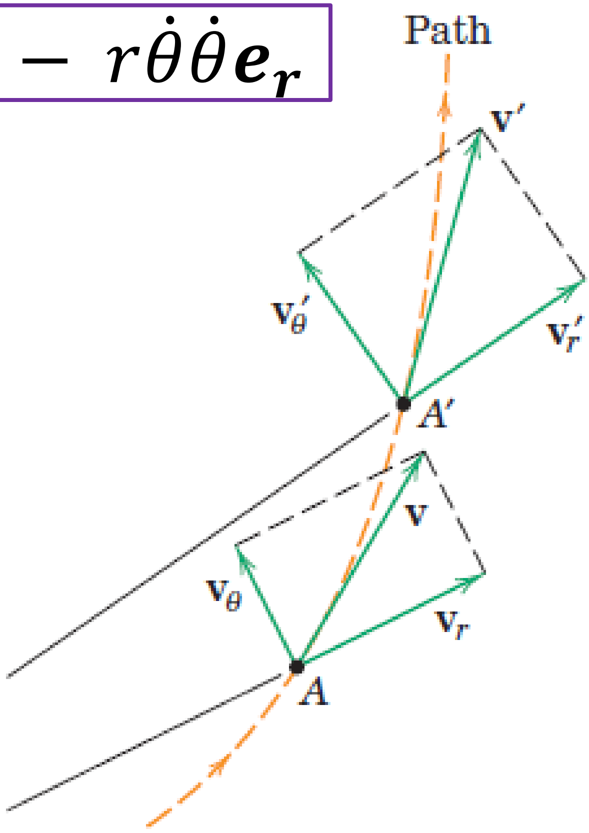
$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} - r\dot{\theta}\dot{\theta}\boldsymbol{e_r}}$$

$$\boxed{\boldsymbol{a} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\boldsymbol{e_\theta}}$$

$$\boxed{a_r = (\ddot{r} - r\dot{\theta}^2)}$$

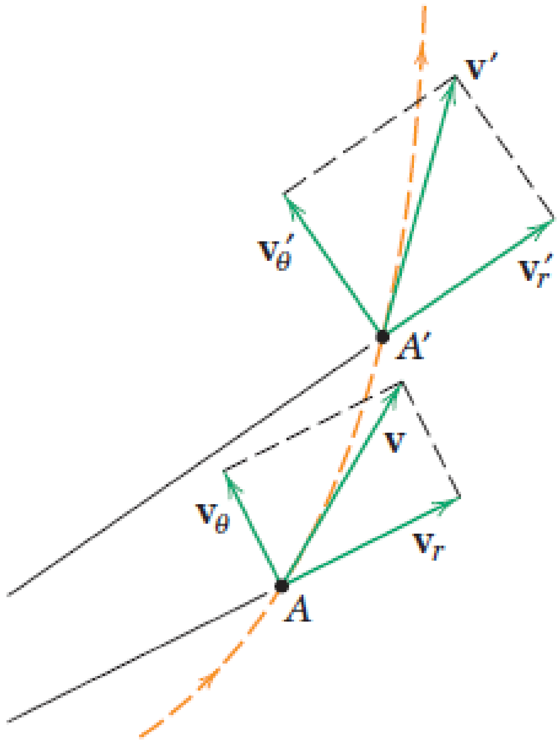
$$\boxed{a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})}$$

$$\boxed{a_\theta = \frac{1}{r}\frac{d(r^2\dot{\theta})}{dt}}$$

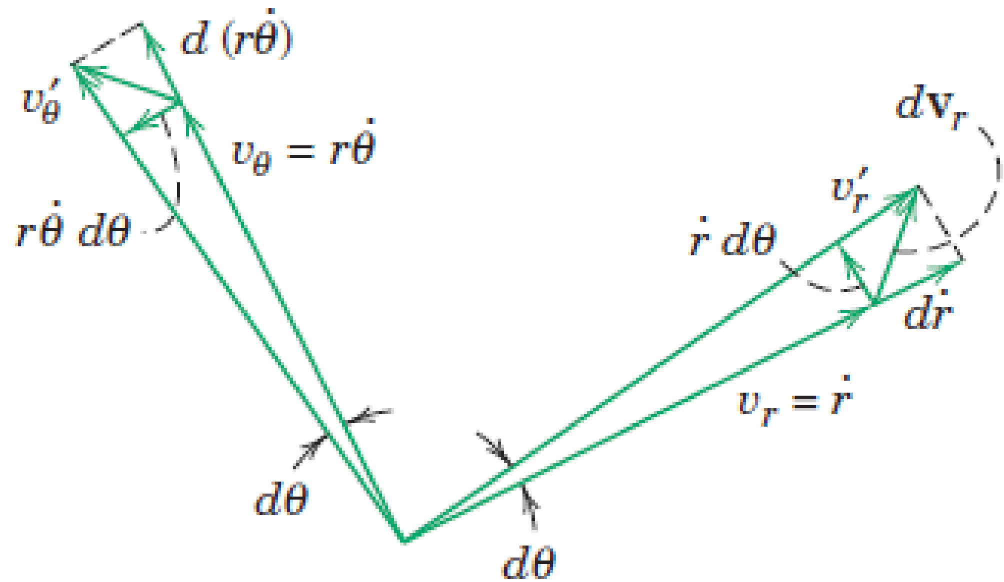


(a)

Polar Coordinates (r-θ)



(a)

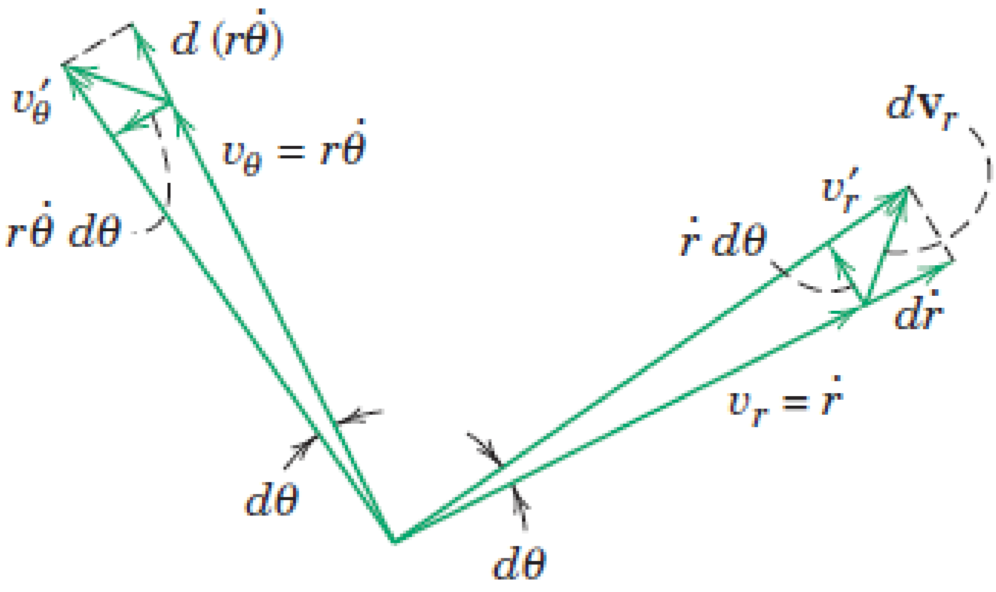


(b)

Polar Coordinates (r-θ)

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = (\ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\boldsymbol{e}}_{\boldsymbol{r}}) + \left(\dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} + r\dot{\theta}\dot{\boldsymbol{e}}_{\boldsymbol{\theta}}\right)}$$

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} - \; r\dot{\theta}\dot{\theta}\boldsymbol{e_r}}$$



(b)

Magnitude change of v_r: $d\dot{r}$

Acceleration contribution:

$\dfrac{d\dot{r}}{dt} = \ddot{r}$ in e_r direction

Direction change of v_r: $\dot{r}d\theta$

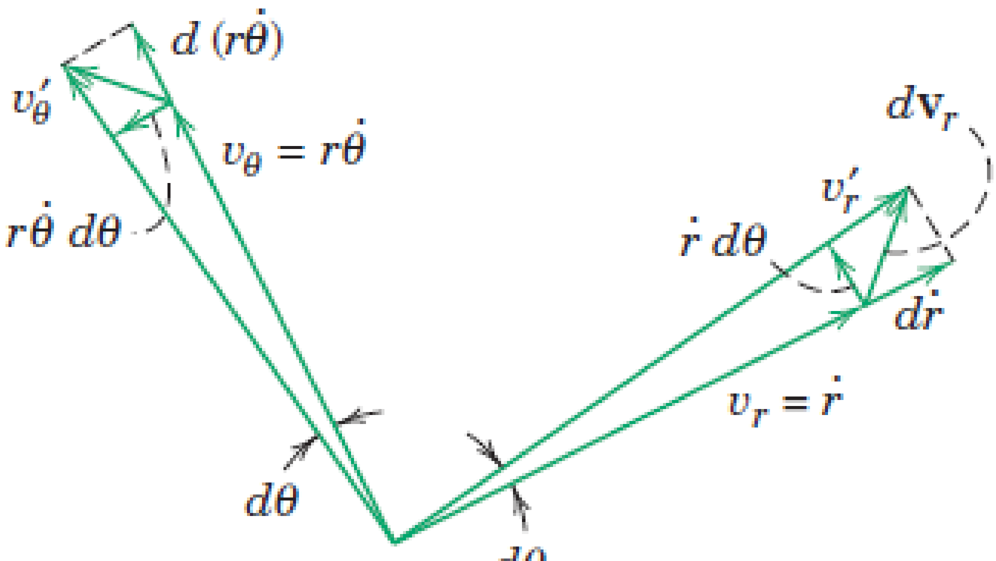
Acceleration contribution:

$\dfrac{\dot{r}d\theta}{dt} = \dot{r}\dot{\theta}$ in e_θ direction

Polar Coordinates (r-θ)

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = (\ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\boldsymbol{e}}_{\boldsymbol{r}}) + \left(\dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} + r\dot{\theta}\dot{\boldsymbol{e}}_{\boldsymbol{\theta}}\right)}$$

$$\boxed{\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{e_r} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + \dot{r}\dot{\theta}\boldsymbol{e_\theta} + r\ddot{\theta}\boldsymbol{e_\theta} - r\dot{\theta}\dot{\theta}\boldsymbol{e_r}}$$



(b)

Magnitude change of v_θ: $d(r\dot{\theta})$

Acceleration contribution:

$$\frac{d(r\dot{\theta})}{dt} = r\ddot{\theta} + \dot{r}\dot{\theta} \text{ in } e_\theta \text{ direction}$$

Direction change of v_θ: $r\dot{\theta}d\theta$

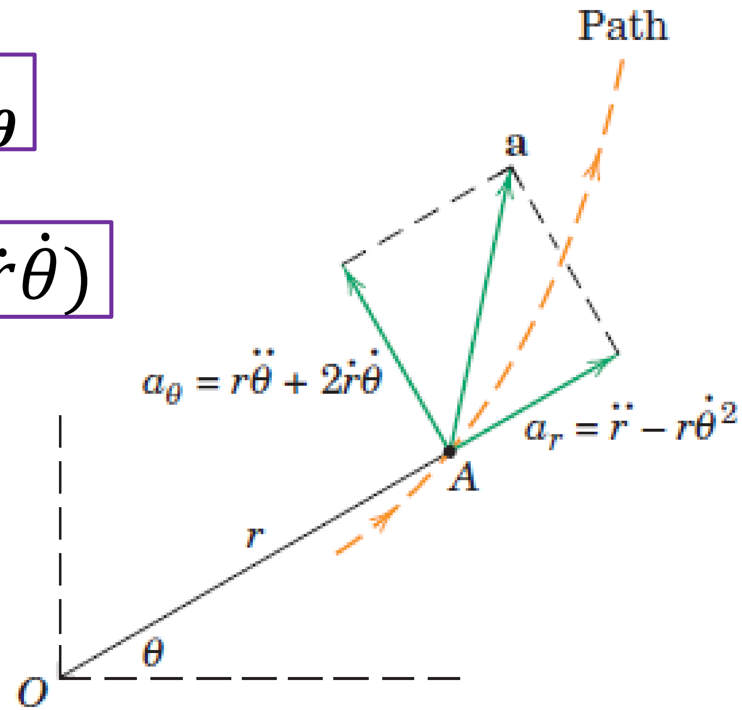
Acceleration contribution:

$$\frac{r\dot{\theta}d\theta}{dt} = r\dot{\theta}\dot{\theta} \text{ in } -e_r \text{ direction}$$

Polar Coordinates (r-θ)

$$\boxed{\boldsymbol{a} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\boldsymbol{e_\theta}}$$

$$\boxed{a_r = (\ddot{r} - r\dot{\theta}^2)} \qquad \boxed{a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})}$$



- $\ddot{r}$: acceleration in r-direction if θ is constant
- $-r\dot{\theta}^2$: acceleration in r-direction if r is constant as in circular motion
- $r\ddot{\theta}$: acceleration in θ-direction if r is constant as in circular motion
- $2\dot{r}\dot{\theta}$: acceleration in θ-direction due to change in direction of v_r (if r is not constant) and also due to change in magnitude of v_θ

Polar Coordinates (r-θ)

- **Circular Motion**

$$\boxed{\boldsymbol{v} = \dot{r}\boldsymbol{e_r} + r\dot{\theta}\boldsymbol{e_\theta}}$$

$$\boxed{\boldsymbol{a} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\boldsymbol{e_\theta}}$$

$$\boldsymbol{v} = \dot{r}\boldsymbol{e_r} + r\dot{\theta}\boldsymbol{e_\theta}$$

$$v_r = \dot{r} = 0$$

$$v_\theta = r\dot{\theta}$$

$$\boxed{\boldsymbol{a} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\boldsymbol{e_\theta}}$$

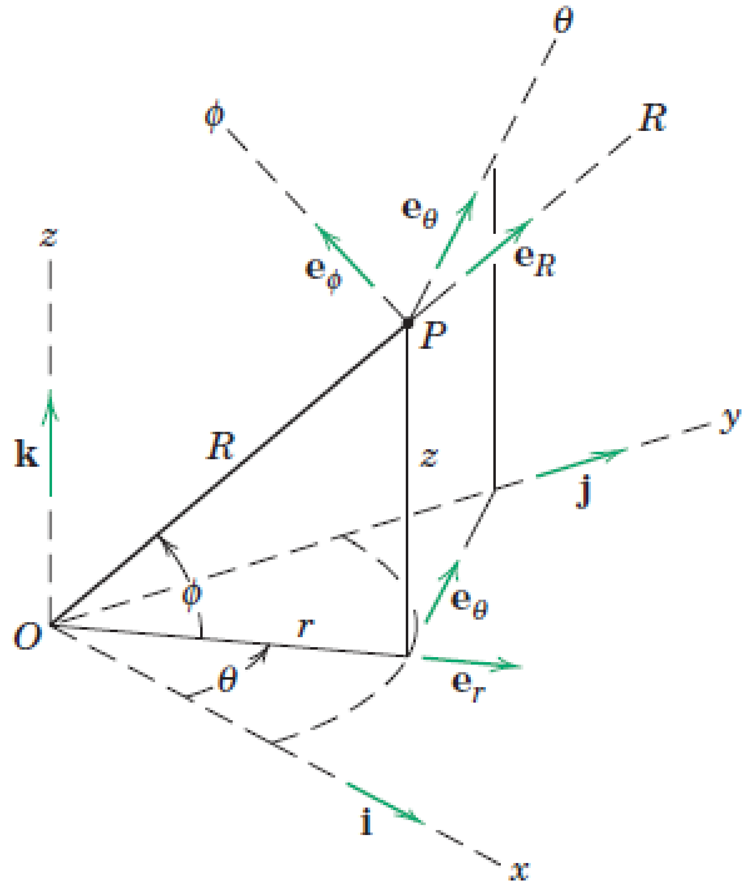
$$\boxed{\ddot{r} = 0 \qquad \dot{r} = 0}$$

$$\boxed{\boldsymbol{a} = (-r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta})\boldsymbol{e_\theta}}$$

$$\boxed{a_\theta = r\ddot{\theta}} \qquad \boxed{a_r = -r\dot{\theta}^2}$$

Space Curvilinear Motion

- Rectangular Coordinates (x-y-z)
- Cylindrical Coordinates (r-θ-z)
- Spherical Coordinates (r-θ-ϕ)



Space Curvilinear Motion

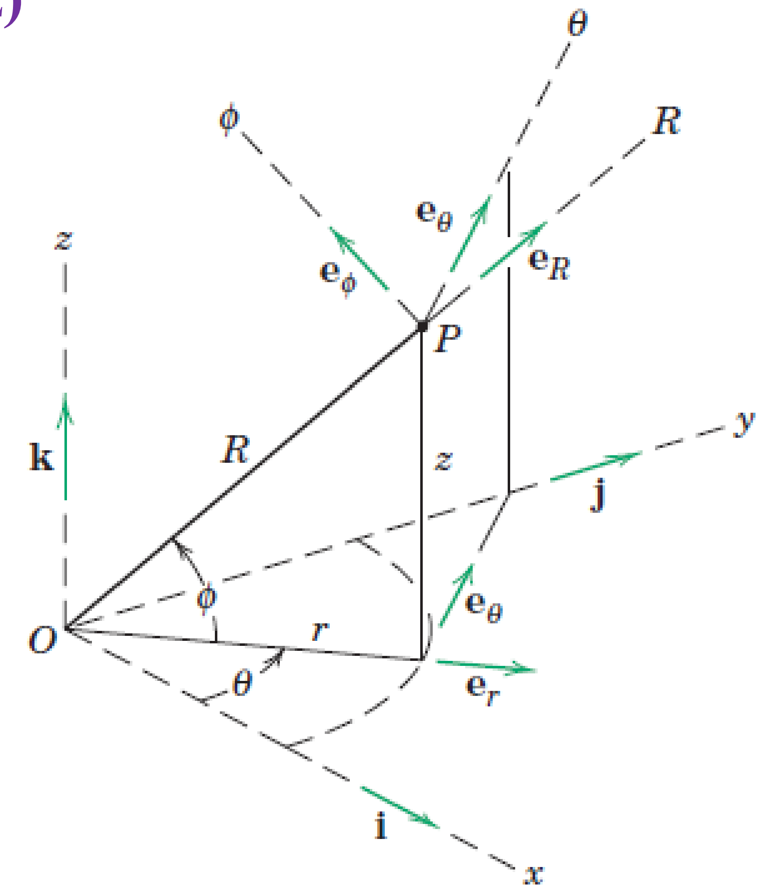
- **Rectangular Coordinates (x-y-z)**

$$R = xi + yj + zk$$

$$v = \dot{R} = \dot{x}i + \dot{y}j + \dot{z}k$$

$$a = \dot{v} = \ddot{R} = \ddot{x}i + \ddot{y}j + \ddot{z}k$$

R* for position vector in place of *r



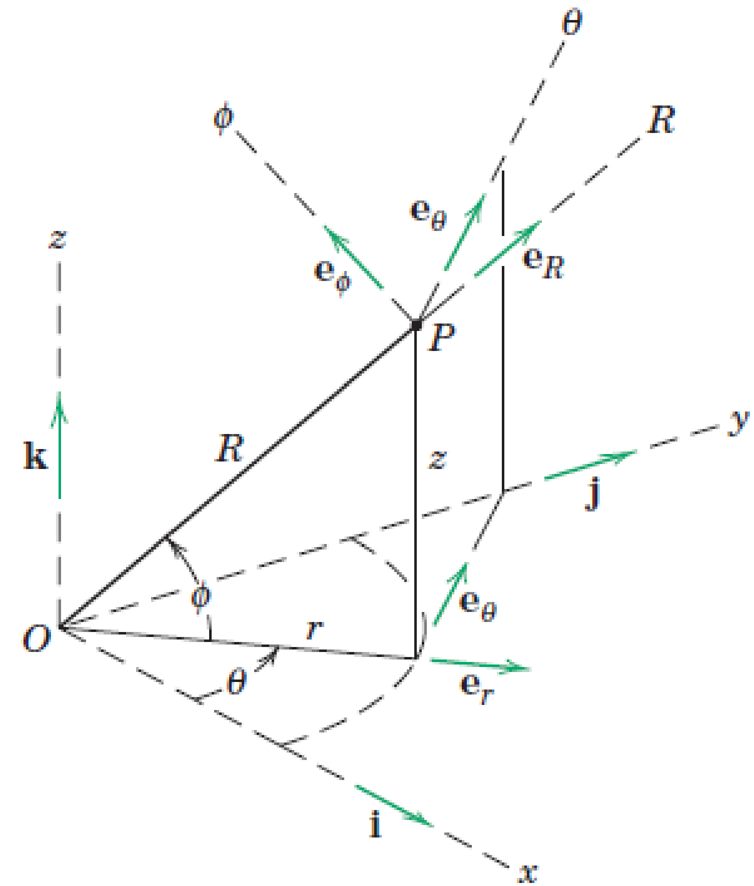
Space Curvilinear Motion

- **Cylindrical Coordinates (r-θ-z)**

$$R = \mathrm{r}e_r + zk$$

$$v = \dot{r}e_r + r\dot{\theta}e_\theta + \dot{z}k$$

$$a = \left(\ddot{r} - r\dot{\theta}^2\right)e_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)e_\theta + \ddot{z}k$$



$$v_r = \dot{r} \qquad a_r = \left(\ddot{r} - r\dot{\theta}^2\right)$$

$$v_\theta = r\dot{\theta} \qquad a_\theta = \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)$$

$$v_z = \dot{z} \qquad a_z = \ddot{z}$$

$$v^2 = {v_r}^2 + {v_\theta}^2 + {v_z}^2 \qquad a^2 = {a_r}^2 + {a_\theta}^2 + {a_z}^2$$

e_r and e_θ have nonzero time derivatives, e_z has zero time derivative

Space Curvilinear Motion

- **Spherical Coordinates (R-θ-ϕ)**

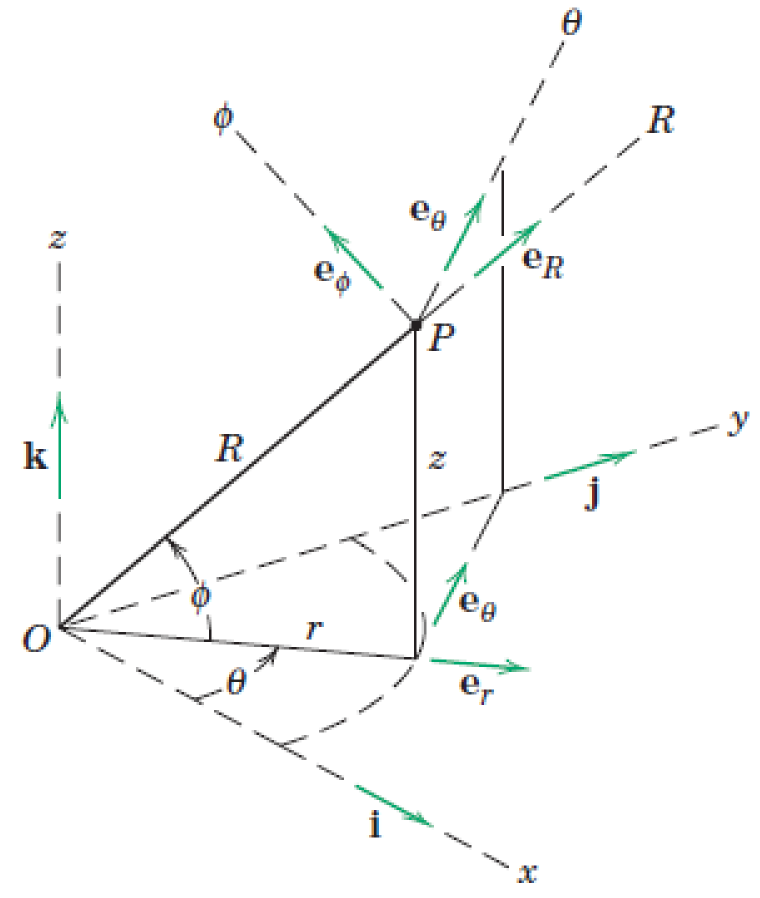
$$v = v_R e_R + v_\theta e_\theta + v_\varphi e_\varphi$$

$$v_R = \dot{R}$$

$$v_\theta = R\dot{\theta}\cos\varphi$$

$$v_\varphi = R\dot{\varphi}$$

$$v^2 = {v_R}^2 + {v_\theta}^2 + {v_\varphi}^2$$



Space Curvilinear Motion

- **Spherical Coordinates (R-θ-ϕ)**

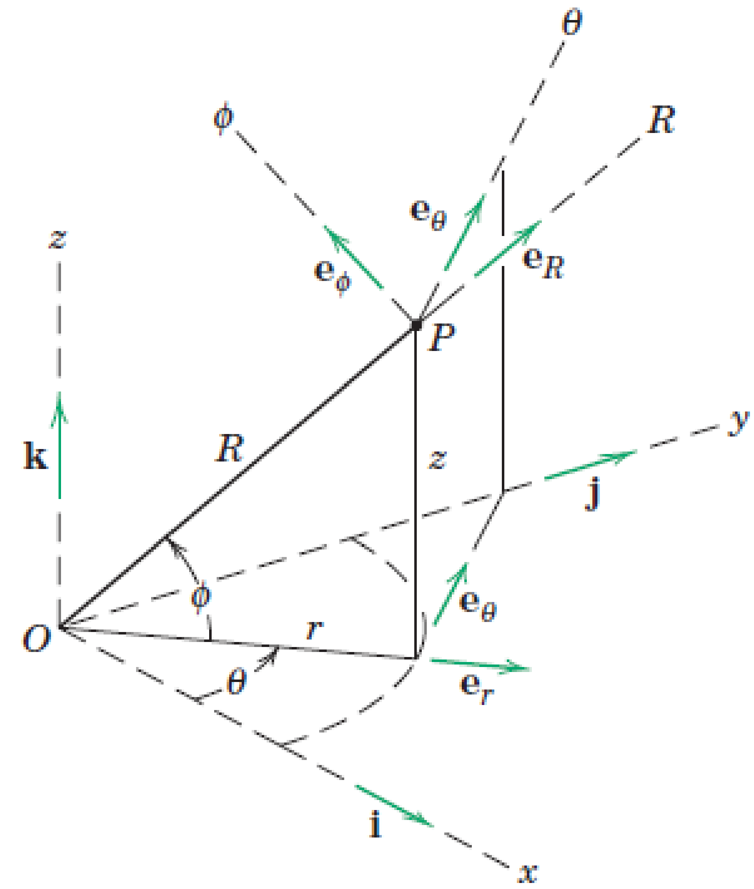
$$a = a_R e_R + a_\theta e_\theta + a_\varphi e_\varphi$$

$$a_R = \ddot{R} - R\dot{\varphi}^2 - R\dot{\theta}^2\,(\cos\varphi)^2$$

$$a_\theta = \frac{\cos\varphi}{R}\frac{d(R^2\dot{\theta})}{dt} - 2R\dot{\theta}\dot{\varphi}\sin\varphi$$

$$a_\varphi = \frac{1}{R}\frac{d(R^2\dot{\varphi})}{dt} + R\dot{\theta}^2\sin\varphi\cos\varphi$$

$$a^2 = {a_r}^2 + {a_\theta}^2 + {a_z}^2$$



Space Curvilinear Motion

- **Which coordinate system should be used?**
  - **Rectangular Coordinates (x-y-z)**
  - **Cylindrical Coordinates (r-θ-z)**
  - **Spherical Coordinates (r-θ-ϕ)**

A portion of the track of this amusement-park ride is in the shape of a helix whose axis is horizontal.

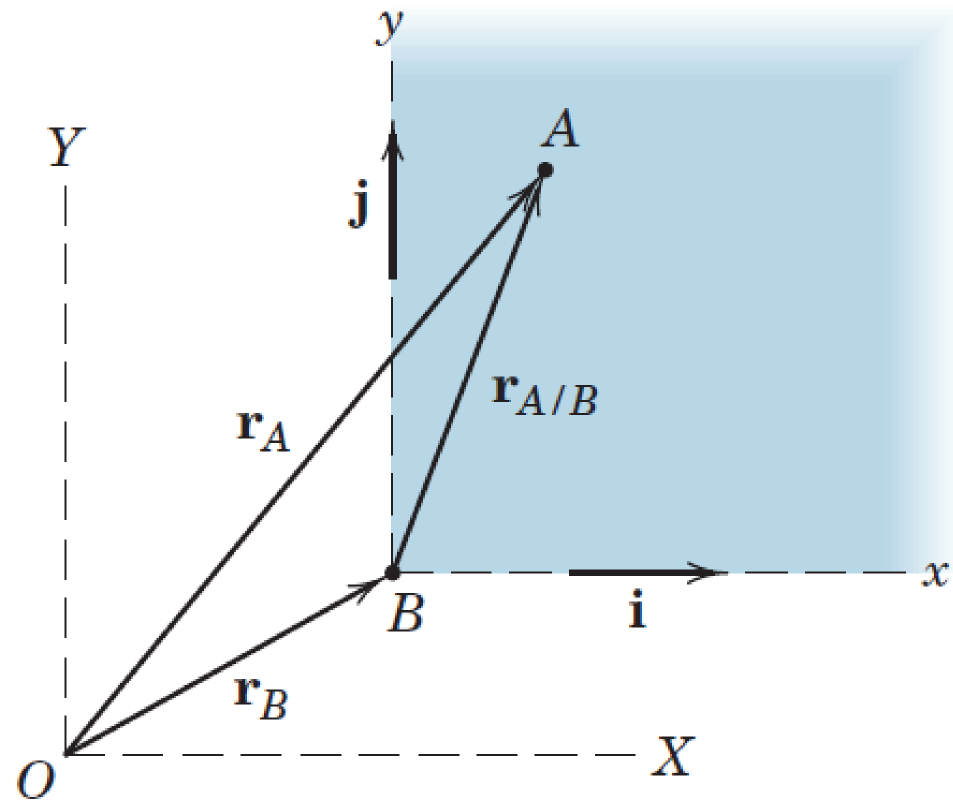
Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

Position vector of *A* as measured relative to the frame *x*-*y*

$$\boldsymbol{r}_{\boldsymbol{A/B}} = xi + yj$$

"***A*/*B***" means "***A* relative to *B***" or "***A* with respect to *B***"



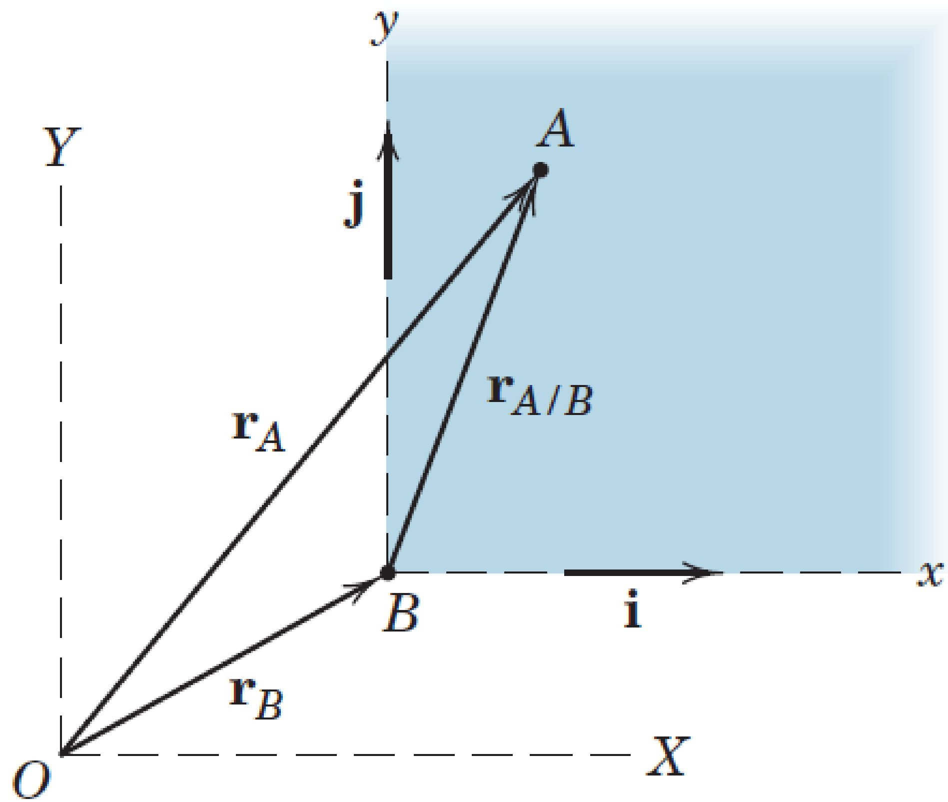
Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

$$r_{A/B} = xi + yj$$
$$r_A = r_B + r_{A/B}$$

$$\dot{r}_A = \dot{r}_B + \dot{r}_{A/B}$$
$$v_A = v_B + v_{A/B}$$

$$\ddot{r}_A = \ddot{r}_B + \ddot{r}_{A/B}$$
$$a_A = a_B + a_{A/B}$$



THANK YOU

Dynamics of Particles

Dr. Amrita Puri
Assistant Professor
Mechanical Department

Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Kinetics of Particles

- **Kinetics** is the **study of the relations between unbalanced forces and the resulting changes in motion**.

- Three Approaches to solve Kinetics problems:
  - Direct application of Newton's second law (called the force-mass acceleration method)
  - Work and energy principles
  - Impulse and momentum methods

Primary Inertial System

Primary inertial system (astronomical frame of reference):

Imaginary set of reference axes which are assumed to have no translation or rotation in space.

Newton's Second Law

Experiment performed in primary inertial system

1. A particle of mass, m is subjected to Force F_1, acceleration a_1 is measured.

2. The particle is subjected to Force F_2, acceleration a_2 is measured.

Experiment repeated multiple times

$$\frac{F_1}{a_1} = \frac{F_2}{a_2} = \frac{F_3}{a_3} = \quad \dots\dots\dots = C$$

C is a constant and depends upon inertia property of the system.

$$C = km \qquad\qquad F = kma$$

Newton's second law

$$F = kma$$

Acceleration is always in the direction of the applied force.

Actual experiment cannot be performed in the ideal manner, the same conclusions have been drawn from countless accurately performed experiments

One of the most accurate checks is given by the precise prediction of the motions of planets based on $F=kma$

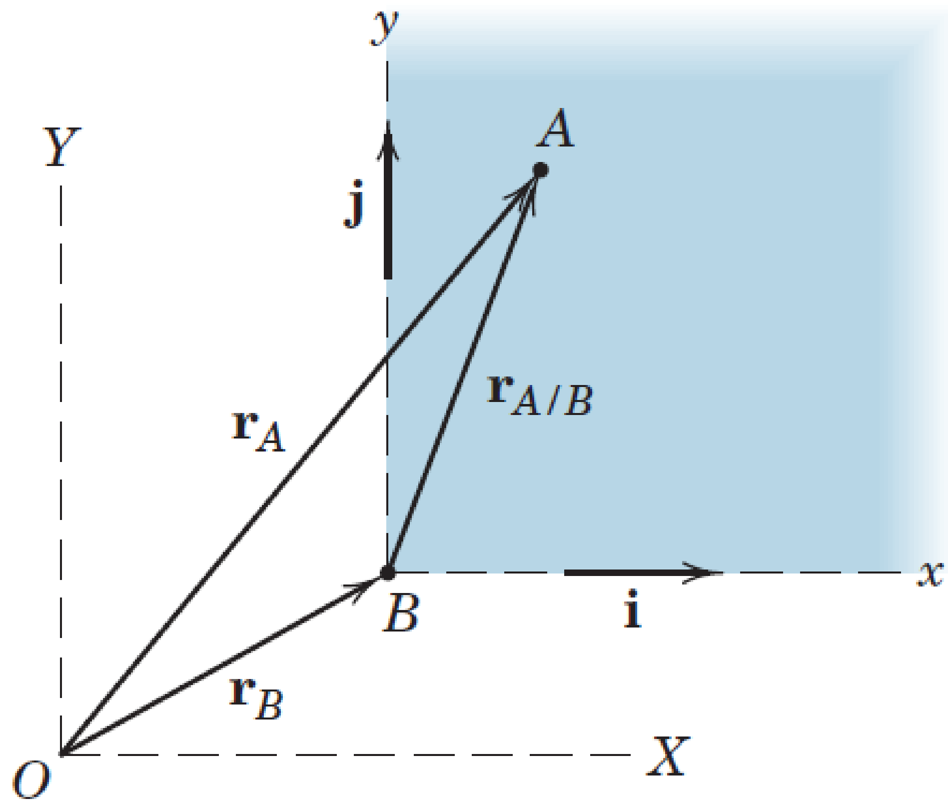
Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

Position vector of *A* as measured relative to the frame *x*-*y*

$$\boldsymbol{r}_{\boldsymbol{A/B}} = xi + yj$$

"***A*/*B***" means "***A* relative to *B***" or "***A* with respect to *B***"



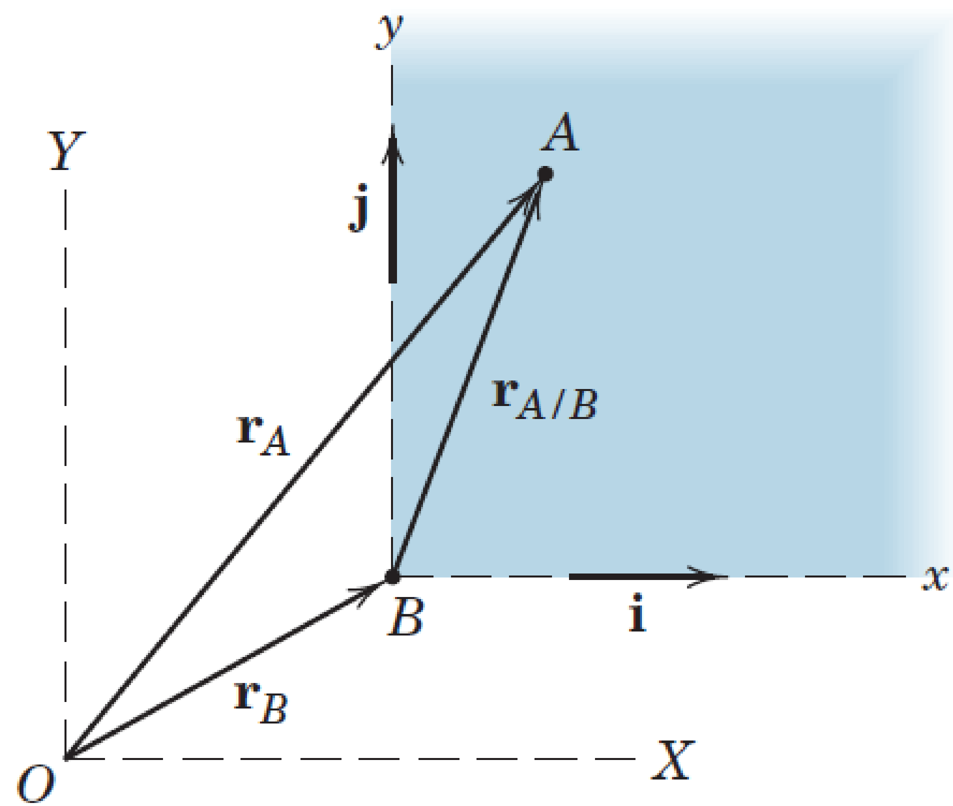
Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

$$r_{A/B} = xi + yj$$
$$r_A = r_B + r_{A/B}$$

$$\dot{r}_A = \dot{r}_B + \dot{r}_{A/B}$$
$$v_A = v_B + v_{A/B}$$

$$\ddot{r}_A = \ddot{r}_B + \ddot{r}_{A/B}$$
$$a_A = a_B + a_{A/B}$$



Inertial System

- Results of the ideal experiment are obtained for measurements made relative to the "fixed" primary inertial system,

  Equally valid for measurements made with respect to any nonrotating reference system which translates with a **constant velocity with respect to the primary system**.

- Acceleration measured in a system translating with no acceleration is the same as that measured in the primary system.

- **Newton's second law holds equally well in a nonaccelerating system, so that we may define an inertial system as any system in which *F=ma* is valid.**

Task 1

Write an article about ‘Newtonian Mechanics and Theory of relativity’.

System of Units

Kinetic System of Units

k is taken as unity

$$F = ma$$

In kinetic system of units, units for force, mass, and acceleration are not independent.

In SI system of units, units of force (Newton) is derived from units of mass (kg) and units of acceleration (m/sec^2).

Absolute system as units for force depends upon absolute value of mass.

System of Units

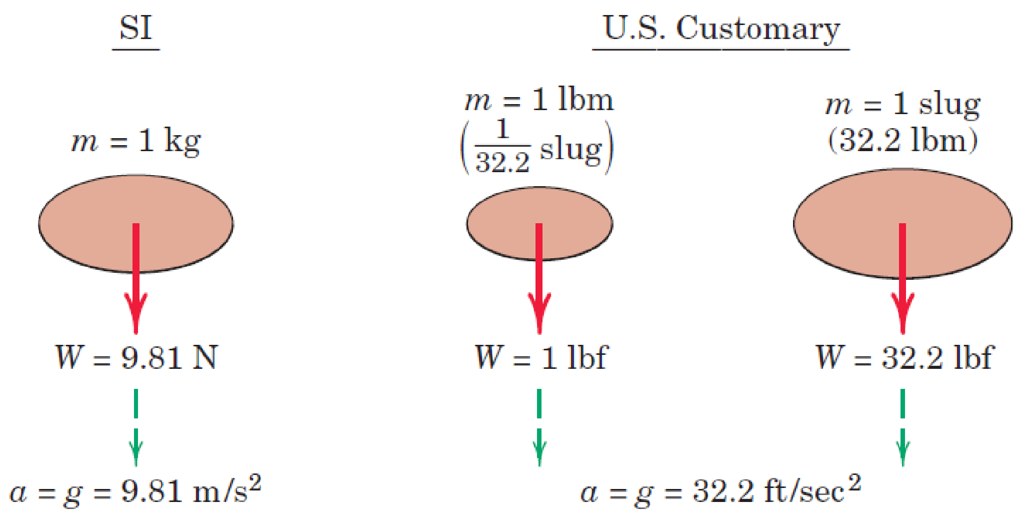
Gravitation system (U.S. Customary unit system)

Units of mass (slugs) is derived from units of force (pound-force, lb) divided by units of acceleration (feet per second squared, ft/sec^2).

Mass is derived from force as determined from gravitational attraction.

Force and Mass Units

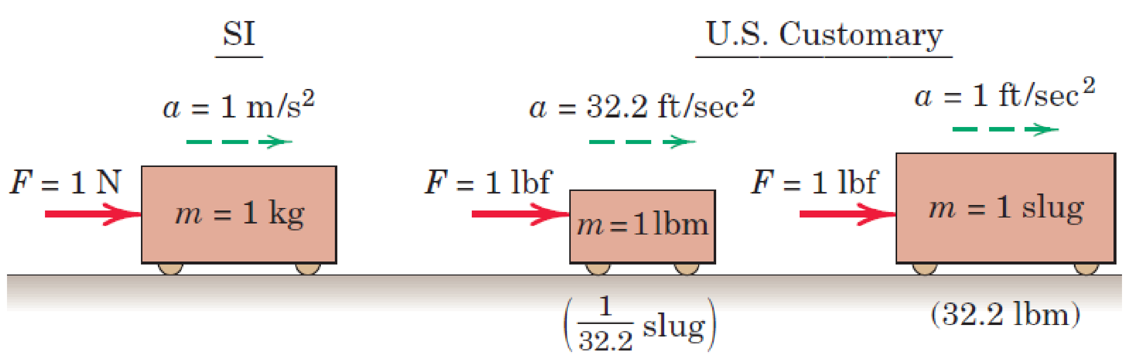
SI system of units	US customary system of units
Mass=1 kg, Weight=9.81 N, a=g=9.81 m/sec^2	Mass=1 lbm (1/32.2 slug), Weight=1 lbf, a=g=32.2 ft/sec^2
	Mass=1 slug (32.2 lbm), Weight=32.2 lbf, a=g=32.2 ft/sec^2



(a) Gravitational Free-Fall

Force and Mass Units

SI system of units	US customary system of units
Mass=1 kg, Force=1 N, a=g=1 m/sec^2	Mass=1 lbm (1/32.2 slug), Force=1 lbf, a=g=32.2 ft/sec^2
	Mass=1 slug (32.2 lbm) Force=1 lbf a=g=1 ft/sec^2



(b) Newton's Second Law

Equations of motion

A particle of mass *m* is subjected to the action of concurrent forces F_1, F_2, F_3 , . . . whose vector sum is $\sum F$

$$\boxed{\sum \boldsymbol{F} = \boldsymbol{ma}}$$

This equation or any of its component form is called **equation of motion**. This gives **instantaneous value of acceleration** for **instantaneous value of force** acting on it.

Unconstrained motion and constrained motion

- **Unconstrained motion:** The particle is free of mechanical guides and follows a path determined by its initial motion and by the forces which are applied to it from external sources.

  Example: Airplane or rocket in flight and an electron moving in a charged field
- **Constrained motion:** Path of the particle is partially or totally determined by restraining guides.

  Example: Train moving along its track and a collar sliding along a fixed shaft

Active Forces: External

Reactive Forces: Reaction forces due to constraining guides.

All the forces, applied and reactive, need to be considered to apply Newton's second law of motion.

Degrees of Freedom

- **Three degrees of freedom:** When the particle moves freely in 3-D space, 3 position coordinates are required to specify its position in space, 3 degrees of freedom.

  Example: Airplane or rocket in free flight
- **Two degrees of freedom:** Two coordinates are needed to specify its position in space.

  Example: Marble sliding on the surface of a bowl
- **One degree of freedom:** Only one coordinate is required to specify its position in space.

  Example: collar moving along a shaft.

Rectilinear motion

- Interested in motion of mass centre of the body
- Body can be assumed as a particle
- Forces can be treated as concurrent through a mass centre
- When motion of the particle can be chosen along one coordinate direction, say, x-direction

$$\sum F_x = m\, a_x$$

$$\sum F_y = 0$$

$$\sum F_z = 0$$

Rectilinear motion

- Motion of the particle can not be chosen along one coordinate direction,

$$\sum F_x = m\,a_x$$

$$\sum F_y = m\,a_y$$

$$\sum F_z = m\,a_z$$

$$a^2 = a_x^2 + a_y^2 + a_Z^2$$

$$F^2 = F_x^2 + F_y^2 + F_Z^2$$

Curvilinear motion

- **Rectangular Coordinates**

$$\sum \boldsymbol{F_x} = \boldsymbol{m a_x} \qquad \sum \boldsymbol{F_y} = \boldsymbol{m a_y}$$

$$\boldsymbol{a_x} = \ddot{\boldsymbol{x}} \qquad \boldsymbol{a_y} = \ddot{\boldsymbol{y}}$$

- **Normal-tangential Coordinates**

$$\sum \boldsymbol{F_n} = \boldsymbol{m a_n} \qquad \sum \boldsymbol{F_t} = \boldsymbol{m a_t}$$

$$\boldsymbol{a_n} = v\dot{\beta} = \rho\dot{\beta}^2 = \frac{v^2}{\rho} \qquad \boldsymbol{a_t} = \dot{v} = \ddot{s} \qquad v = \rho\dot{\beta}$$

- **Polar Coordinates**

$$\sum \boldsymbol{F_r} = \boldsymbol{m a_r} \qquad \sum \boldsymbol{F_\theta} = \boldsymbol{m a_\theta}$$

$$\boldsymbol{a_r} = \ddot{\boldsymbol{r}} - \boldsymbol{r}\dot{\boldsymbol{\theta}}^2 \qquad \boldsymbol{a_\theta} = \boldsymbol{r}\ddot{\boldsymbol{\theta}} + 2\dot{\boldsymbol{r}}\dot{\boldsymbol{\theta}}$$

Three Approaches

- **Newton's second law of motion**

$$\boldsymbol{F} = \boldsymbol{ma}$$

- **Integration with respect to displacement gives work-energy equation.**

- **Integration with respect to time gives impulse-momentum equation.**

Work

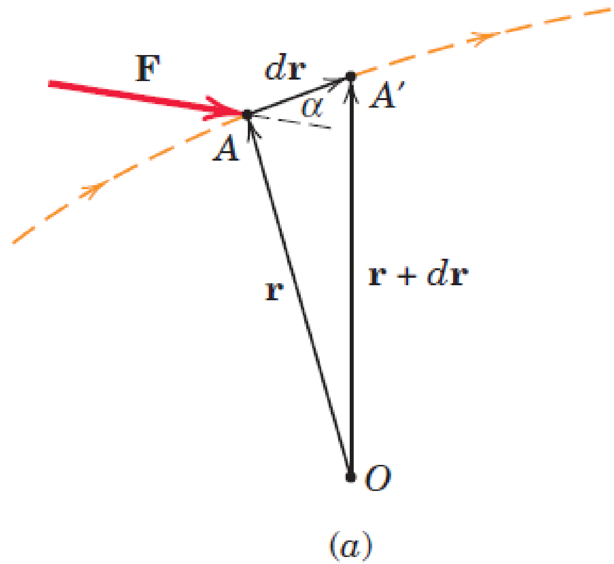
Work done by force during the displacement $\boldsymbol{dr}$

$$dU = \boldsymbol{F}.d\boldsymbol{r}$$

$$dU = F\ ds\ cos\alpha$$

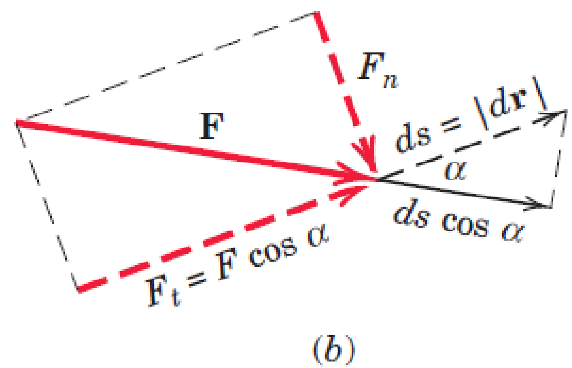
α: angle between $\boldsymbol{F}$ and $d\boldsymbol{r}$

ds is magnitude of $d\boldsymbol{r}$



Interpretations of $dU = F\ ds\ cos\alpha$

- Component of $\boldsymbol{F}$ in the direction of displacement $(Fcos\alpha)$multiplied by magnitude of displacement
- Component of displacement in the direction of force $(ds\ cos\alpha)$ multiplied by magnitude of force.



Work

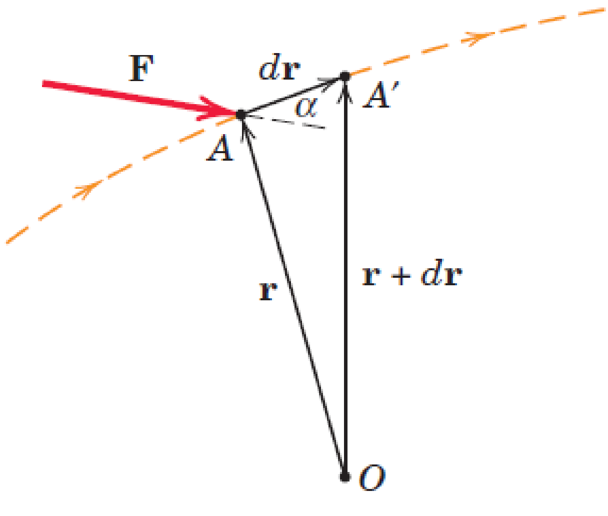
Normal component of the force $\boldsymbol{F_n}$ does no work.

Work done by force during the displacement $\boldsymbol{dr}$

$$dU = F_t \, ds$$

- Work is positive if F_t is in the direction of ds
- Work is negative if F_t is in the opposite of ds

- **Active Forces**: Forces which do work
- **Reactive Forces**: Constraint forces which do no work



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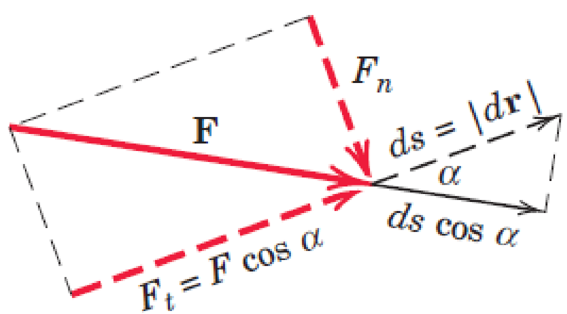


(b)

Units of Work

1 Joule:

work done by a force of 1 N acting through a distance of 1 m in the direction of the force

Units of **Moment or Torque: N-m**

Work is **scalar quantity**.

Moment or **Torque** is a **vector quantity**.

In U.S. Customary units

Work (foot-pounds, ft-lb)

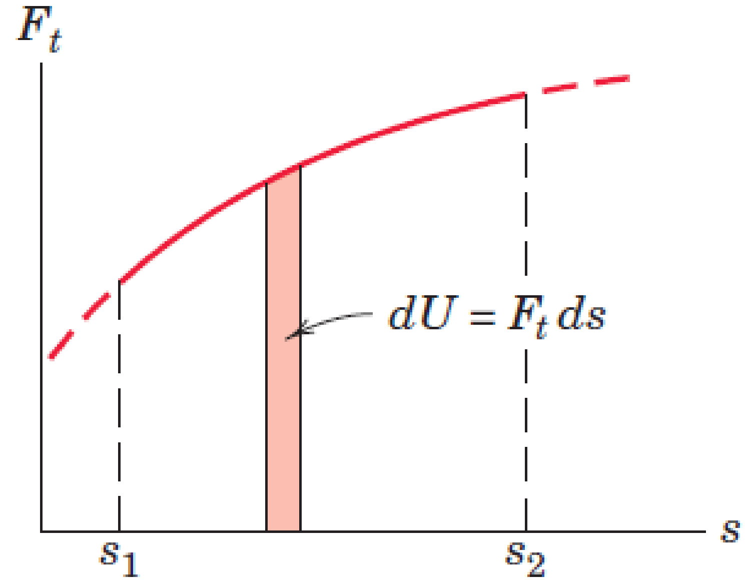
Moment (Pound-feet, lb-ft)

Calculation of Work

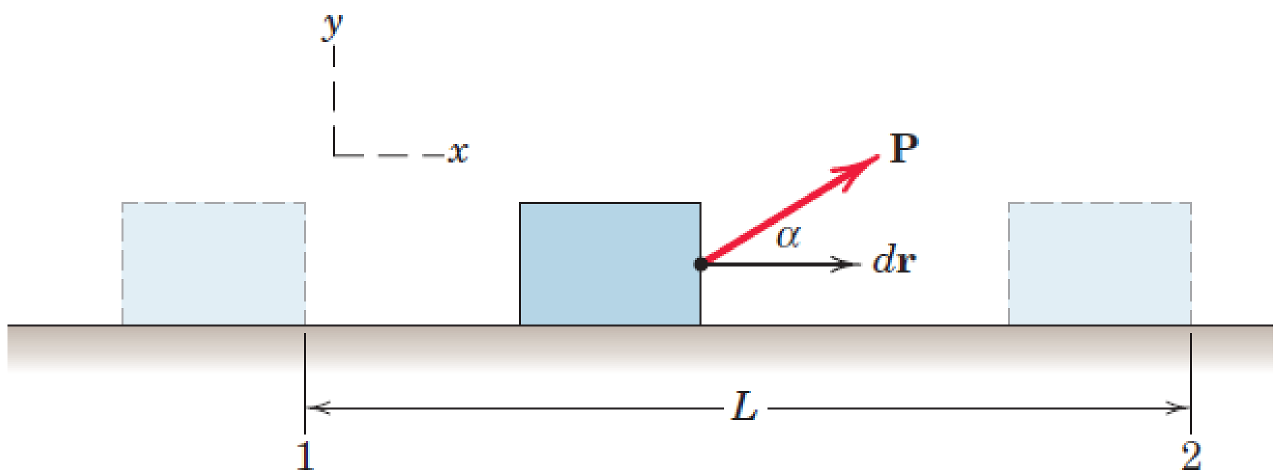
$$U = \int_1^2 \boldsymbol{F}.\boldsymbol{dr} = \int_1^2 \left(F_x\,dx + F_y\,dy + F_z\,dz\right)$$

$$U = \int_1^2 F_t.\,ds$$

Work done: Area under $F_t - s$ curve



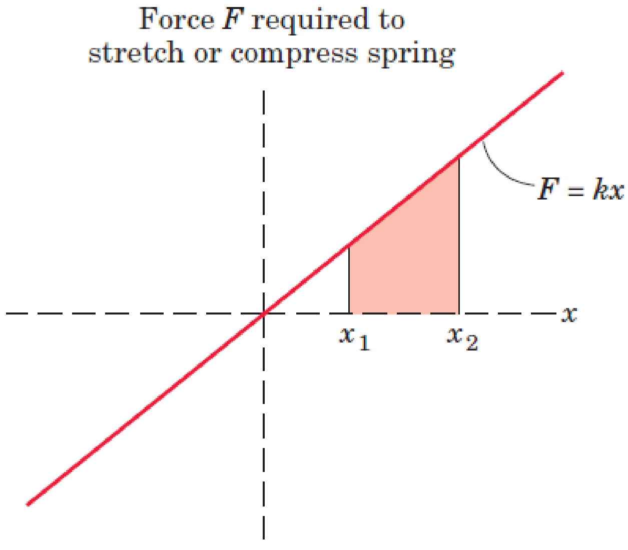
Work Done by a constant force



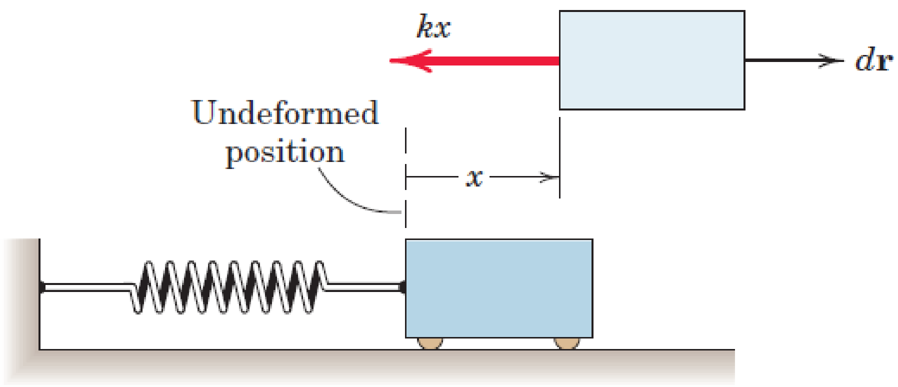
$$U_{1-2} = \int_1^2 \boldsymbol{F}.d\boldsymbol{r} = \int_1^2 [(Pcos\ \alpha)\mathbf{i} + (Psin\alpha)\mathbf{j}].dx\ \mathbf{i}$$

$$U_{1-2} = \int_1^2 (Pcos\ \alpha)dx\ = (Pcos\ \alpha)\ (x_2 - x_1) = (Pcos\ \alpha)L$$

Work Done by a spring force



(a)



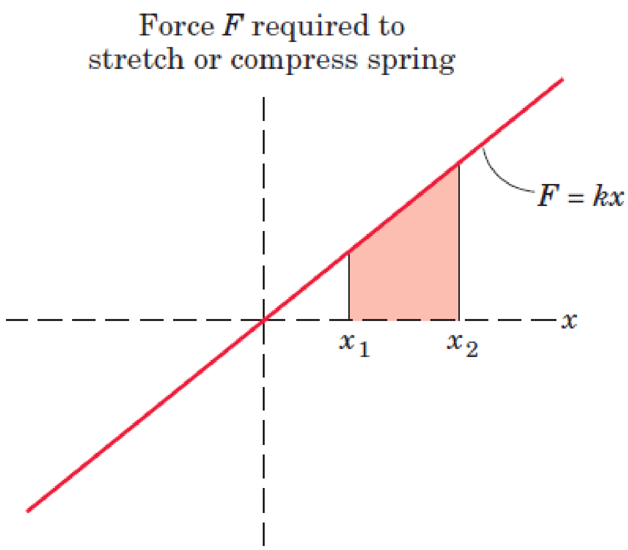
(b)

- $F = kx$ is a **linear static relationship**.

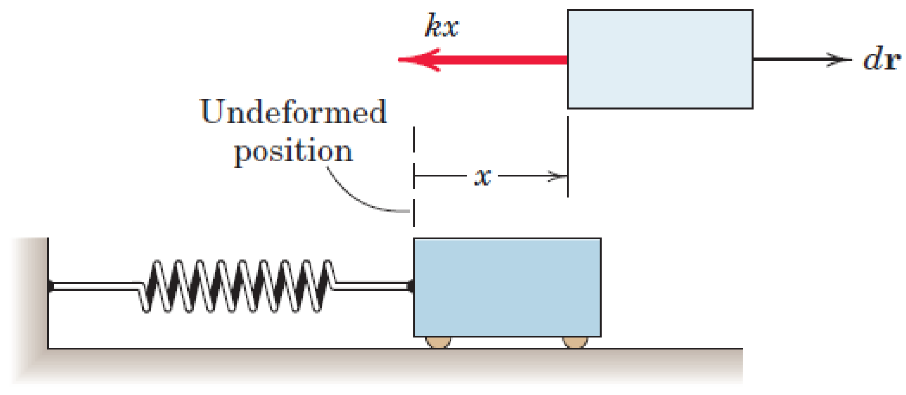
- This is valid assumption if **mass of spring is negligible as compared to mass of other parts** in the system.

- Magnitude of work is equal to area under *F*-*x* curve.

Work Done by a spring force



(a)

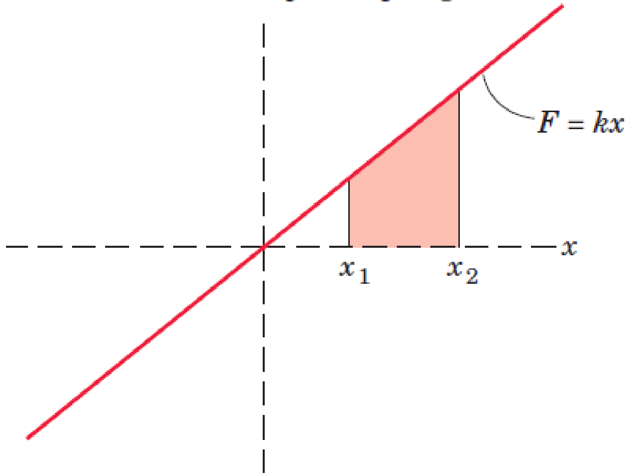


(b)

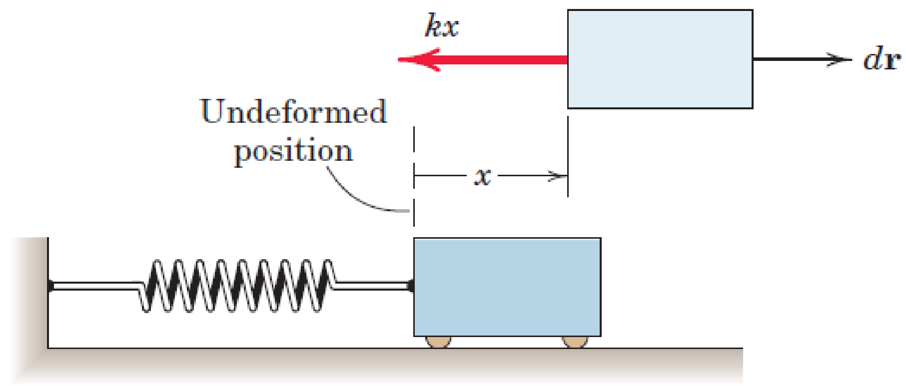
Linear spring of stiffness k : Force required to stretch or compress the spring is proportional to the deformation x

$$\boxed{U_{1-2} = \int_1^2 F.dr = \int_1^2 (-kxi).dxi = \frac{1}{2}k({x_1}^2 - {x_2}^2)}$$

Work Done by a spring force



(a)



(b)

Work done is negative when body moves from undeformed position

Work done is positive when body moves towards undeformed position

$$U_{1-2} = \int_1^2 F.dr = \int_1^2 (-kxi).dxi = \frac{1}{2}k({x_1}^2 - {x_2}^2)$$

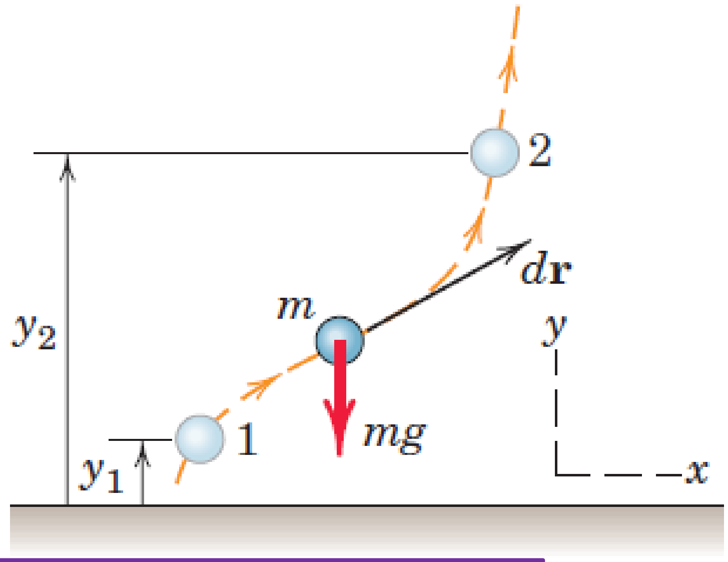
Work Done by Weight

Case (a) Constant Gravity

Altitude variation is sufficiently small.

If body rises, work done is negative.

If body falls, work done is positive.



$$U_{1-2} = \int_1^2 \boldsymbol{F}.d\boldsymbol{r} = \int_1^2 (-mg\mathbf{i}).(dx\mathbf{i} + dy\mathbf{j})$$
$$= -mg\int_{y_1}^{y_2} dy = -mg(y_2 - y_1)$$

- Horizontal movement does not contribute to this work

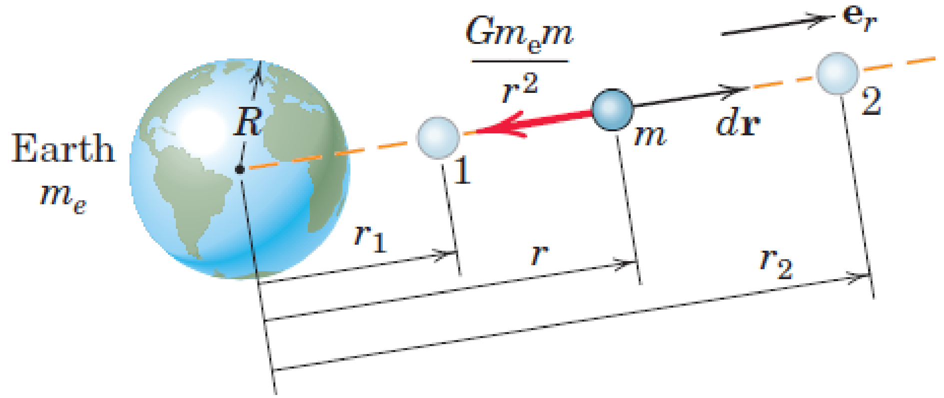
Work Done by Weight

Case (b) Gravity Not Constant

Altitude variation is NOT small.

If body rises, work done is negative.

If body falls, work done is positive.



$$U_{1-2} = \int_1^2 F \cdot dr = \int_1^2 \frac{-Gm_em}{r^2} \boldsymbol{e}_r \cdot dr\, \boldsymbol{e}_r$$

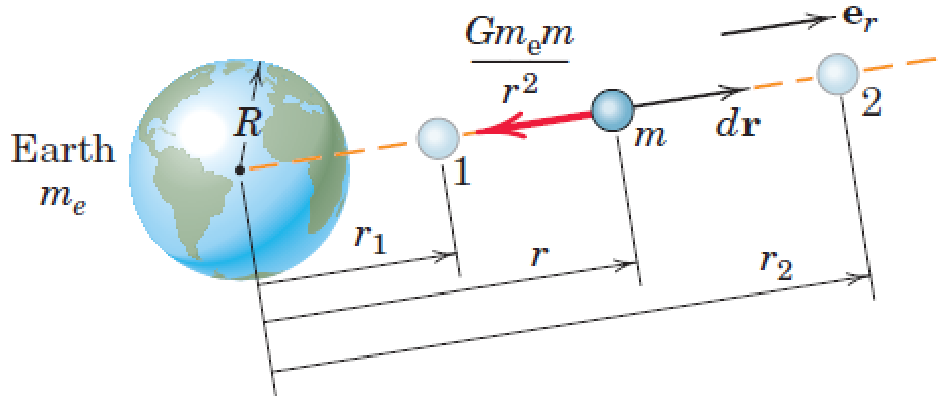
$$= \int_{r_1}^{r_2} \frac{-Gm_em}{r^2} dr \ = mGm_e\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = mgR^2\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

- Transverse movement (perpendicular to radial direction) does not contribute to work

Work Done by Weight

Case (b) Gravity Not Constant

- r represents a radial distance from the center of the earth and not an altitude $h = r - R$ above the surface of the earth



(b)

$$\boxed{U_{1-2} = mgR^2\left(\frac{1}{r_2} - \frac{1}{r_1}\right)}$$

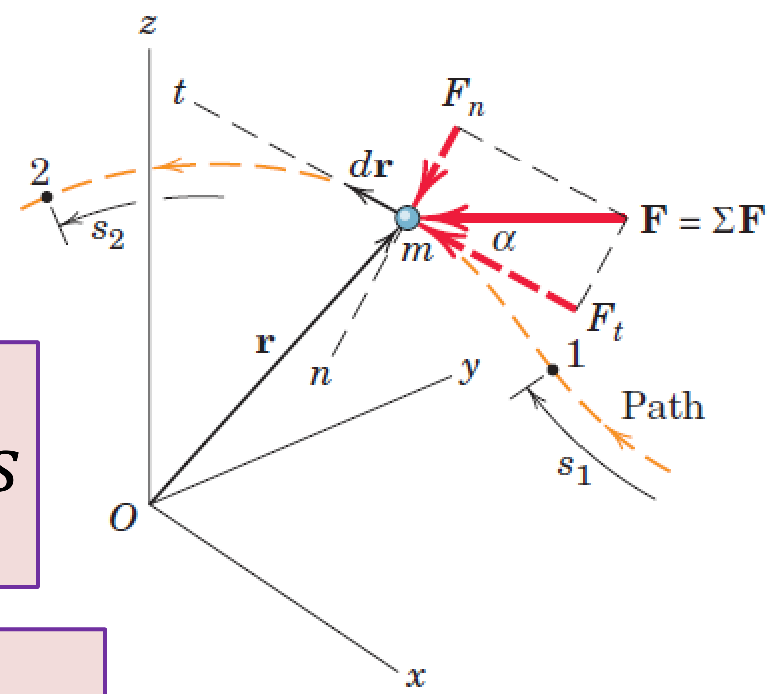
- Transverse movement (perpendicular to radial direction) does not contribute to work.

Work and Curvilinear motion

F resultant of all the forces

F acts on the particles of mass m

Particles moves from point 1 to point 2



$$U_{1-2} = \int_1^2 \boldsymbol{F} \cdot d\boldsymbol{r} = \int_{s_1}^{s_2} F_t ds$$

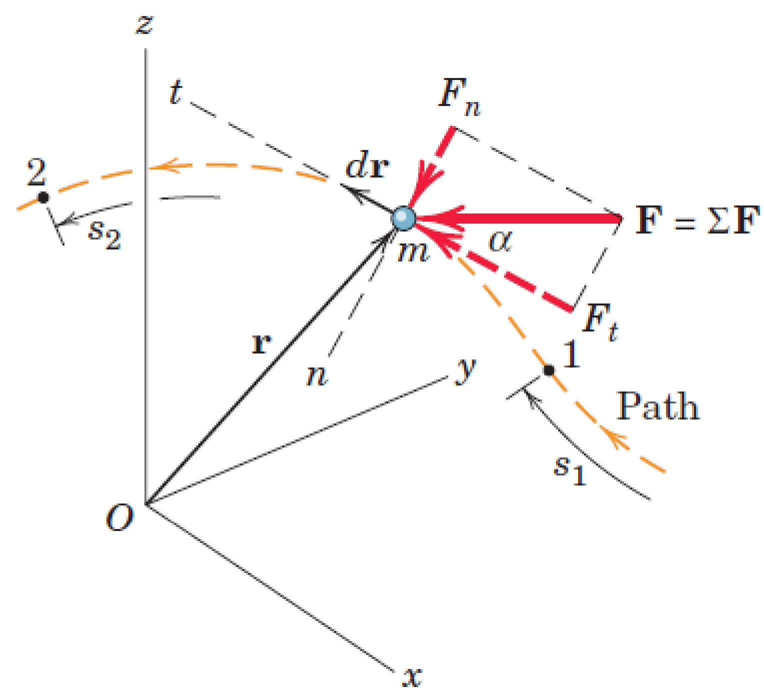
$$U_{1-2} = \int_1^2 \boldsymbol{F} . d\boldsymbol{r} = \int_1^2 m\boldsymbol{a} \cdot d\boldsymbol{r}$$

Work and Curvilinear motion

$$U_{1-2} = \int_1^2 \boldsymbol{F}.d\boldsymbol{r} = \int_1^2 m\boldsymbol{a} \cdot d\boldsymbol{r}$$

$$\boldsymbol{a} \cdot d\boldsymbol{r} = a_t ds$$

$$a_t ds = v dv$$



$$\boxed{U_{1-2} = \int_1^2 \boldsymbol{F} \cdot d\boldsymbol{r} = \int_{v_1}^{v_2} mv\, dv = \frac{1}{2}m({v_2}^2 - {v_1}^2)}$$

Kinetic Energy

$$U_{1-2} = \int_1^2 \boldsymbol{F} \cdot d\boldsymbol{r} = \int_{v_1}^{v_2} mv\, dv = \frac{1}{2} m({v_2}^2 - {v_1}^2)$$

Kinetic Energy

$$T = \frac{1}{2} m\, v^2$$

Total work which must be done on the particle to bring it from a state of rest to a velocity v.

Units: Joule (N-m)

Work Energy Equation

$$U_{1-2} = \int_{1}^{2} \boldsymbol{F} \cdot d\boldsymbol{r} = \int_{v_1}^{v_2} mv\, dv = \frac{1}{2} m({v_2}^2 - {v_1}^2)$$

$$U_{1-2} = \frac{1}{2} m({v_2}^2 - {v_1}^2) = T_2 - T_1 = \Delta T$$

Total work done by all forces acting on a particle as it moves from point 1 to point 2 equals the corresponding **change in kinetic energy** of the particle

$$T_1 + U_{1-2} = T_2$$

Advantages of Work Energy Equation

$$\boxed{U_{1-2} = \frac{1}{2}m({v_2}^2 - {v_1}^2) = T_2 - T_1 = \Delta T}$$

$$\boxed{T_1 + U_{1-2} = T_2}$$

- **Avoid the necessity of computing the acceleration** and leads directly to the velocity changes as functions of the forces which do work.
- Only active forces which do work are considered.

Advantages of Work Energy Equation

$$\boxed{U_{1-2} = \frac{1}{2}m({v_2}^2 - {v_1}^2) = T_2 - T_1 = \Delta T}$$

$$\boxed{T_1 + U_{1-2} = T_2}$$

- A system of two particles joined together by a connection which is frictionless and incapable of any deformation
- Internal forces equal and opposite, work done by these forces is equal and cancel out.
- **Only work done by external forces needs to be considered**.

Advantages of Work Energy Equation

$$\boxed{U_{1-2} = \frac{1}{2}m({v_2}^2 - {v_1}^2) = T_2 - T_1 = \Delta T}$$

$$\boxed{T_1 + U_{1-2} = T_2}$$

- U_{1-2} is the total or net work done on the system by forces external to it and ΔT , $T_2 - T_1$, is change in the total kinetic energy of the system.
- The total kinetic energy is the sum of the kinetic energies of both elements of the system.
- It enables us to **analyze a system of particles** joined in the manner described without **dismembering the system**.

Power

Power: Time rate of doing work

Power P developed by a force F which does an amount of work U is,

$$\boxed{P = \frac{dU}{dt} = \frac{\boldsymbol{F}.d\boldsymbol{r}}{dt} = \boldsymbol{F}.\boldsymbol{v}}$$

$\frac{d\boldsymbol{r}}{dt}$ is the velocity $\boldsymbol{v}$ of the point of application of the force.

Units of Power

Units of Power: N-m/sec =Joule/sec=Watt

1 Watt equals **one joule per second** (J/s).

In U.S. customary units,

Unit for mechanical power is ***horsepower*** **(hp).**

1 Watt = 1 J/s

1 horsepower = 550 ft-lb/sec = 33000 ft-lb/min

1 horsepower = 746 W = 0.746 kW

Efficiency

Mechanical efficiency $\boldsymbol{e_m}$ of the machine:

Ratio of the work done *by* a machine to the work done *on* the machine during the same time interval

$$e_m = \frac{P_{output}}{P_{input}}$$

$$e = e_m e_t e_e$$

e: Overall efficiency

e_t: thermal efficiency

e_t: electrical efficiency

Gravitational Potential Energy

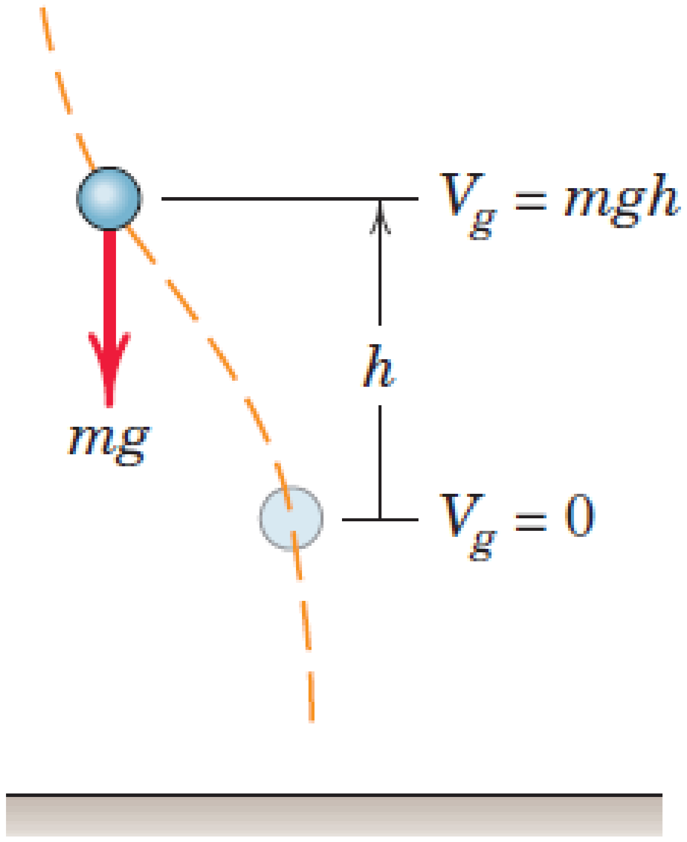
Gravitational potential energy, V_g, of the particle:

Work mgh done against the gravitational field to elevate the particle a **distance h above some arbitrary reference plane** (called a datum), where V_g is taken to be zero

$$V_g = mgh$$

In going from one level at $h = h_1$ to a higher level at $h = h_2$, the *change* in potential energy becomes

$$\boxed{\Delta V_g = mg(h_2 - h_1) = mg\Delta h}$$



Gravitational Potential Energy

- In going from one level at $h = h_1$ to a higher level at $h = h_2$, the *change* in potential energy becomes

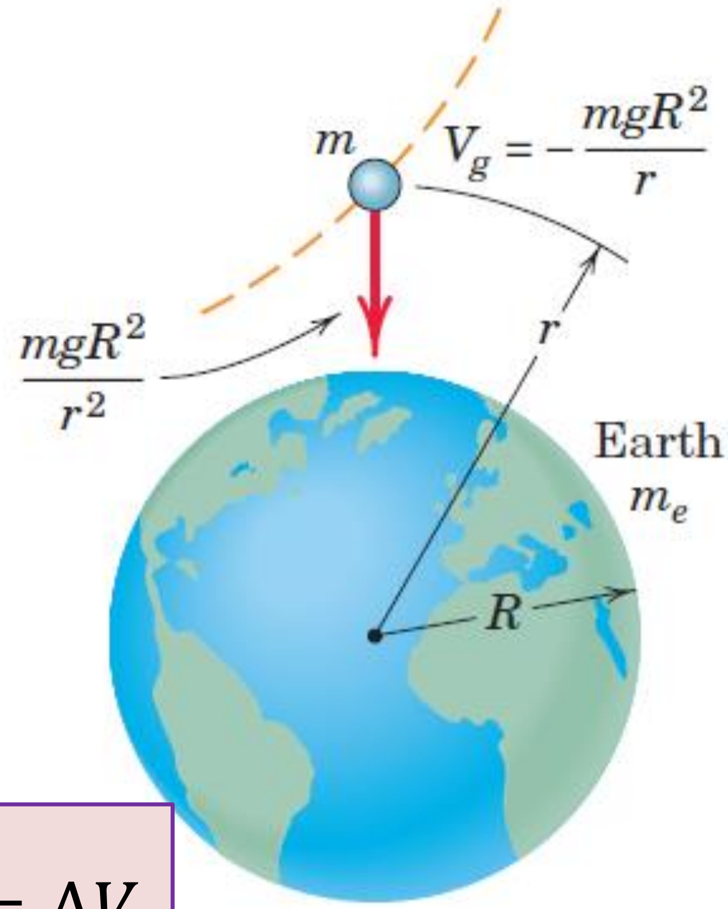
$$\Delta V_g = mg(h_2 - h_1) = mg\Delta h$$

- The corresponding **work done *by* the gravitational force on the particle** is $-mg\Delta h$.

- The **work done by the gravitational force** is the **negative of the change in potential energy**.

Gravitational Potential Energy

$$F = \frac{Gmm_e}{r^2} = \frac{mgR^2}{r^2}$$

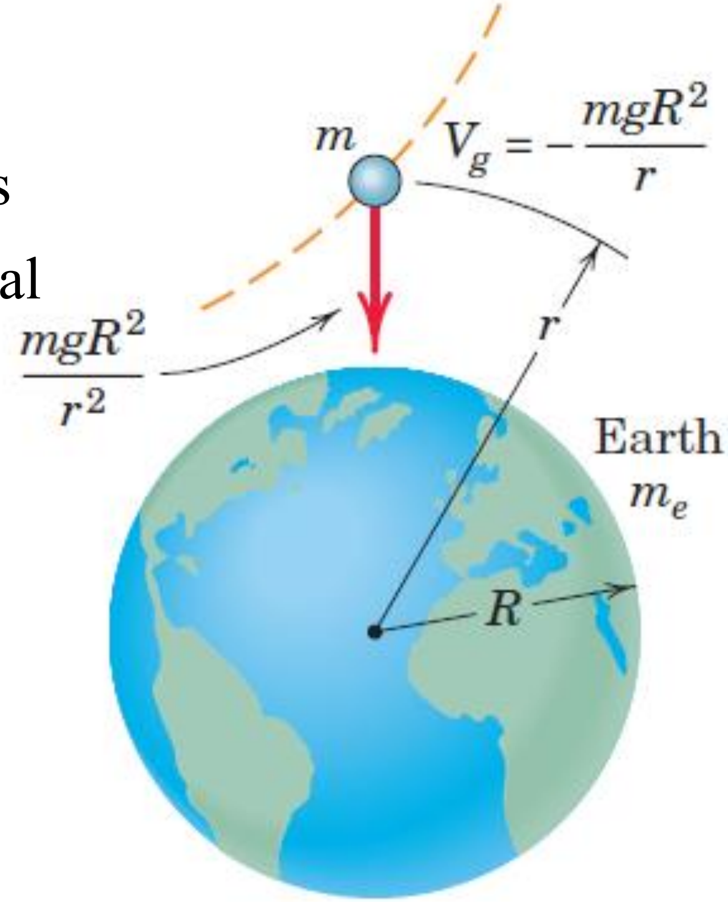


Work done *against* this force to change the radial position of the particle from r_1 to r_2 is the change $[(V_g)_2 - (V_g)_1]$ in gravitationa potential energy

$$\boxed{\int_{r_1}^{r_2} mgR^2 \frac{dr}{r^2} = mgR^2\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \Delta V_g}$$

Gravitational Potential Energy

Work done *against* this force to change the radial position of the particle from r_1 to r_2 is the change $[(V_g)_2 - (V_g)_1]$ in gravitational potential energy.



$$\Delta V_g = mgR^2\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

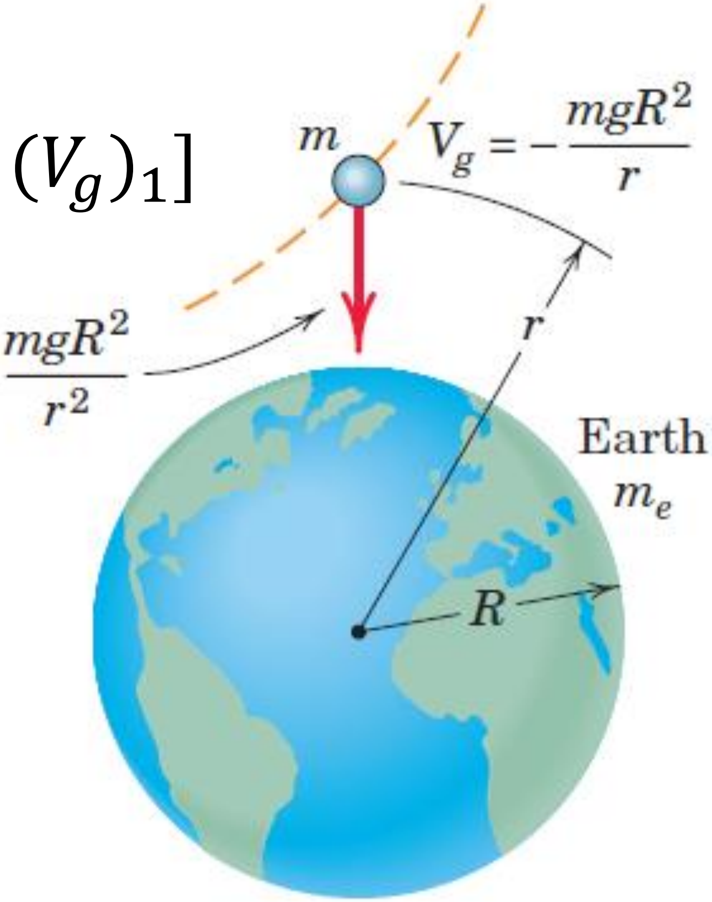
ΔV_g is the ***negative* of the work done *by* the gravitational force**.

Gravitational Potential Energy

$$\Delta V_g = mgR^2\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = [(V_g)_2 - (V_g)_1]$$

$(V_g)_2 = 0$ when r_2 tends to infinity,

$$V_g = -\left(\frac{mgR^2}{r}\right)$$

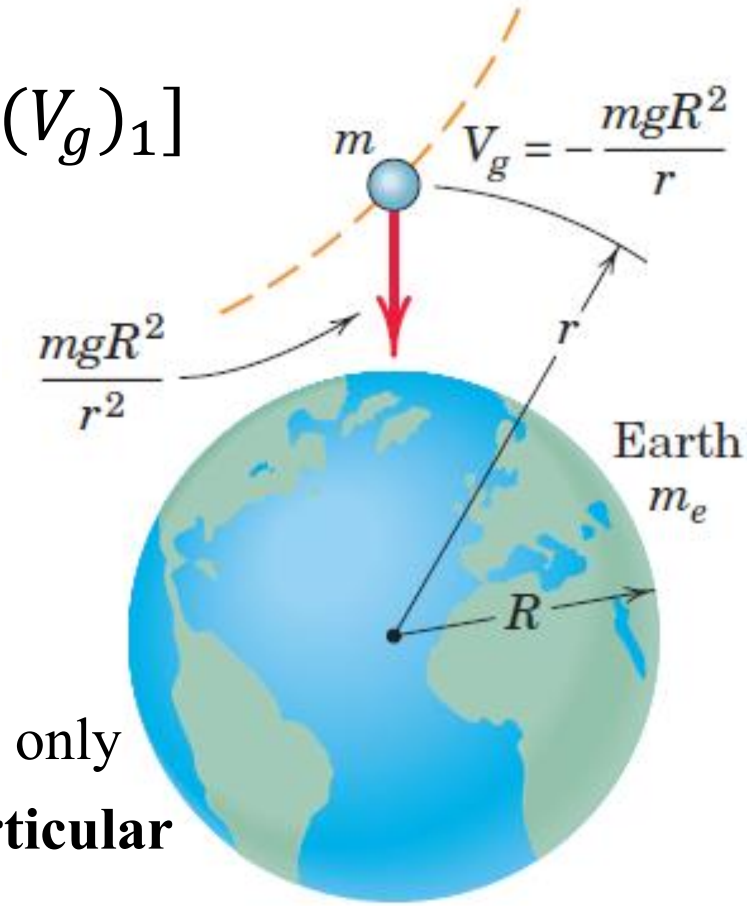


Gravitational Potential Energy

$$\Delta V_g = mgR^2\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = [(V_g)_2 - (V_g)_1]$$

$(V_g)_2 = 0$ when r_2 tends to infinity,

$$V_g = -\left(\frac{mgR^2}{r}\right)$$



Potential energy of a given particle depends only on **its position**, *h* or *r*, and **NOT on the particular path** it followed in reaching that position.

Elastic Potential Energy

- **Work done, on the spring to deform it**, is stored in the spring and is called its **elastic potential energy** $\boldsymbol{V_e}$
- During the release of the deformation of the spring, energy is recoverable in the form of work done by the spring on the body attached to its movable end

$$V_e = \int_0^x kx\,dx = \frac{1}{2}kx^2$$

$$\Delta V_e = \frac{1}{2}k({x_2}^2 - {x_1}^2)$$

Elastic Potential Energy

$$\Delta V_e = \frac{1}{2}k({x_2}^2 - {x_1}^2)$$

- **Deformation**, either tensile or compressive, of a spring **increases** from x_1 to x_2 during the motion, **change in potential energy of the spring is positive**

- **Deformation**, either tensile or compressive, of a spring **decreases** from x_1 to x_2 during the motion, **change in potential energy of the spring is negative**

Elastic Potential Energy

- **Force exerted on the spring by the moving body** is equal and opposite to the **force F exerted by the spring on the body**
- **Work done on the spring is the negative of the work done on the body**.

- **Work U done by the spring on the body**, the **negative of the potential energy change for the spring $-\Delta V_e$**

$$U = -\Delta V_e$$

Work-Energy Equation

$$\boxed{U_{1-2} = T_2 - T_1 = \Delta T}$$

$$U'_{1-2} + \left(-\Delta V_g\right) + \left(-\Delta V_e\right) = \Delta T$$

$$U'_{1-2} = \Delta T + \Delta V$$

$$\Delta V = \left(\Delta V_g\right) + \left(\Delta V_e\right)$$

ΔV_g depends upon end point positions of the particle
ΔV_e depends upon end point lengths of elastic spring

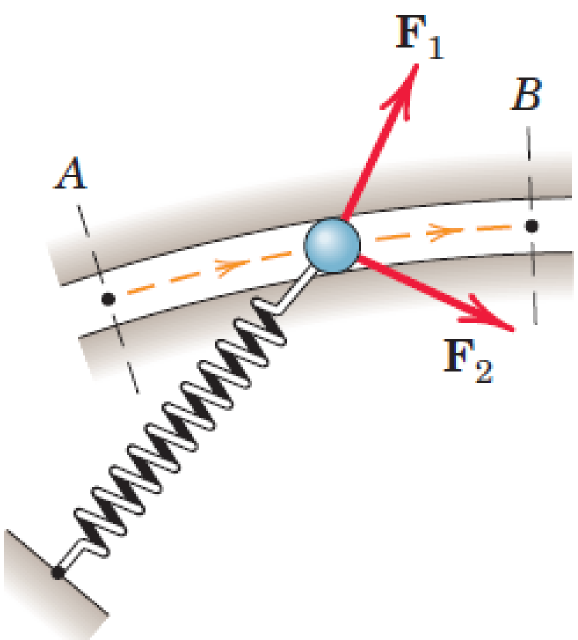
$$T_1 + V_1 + U'_{1-2} = T_2 + V_2$$

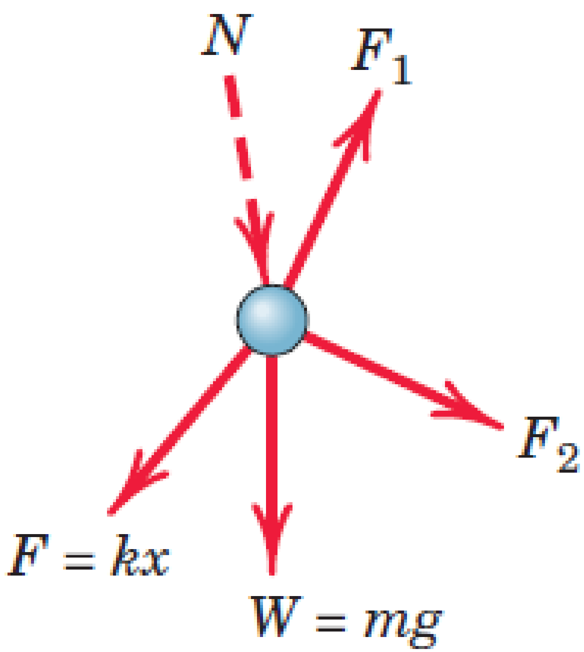
Work-Energy Equation

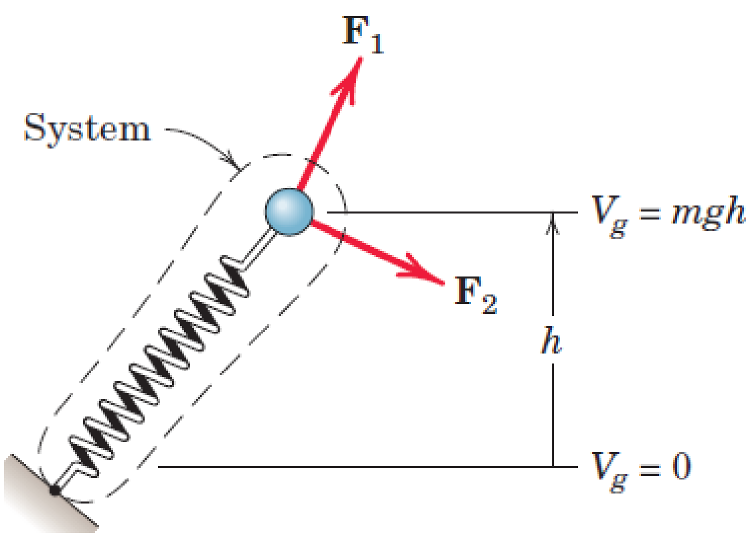
$$\boxed{U_{1-2} = T_2 - T_1 = \Delta T}$$

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

$$\boxed{U_{1-2} = \Delta T} \qquad \boxed{U'_{1-2} = \Delta T + \Delta V}$$

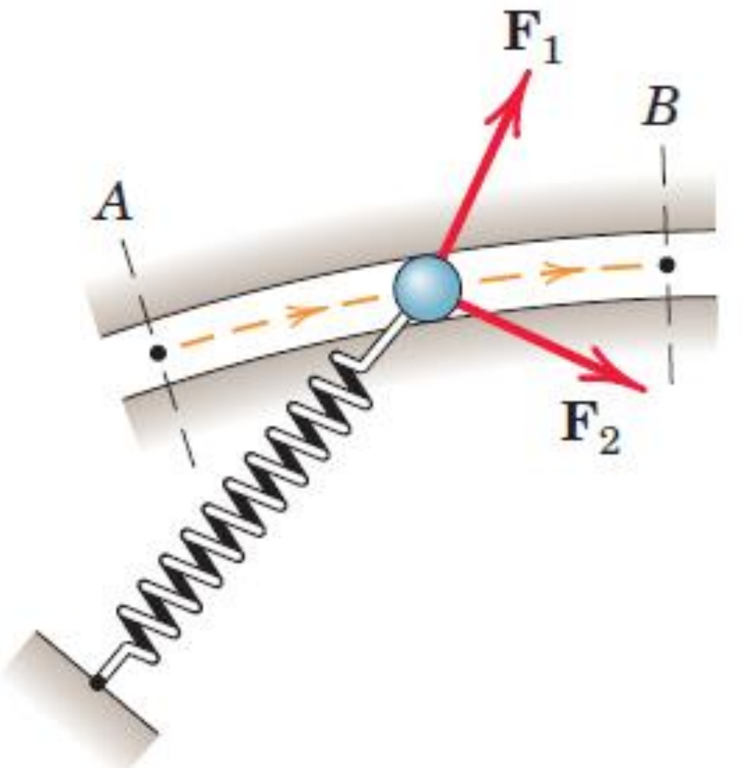


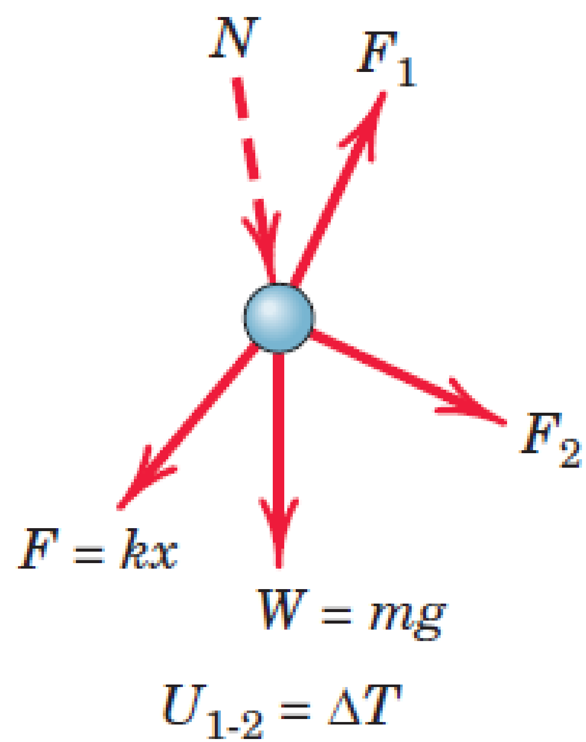




Work-Energy Equation

$$U_{1-2} = T_2 - T_1 = \Delta T$$

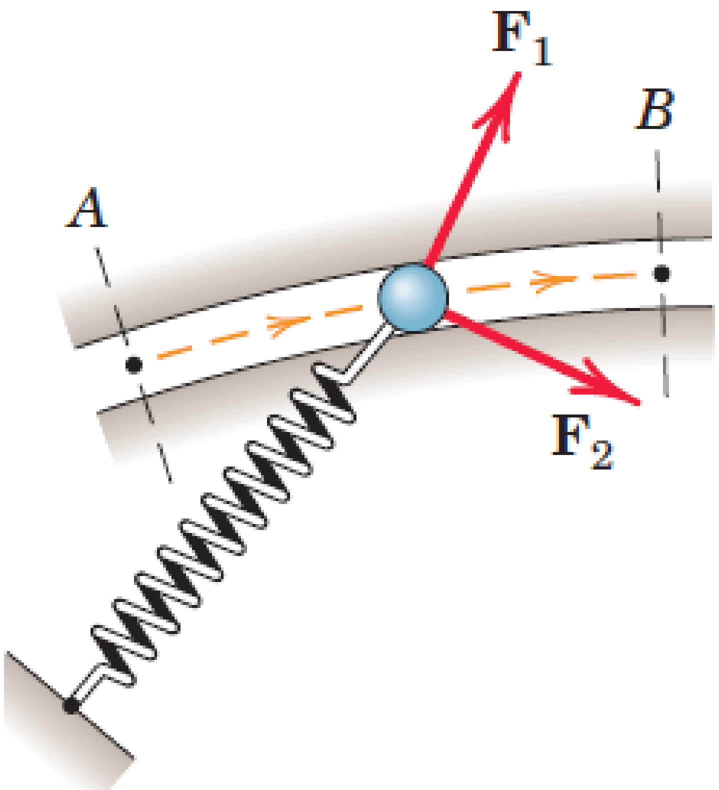


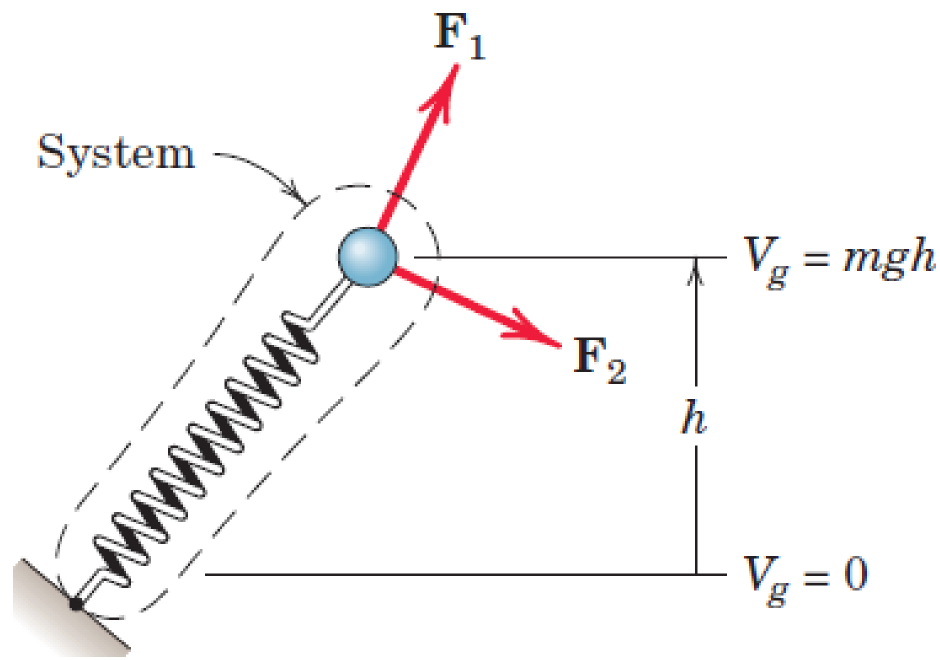


Work-Energy Equation

$$U'_{1-2} + (-\Delta V_g) + (-\Delta V_e) = \Delta T$$

$$\boxed{U'_{1-2} = \Delta T + \Delta V}$$





Law of conservation of dynamical energy

$$U'_{1-2} + \left(-\Delta V_g\right) + \left(-\Delta V_e\right) = \Delta T$$

$$\boxed{U'_{1-2} = \Delta T + \Delta V}$$

NO external force act on the particle

If particle is subjected to only spring force, gravitational force and constraint forces, $U'_{1-2} = 0$

$$\boxed{\Delta T + \Delta V = 0}$$

$$\boxed{T_2 - T_1 + V_2 - V_1 = 0} \qquad \boxed{T_1 + V_1 = T_2 + V_2}$$

$$\boxed{E_1 = E_2}$$

Law of conservation of dynamical energy

$$U'_{1-2} + \left(-\Delta V_g\right) + \left(-\Delta V_e\right) = \Delta T$$

$$\boxed{U'_{1-2} = \Delta T + \Delta V}$$

$$\boxed{T_1 + V_1 = T_2 + V_2} \qquad \boxed{E_1 = E_2}$$

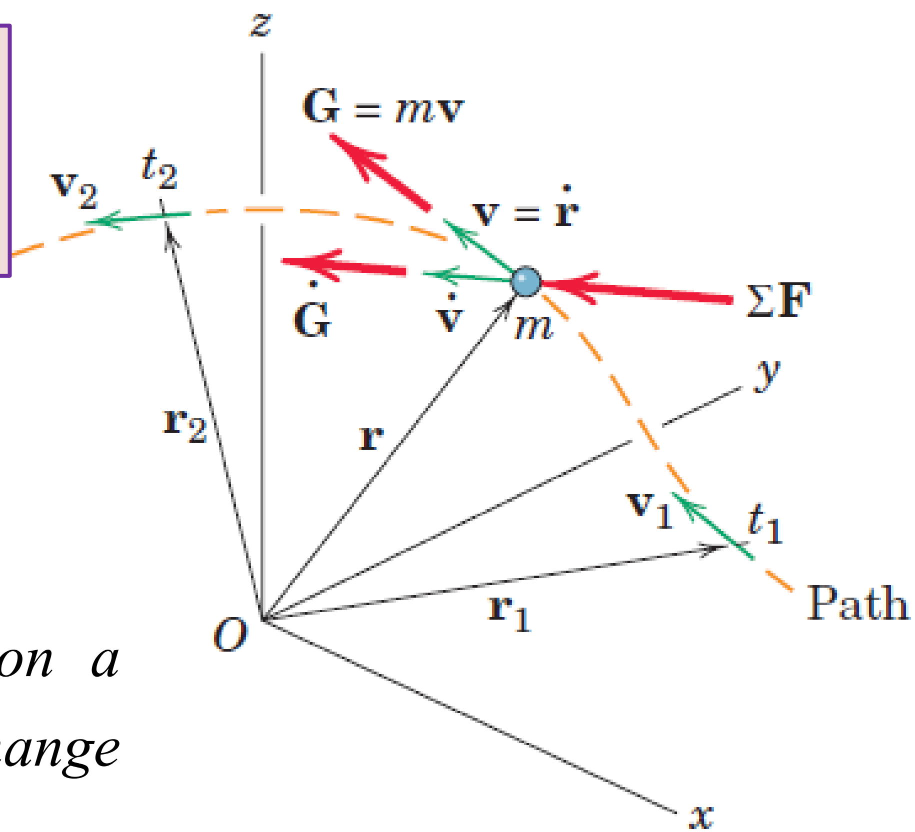
When E is constant, transfers of energy between kinetic and potential may take place as long as the total mechanical energy $T + V$ does not change

Linear Momentum

Linear momentum of the particle, ***G***: Product of the mass and velocity

$$\sum F = m\dot{v} = \frac{d}{dt}(mv)$$

$$\sum F = \dot{G}$$



Resultant of all forces acting on a particle equals its time rate of change of linear momentum.

Units of Linear Momentum

SI System: kg. m-sec, N-sec

US Customary units: [lb/(ft/sec^2)] [ft/sec]: lb-sec

$$\boxed{\sum F_x = \dot{G}_x}$$

$$\boxed{\sum F_y = \dot{G}_y}$$

$$\boxed{\sum F_z = \dot{G}_z}$$

Linear Impulse and Momentum Principle

$$\boxed{\sum F = m\dot{v} = \frac{d}{dt}(mv)}$$

$$\boxed{\sum F = \dot{G}}$$

$$\boxed{\int_{t_1}^{t_2} \sum F \, dt = G_2 - G_1 = \Delta G}$$

- **Linear impulse of the force**: Product of force and time
- **Total linear impulse on *m*** equals the **corresponding change in linear momentum of *m***

Impulse-Momentum Principle

Initial linear momentum of the body plus the linear impulse applied to it equals its final linear momentum

$$G_1 + \int_{t_1}^{t_2} \sum F \, dt = G_2$$

$$mv_{1x} + \int_{t_1}^{t_2} \sum F_x \, dt = mv_{2x}$$

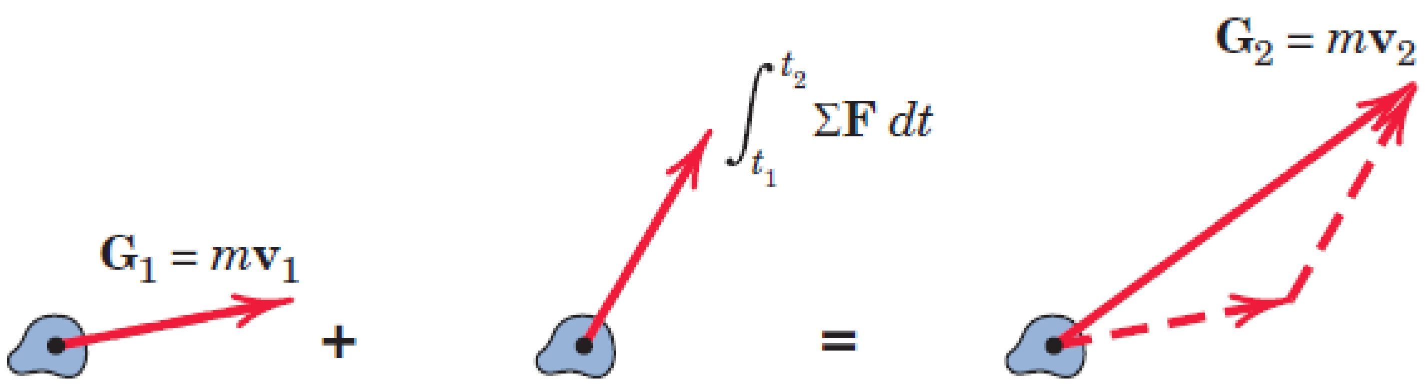
$$mv_{1y} + \int_{t_1}^{t_2} \sum F_y \, dt = mv_{2y}$$

$$mv_{1z} + \int_{t_1}^{t_2} \sum F_z \, dt = mv_{2z}$$

Impulse-Momentum Diagram

$$G_1 + \int_{t_1}^{t_2} \sum F\, dt = G_2$$

Initial linear momentum of the body plus the linear impulse applied to it equals its final linear momentum



Impulsive and Non-impulsive Forces

Impulsive Forces: Forces are very large and of short duration

Example: Forces of short impact

Non-impulsive Forces

Example: Weight of ball

In many cases, non-impulsive forces may be negligible as compared to impulsive forces

Weight of the ball (about 5 oz) is small compared with the force (which could be several hundred pounds in magnitude) exerted on the ball by the bat

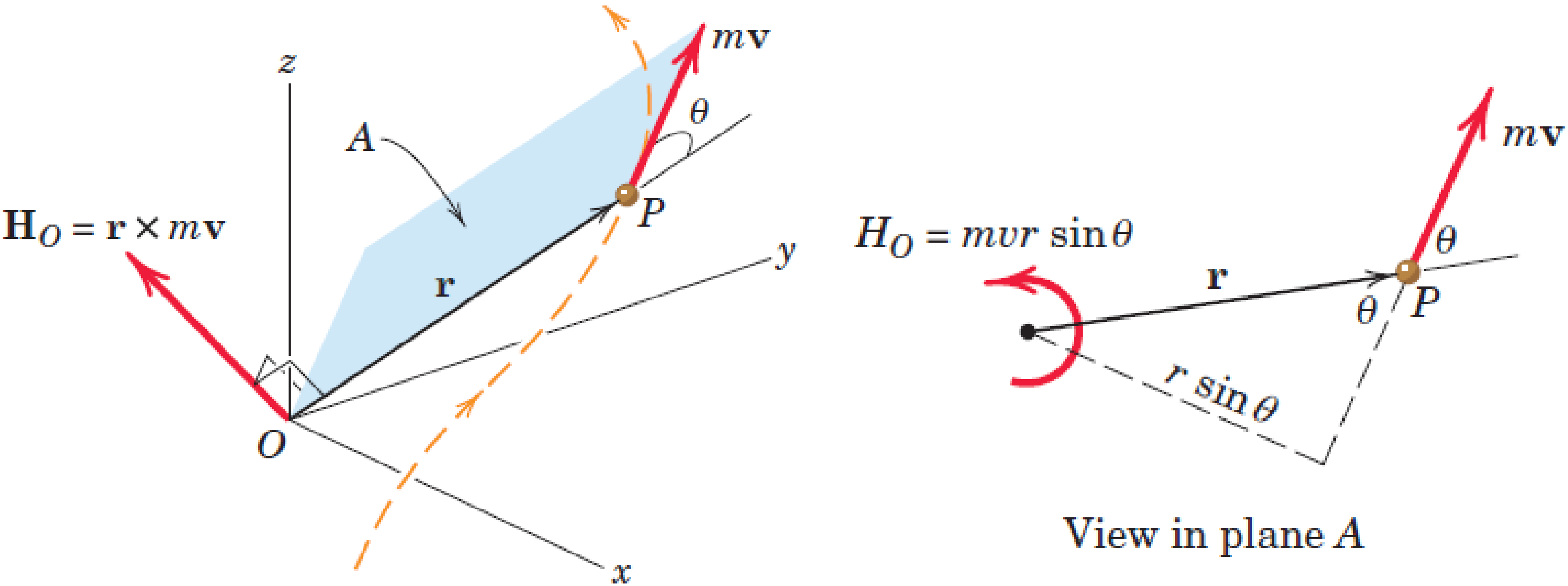
Conservation of Linear Momentum

$$\int_{t_1}^{t_2} \sum F\, dt = G_2 - G_1 = \Delta G = 0$$

$$G_1 = G_2$$

- If the **resultant force** on a particle is **zero** during an interval of time, its **linear momentum** *G* remains **constant**
- Linear momentum of the particle is said to be conserved
- Total linear impulse on the particle in **a particular direction** is zero, the **corresponding linear momentum** is unchanged (conserved) in that **direction**.

Angular Momentum



Angular Momentum: **moment of linear momentum**

The angular momentum then is a vector perpendicular to the plane A defined by **r** and **v**.

$$\mathrm{H_o} = \mathrm{r} \times mv \qquad H_o = mvr\,sin\,\theta$$

Angular Momentum

Angular Momentum: moment of linear momentum

$$\mathrm{H_o} = \mathrm{r} \times m\mathrm{v}$$

$$\mathrm{H_o} = m \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ v_x & v_y & v_z \end{bmatrix}$$

$$\boxed{\mathrm{H_o} = m\,(yv_z - zv_y)\mathbf{i} + m(zv_x - xv_z)\mathbf{j} + m\left(xv_y - yv_x\right)\mathbf{k}}$$

Angular Momentum

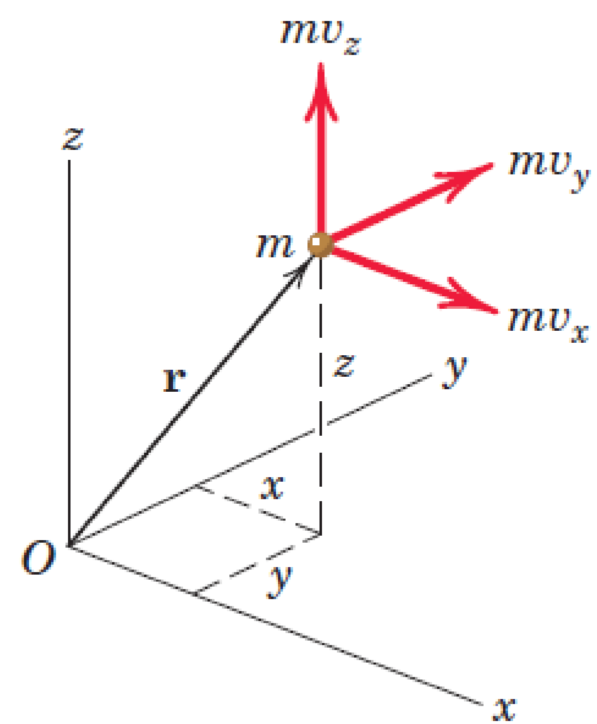
Angular Momentum: moment of linear momentum

$$\boxed{H_o = m\,(yv_z - zv_y)\mathbf{i} + m(zv_x - xv_z)\mathbf{j} + m\left(xv_y - yv_x\right)\mathbf{k}}$$

$$\boxed{H_x = m\,(yv_z - zv_y)}$$

$$\boxed{H_y = m(zv_x - xv_z)}$$

$$\boxed{H_z = m\left(xv_y - yv_x\right)}$$



Units of Angular Momentum

SI Units of angular momentum

(Kg.m/s).m= N.m.s

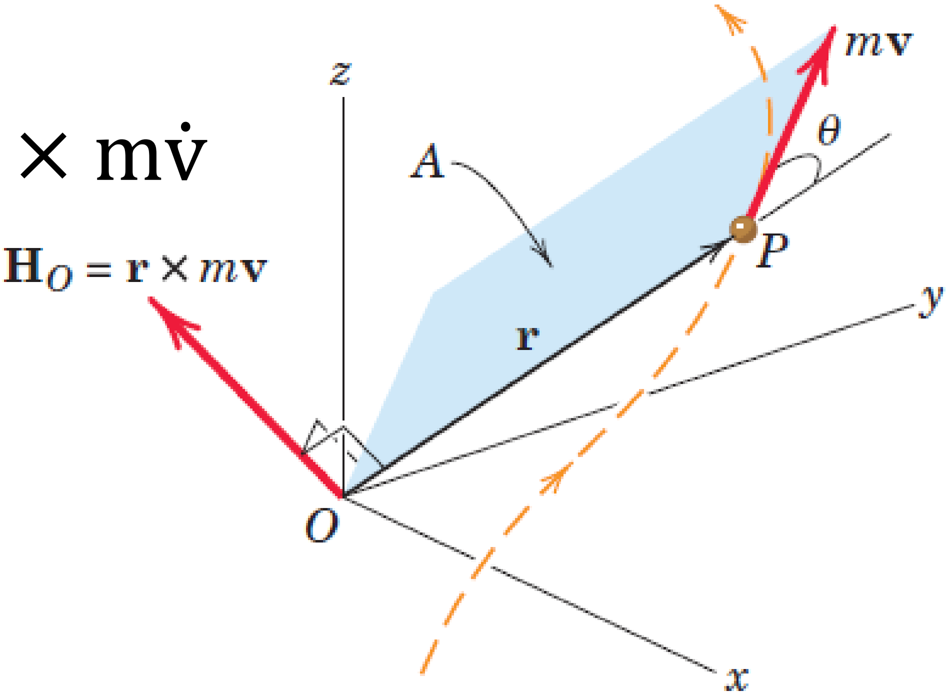
$$\mathrm{H_o} = \mathrm{r} \times m\mathrm{v}$$

US Customary Units

[lb/(ft/sec^2)][ft/sec][ft]=lb-ft-sec

Rate of change of Angular Momentum

$$\sum M_o = r \times \sum F = r \times m\dot{v}$$



$$\dot{H}_o = \dot{r} \times mv + r \times m\dot{v}$$

$$\dot{H}_o = v \times mv + r \times m\dot{v} = 0 + r \times m\dot{v} = \sum M_o$$

Rate of change of Angular Momentum

$$\sum \mathrm{M_o} = \mathrm{r} \times \sum \mathrm{F} = \mathrm{r} \times \mathrm{m}\dot{\mathrm{v}}$$

$$\dot{\mathrm{H}}_{\mathrm{o}} = \dot{\mathrm{r}} \times m\mathrm{v} + \mathrm{r} \times m\dot{\mathrm{v}}$$

$$\dot{\mathrm{H}}_{\mathrm{o}} = \mathrm{v} \times m\mathrm{v} + \mathrm{r} \times m\dot{\mathrm{v}} = 0 + \mathrm{r} \times m\dot{\mathrm{v}} = \sum M_o$$

$$\boxed{\dot{\mathrm{H}}_{\mathrm{o}} = \sum M_o}$$

Moment about the fixed point O of all forces acting on m <u>equals</u> **the time rate of change of angular momentum of m about O**.

Rate of change of Angular Momentum

$$\sum \boldsymbol{M}_o = \dot{\boldsymbol{H}}_{\mathrm{o}}$$

$$\sum M_{ox} = \dot{H}_{ox}$$

$$\sum M_{oy} = \dot{H}_{oy}$$

$$\sum M_{oz} = \dot{H}_{oz}$$

Moment about the fixed point O of all forces acting on m <u>equals</u> **the time rate of change of angular momentum of m about O**.

Angular Impulse-Momentum Principle

Angular impulse: Product of moment and time

$$\sum M_o = \dot{H}_o \qquad\qquad \sum M_o\,dt = dH_o$$

$$\int_{t_1}^{t_2} \sum M_o\,dt = \Delta H_o = (H_o)_2 - (H_o)_1$$

Total angular impulse on *m* about the fixed point O equals the **corresponding change in angular momentum of m about O**.

$$(\mathrm{H_o})_2 = \mathrm{r} \times m\mathrm{v}_2 \qquad\qquad (\mathrm{H_o})_1 = \mathrm{r} \times m\mathrm{v}_1$$

Angular Impulse-Momentum Principle

Initial angular momentum of the particle plus the angular impulse applied to it equals its final angular momentum

$$\boxed{(H_o)_1 + \int_{t_1}^{t_2} \sum M_o \, dt = (H_o)_2}$$

$$(H_{ox})_1 + \int_{t_1}^{t_2} \sum M_{ox} \, dt = (H_{ox})_2$$

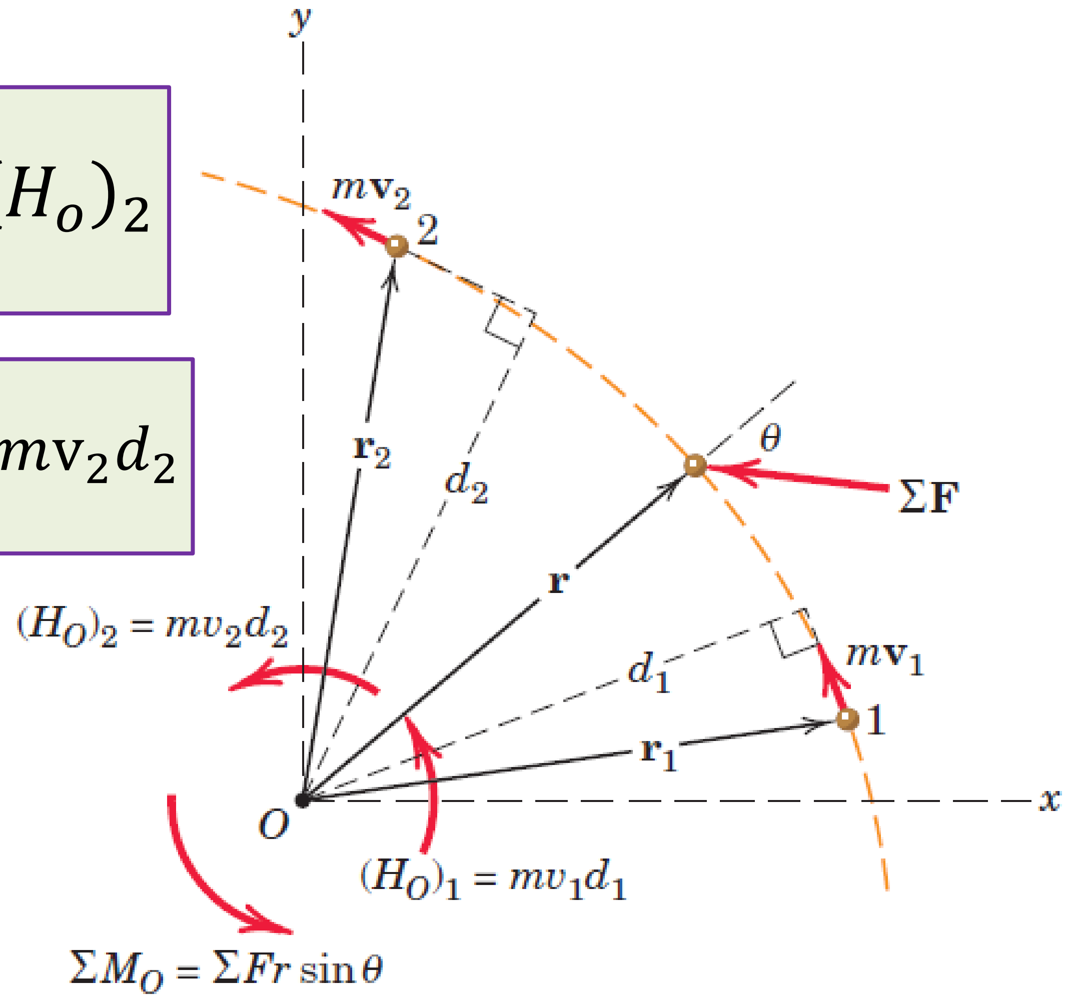
$$\left(H_{oy}\right)_1 + \int_{t_1}^{t_2} \sum M_{oy} \, dt = \left(H_{oy}\right)_2$$

$$(H_{oz})_1 + \int_{t_1}^{t_2} \sum M_{oz} \, dt = (H_{oz})_2$$

Plane Motion Application of Angular Impulse-Momentum Principle

$$(H_o)_1 + \int_{t_1}^{t_2} \sum M_o \, dt = (H_o)_2$$

$$m\text{v}_1 d_1 + \int_{t_1}^{t_2} \sum Fr\sin\theta \, dt = m\text{v}_2 d_2$$



Conservation of Angular Momentum

$$\int_{t_1}^{t_2} \sum M_o \, dt = \Delta H_o = (H_o)_2 - (H_o)_1$$

Resultant moment about a fixed point O of all forces acting on a particle is **zero** during an interval of time

$$\boxed{\Delta H_o = 0}$$

$$\int_{t_1}^{t_2} \sum M_o \, dt = 0$$

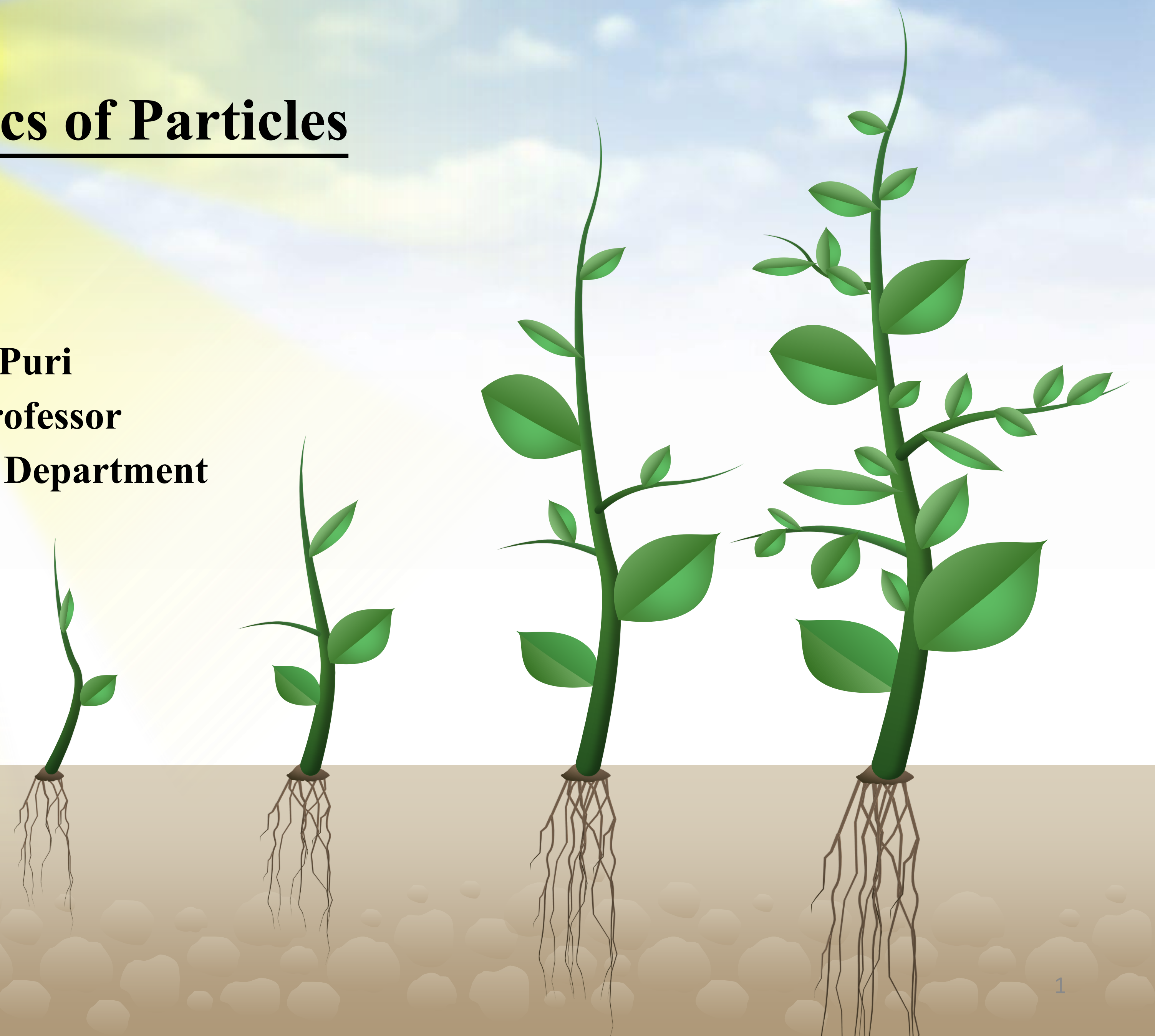
$$\boxed{(H_o)_1 = (H_o)_2}$$

Resultant moment about a fixed point O of all forces acting on a particle is **zero** during an interval of time, its **angular momentum** H_o about that point remain **constant**

THANK YOU

Dynamics of Particles

Dr. Amrita Puri
Assistant Professor
Mechanical Department



Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

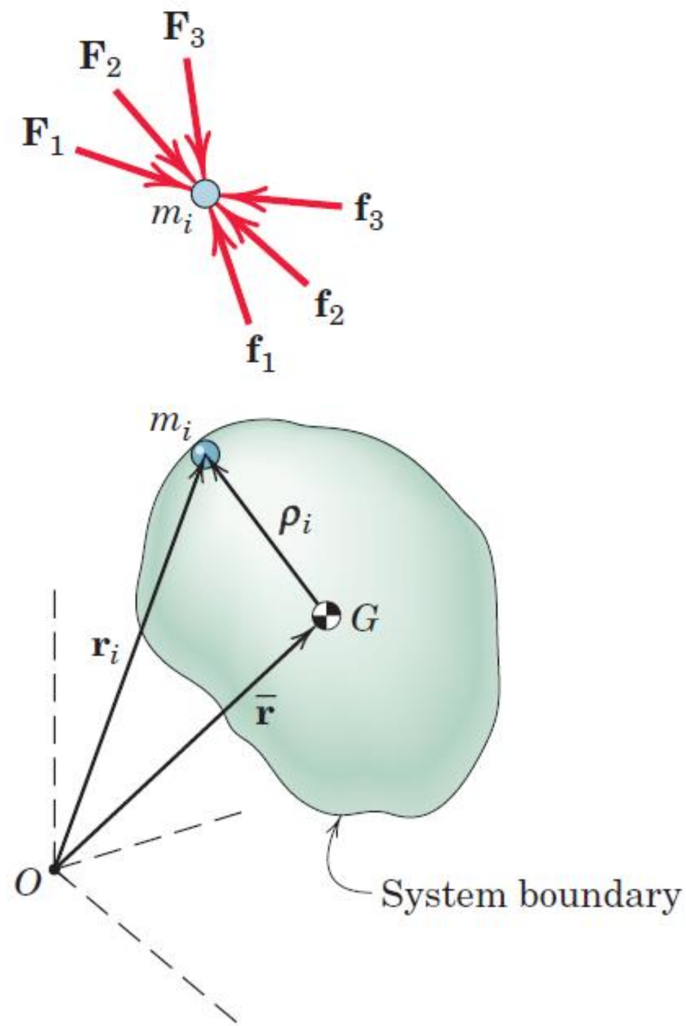
Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations - single particle/system of particles

Generalized Newton's Second Law

General Mass System: A system of *n* mass particles enveloped by a closed surface in space

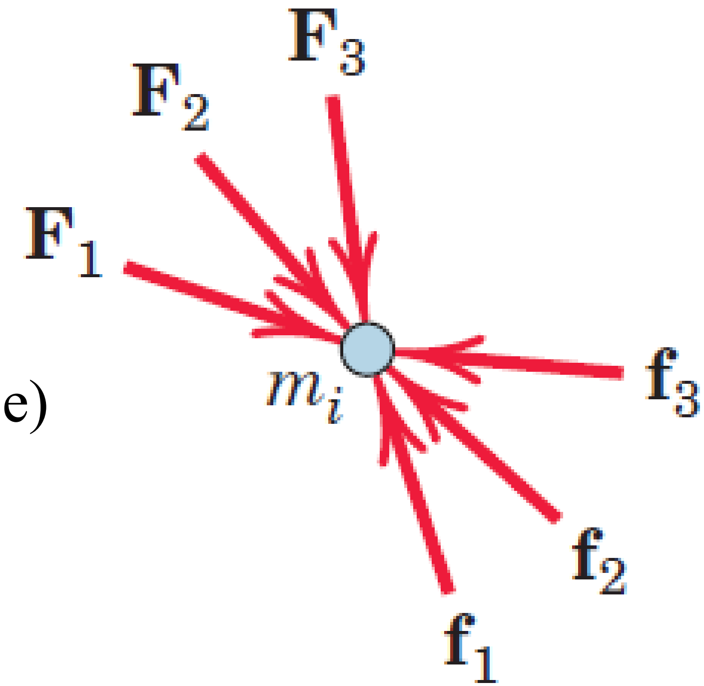
- Exterior surface of a given rigid body
- Bounding surface of an arbitrary portion of the body
- Exterior surface of a rocket containing both rigid and flowing particles
- A particular volume of fluid particles



Generalized Newton's Second Law

External forces (outside the envelope) acting on particle m_i

$F_1, F_2, F_3, F_4 \ldots\ldots\ldots\ldots\ldots$



Internal reaction forces (inside the envelope) acting on particle m_i

$f_1, f_2, f_3, f_4 \ldots\ldots\ldots\ldots\ldots$

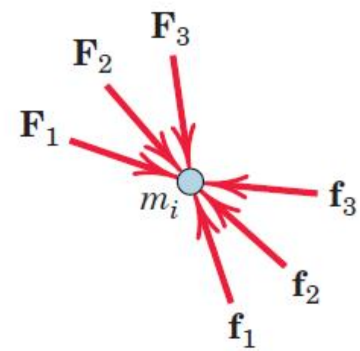
Newton's second law applied on m_i

$$F_1 + F_2 + F_3 + \cdots + f_1 + f_2 + f_3 = m_i \ddot{r}_i$$

Generalized Newton's Second Law

Position Vector of Mass Centre *G*

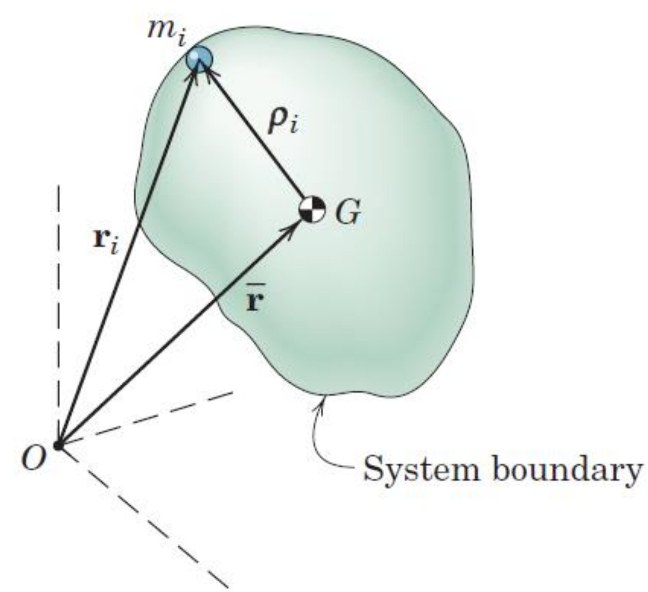
$$m\bar{r} = \sum_{i=1}^{n} m_i r_i \qquad m = \sum_{i=1}^{n} m_i$$



Sum for all *n* mass particles

$$\sum F + \sum f = \sum_{i=1}^{n} m_i \ddot{r}_i$$

$$\sum F = \sum_{i=1}^{n} m_i \ddot{r}_i$$



Generalized Newton's Second Law

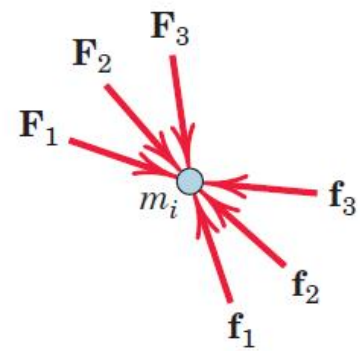
Position Vector of Mass Centre G

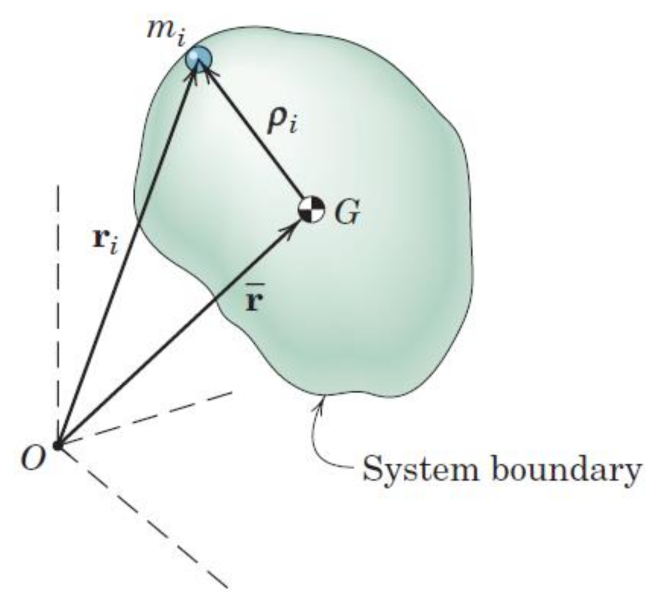
$$m\bar{r} = \sum_{i=1}^{n} m_i r_i$$

$$m\ddot{\bar{r}} = \sum_{i=1}^{n} m_i \ddot{r}_i$$

Sum for all n mass particles

$$\boxed{\sum F = \sum_{i=1}^{n} m_i \ddot{r}_i = m\ddot{\bar{r}} = m\bar{a}}$$



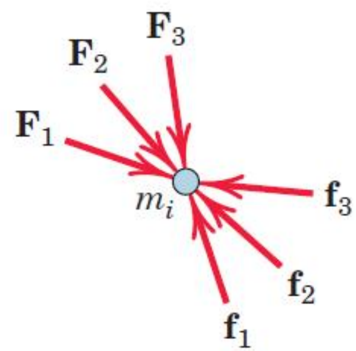


$\bar{a}$ is acceleration of mass center of the body

Generalized Newton's Second Law

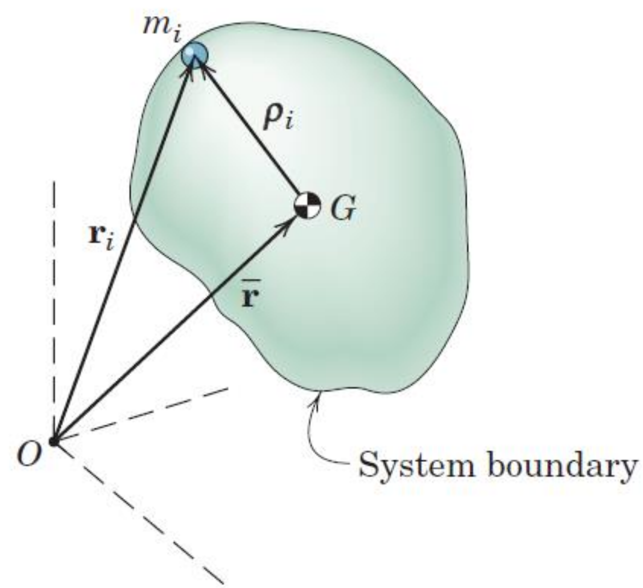
Sum for all *n* mass particles

$$\sum F = \sum_{i=1}^{n} m_i \ddot{r}_i = m\ddot{\bar{r}} = m\bar{a}$$



Instantaneous relationship

$\bar{a}$ is acceleration of mass center of the body

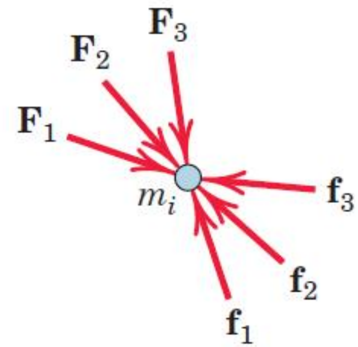


Resultant of the external forces on *any* system of masses equals the **total mass of the system times the acceleration of the center of mass**

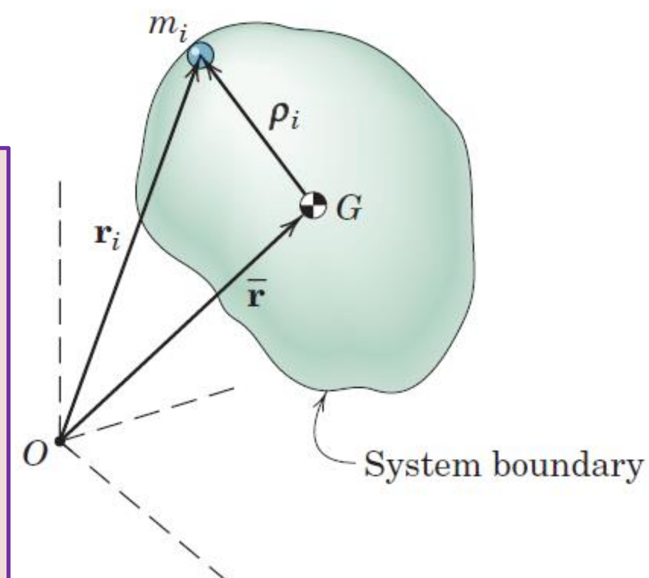
Generalized Newton's Second Law

Sum for all n mass particles

$$\sum F = m\bar{a}$$



$\bar{a}$ is the acceleration of the mathematical point which represents **instantaneously** the position of the mass center for the given n particles



- Acceleration vector have the same direction as the resultant external force $\sum F$
- $\sum F$ NOT necessarily passes through G

Work-Energy

Work energy equation for i^{th} particle

$$\mathrm{U}_{1-2_{\mathrm{i}}} = \Delta \mathrm{T}_{\mathrm{i}}$$

$\mathrm{U}_{1-2_{\mathrm{i}}}$ is work done by external and internal forces.

Sum of work energy equation for all the particles

$$\sum_{i=1}^{n} \mathrm{U}_{1-2_{\mathrm{i}}} = \sum_{i=1}^{n} \Delta \mathrm{T}_{\mathrm{i}}$$

Work done by internal forces is zero. $\sum_{i=1}^{n} \mathrm{U}_{1-2_{\mathrm{i}}}$ includes work done by external forces only

$\sum_{i=1}^{n} \Delta \mathrm{T}_{\mathrm{i}}$: Change in total kinetic energy of the system

Work-Energy

Sum of work energy equation for all the particles

$$U_{1-2}{}' = \Delta E = \Delta \mathrm{T} + \Delta \mathrm{V}$$

$$\Delta \mathrm{V} = V_g + V_e$$

V_g: Gravitational potential energy

V_e: Elastic potential energy stored in spring

E: Total Mechanical Energy

$$U_{1-2}{}' + T_1 + V_1 = T_2 + V_2$$

Kinetic Energy

Sum for all *n* mass particles

$$T = \sum \frac{1}{2} m_i {v^2}_i \qquad \mathbf{v}_{\boldsymbol{i}} = \bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}$$

$\mathbf{v}_{\boldsymbol{i}}$ is the velocity of i^{th} particle. $\bar{\mathbf{v}}$ is velocity of mass center *G*.

$\dot{\boldsymbol{\rho}}_{\boldsymbol{i}}$ is the velocity of i^{th} particle with respect to translating frame of reference moving with mass center *G*.

$${v^2}_i = \mathbf{v}_\mathbf{i} \cdot \mathbf{v}_\mathbf{i}$$

$$T = \sum \frac{1}{2} m_i \mathbf{v}_\mathbf{i} \cdot \mathbf{v}_\mathbf{i} = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}) \cdot (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}})$$

$$T = \sum \frac{1}{2} m_i (\bar{\mathbf{v}}.\bar{\mathbf{v}} + \mathbf{2}\,\dot{\boldsymbol{\rho}}_{\boldsymbol{i}}\,.\bar{\mathbf{v}} \ + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}.\dot{\boldsymbol{\rho}}_{\boldsymbol{i}})$$

Kinetic Energy

$$T = \sum \frac{1}{2} m_i \mathbf{v_i} \cdot \mathbf{v_i} = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}) \cdot (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}})$$

$$T = \sum \frac{1}{2} \left(m_i \bar{\mathbf{v}}.\bar{\mathbf{v}} + \mathbf{2} m_i \dot{\boldsymbol{\rho}}_{\boldsymbol{i}} \,.\, \bar{\mathbf{v}} + m_i \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}.\dot{\boldsymbol{\rho}}_{\boldsymbol{i}}\right)$$

$$\sum (m_i \boldsymbol{\rho}_{\boldsymbol{i}}) = 0$$

$$\bar{\mathbf{V}}.\frac{d}{dt}\left(\sum (m_i \boldsymbol{\rho}_{\boldsymbol{i}})\right) = 0$$

$$T = \sum \frac{1}{2} \left(m_i \bar{v}^2 + m_i |\dot{\rho}_i|^2\right)$$

Kinetic Energy

$$T = \sum \frac{1}{2} m_i \mathbf{v_i} \cdot \mathbf{v_i} = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}) \cdot (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}})$$

$$T = \sum \frac{1}{2} \left(m_i \bar{\mathbf{v}}.\bar{\mathbf{v}} + \mathbf{2} m_i \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}\,.\bar{\mathbf{v}} \; + m_i \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}.\dot{\boldsymbol{\rho}}_{\boldsymbol{i}}\right)$$

$$T = \sum \frac{1}{2} \left(m_i \bar{v}^2 + m_i |\dot{\rho}_i|^2\right) = \frac{1}{2} \left(m\bar{v}^2 + \sum m_i |\dot{\rho}_i|^2\right)$$

Total kinetic energy of a mass system *equals* the **kinetic energy of mass-center** translation of the system as a whole plus the **kinetic energy due to motion of all particles relative to the mass center**.

Linear Impulse-Momentum

Momentum of i^{th} particle of mass

$$\mathbf{G_i} = m_i \mathbf{v_i}$$

Linear momentum of the system: Vector sum of the linear momenta of all of its particles

$$\mathbf{G} = \sum m_i \mathbf{v_i} \qquad\qquad \mathbf{v}_{\boldsymbol{i}} = \bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}$$

$$\mathbf{G} = \sum m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}}) = \sum m_i \bar{\mathbf{v}} + \sum m_i \dot{\boldsymbol{\rho}}_{\boldsymbol{i}} = m\bar{\mathbf{v}} + 0$$

Linear momentum of any system of constant mass is the product of the mass and the velocity of its center of mass.

Linear Impulse-Momentum

$$\mathbf{G} = m\bar{\mathbf{v}}$$

$$\dot{\mathbf{G}} = m\dot{\bar{\mathbf{v}}} = ma = \sum F$$

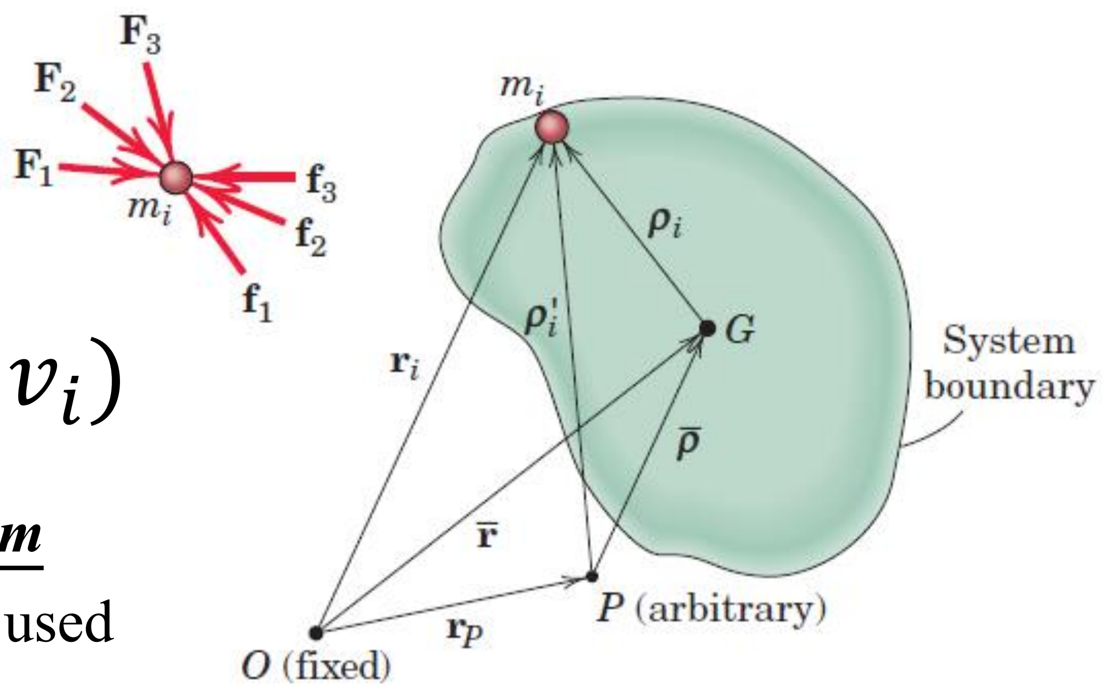
- The resultant of the external forces on any mass system equals the time rate of change of the linear momentum of the system

- $\sum F$, in general, does not pass through the mass center G.

- The equation does not apply to systems whose mass changes with time.

Angular Momentum about Fixed Point

The angular momentum of the mass system about the point O, fixed in the Newtonian reference system, is defined as the vector sum of the moments of the linear momenta about O of all particles of the system



$$H_o = \sum (r_i \times m_i v_i)$$

<u>Absolute angular momentum</u>

because absolute velocity is used

Angular Momentum about Fixed Point

$$\dot{H}_o = \sum (\dot{r}_i \times m_i v_i) + (r_i \times m_i \dot{v}_i)$$

$$\dot{H}_o = \sum (r_i \times m_i \dot{v}_i) = \sum (r_i \times F_i)$$

$$\dot{H}_o = \sum M_o$$

The moment sum $\sum M_0$ represents only the moments of forces external to the system, since the internal forces cancel one another and their moments add up to zero

Angular Momentum about Fixed Point

$$\dot{H}_o = \sum (r_i \times m_i \dot{v}_i) = \sum (r_i \times F_i) = \sum M_o$$

- The moment sum $\sum M_0$ represents only the moments of forces external to the system, since the internal forces cancel one another and their moments add up to zero

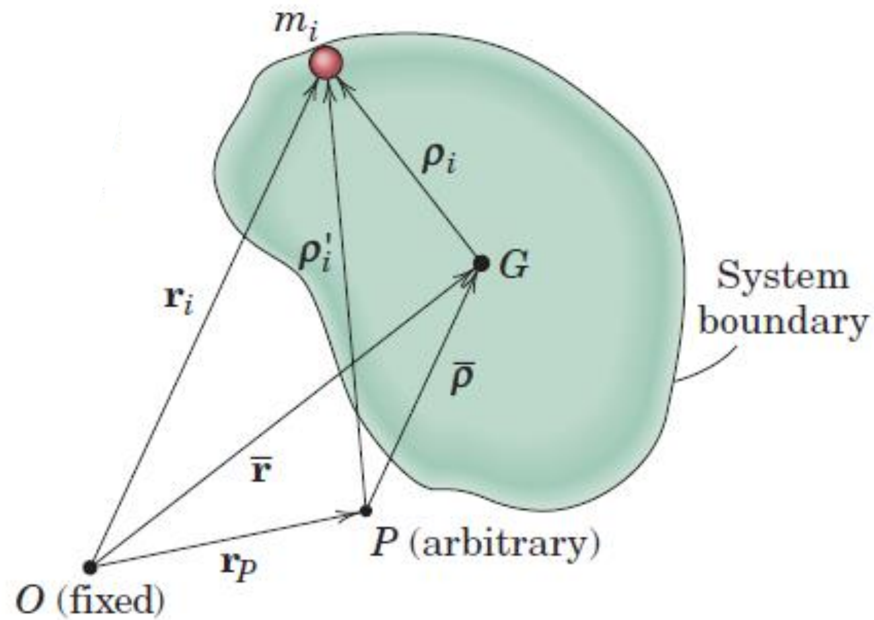
- The **resultant vector moment about any fixed point of all external forces** on any system of mass equals the **time rate of change of angular momentum** of the system about the fixed point

Angular Momentum about Mass Center, *G*

Sum of the moments of the linear momenta about *G* of all particles

$$H_G = \sum \rho_i \times m_i \dot{r}_i$$

$$\dot{r}_i = (\dot{\bar{r}} + \dot{\rho}_i)$$



$$H_G = \sum \rho_i \times m_i (\dot{\bar{r}} + \dot{\rho}_i)$$

$$H_G = \sum \rho_i \times m_i (\dot{\bar{r}} + \dot{\rho}_i)$$

Angular Momentum about Mass Center, *G*

Sum of the moments of the linear momenta about *G* of all particles

$$H_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{r}}_i \qquad \dot{\boldsymbol{r}}_{\boldsymbol{i}} = (\dot{\bar{\boldsymbol{r}}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}})$$

$$H_G = \sum \boldsymbol{\rho}_i \times m_i (\dot{\bar{\boldsymbol{r}}} + \dot{\boldsymbol{\rho}}_{\boldsymbol{i}})$$

$$H_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\bar{\boldsymbol{r}}} + \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$$H_G = \sum -\dot{\bar{\boldsymbol{r}}} \times m_i \boldsymbol{\rho}_i + \sum \boldsymbol{\rho}_i \times \frac{d m_i \boldsymbol{\rho}_i}{dt}$$

Angular Momentum about Mass Center, *G*

Sum of the moments of the linear momenta about *G* of all particles

$$H_G = -\dot{\bar{\boldsymbol{r}}} \times \sum m_i \boldsymbol{\rho}_i + \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$$H_G = 0 + \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$$H_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

Relative angular momentum because the relative velocity is used

Angular Momentum about Mass Center, *G*

Sum of the moments of the linear momenta about *G* of all particles

$$\boldsymbol{H}_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{r}}_i$$

$$\dot{\boldsymbol{r}}_i = (\dot{\bar{\boldsymbol{r}}} + \dot{\boldsymbol{\rho}}_i) \qquad H_G = \sum \boldsymbol{\rho}_i \times m_i(\dot{\bar{\boldsymbol{r}}} + \dot{\boldsymbol{\rho}}_i)$$

$$\dot{\boldsymbol{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i \dot{\boldsymbol{r}}_i + \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i$$

$$\dot{\boldsymbol{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i(\dot{\bar{\boldsymbol{r}}} + \dot{\boldsymbol{\rho}}_i) + \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i$$

Angular Momentum about Mass Center, *G*

Sum of the moments of the linear momenta about *G* of all particles

$$\dot{\boldsymbol{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i(\dot{\bar{\boldsymbol{r}}} + \dot{\boldsymbol{\rho}}_i) + \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i$$

$$\dot{\boldsymbol{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i \dot{\bar{\boldsymbol{r}}} + \sum \dot{\boldsymbol{\rho}}_i \times m_i \dot{\boldsymbol{\rho}}_i + \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i$$

$$\dot{\boldsymbol{H}}_G = -\dot{\bar{\boldsymbol{r}}} \times \frac{d \sum (m_i \rho_i)}{dt} + 0 + \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i$$

$$\dot{\boldsymbol{H}}_G = 0 + 0 + \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i$$

Angular Momentum about Mass Center, *G*

Sum of the moments of the linear momenta about *G* of all particles

$$\dot{\boldsymbol{H}}_G = \sum \boldsymbol{\rho}_i \times m_i \ddot{\boldsymbol{r}}_i = \sum \boldsymbol{\rho}_i \times (F_i + f_i) = \sum M_G$$

$$\dot{\boldsymbol{H}}_G = \sum \boldsymbol{\rho}_i \times (F_i + f_i) = \sum \boldsymbol{\rho}_i \times F_i + \sum \boldsymbol{\rho}_i \times f_i = \sum M_G$$

$$\dot{\boldsymbol{H}}_G = \sum \boldsymbol{\rho}_i \times F_i + \sum \boldsymbol{\rho}_i \times f_i = \sum \boldsymbol{\rho}_i \times F_i + 0 = \sum M_G$$

$$\dot{\boldsymbol{H}}_G = \sum \boldsymbol{\rho}_i \times F_i + 0 = \sum M_G$$

Rate of change of angular momentum about point *G* is **sum of all external moments about point *G***

Angular Momentum about Arbitrary Point, *P*

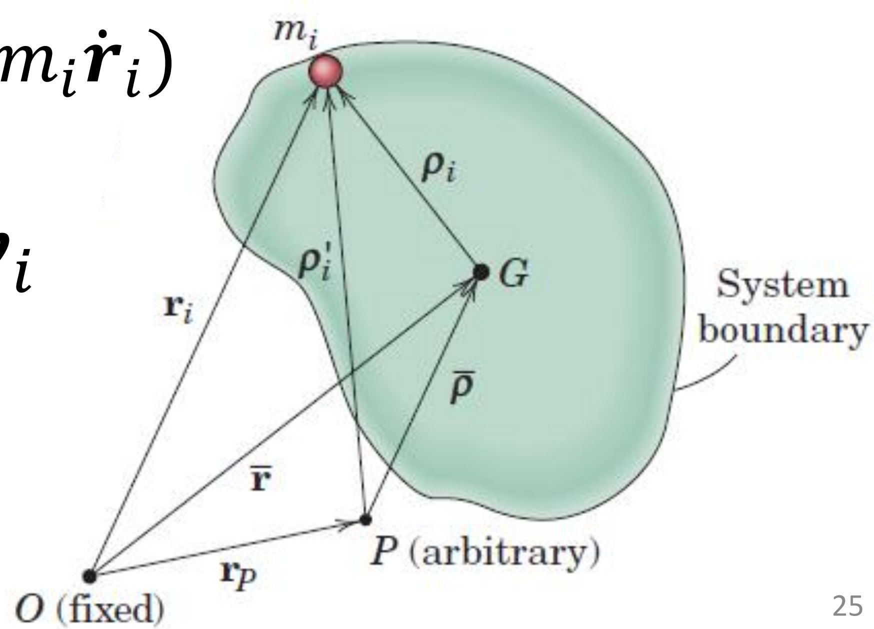
Sum of the moments of the linear momenta about *P* (which may have acceleration) of all particles

$$\boldsymbol{H}_P = \sum \boldsymbol{\rho}'_i \times m_i \dot{\boldsymbol{r}}_i = \sum (\overline{\boldsymbol{\rho}} + \boldsymbol{\rho}_i) \times m_i \dot{\boldsymbol{r}}_i$$

$$\boldsymbol{H}_P = \sum (\overline{\boldsymbol{\rho}} \times m_i \dot{\boldsymbol{r}}_i + \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{r}}_i)$$

$$\boldsymbol{H}_P = \boldsymbol{H}_G + \overline{\boldsymbol{\rho}} \times \sum m_i \boldsymbol{v}_i$$

$$\boldsymbol{H}_P = \boldsymbol{H}_G + \overline{\boldsymbol{\rho}} \times m \overline{\boldsymbol{v}}$$

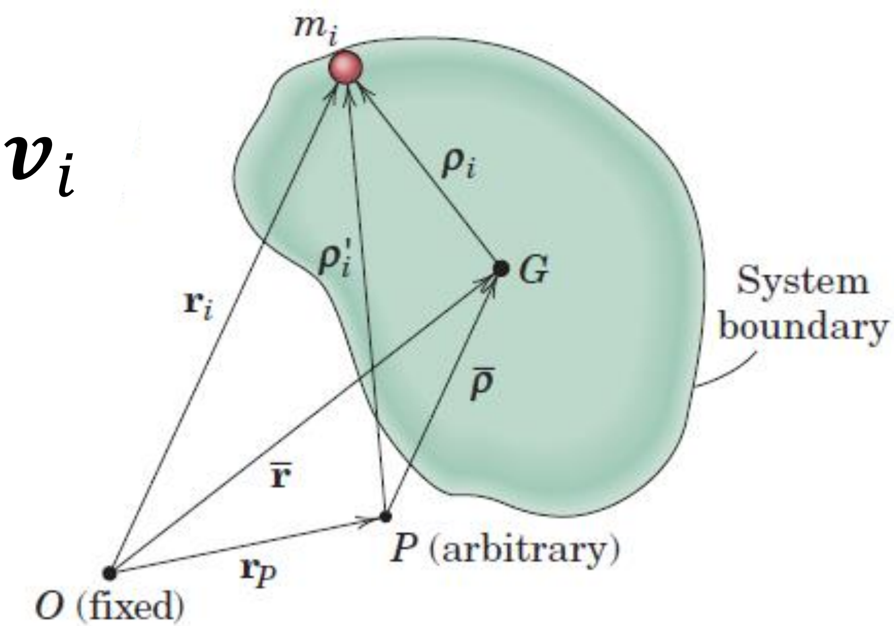


Angular Momentum about Arbitrary Point, *P*

Sum of the moments of the linear momenta about *P* of all particles

$$\boldsymbol{H}_P = \boldsymbol{H}_G + \overline{\boldsymbol{\rho}} \times \sum m_i \boldsymbol{v}_i$$

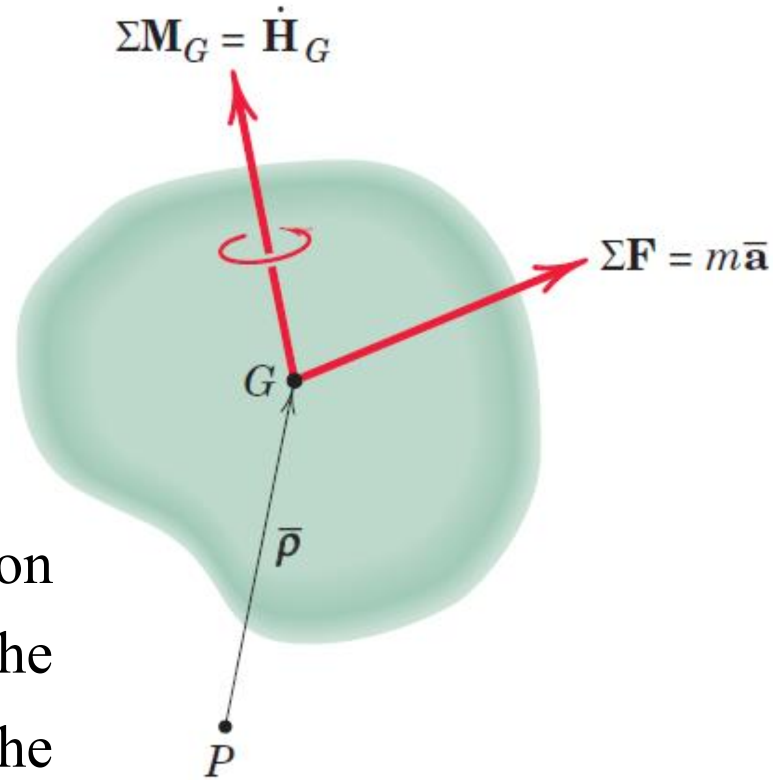
$$\boldsymbol{H}_P = \boldsymbol{H}_G + \overline{\boldsymbol{\rho}} \times m\overline{\boldsymbol{v}}$$



Absolute angular momentum about any point *P* equals the **angular momentum about *G*** plus the **moment about *P* of the linear momentum of the system considered concentrated at *G***

Angular Momentum about Arbitrary Point, *P*

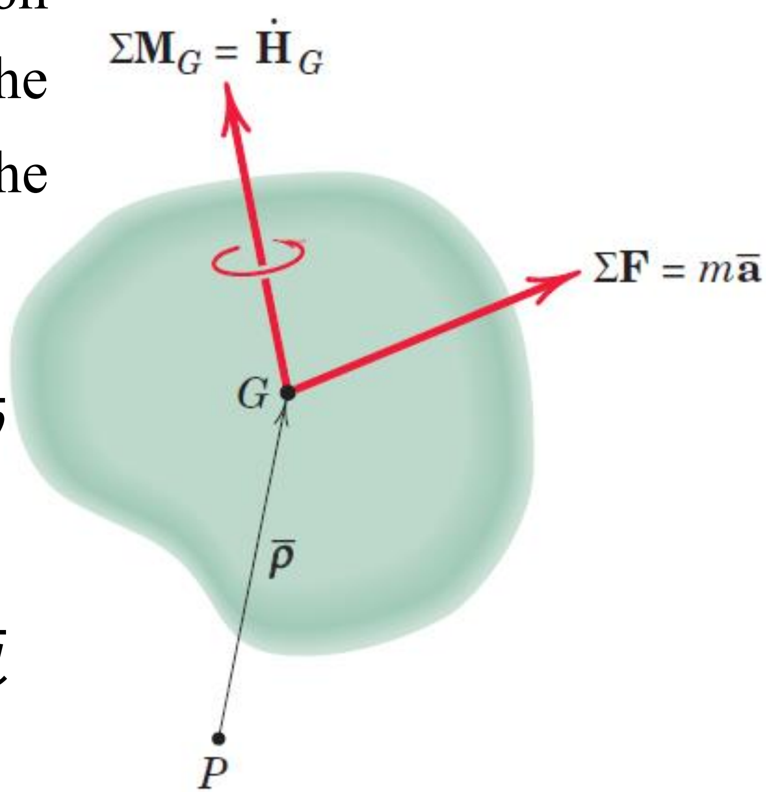
Representing a force system by a resultant force through any point, such as *G*, and a corresponding couple



Resultant of the external forces acting on the system expressed in terms of the resultant force $\sum \boldsymbol{F}$ through *G* and the corresponding couple $\sum M_G$

Angular Momentum about Arbitrary Point, *P*

Resultant of the external forces acting on the system expressed in terms of the resultant force $\sum \boldsymbol{F}$ through G and the corresponding couple $\sum M_G$



$$\sum \boldsymbol{M}_P = \sum \boldsymbol{M}_G + \overline{\boldsymbol{\rho}} \times m\dot{\overline{\boldsymbol{v}}}$$

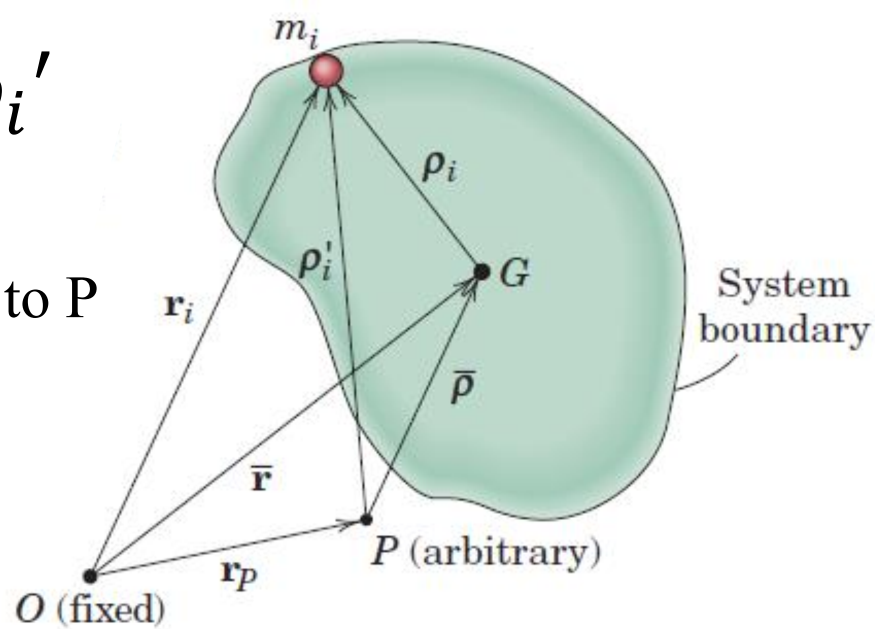
$$\sum \boldsymbol{M}_P = \sum \boldsymbol{M}_G + \overline{\boldsymbol{\rho}} \times m\overline{\boldsymbol{a}}$$

$$\sum \boldsymbol{M}_P = \dot{\boldsymbol{H}}_G + \overline{\boldsymbol{\rho}} \times m\overline{\boldsymbol{a}}$$

Relative Angular Momentum about Arbitrary Point, *P*

$$(\mathbf{H}_P)_{rel} = \sum \rho_i{}' \times m_i \dot{\rho}_i{}'$$

$\dot{\rho}_i{}'$ is velocity of m_i with respect to P



$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i (\dot{\bar{\rho}} + \dot{\rho}_i)$$

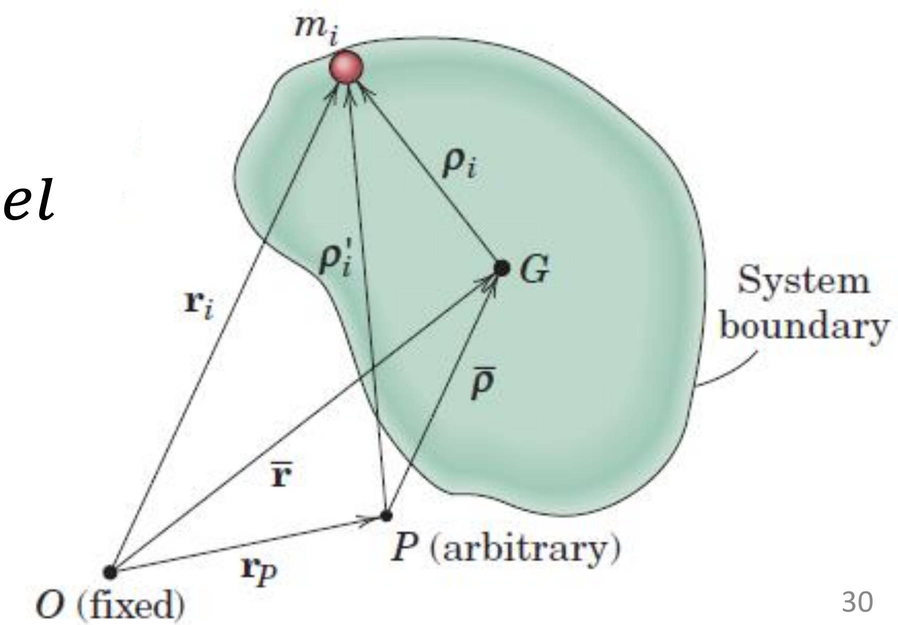
Relative Angular Momentum about Arbitrary Point, *P*

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i(\dot{\bar{\rho}} + \dot{\rho}_i)$$

$$(\mathbf{H}_P)_{rel}$$
$$= \sum (\bar{\rho} \times m_i\dot{\bar{\rho}} + \bar{\rho} \times m_i\dot{\rho}_i + \rho_i \times m_i\dot{\bar{\rho}} + \rho_i \times m_i\dot{\rho}_i)$$

$$\sum \bar{\rho} \times m_i\dot{\bar{\rho}} = \bar{\rho} \times m v_{rel}$$

$$\sum \bar{\rho} \times \frac{d(m_i\rho_i)}{dt} = 0$$



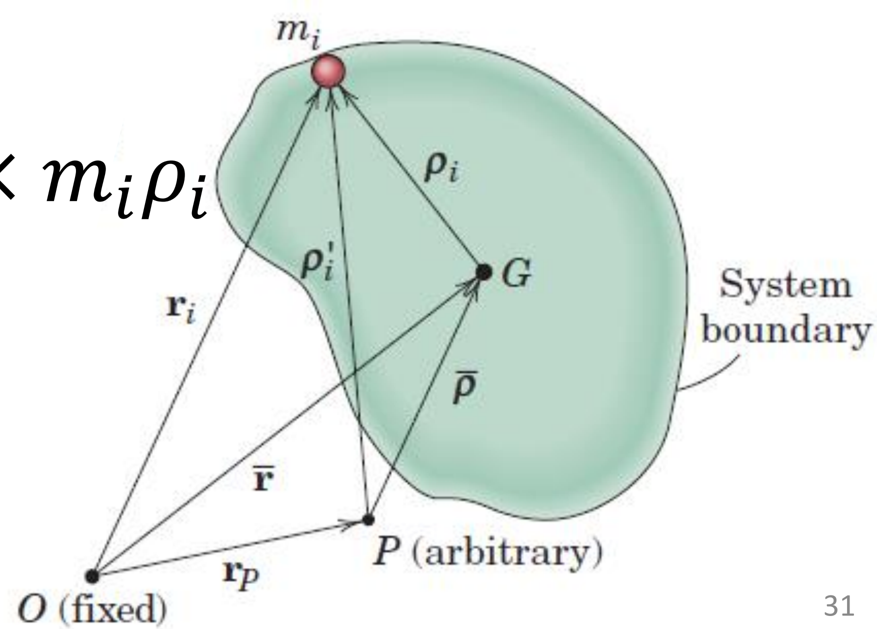
Relative Angular Momentum about Arbitrary Point, *P*

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i(\dot{\bar{\rho}} + \dot{\rho}_i)$$

$$(\mathbf{H}_P)_{rel}$$
$$= \sum (\bar{\rho} \times m_i\dot{\bar{\rho}} + \bar{\rho} \times m_i\dot{\rho}_i + \rho_i \times m_i\dot{\bar{\rho}} + \rho_i \times m_i\dot{\rho}_i)$$

$$\sum \rho_i \times m_i\dot{\bar{\rho}} = \sum -\dot{\bar{\rho}} \times m_i\rho_i$$

$$-\dot{\bar{\rho}} \times \sum m_i\rho_i = 0$$



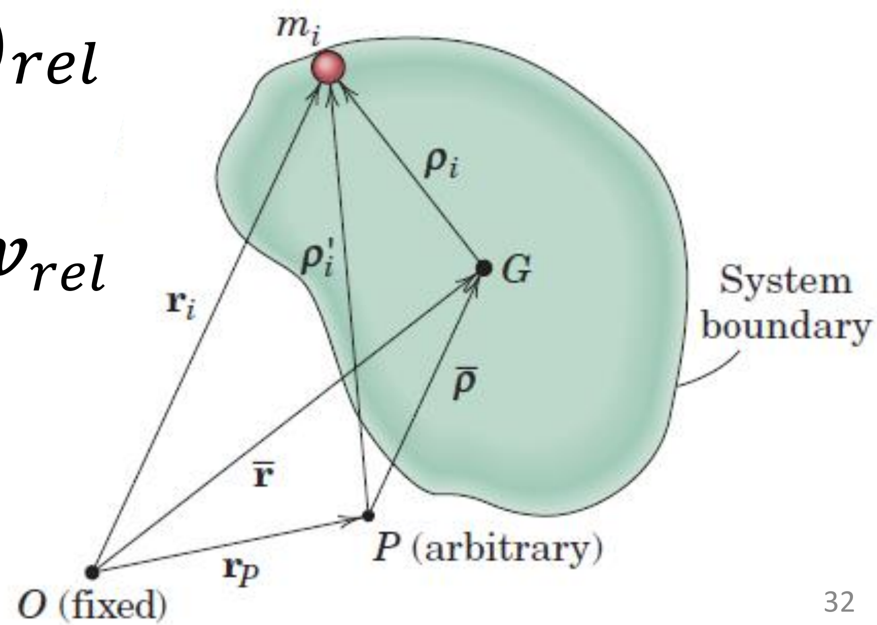
Relative Angular Momentum about Arbitrary Point, *P*

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i(\dot{\bar{\rho}} + \dot{\rho}_i)$$

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} \times m_i\dot{\bar{\rho}} + \bar{\rho} \times m_i\dot{\rho}_i + \rho_i \times m_i\dot{\bar{\rho}} + \rho_i \times m_i\dot{\rho}_i)$$

$$\sum \rho_i \times m_i\dot{\rho}_i = (H_G)_{rel}$$

$$(\mathbf{H}_P)_{rel} = (\mathbf{H}_G)_{rel} + \bar{\boldsymbol{\rho}} \times m\boldsymbol{v}_{rel}$$



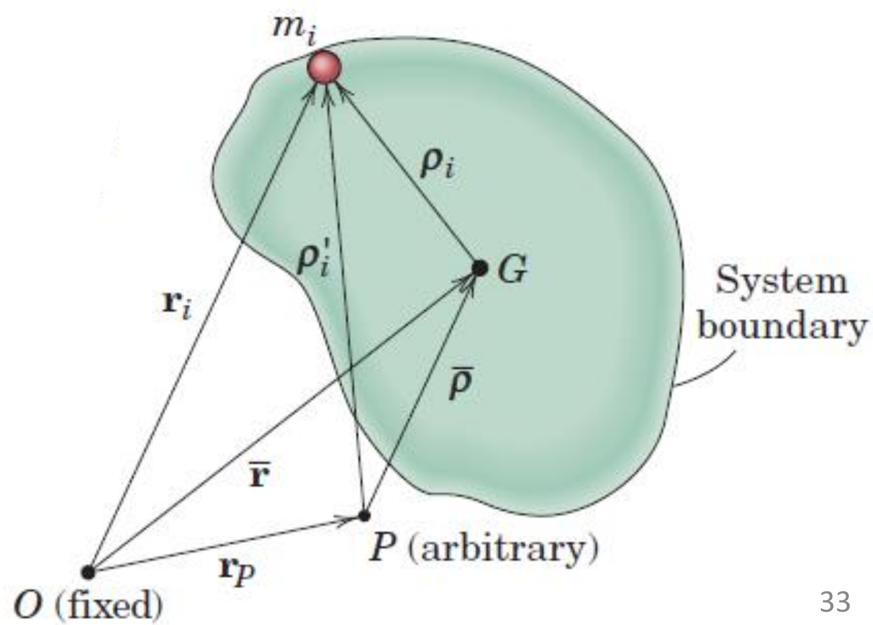
Relative Angular Momentum about Arbitrary Point, *P*

$$(\mathbf{H}_P)_{rel} = \sum (\rho_i{}') \times m_i(\dot{\rho}_i{}')$$

$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum (\dot{\rho}_i{}') \times m_i(\dot{\rho}_i{}') + (\rho_i{}') \times m_i(\ddot{\rho}_i{}')$$

$$r_i = r_p + (\rho_i{}')$$

$$\ddot{r}_i = \ddot{r}_p + (\ddot{\rho}_i{}')$$

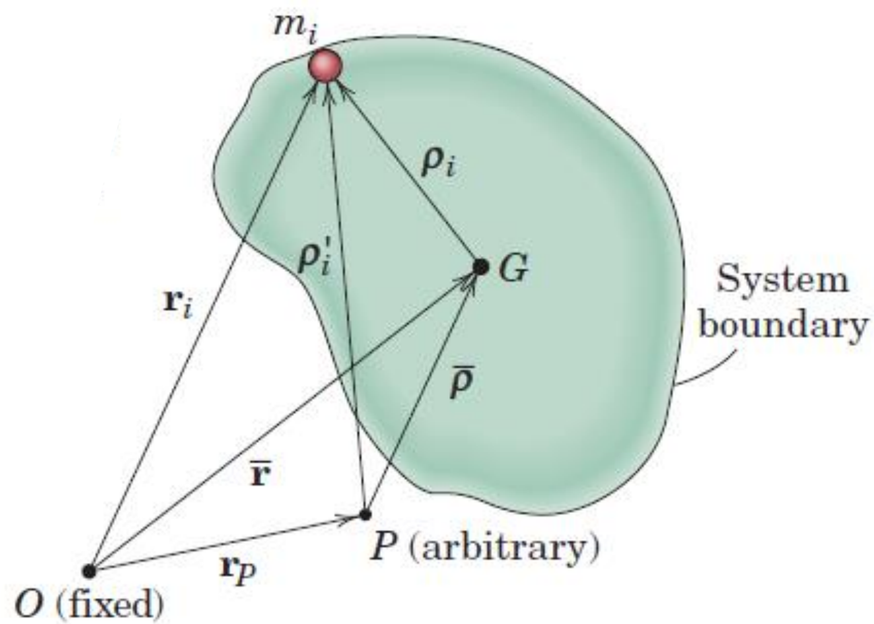


Relative Angular Momentum about Arbitrary Point, *P*

$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum(\dot{\rho}_i{}') \times m_i(\dot{\rho}_i{}') + (\rho_i{}') \times m_i(\ddot{\rho}_i{}')$$

$$r_i = r_p + (\rho_i{}')$$

$$(\ddot{\rho}_i{}') = \ddot{r}_i - \ddot{r}_p$$

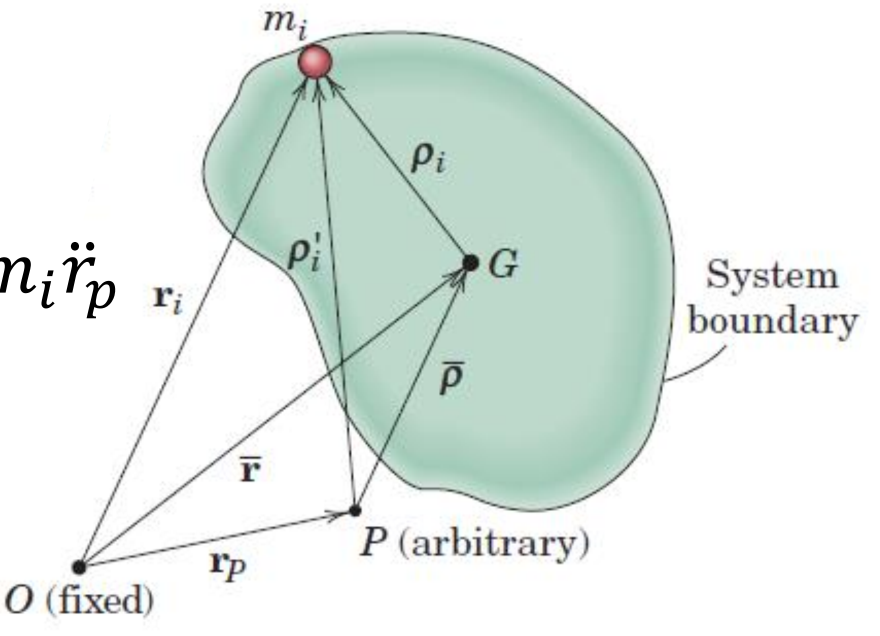


$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum(\rho_i{}') \times m_i\left(\ddot{r}_i - \ddot{r}_p\right)$$

Relative Angular Momentum about Arbitrary Point, *P*

$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum (\rho_i{}') \times m_i(\ddot{r}_i - \ddot{r}_p)$$

$$\left(\dot{\mathbf{H}}_P\right)_{rel}$$
$$= \sum (\rho_i{}') \times m_i\ddot{r}_i - (\rho_i{}') \times m_i\ddot{r}_p$$



$$\left(\dot{\mathbf{H}}_P\right)_{rel}$$
$$= \sum \boldsymbol{M}_P + \sum -(\rho_i{}') \times m_i\ddot{r}_p$$

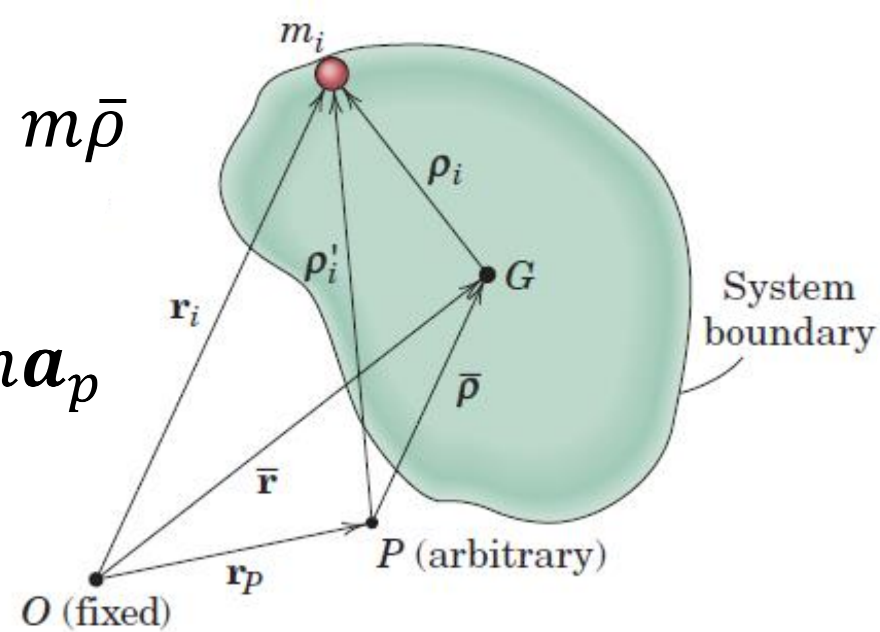
Relative Angular Momentum about Arbitrary Point, *P*

$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum \boldsymbol{M}_P + \sum \ddot{r}_p \times m_i\,(\boldsymbol{\rho}_i{}')$$

$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum \boldsymbol{M}_P + \boldsymbol{a}_p \times \sum m_i\,(\boldsymbol{\rho}_i{}')$$

$$\left(\dot{\mathbf{H}}_P\right)_{rel} = \sum \boldsymbol{M}_P + \boldsymbol{a}_p \times m\bar{\rho}$$

$$\sum \boldsymbol{M}_P = \left(\dot{\mathbf{H}}_P\right)_{rel} + \bar{\rho} \times m\boldsymbol{a}_p$$



Relative Angular Momentum about Arbitrary Point, *P*

$$\sum \boldsymbol{M}_P = \left(\dot{\mathbf{H}}_P\right)_{rel} + \bar{\rho} \times m\boldsymbol{a}_p$$

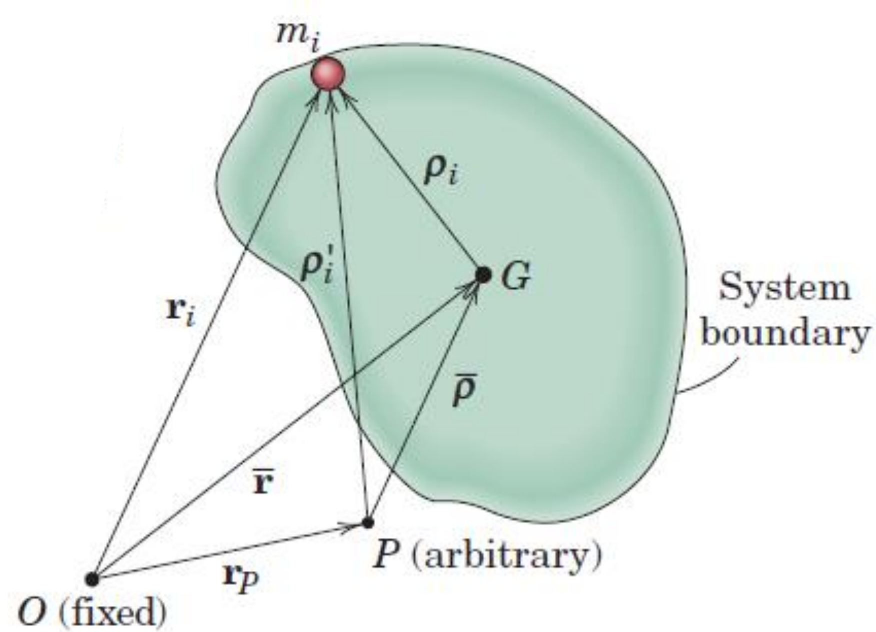
If

(a) $\bar{\rho} = 0$

(b) $\boldsymbol{a}_p = 0$

(c) $\bar{\rho}$ and $\boldsymbol{a}_p$ are parallel

$$\sum \boldsymbol{M}_P = \left(\dot{\mathbf{H}}_P\right)_{rel}$$



Law of Conservation of Dynamical Energy

- A mass system is said to be *conservative* if it does not lose energy by virtue of internal friction forces which do negative work or by virtue of inelastic members which dissipate energy upon cycling.
- If no work is done on a conservative system during an interval of motion by external forces (other than gravity or other potential forces), then none of the energy of the system is lost.

$$\boxed{U'_{1-2} = 0}$$

$$\boxed{\Delta T + \Delta V = 0}$$

$$\boxed{T_1 + V_1 = T_2 + V_2}$$

This law holds only in the ideal case where internal kinetic friction is sufficiently small to be neglected

Principle of Conservation of Linear Momentum

Resultant external force $\Sigma\mathbf{F}$ acting on a conservative or nonconservative mass system is zero,

$$\dot{\mathbf{G}} = \sum F = 0$$

$$\mathbf{G_1} = \mathbf{G_2}$$

In the absence of an external impulse, the linear momentum of a system remains unchanged

Principle of Conservation of Angular Momentum

Resultant moment about a fixed point O or about the mass center G of all external forces on any mass system is zero,

$$\dot{\mathbf{H}}_{\boldsymbol{o}} = \sum M_o = 0 \qquad \dot{\mathbf{H}}_{\boldsymbol{G}} = \sum M_G = 0$$

$$\mathbf{H}_{\mathbf{o}_1} = \mathbf{H}_{\mathbf{o}_2} \qquad \mathbf{H}_{\mathbf{G}_1} = \mathbf{H}_{\mathbf{G}_2}$$

There is no angular impulse about a fixed point (or about the mass center), the angular momentum of the system about the fixed point (or about the mass center) remains unchanged.

Application in a translating frame of reference

- The basic laws of Newtonian mechanics hold for measurements made relative to a set of axes which translate with a constant velocity.

- Thus, all equations, mentioned in previous slides, are valid provided all quantities are expressed relative to the translating axes.

Steady Mass Flow

- **Steady Flow Condition**: Rate at which mass enters a given volume equals the rate at which mass leaves the same volume

- The volume may be enclosed by a **rigid container**, fixed or moving, such as the **nozzle of a jet aircraft or rocket**, the **space between blades in a gas turbine**, the **volume within the casing of a centrifugal pump**, the **volume within the bend of a pipe through which a fluid is flowing at a steady rate**

- Design of such fluid machines depends on the analysis of the forces and moments associated with the corresponding momentum changes of the flowing mass

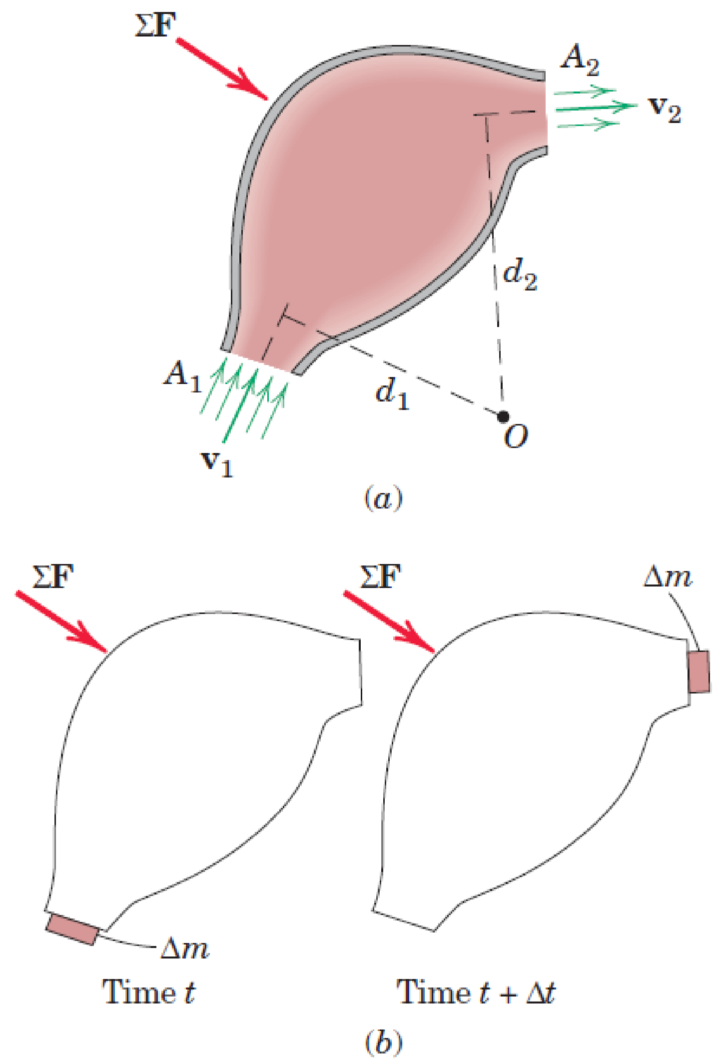
Steady Mass Flow through a Rigid Container

- If ρ_1 and ρ_2 are the respective densities of the two streams, conservation of mass requires that

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 = m'$$

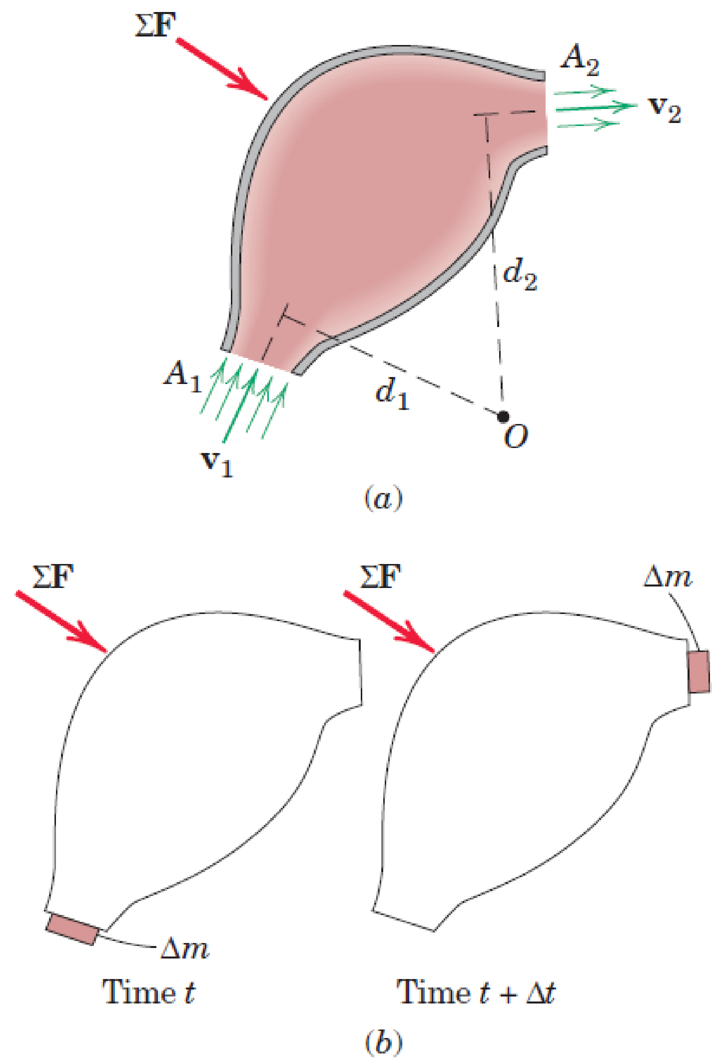
- Two approaches to isolate the system

1. Isolate the mass of fluid within the container (if the forces between the container and the fluid were to be described)
2. Entire container and the fluid within it (when the forces external to the container are desired)



Steady Mass Flow through a Rigid Container

- Account for all forces applied externally to this system, $\sum F$
  - Forces exerted on the container at points of its attachment to other structures, including attachments at A_1 and A_2, if present
  - Forces acting on the fluid within the container at A_1 and A_2 due to any static pressure which may exist in the fluid at these positions
  - Weight of the fluid and structure if appreciable

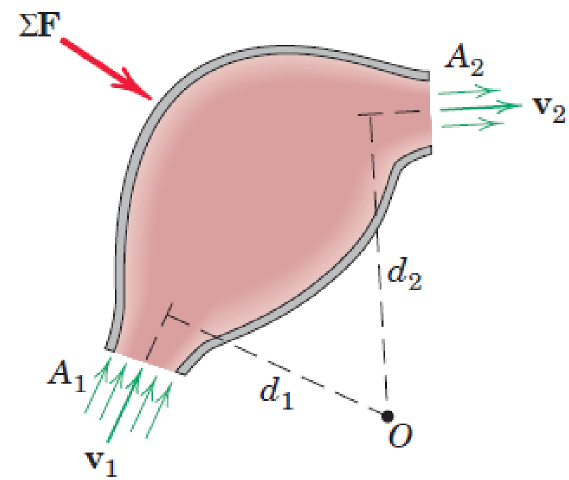


Steady Mass Flow through a Rigid Container

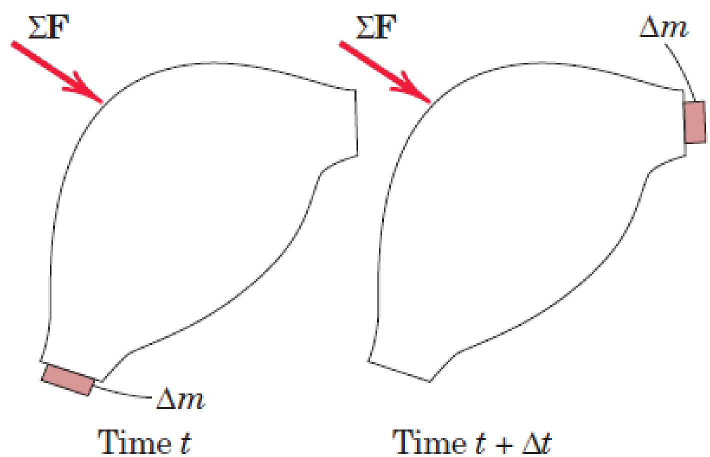
- All forces applied externally to this system, $\sum F$

$$\sum F = \dot{G}$$

- **The resultant of all of these external forces must equal the time rate of change of the linear momentum of the isolated system**



(a)

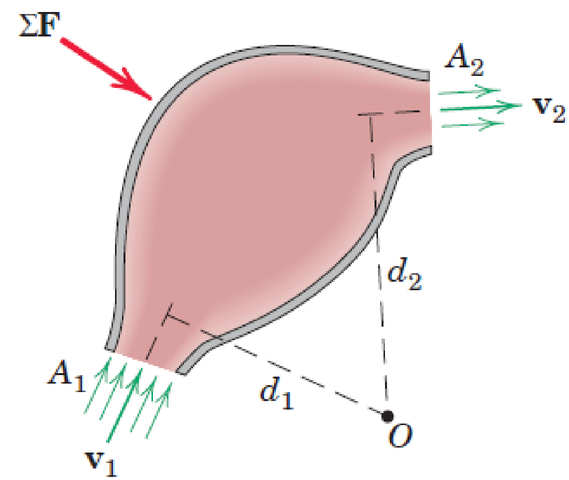


(b)

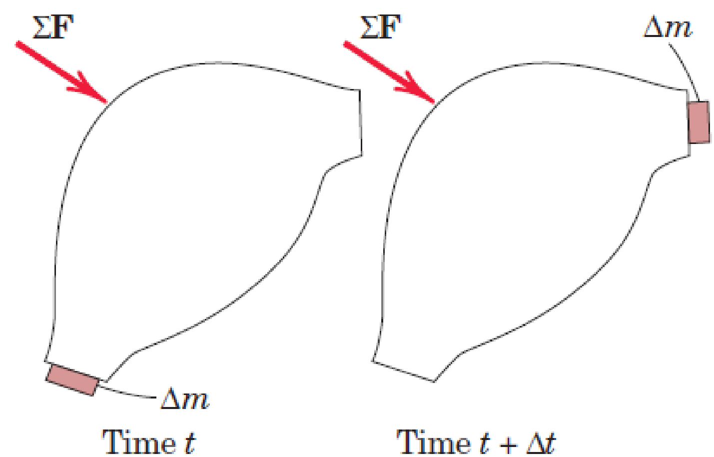
Steady Mass Flow through a Rigid Container

Incremental Analysis

- The system at time t when the system mass is that of the container, the mass within it, and an increment Δm about to enter during time t. At time $t + \Delta t$ the same total mass is that of the container, the mass within it, and an equal increment Δm which leaves the container in time Δt.
- The linear momentum of the container and mass within it between the two sections A_1 and A_2 remains unchanged during Δt so that the change in momentum of the system in time Δt



(a)



(b)

Steady Mass Flow through a Rigid Container

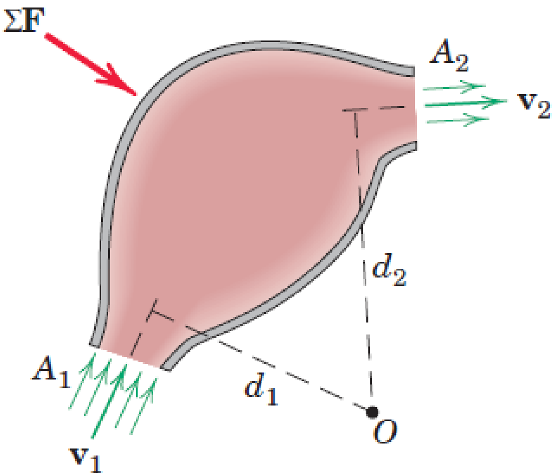
Incremental Analysis

- The change in linear momentum of the system in time Δt

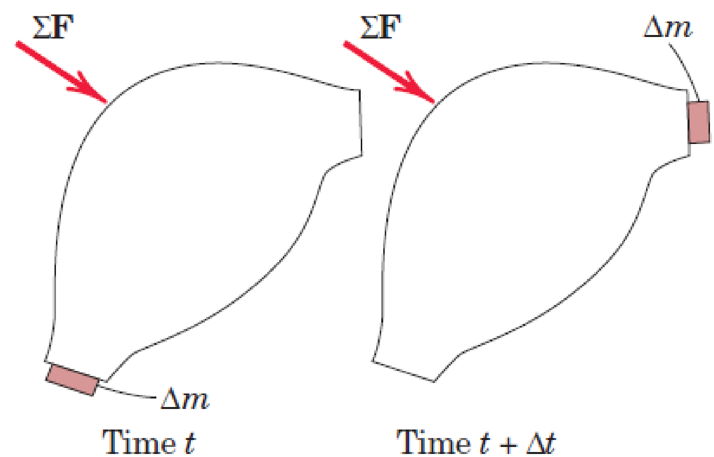
$$\Delta G = (\Delta m)v_2 - (\Delta m)v_1$$
$$= \Delta m(v_2 - v_1)$$

$$m' = \lim_{\Delta t \to 0}\left(\frac{\Delta m}{\Delta t}\right) = \frac{\mathrm{d}m}{\mathrm{d}t}$$

$$\sum F = m'\Delta v = \dot{G}$$



(a)



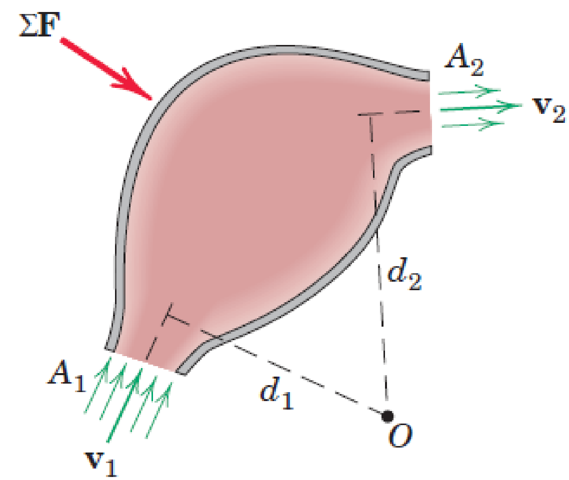
(b)

Steady Mass Flow through a Rigid Container

Incremental Analysis

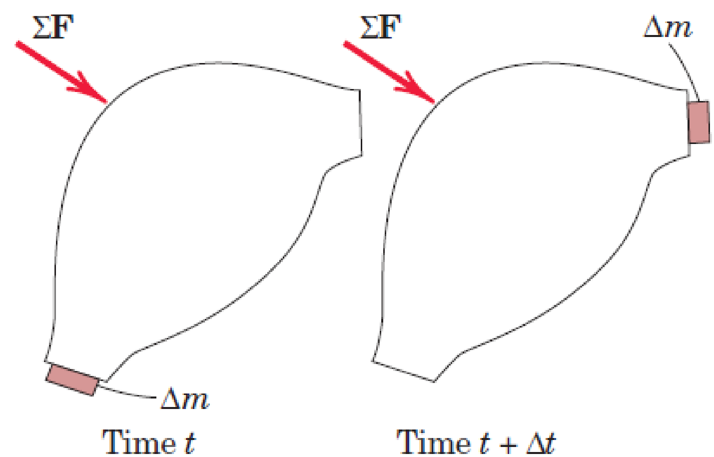
- The change in linear momentum of the system in time Δt

$$\sum F = m' \Delta v = \dot{G}$$



(*a*)

Relation between the resultant force on a steady-flow system and the corresponding mass flow rate and vector velocity increment



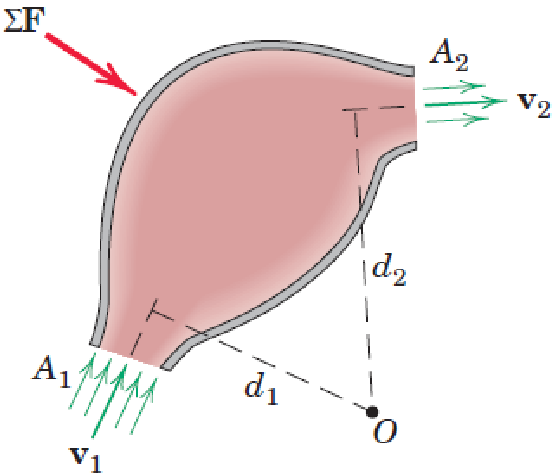
(*b*)

Steady Mass Flow through a Rigid Container

Incremental Analysis

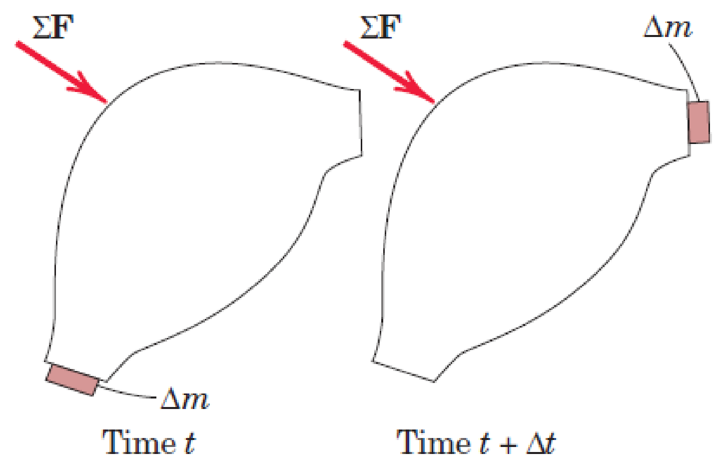
- The change in linear momentum of the system in time Δt

$$\sum F = m' \Delta v = \dot{G}$$



(a)

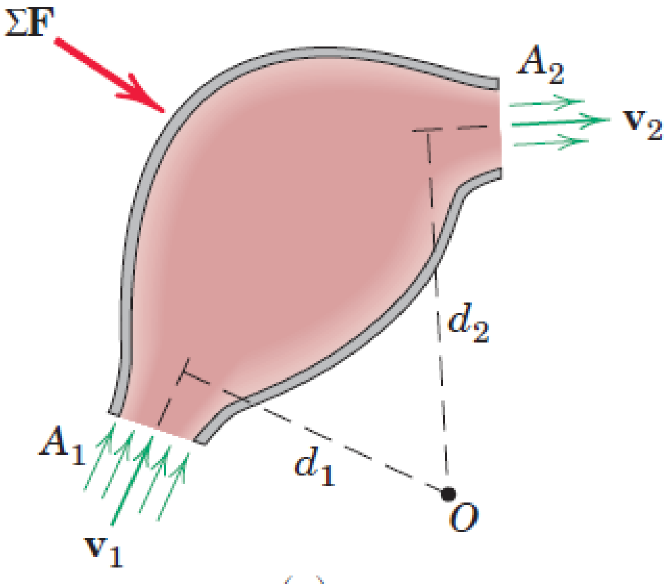
Time rate of change of linear momentum is the vector difference between the rate at which linear momentum leaves the system and the rate at which linear momentum enters the system



(b)

Angular Momentum in Steady Mass Flow System

- The resultant moment of all external forces about some fixed point O (which can on or off the system) equals the time rate of change of angular momentum of the system about O



$$\sum M_0 = m'(v_2 d_2 - v_1 d_1)$$

$$\sum \boldsymbol{M}_0 = m'(\boldsymbol{d}_2 \times \boldsymbol{v}_2 - \boldsymbol{d}_1 \times \boldsymbol{v}_1)$$

THANK YOU

Dynamics of Particles

Dr. Amrita Puri
Assistant Professor
Mechanical Department

Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

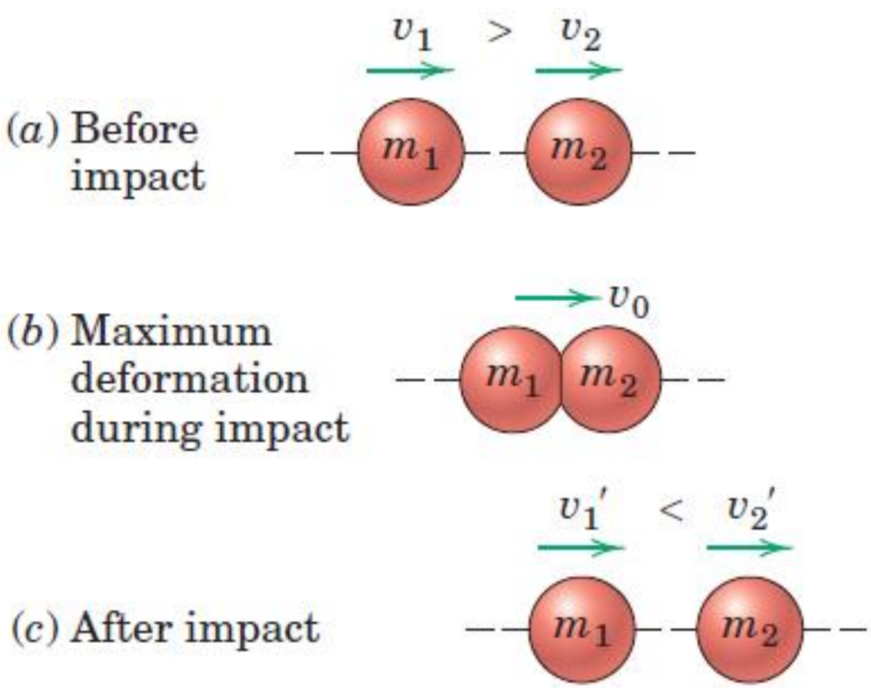
Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Impact

Impact refers to the collision between two bodies and is characterized by the generation of relatively large contact forces which act over a very short interval of time

Direct Central Impact

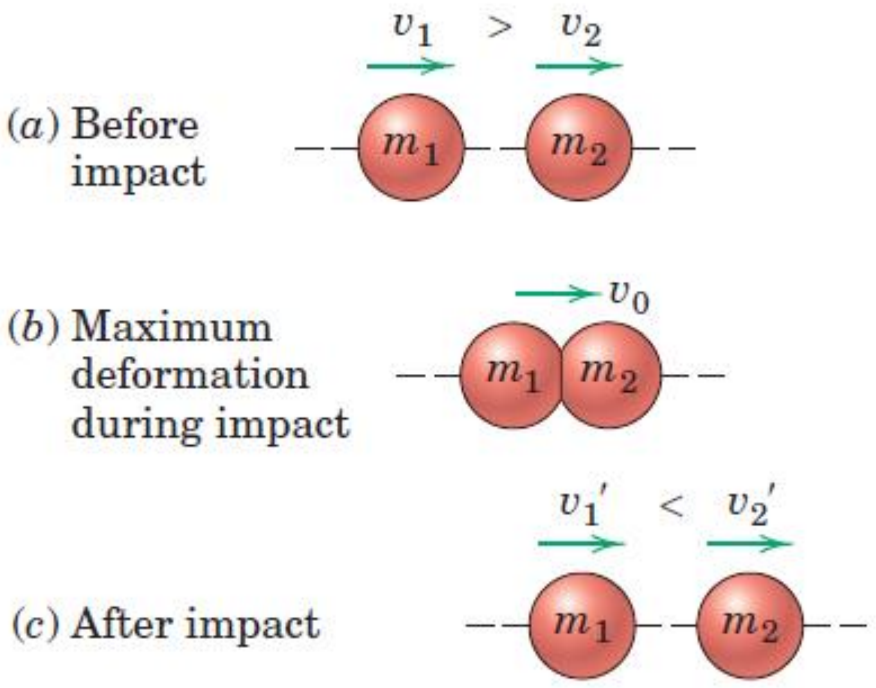
- Collinear motion of two spheres of masses m_1 and m_2
- v1 is greater than v2, collision occurs with the contact forces directed along the line of centers
- Following initial contact, a short period of increasing deformation takes place until the contact area between the spheres ceases to increase.



Impact

Direct Central Impact

- Following initial contact, a short period of increasing deformation takes place until the contact area between the spheres ceases to increase. At this instant, both spheres, are moving with the same velocity v_0.
- During the remainder of contact, a period of restoration occurs during which the contact area decreases to zero. In the final condition, the spheres now have new velocities v_1', v_2'.



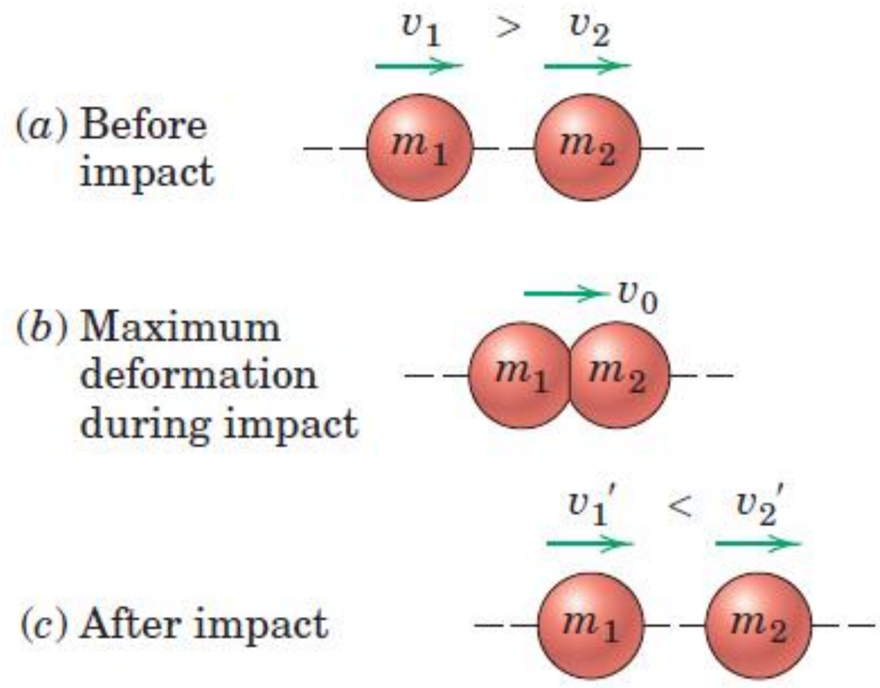
Impact

Direct Central Impact

- Because the contact forces are equal and opposite during impact, the linear momentum of the system remains unchanged

Law of conservation of linear momentum

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$



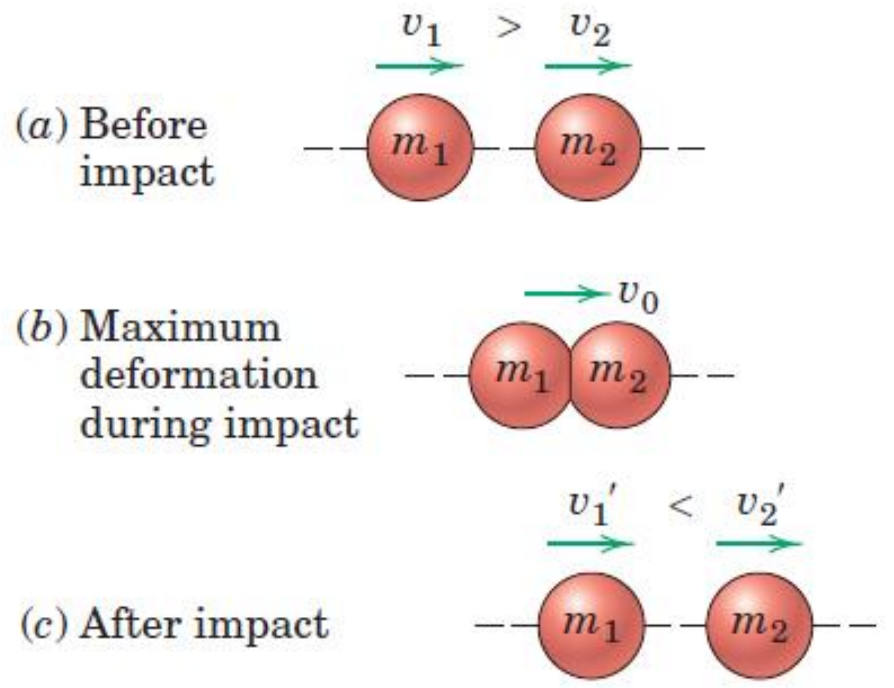
Coefficient of Restitution

Law of conservation of linear momentum

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$

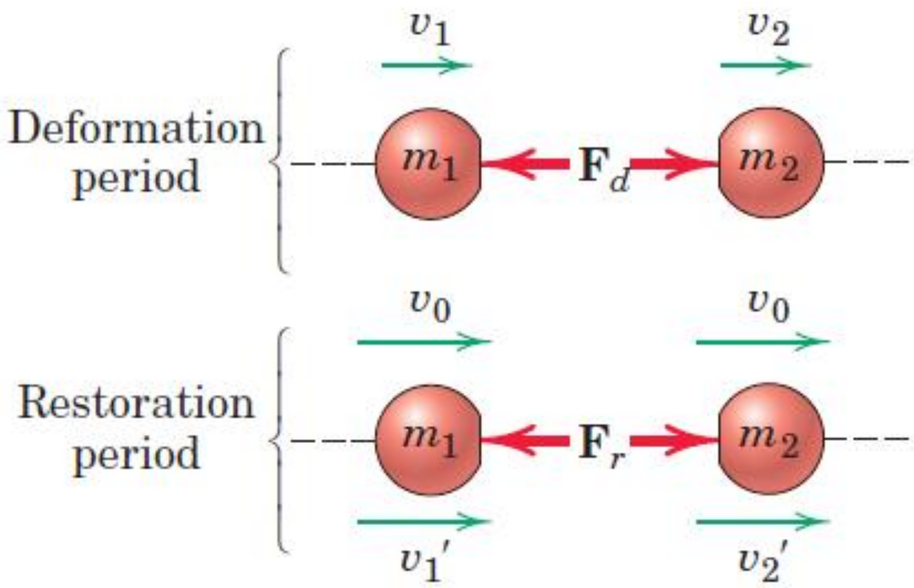
Coefficient of restitution: Ratio e of the magnitude of the restoration impulse to the magnitude of the deformation impulse.

F_r and F_d are magnitudes of the contact forces during restoration and deformation periods.



Coefficient of Restitution

Coefficient of restitution: Ratio e of the magnitude of the restoration impulse to the magnitude of the deformation impulse.



F_r and F_d are magnitudes of the contact forces during restoration and deformation periods.

$$e = \frac{\int_{t_0}^{t} F_r \, dt}{\int_{0}^{t_0} F_d \, dt} = \frac{m_1[-v_1' - (-v_0)]}{m_1[-v_0 - (-v_1)]} = \frac{v_0 - v_1'}{v_1 - v_0}$$

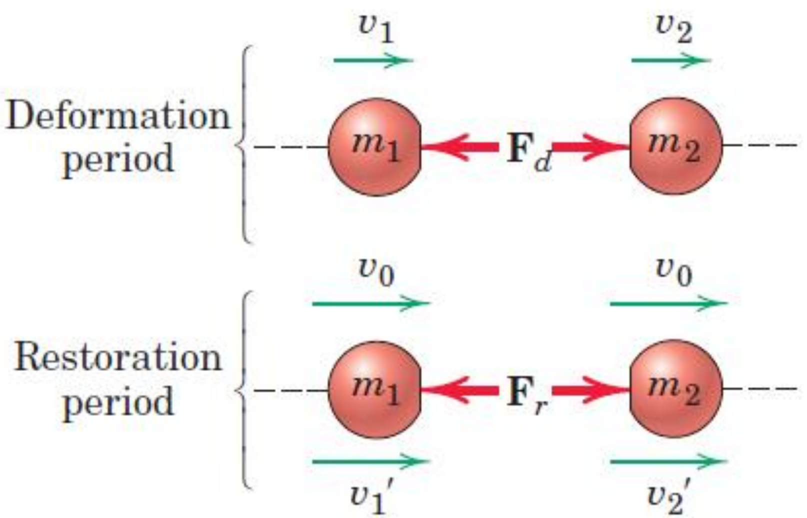
Coefficient of Restitution

For Particle 1

$$e = \frac{\int_{t_0}^{t} F_r \, dt}{\int_0^{t_0} F_d \;\, dt} = \frac{m_1[-v_1' - (-v_0)]}{m_1[-v_0 - (-v_1)]} = \frac{v_0 - v_1'}{v_1 - v_0}$$

For Particle 2

$$e = \frac{\int_{t_0}^{t} F_r \, dt}{\int_0^{t_0} F_d \;\, dt} = \frac{m_2[v_2' - (v_0)]}{m_2[v_0 - (v_2)]} = \frac{v_2' - v_0}{v_0 - v_2}$$



Express the change of momentum (and therefore v) in the same direction as the impulse

Coefficient of Restitution

For Particle 1

$$e = \frac{\int_{t_0}^{t} F_r \, dt}{\int_{0}^{t_0} F_d \, dt} = \frac{m_1[-v_1' - (-v_0)]}{m_1[-v_0 - (-v_1)]} = \frac{v_0 - v_1'}{v_1 - v_0}$$

For Particle 2

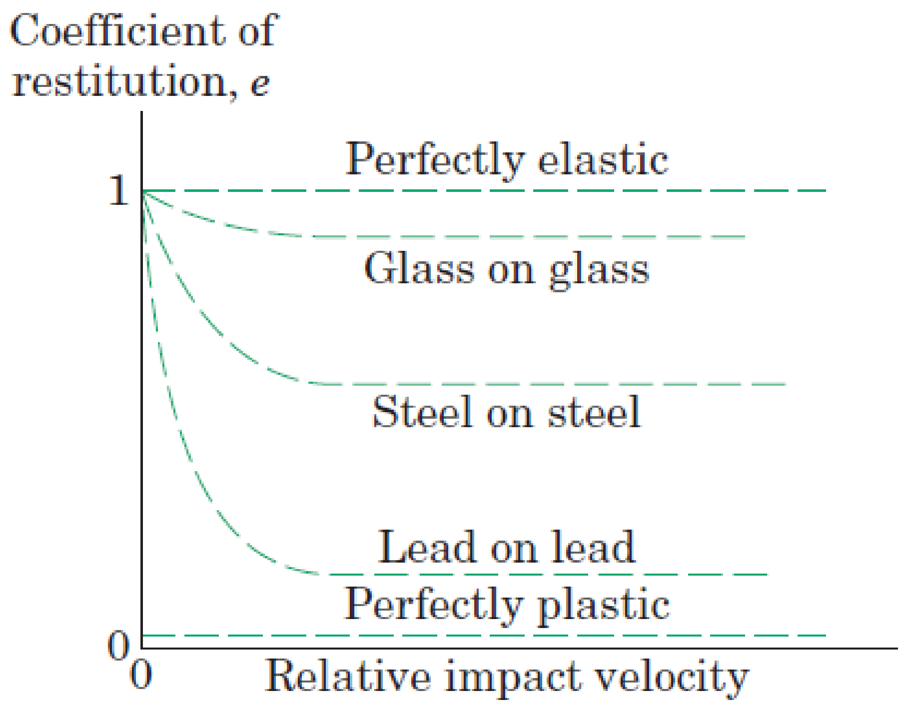
$$e = \frac{\int_{t_0}^{t} F_r \, dt}{\int_{0}^{t_0} F_d \, dt} = \frac{m_2[v_2' - (v_0)]}{m_2[v_0 - (v_2)]} = \frac{v_2' - v_0}{v_0 - v_2}$$

$$\frac{v_0 - v_1'}{v_1 - v_0} = \frac{v_2' - v_0}{v_0 - v_2}$$

$$e = \frac{v_2' - v_1'}{v_1 - v_2} = \frac{|relative\ velocity\ of\ separation|}{|relative\ velocity\ of\ approach|}$$

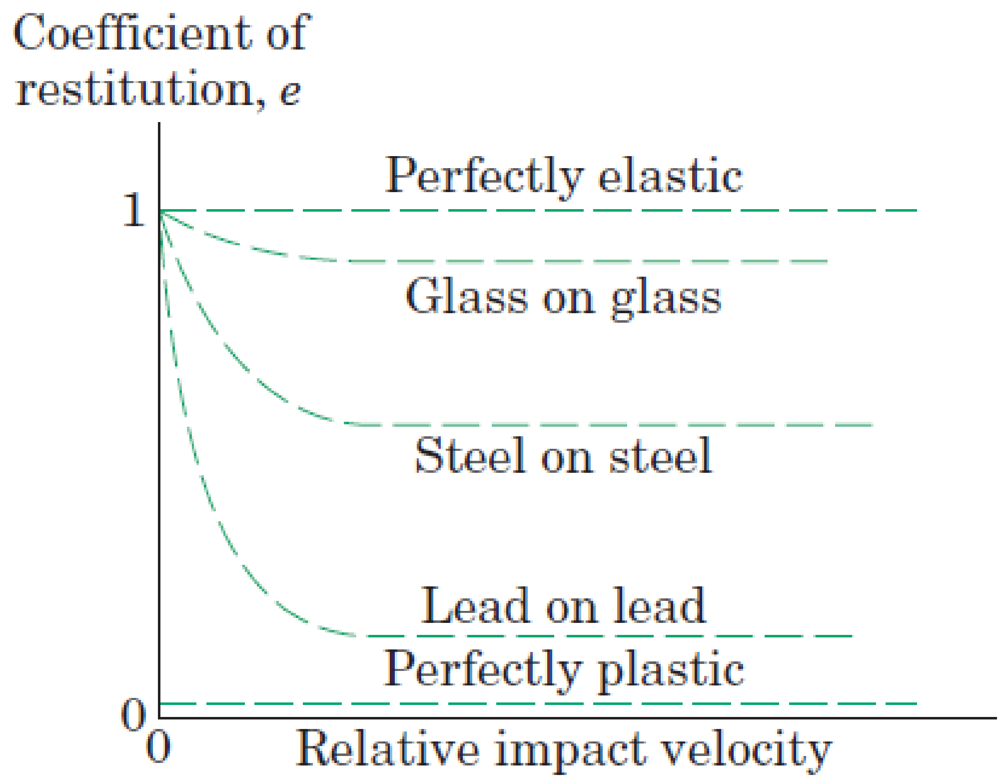
Energy Loss During Impact

- Energy loss =KE before impact-KE after impact
- Value e=1 means that the capacity of the two particles to recover equals their tendency to deform. This condition is one of *elastic impact* with no energy loss.
- Value e=0 describes *inelastic* or *plastic impact* where the particles cling together after collision and the loss of energy is a maximum.
- All impact conditions lie somewhere between these two extremes.

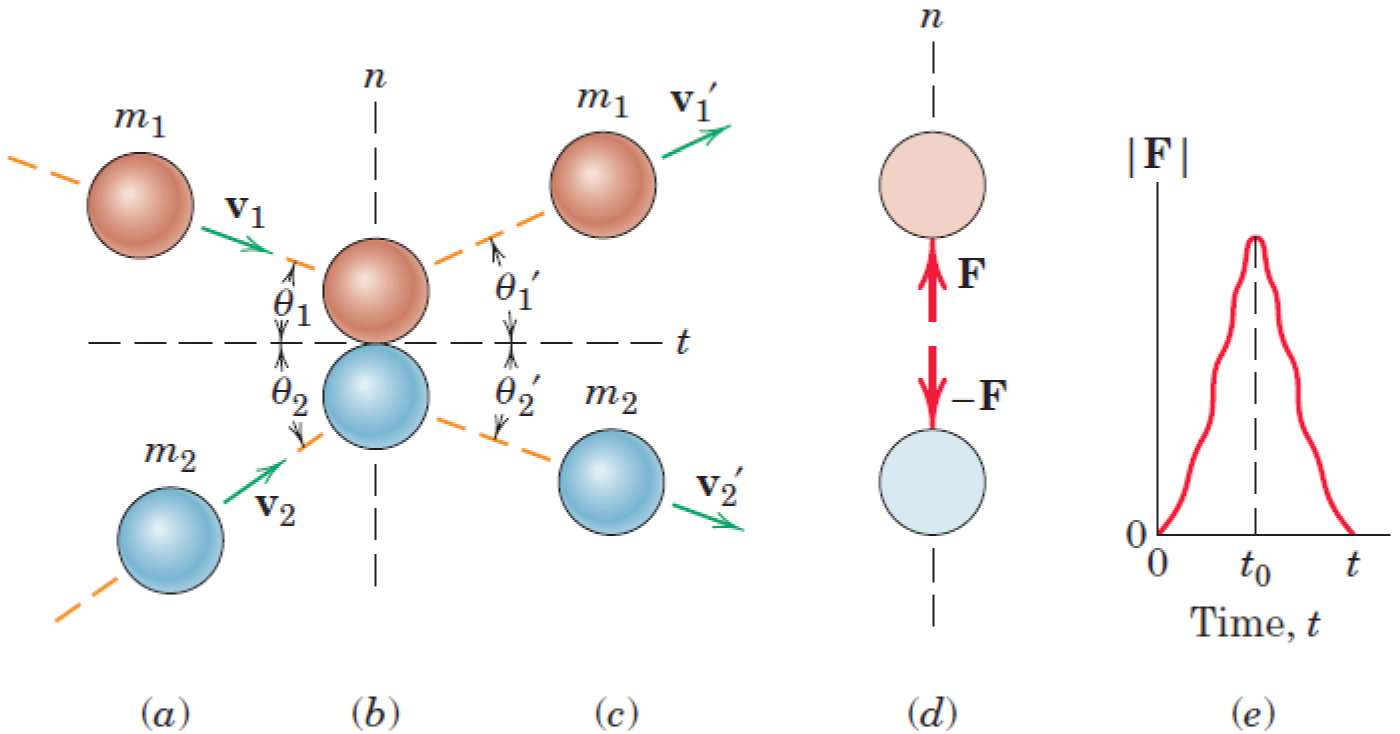


Energy Loss During Impact

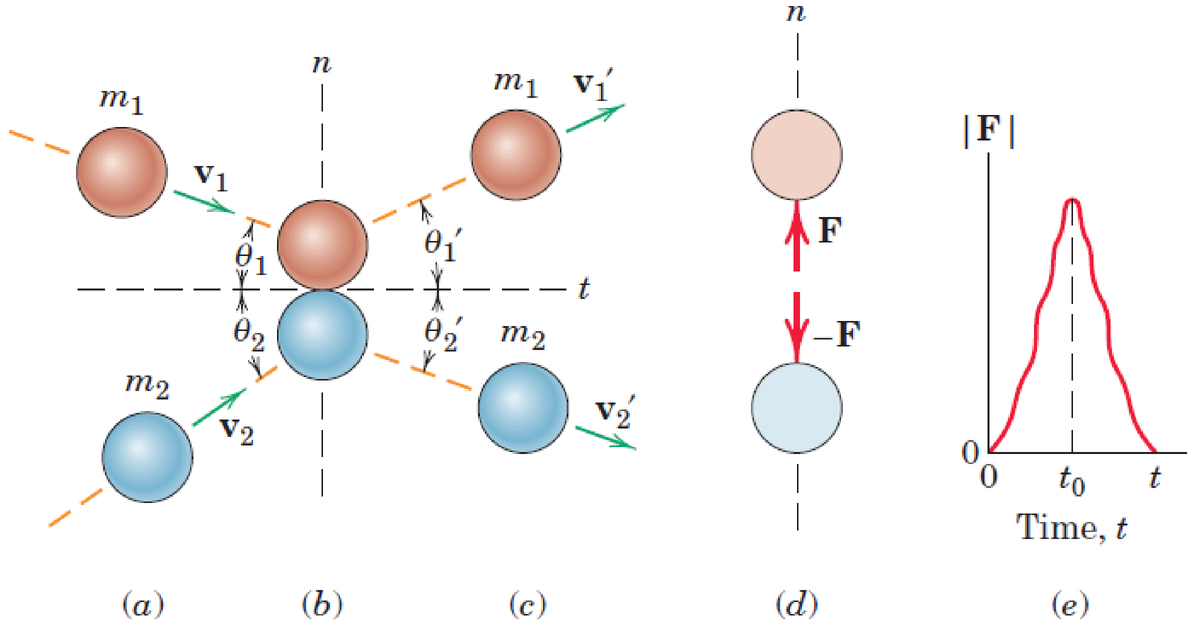
- Coefficient of restitution is associated with a pair of contacting bodies.
- Coefficient of restitution depends upon given geometries and a given combination of contacting materials, and also on the impact velocity
- It approaches unity as the impact velocity approaches zero



Oblique Central Impact

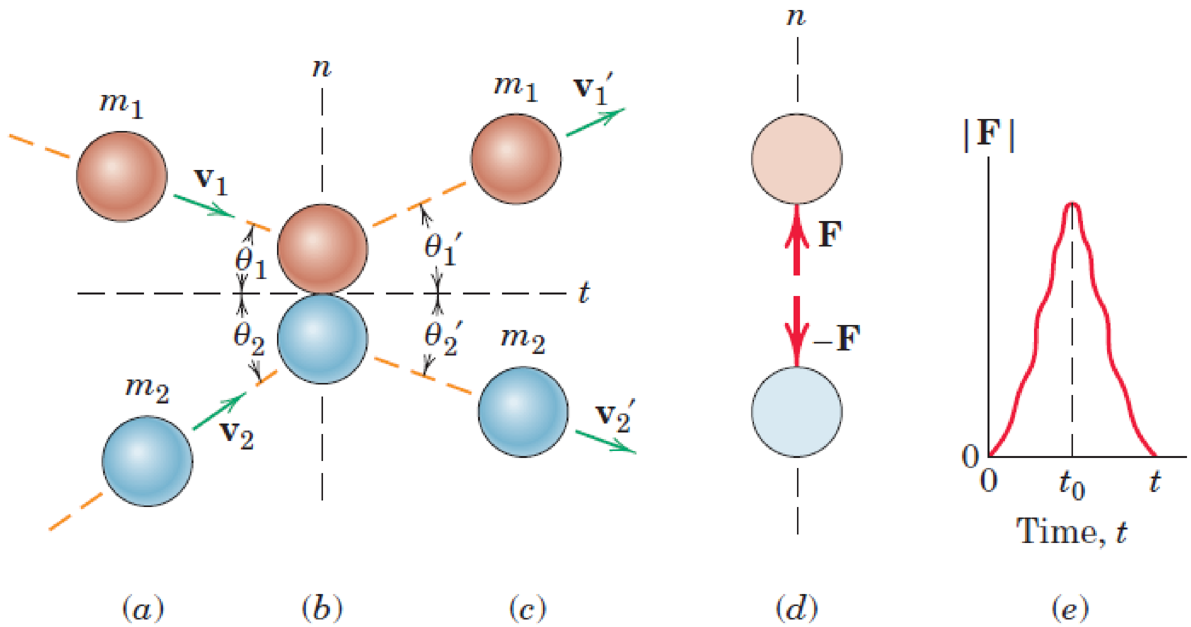


Oblique Central Impact



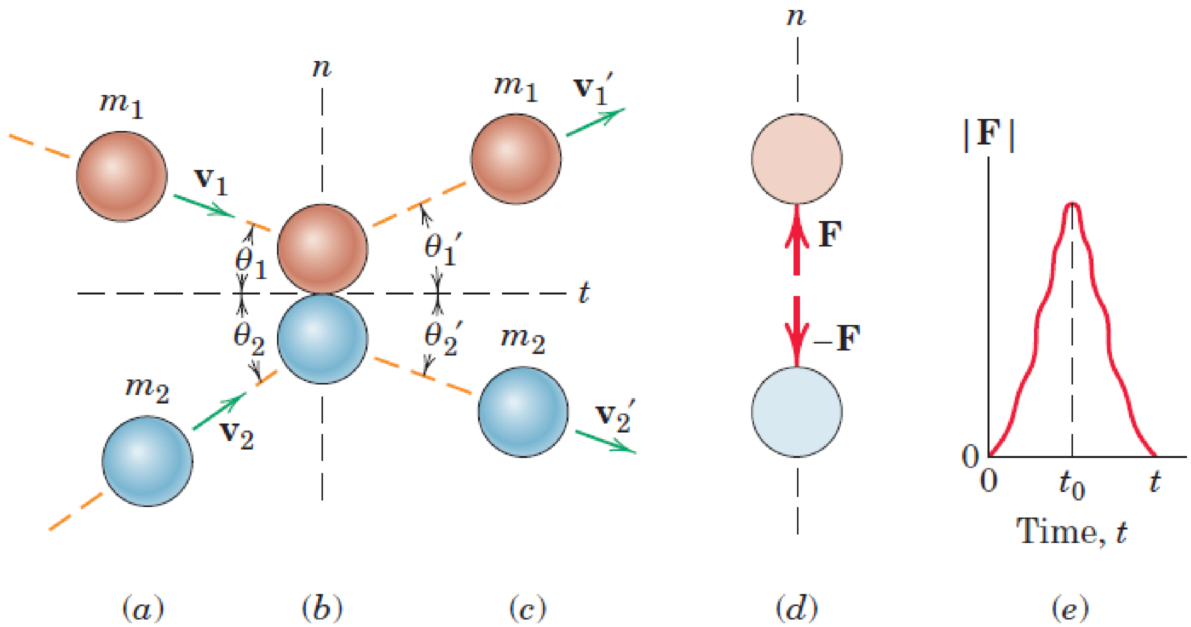
- Spherical particles of mass m_1 and m_2 have initial velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ in the same plane and approach each other on a collision. The directions of the velocity vectors are measured from the direction tangent to the contacting surfaces

Oblique Central Impact



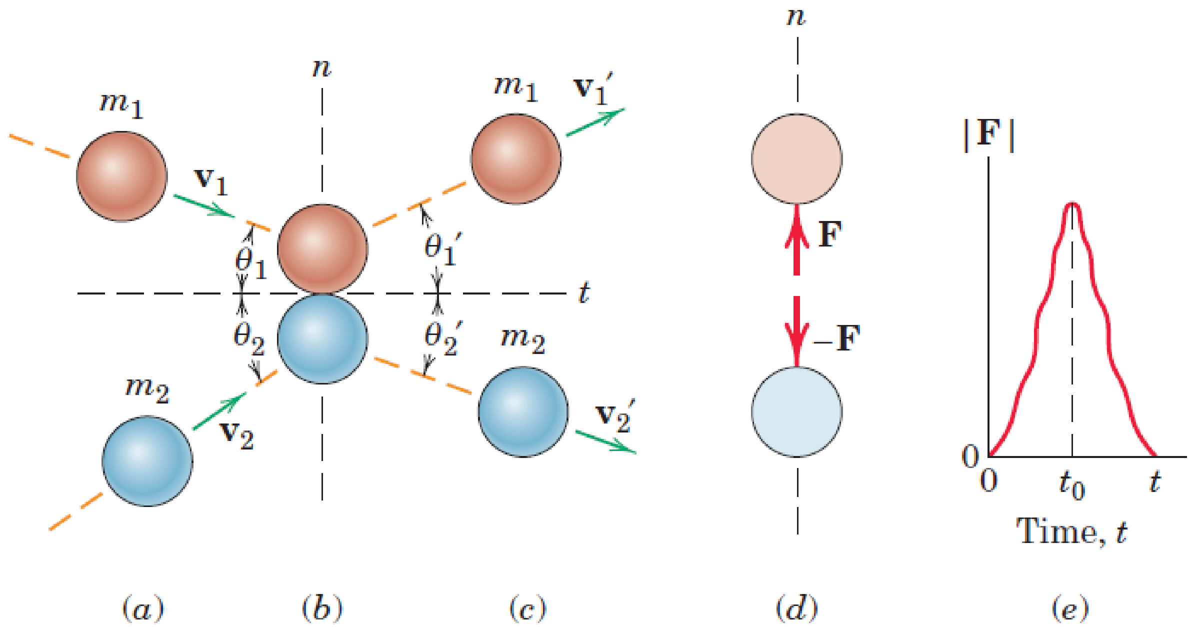
- The impact forces are **F** and **-F**
- They vary from zero to their peak value during the deformation portion of the impact and back again to zero during the restoration period

Oblique Central Impact



- Momentum of the system is conserved in the *n*-direction
- The momentum for each particle is conserved in the *t*-direction since there is no impulse on either particle in the *t*-direction
- The coefficient of restitution, as in the case of direct central impact, is the positive ratio of the recovery impulse to the deformation impulse

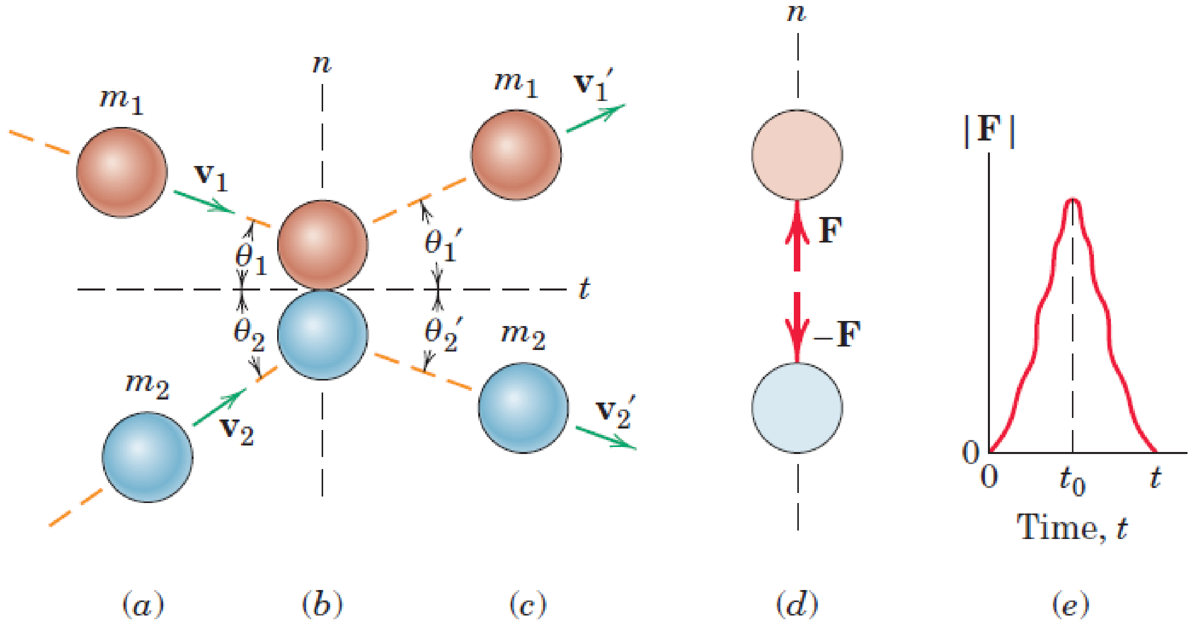
Oblique Central Impact



➢ Momentum of the system is conserved in the *n*-direction

$$m_1 v_{1n} + m_2 v_{2n} = m_1 v'_{1n} + m_2 v'_{2n}$$

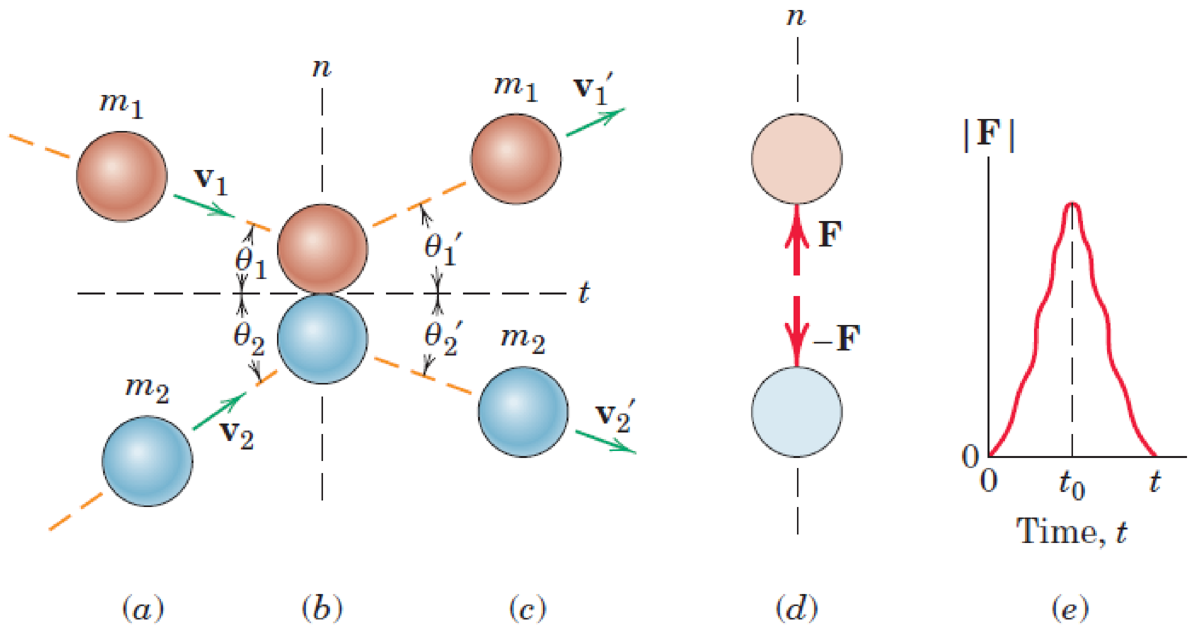
Oblique Central Impact



➢ The momentum for each particle is conserved in the *t*-direction since there is no impulse on either particle in the *t*-direction

$$m_1 v_{1t} = m_1 v'_{1t} \qquad\qquad m_2 v_{2t} = m_2 v'_{2t}$$

Oblique Central Impact



- The coefficient of restitution, as in the case of direct central impact, is the positive ratio of the recovery impulse to the deformation impulse

$$e = \frac{v_{2n}' - v_{1n}'}{v_{1n} - v_{2n}}$$

Relative Motion

x-y-z are parallel to X-Y-Z

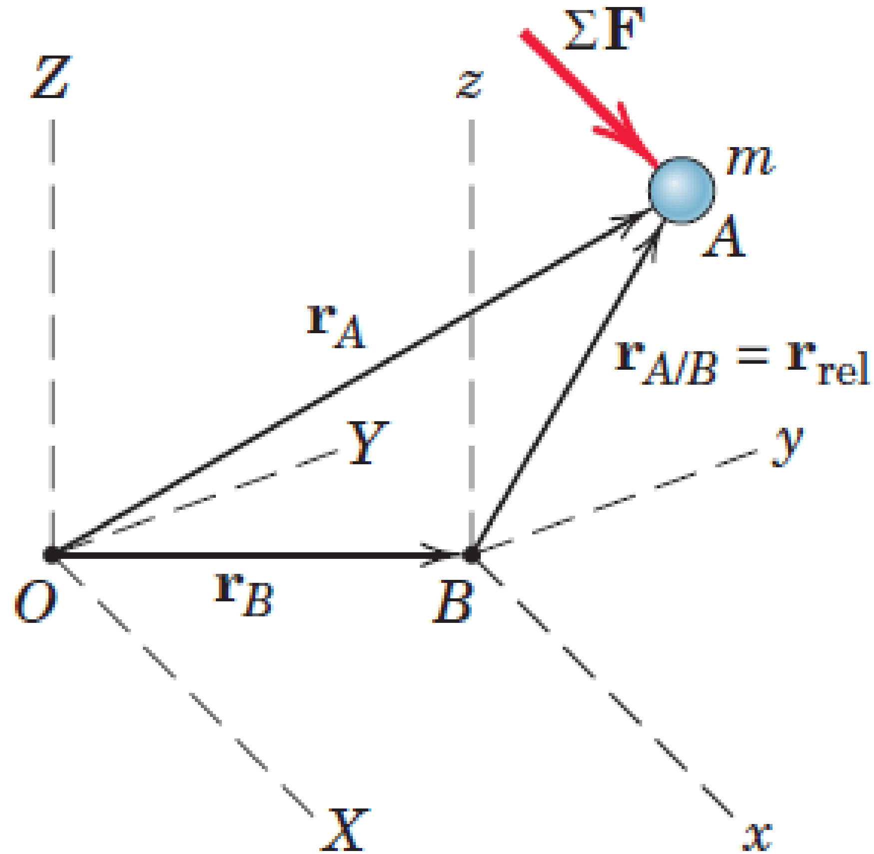
Point B is accelerating with a_B

$$a_{A/B} = \ddot{r}_{A/B}$$

$$a_A = a_B + a_{A/B}$$

Newton's second law of motion

$$\sum F = ma_A$$



D'Alembert's Principle

Newton's second law of motion $$\sum F = ma_A$$

- The system x-y-z attached to the particle, the particle necessarily appears to be at rest or in equilibrium in x-y-z
- The observer who is accelerating with x-y-z concludes that a force $-m\boldsymbol{a}$ acts on the particle to balance $\sum \boldsymbol{F}$

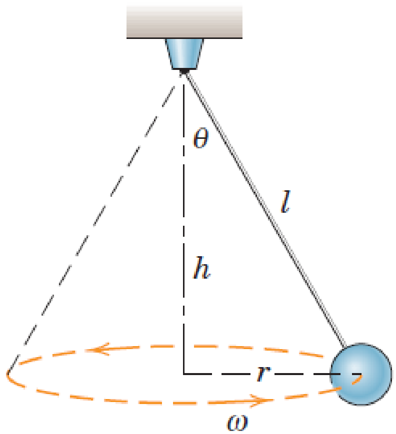
$$\sum F - ma_A = 0$$

This fictitious force is known as the ***inertia force***, and the artificial state of equilibrium created is known as ***dynamic equilibrium***.

The apparent transformation of a problem in dynamics to one in statics has become known as ***D'Alembert's principle***

D'Alembert's Principle

$$\sum F = ma_n$$

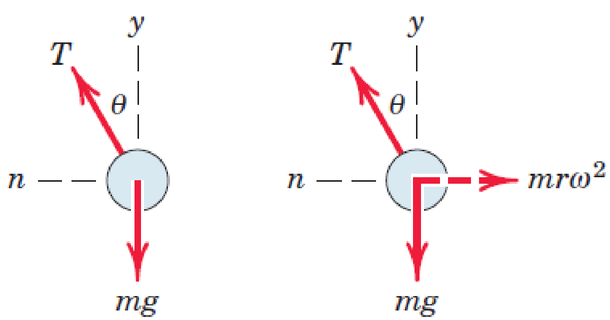


(a)

$$T sin\theta = mr\omega^2 \qquad T cos\theta = mg$$

Adding $mr\omega^2$ opposite to the direction of acceleration

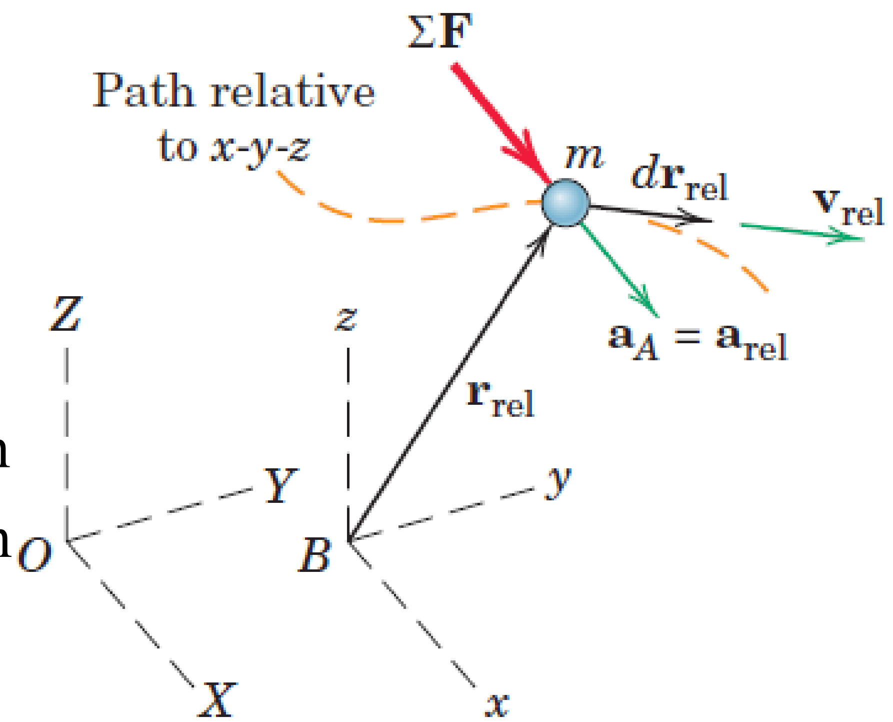
$$T sin\theta - mr\omega^2 = 0$$



(b) (c)

Constant velocity Non-rotating system

$$\sum \boldsymbol{F} = m\boldsymbol{a} = m\boldsymbol{a_{rel}}$$



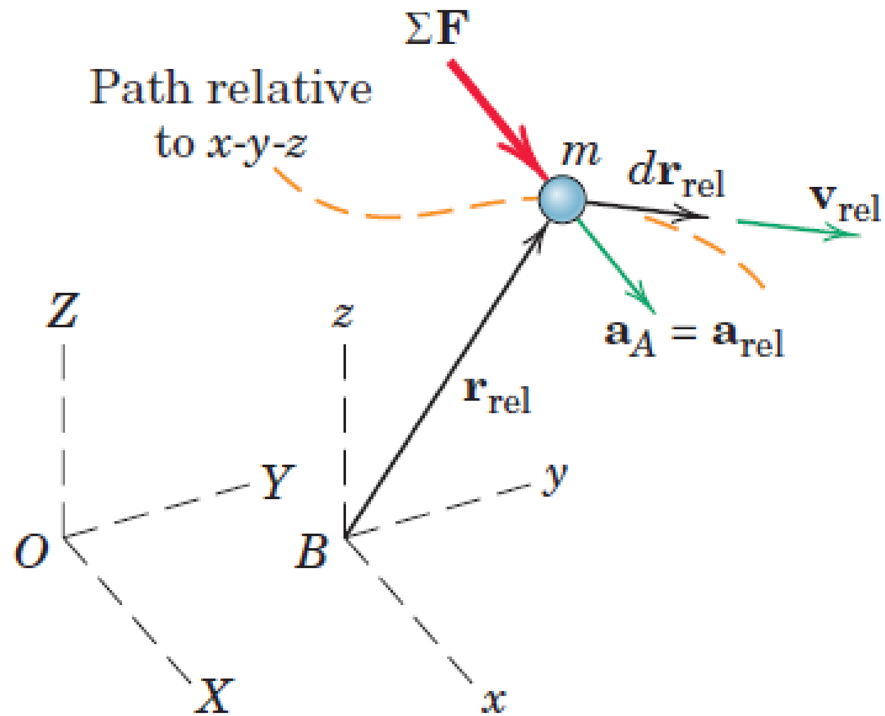
Validity of the work-energy equation and the impulse-momentum equation relative to a constant-velocity, nonrotating system

Work-Energy Equation in Constant velocity Non-rotating system

Work done by $\sum F$ relative to *x-y-z*

$$dU_{rel} = \sum \boldsymbol{F} . \boldsymbol{dr_{rel}}$$

$$= m\boldsymbol{a}. \boldsymbol{dr_{rel}} = m\boldsymbol{a}_{rel}. \boldsymbol{dr_{rel}}$$



$$\boldsymbol{a}_{rel}. \boldsymbol{dr_{rel}} = \boldsymbol{v_{rel}}. \boldsymbol{dv_{rel}}$$

$$dU_{rel} = m\boldsymbol{a}_{rel}. \boldsymbol{dr_{rel}}$$
$$= m\boldsymbol{v_{rel}}. \boldsymbol{dv_{rel}} = mv_{rel}dv_{rel}$$
$$= d\left(\frac{1}{2}mv_{rel}^2\right) = dT_{rel}$$

Work-Energy Equation in Constant velocity Non-rotating system

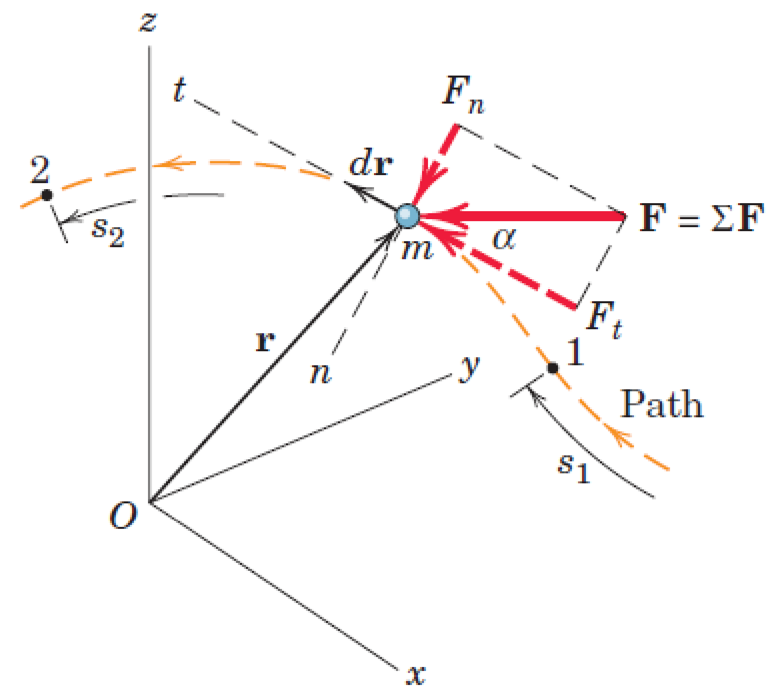
Work done by $\sum F$ relative to *x*-*y*-*z*

$$dU_{rel} = m\boldsymbol{a}_{rel}.\boldsymbol{dr_{rel}} = m\boldsymbol{v_{rel}}.\boldsymbol{dv_{rel}} = mv_{rel}dv_{rel}$$
$$= d\left(\frac{1}{2}mv_{rel}^2\right) = dT_{rel}$$

$$\boldsymbol{a} \cdot d\boldsymbol{r} = a_t ds$$

$$a_t ds = vdv$$

$$a_{t\_rel} ds = v_{rel} dv_{rel}$$



Work-Energy Equation in Constant velocity Non-rotating system

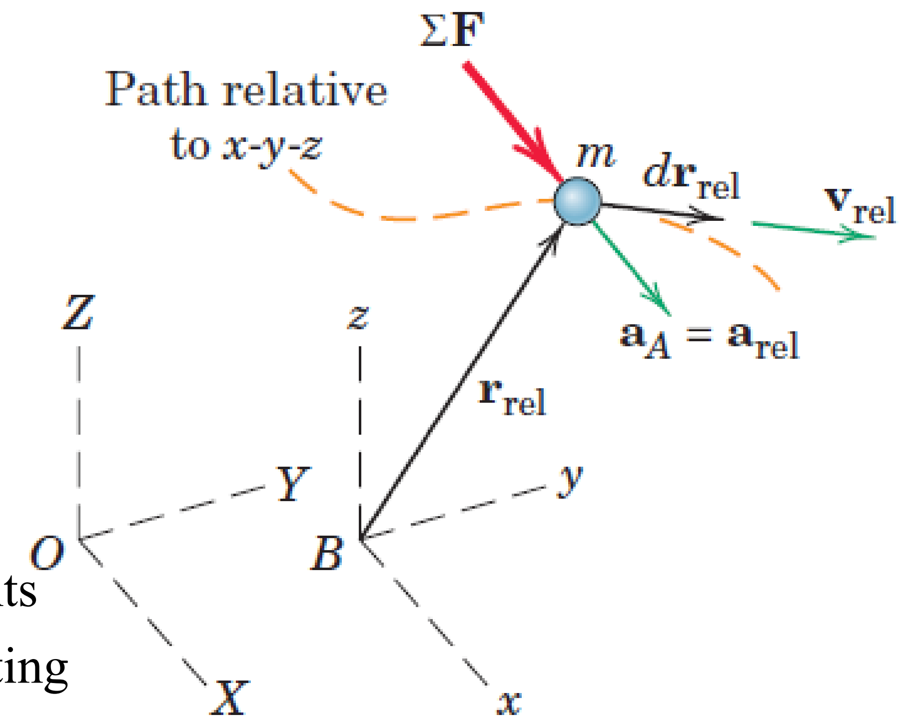
Work done by $\sum F$ relative to *x-y-z*

$$dU_{rel} = m\boldsymbol{a}_{rel}.\boldsymbol{dr}_{\boldsymbol{rel}} = m\boldsymbol{v}_{\boldsymbol{rel}}.\boldsymbol{dv}_{\boldsymbol{rel}} = mv_{rel}dv_{rel}$$

$$= d\left(\frac{1}{2}mv_{rel}^2\right) = dT_{rel}$$

$$dU_{rel} = dT_{rel}$$

$$U_{rel} = \Delta T_{rel}$$



Work-energy equation holds for measurements made relative to a constant-velocity, nonrotating system

Impulse-momentum equation in Constant velocity Non-rotating system

$$\sum \boldsymbol{F} = m\boldsymbol{a} = m\boldsymbol{a_{rel}} \qquad \sum \boldsymbol{F}\,dt = m\boldsymbol{a}dt = m\boldsymbol{a_{rel}}dt$$

$$\sum \boldsymbol{F}\,dt = m\boldsymbol{a_{rel}}dt = md\boldsymbol{v}_{rel} = d(m\boldsymbol{v}_{rel})$$

$$\sum \boldsymbol{F}\,dt = d(m\boldsymbol{v}_{rel}) = d(G_{rel})$$

$$\sum \boldsymbol{F} = (\dot{G}_{rel}) \qquad \int \sum \boldsymbol{F}\,dt = \Delta\mathbf{G}_{\mathrm{rel}}$$

Impulse-momentum equations for a fixed reference system also hold for measurements made relative to a constant-velocity, nonrotating system

Moment-Angular Momentum Equation in Constant velocity Non-rotating system

Relative angular momentum of the particle about a point in *x*-*y*-*z*, such as the origin *B*, as the moment of the relative linear momentum

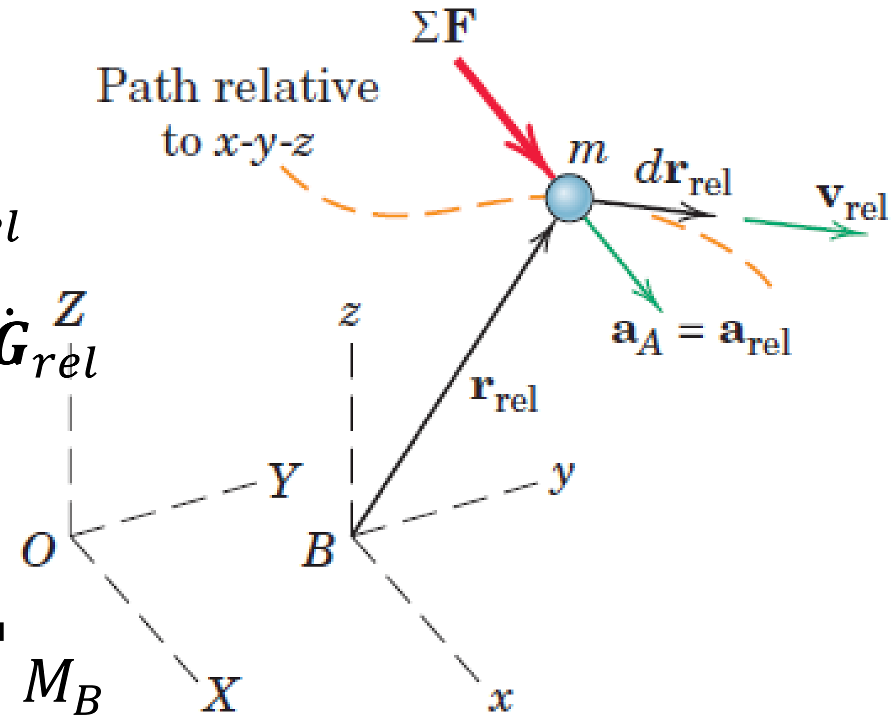
$$(\boldsymbol{H}_B)_{rel} = \boldsymbol{r}_{rel} \times \boldsymbol{G}_{rel}$$

$$\left(\dot{\boldsymbol{H}}_B\right)_{rel} = \dot{\boldsymbol{r}}_{rel} \times \boldsymbol{G}_{rel} + \boldsymbol{r}_{rel} \times \dot{\boldsymbol{G}}_{rel}$$

$$\left(\dot{\boldsymbol{H}}_B\right)_{rel} = \boldsymbol{v}_{rel} \times m\boldsymbol{v}_{rel} + \boldsymbol{r}_{rel} \times \dot{\boldsymbol{G}}_{rel}$$

$$\left(\dot{\boldsymbol{H}}_B\right)_{rel} = 0 + \boldsymbol{r}_{rel} \times \dot{\boldsymbol{G}}_{rel}$$

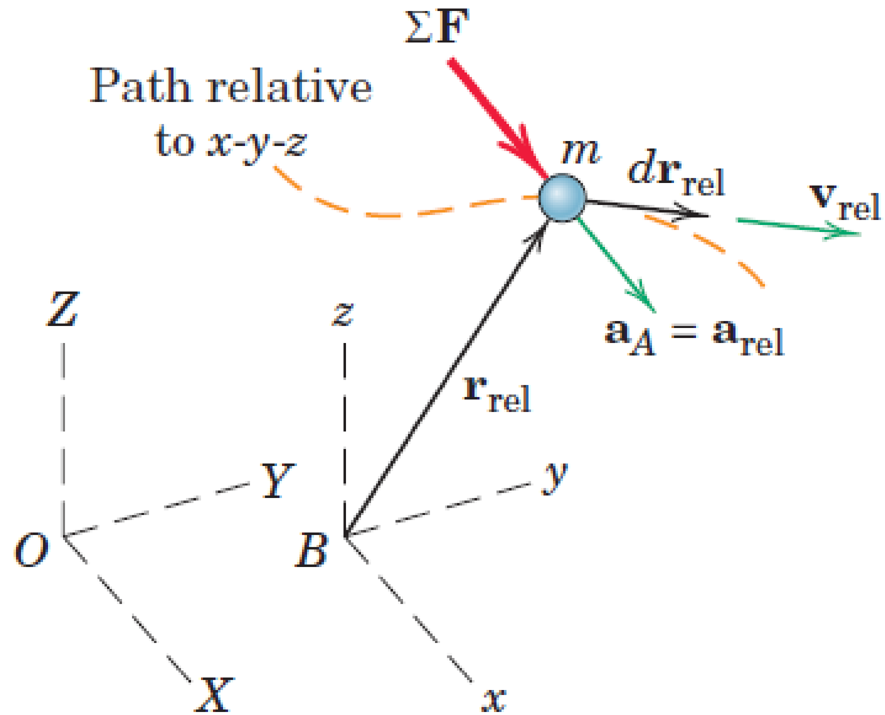
$$\left(\dot{\boldsymbol{H}}_B\right)_{rel} = 0 + \boldsymbol{r}_{rel} \times \sum F = \sum M_B$$



Moment-Angular Momentum Equation in Constant velocity Non-rotating system

Rate of change of angular momentum about point B is equal to sum of the moments about B of all forces on m

$$\left(\dot{\boldsymbol{H}}_B\right)_{rel} = \sum M_B$$



Moment-angular momentum relation holds with respect to a constant-velocity, nonrotating system

Difference between Absolute and Relative Quantities

Although the work-energy and impulse-momentum equations hold relative to a system translating with a constant velocity, the individual expressions for work, kinetic energy, and momentum differ between the fixed and the moving systems

$$\boldsymbol{G} = m v_{\boldsymbol{A}} \qquad \neq \boldsymbol{G}_{rel} = m\boldsymbol{v}_{rel}$$

$$dU = \sum \boldsymbol{F} \,.\, \boldsymbol{dr} \quad \neq \quad \boldsymbol{d}U_{rel} = \sum \boldsymbol{F} \,.\, \boldsymbol{dr}_{rel}$$

$$T = \frac{1}{2} m v^2 \qquad \neq \quad T_{rel} = \frac{1}{2} m v_{rel}^2$$

Central Force Motion

When a particle **moves under the influence of a force directed toward a fixed center of attraction**, the motion is called CENTRAL FORCE MOTION

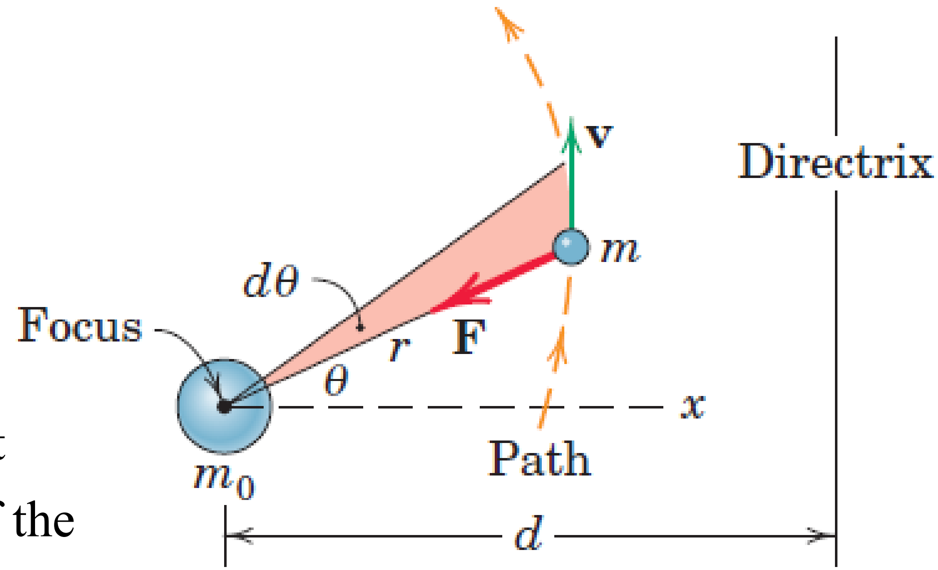
Examples: Orbital movement of planets and satellites

The laws which govern this motion were deduced from observation of the motions of the planets by J. Kepler (1571–1630).

An understanding of central-force motion is required to design high-altitude rockets, earth satellites, and space vehicles

Central Force Motion

$$F = G\frac{mm_0}{r^2}$$



m_0 is the mass of the attracting body and it assumed to be fixed

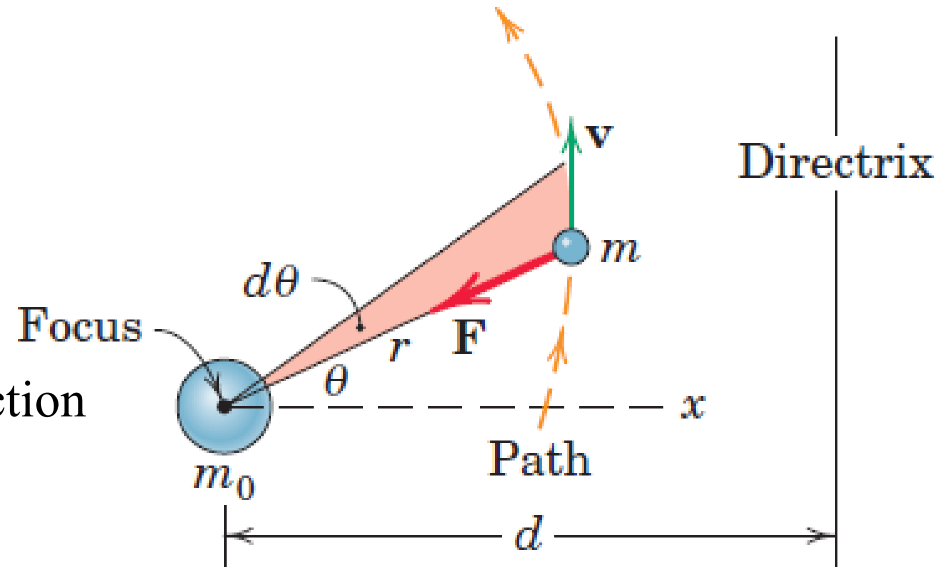
G is the universal gravitational constant

r is the distance between the centers of the masses

The particle of mass m could represent the **earth moving about the sun**, the **moon moving about the earth**, or a **satellite in its orbital motion about the earth above the atmosphere**

Central Force Motion

$$F = G\frac{mm_0}{r^2}$$



In polar coordinates (r-θ coordinates),

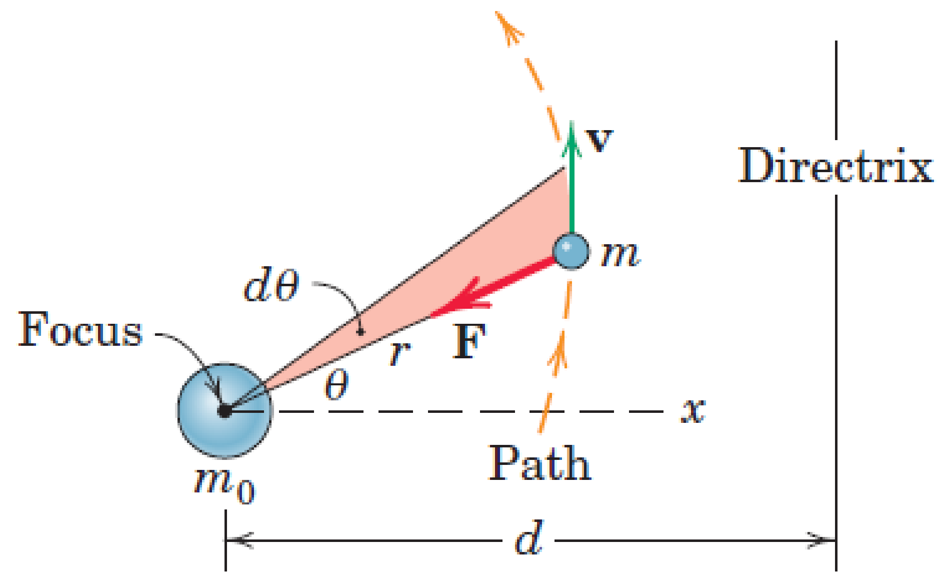
- F will always be in the negative r-direction
- There is no force in the θ-direction

$$-G\frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$0 = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

Central Force Motion

$$-G\frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$



$$0 = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

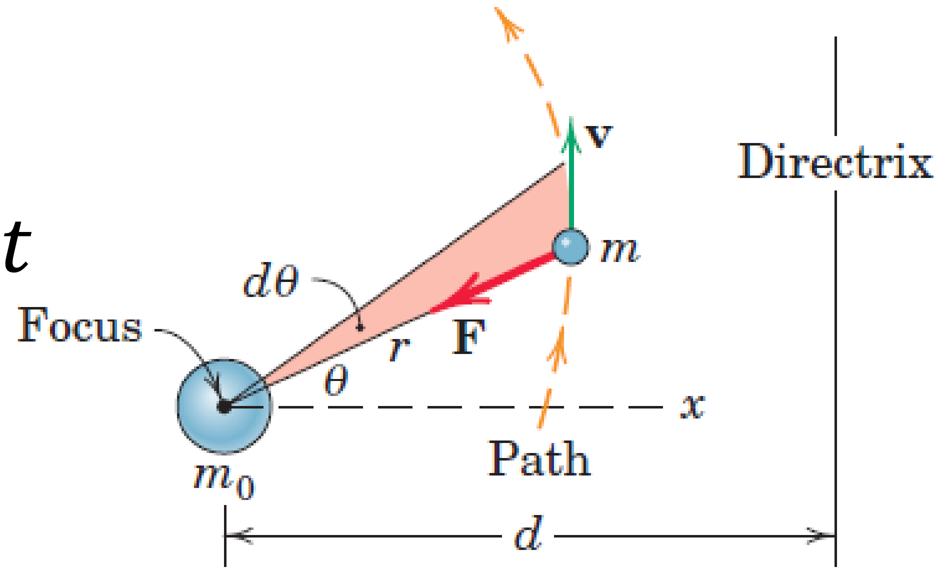
$$0 = \frac{d(r^2\dot{\theta})}{dt} = 2r\dot{r}\dot{\theta} + r^2\ddot{\theta}$$

$$r^2\dot{\theta} = h = constant$$

Central Force Motion

$$r^2\dot{\theta} = h = constant$$

$$mr^2\dot{\theta} = h_1 = constant$$



$$\mathbf{v} = \dot{r}\boldsymbol{e_r} + r\dot{\theta}\boldsymbol{e_\theta}$$

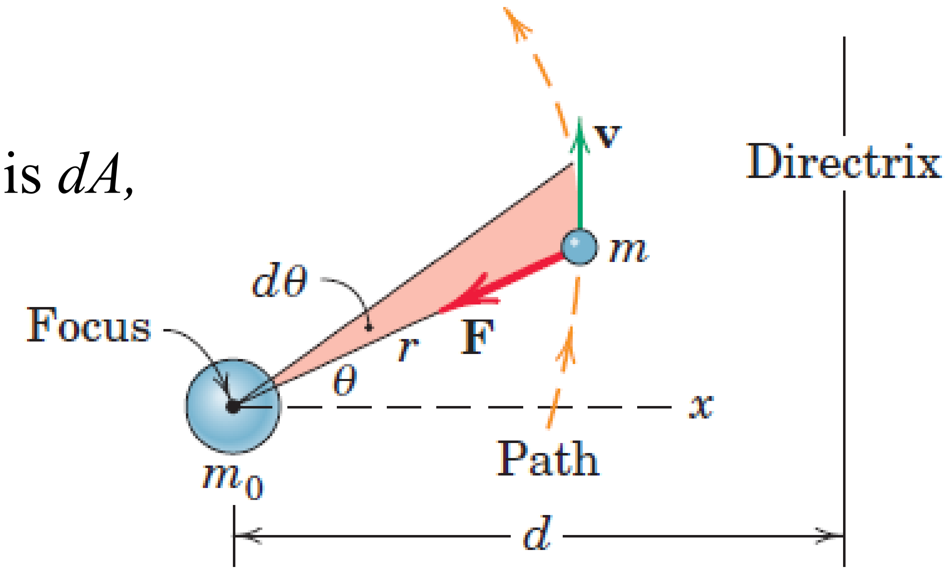
$$\boldsymbol{H} = \boldsymbol{r} \times m\mathbf{v}$$

$$|\boldsymbol{H}| = |\boldsymbol{r} \times m\mathbf{v}| = mr^2\dot{\theta} = h_1 = constant$$

Angular momentum **H** remains constant (is conserved) if there is no moment acting on the particle about a fixed point *O*.

Central Force Motion

In time *dt,* area swept by mass *m* is *dA,*



$$dA = \frac{1}{2} r\,(r d\theta) = \frac{1}{2} r^2 d\theta$$

$$\frac{dA}{dt} = \dot{A} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{1}{2} r^2 \dot{\theta} = constant$$

Kepler's *second law* of planetary motion, which states that the areas swept through in equal times are equal.

Central Force Motion

$$-G\frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$r = \frac{1}{u} \qquad \dot{r} = -\frac{1}{u^2}\dot{u}$$

$$\dot{r} = -r^2\dot{u}$$

$$\dot{r} = -h\frac{\dot{u}}{\dot{\theta}}$$

$$r^2\dot{\theta} = h = constant$$

$$\dot{r} = -h\frac{du}{d\theta}$$

$$\dot{r} = -h\frac{du}{d\theta}$$

$$\ddot{r} = -h\frac{d^2u}{d\theta^2}\dot{\theta} \qquad \ddot{r} = -\frac{h^2}{r^2}\frac{d^2u}{d\theta^2} \qquad \ddot{r} = -h^2u^2\frac{d^2u}{d\theta^2}$$

Using expression for $\ddot{r}$, r, $r^2\dot{\theta} = h$

$$-Gm_0u^2 = -h^2u^2\frac{d^2u}{d\theta^2} - \frac{1}{u}h^2u^4$$

Central Force Motion

$$-G\frac{mm_0}{r^2} = m(\ddot{r} - r\dot{\theta}^2)$$

$$-Gm_0u^2 = -h^2u^2\frac{d^2u}{d\theta^2} - \frac{1}{u}h^2u^4$$

$$\frac{d^2u}{d\theta^2} + u = \frac{Gm_0}{h^2}$$

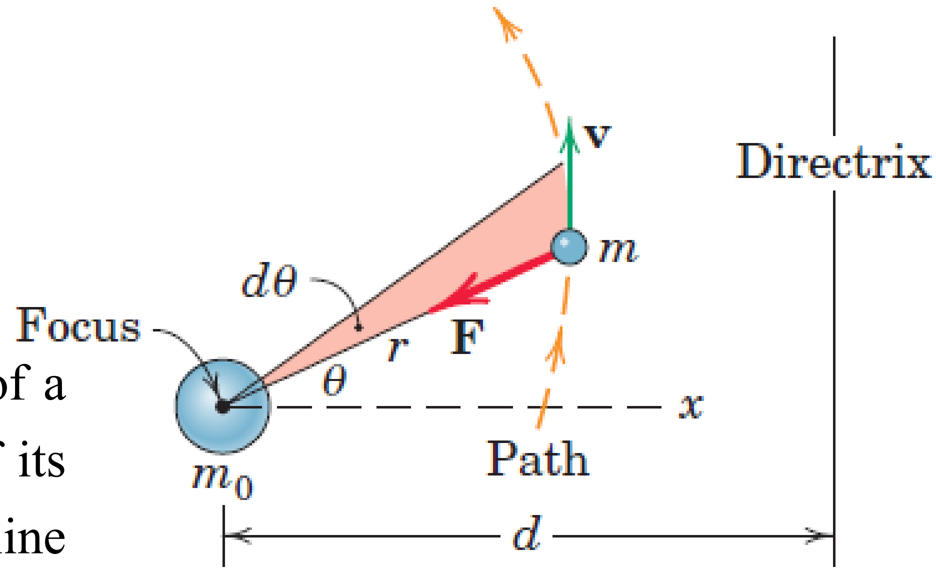
$$u = \frac{1}{r} = C\cos(\theta + \delta) + \frac{Gm_0}{h^2}$$

$$u = C\cos(\theta) + \frac{Gm_0}{h^2}$$

$\delta = 0$ if x-axis is chosen such that *r* is minimum when θ is zero.

Central Force Motion

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$



A conic section is formed by the locus of a point which moves so that the ratio *e* of its distance from a point (focus) to a line (directrix) is constant

$$e = \frac{r}{d - r\cos\theta} \qquad \frac{ed - er\cos\theta}{r} = \frac{ed}{r} - e\cos\theta = 1$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{\cos\theta}{d} \qquad \frac{1}{r} = \frac{e\cos\theta + 1}{ed}$$

Central Force Motion

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

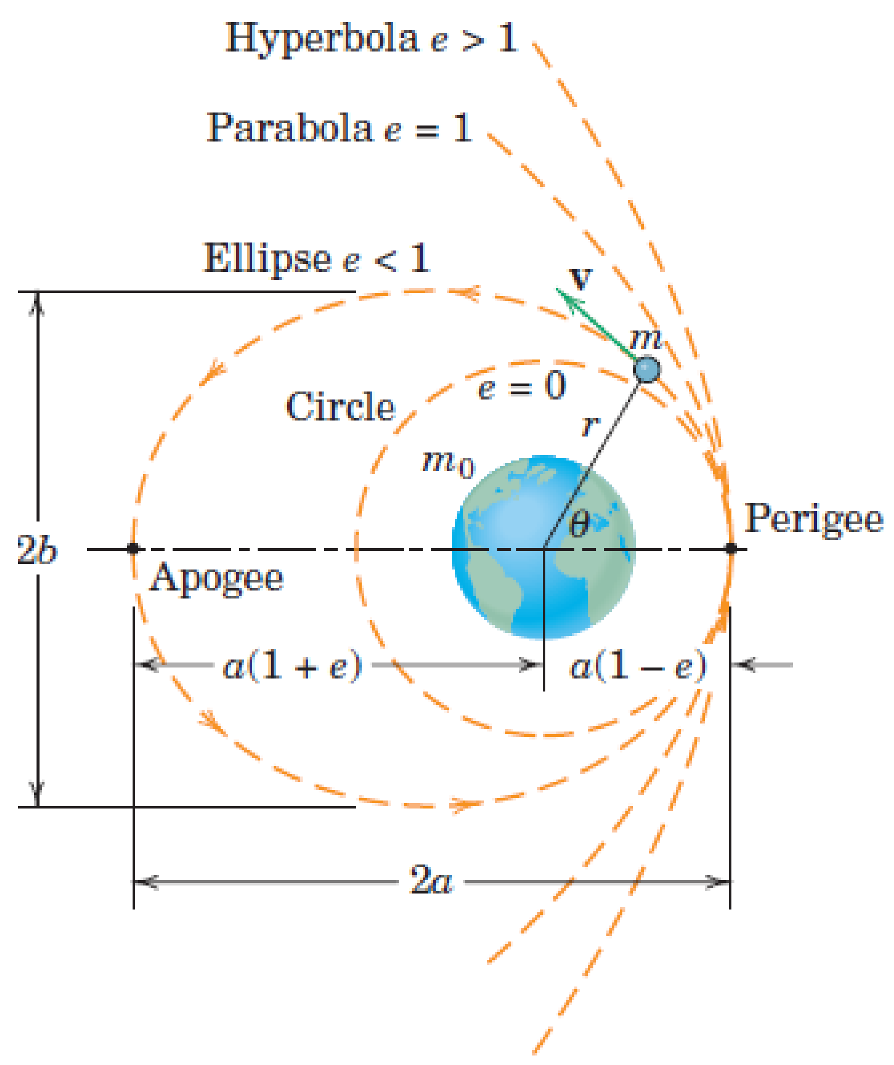
$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

$$C = \frac{1}{d} \qquad ed = \frac{h^2}{Gm_0}$$

$$C = \frac{1}{d} \qquad e = \frac{h^2 C}{Gm_0}$$

$e < 1$ Ellipse $\qquad e = 0$ Parabola

$e > 1$ Hyperbola



Central Force Motion Ellipse, e<1

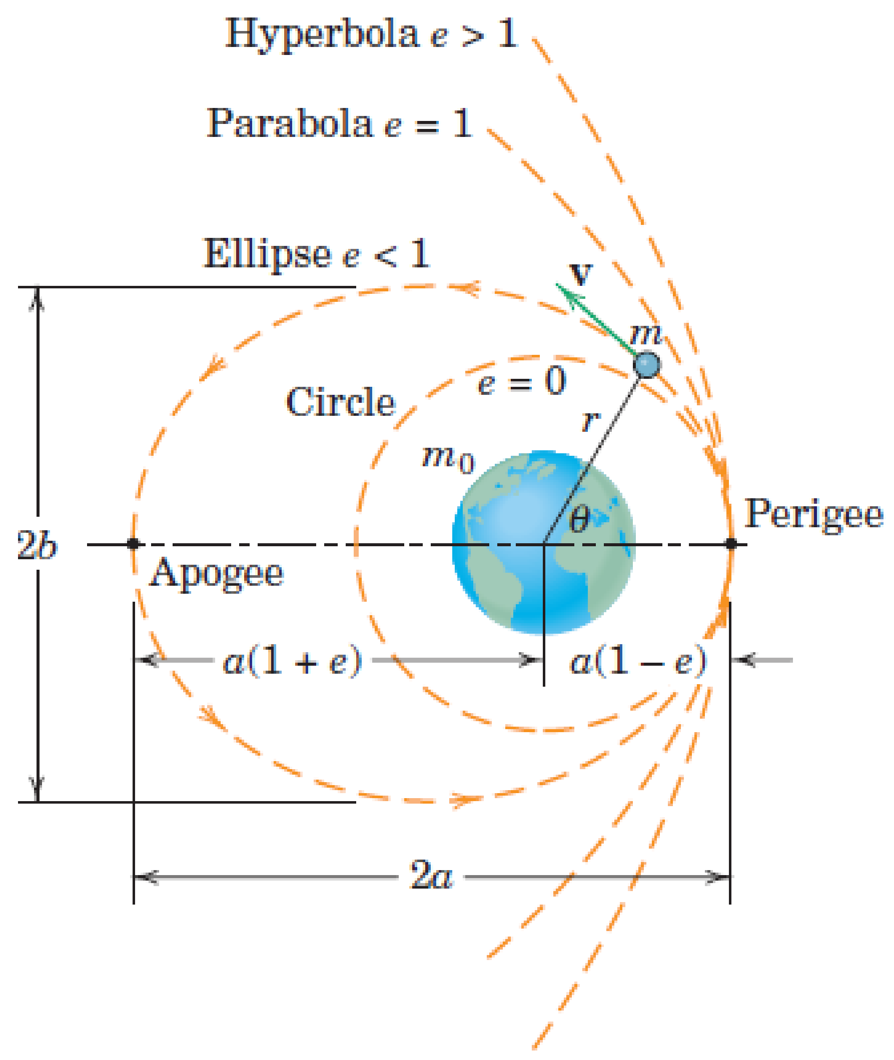
$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

At $\theta = 0$ $$\frac{1}{r_{min}} = \frac{1}{ed} + \frac{1}{d}$$

$$r_{min} = \frac{ed}{1+e}$$

At $\theta = \pi$ $$r_{max} = \frac{ed}{1-e}$$



Central Force Motion Ellipse, e<1

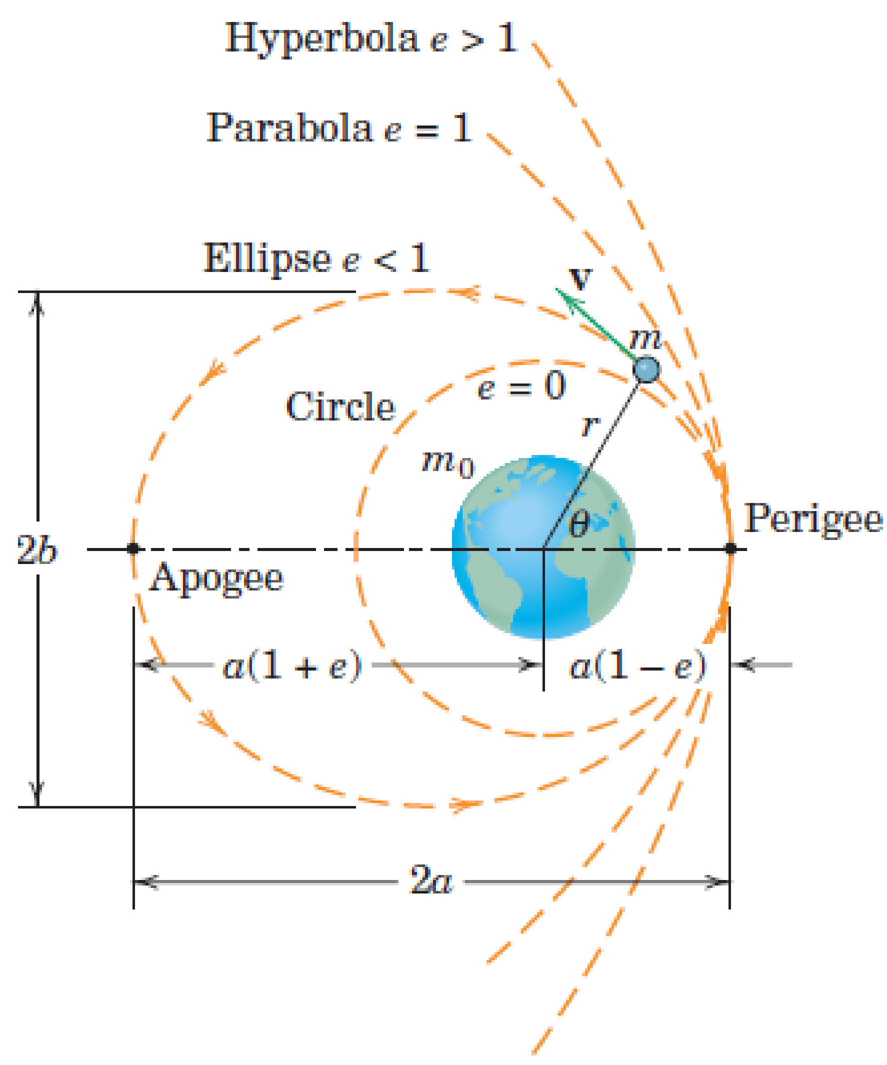
$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

$$2a = r_{min} + r_{max} = \frac{ed}{1+e} + \frac{ed}{1-e} = \frac{ed}{1-e^2}$$

$$2a = \frac{2ed}{1-e^2}$$

$$d = \frac{a(1-e^2)}{e}$$



Central Force Motion Ellipse, e<1

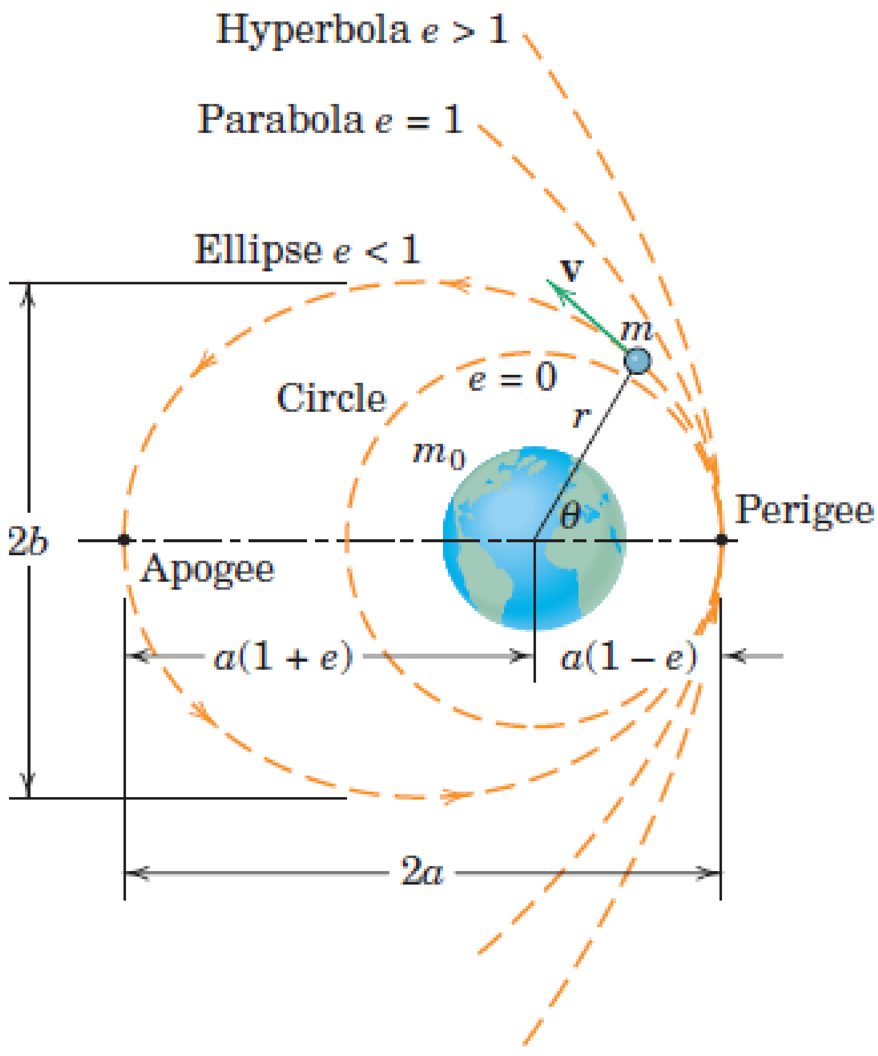
$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

$$\frac{1}{d} = \frac{e}{a(1-e^2)}$$

$$\frac{1}{r} = \frac{1}{a(1-e^2)} + \frac{e\ cos\theta}{a(1-e^2)}$$

$$\frac{1}{r} = \frac{1 + e\ cos\theta}{a(1-e^2)}$$



Central Force Motion Ellipse, e<1

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

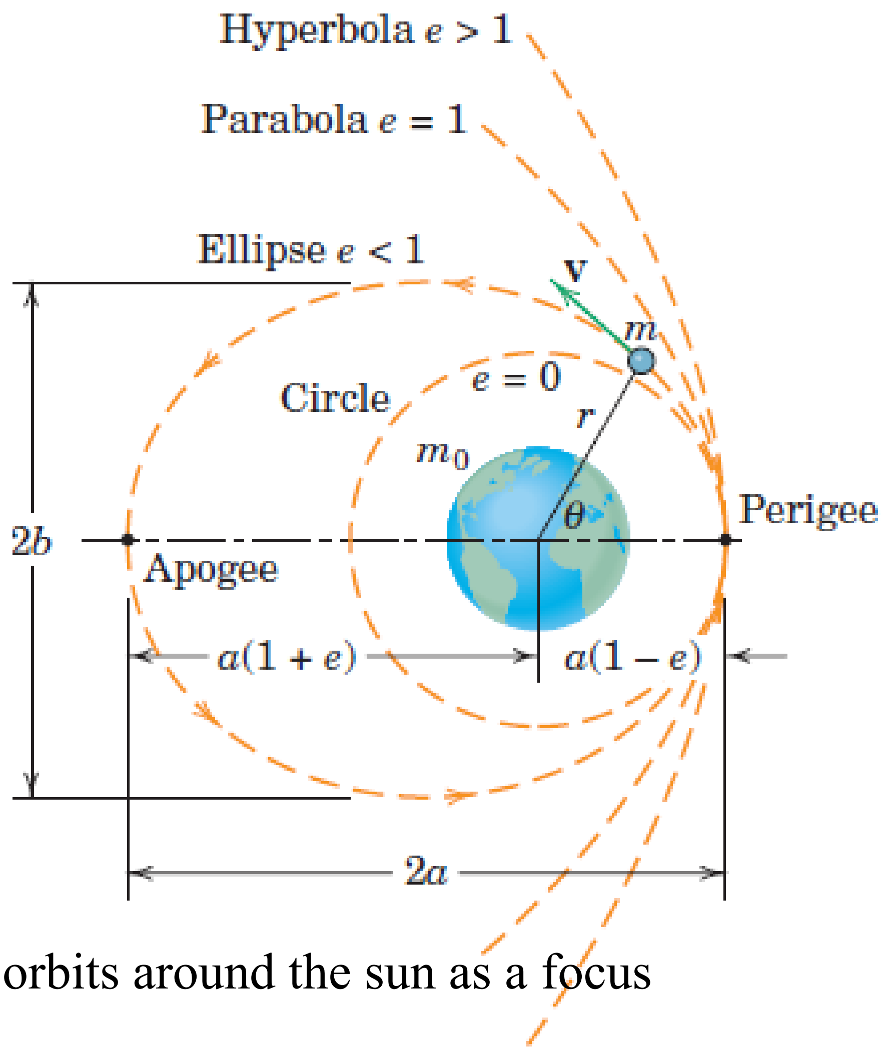
$$\frac{1}{r} = \frac{1 + e\,cos\theta}{a(1 - e^2)}$$

At $\theta = 0$

$$r_{min} = a(1 - e)$$

$$r_{max} = a(1 + e)$$

$$b = a\sqrt{(1 - e^2)}$$

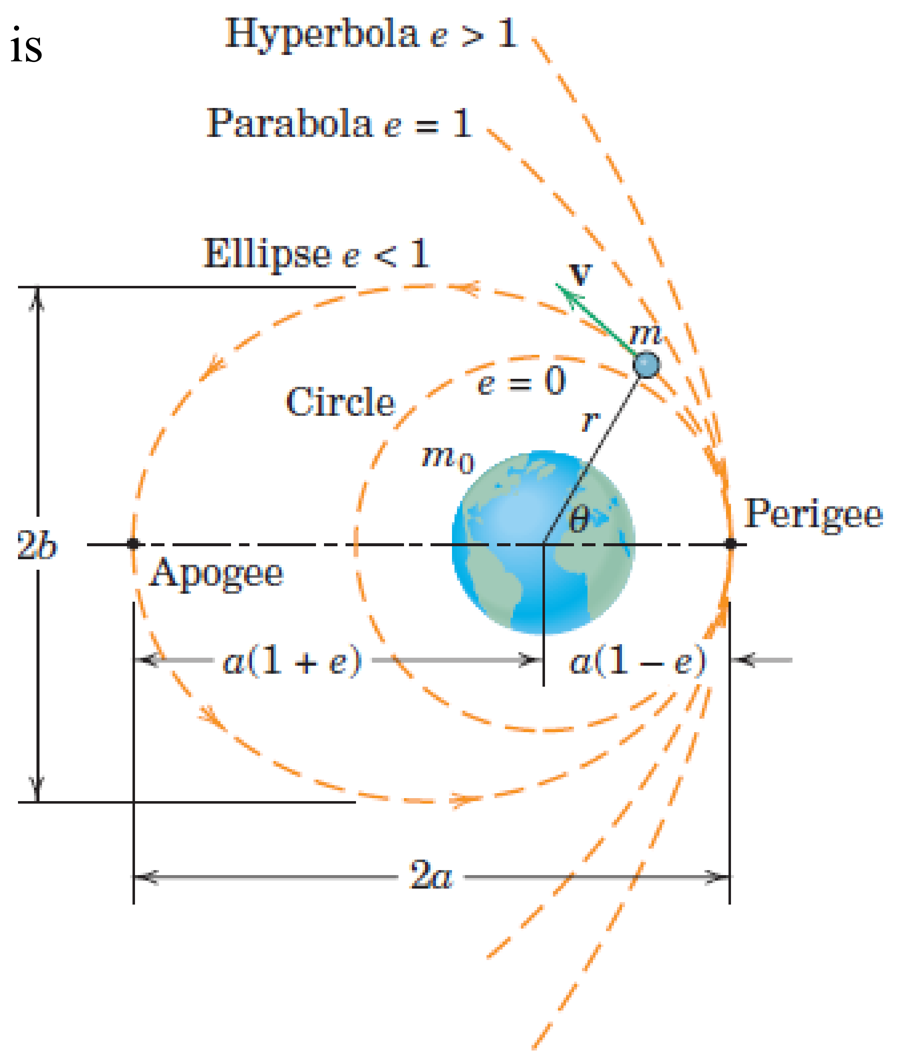


Kepler's *first law*: Planets move in elliptical orbits around the sun as a focus

Central Force Motion Ellipse, e<1

Time period: Period for the elliptical orbit is the total area of the ellipse divided by the constant rate at which the area is swept through

$$\tau = \frac{A}{\dot{A}} = \frac{\pi ab}{\frac{1}{2}r^2\dot{\theta}} = \frac{2\pi ab}{h}$$



Central Force Motion Ellipse, e<1

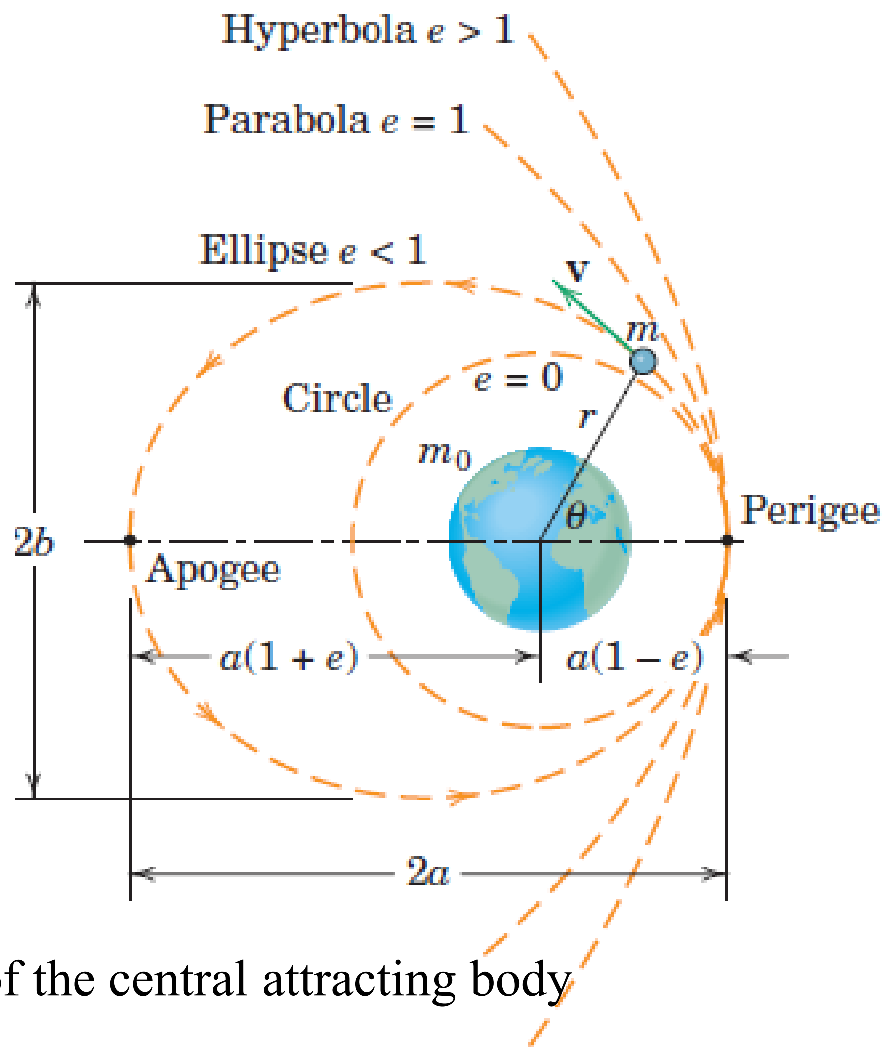
$$\tau = \frac{A}{\dot{A}} = \frac{\pi ab}{\frac{1}{2}r^2\dot{\theta}} = \frac{2\pi ab}{h}$$

$$e = \frac{h^2 C}{G m_0} \qquad C = \frac{1}{d}$$

$$G m_0 = g R^2 \qquad d = \frac{a(1-e^2)}{e}$$

$$b = a\sqrt{(1-e^2)}$$

$$\boxed{\tau = \frac{2\pi a^{\frac{3}{2}}}{R\sqrt{g}}}$$



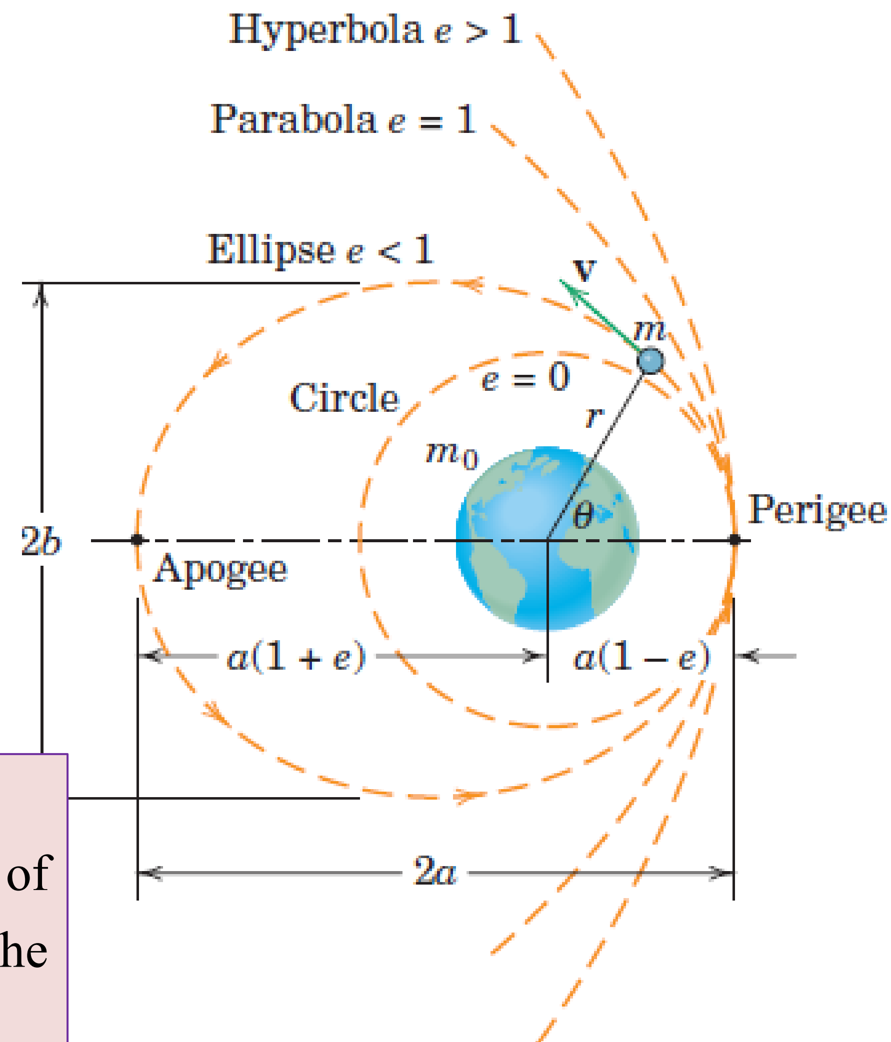
R is the mean radius of the central attracting body

Central Force Motion Ellipse, e<1

$$\tau = \frac{A}{\dot{A}} = \frac{\pi ab}{\frac{1}{2}r^2\dot{\theta}} = \frac{2\pi ab}{h}$$

$$\boxed{\tau = \frac{2\pi a^{\frac{3}{2}}}{R\sqrt{g}}}$$

R is the mean radius of the central attracting body



Kepler's *third law* of planetary motion which states that the square of the period of motion is proportional to the cube of the semimajor axis of the orbit

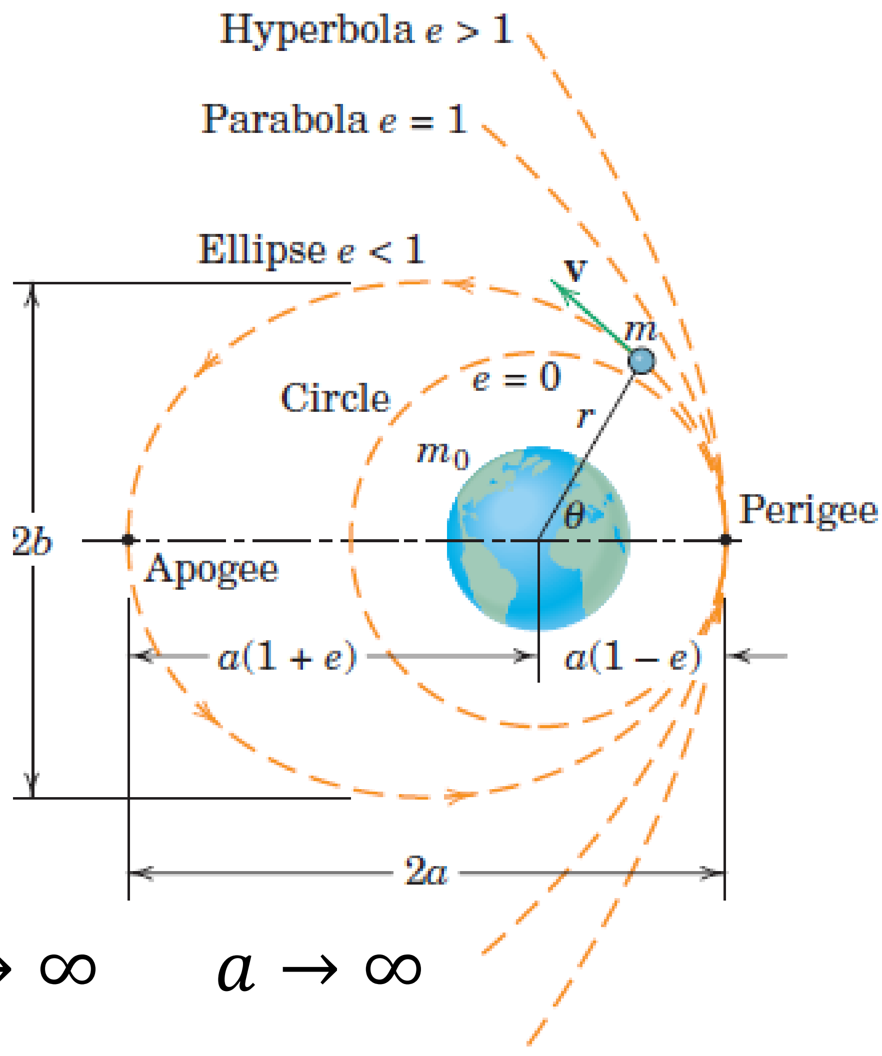
Central Force Motion Parabola, e=1

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

$$C = \frac{1}{d} \qquad ed = \frac{h^2}{Gm_0}$$

$$\frac{1}{r} = \frac{1 + cos\theta}{d} \qquad Gm_0 = h^2 C$$



At $\theta = \pi$ $$\frac{1}{r} = \frac{1-1}{d} \qquad r \to \infty \qquad a \to \infty$$

Central Force Motion Hyperbola, e>1

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

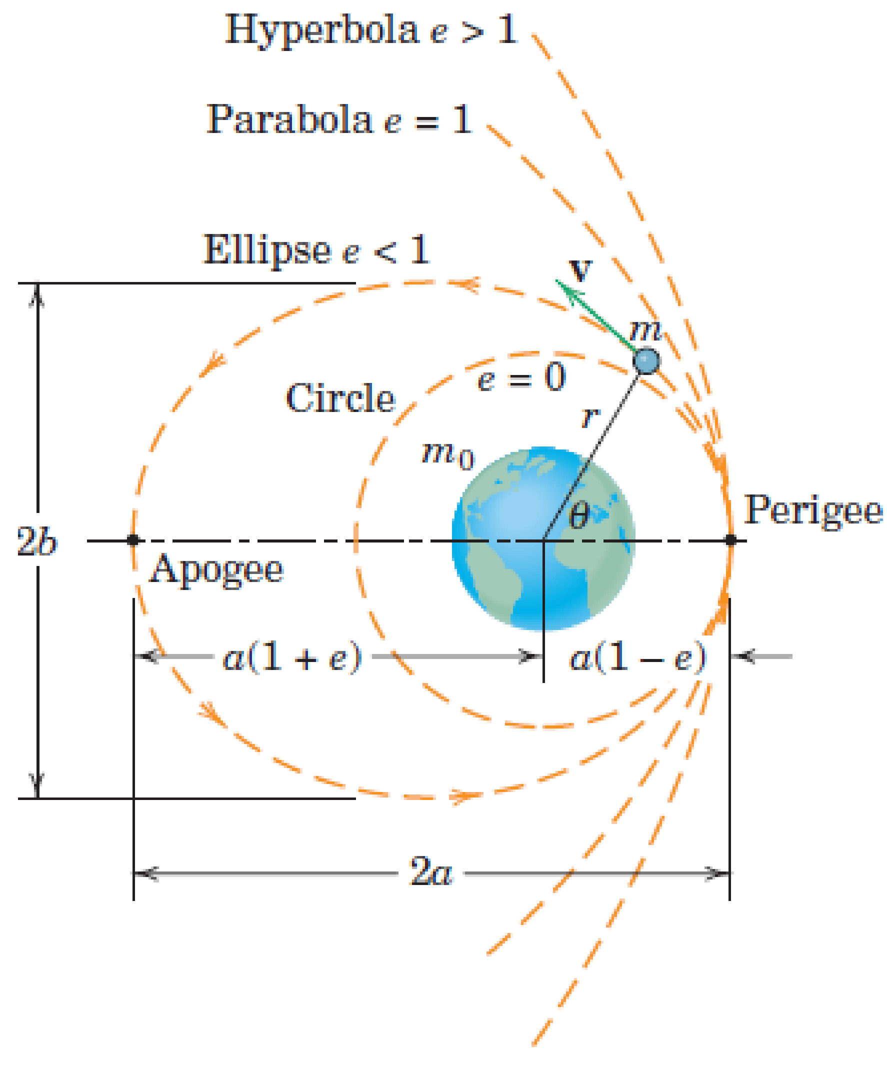
$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

e > 1

$$cos\,\theta_1 = -\frac{1}{e}$$

$$cos\,(-\theta_1) = -\frac{1}{e}$$

$$r \to \infty$$



Central Force Motion Hyperbola, e>1

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

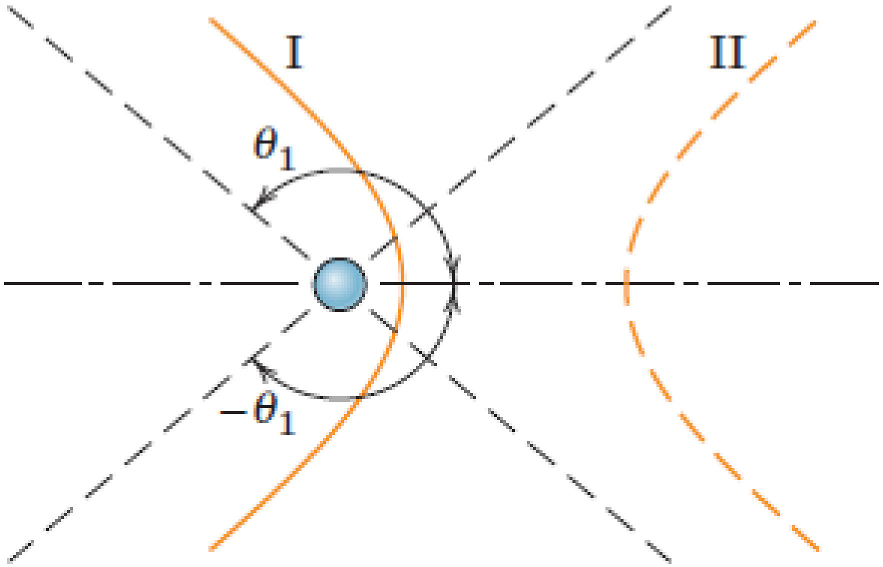
$$\frac{1}{r} = \frac{1}{ed} + \frac{cos\theta}{d}$$

e > 1

$$cos\,\theta_1 = -\frac{1}{e}$$

$$cos\,(-\theta_1) = -\frac{1}{e}$$

$$r \to \infty$$



Branch I exists (between angles $-\theta_1$ and θ_1) and branch II (for remaining angles) will lead to negative values of r.

Energy Analysis: Central Force Motion

Conservative System

$$E = T + V = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{mgR^2}{r}$$

At $\theta = 0 \quad \dot{r} = 0$

$$r^2\dot{\theta}^2 = \left(\frac{h}{r}\right)^2$$

$$u = \frac{1}{r} = C\cos(\theta) + \frac{Gm_0}{h^2}$$

At $\theta = 0$

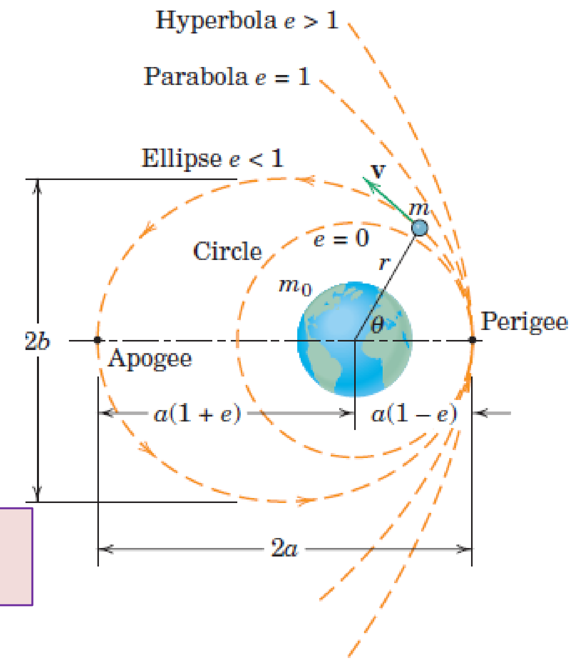
$$\frac{2E}{m} = h^2C^2 - \frac{g^2R^4}{h^2}$$

$$eGm_0 = h^2C$$

$$e = +\sqrt{1 + \frac{2Eh^2}{mg^2R^4}}$$

$$Gm_0 = gR^2$$

$$egR^2 = h^2C$$



Energy Analysis: Central Force Motion

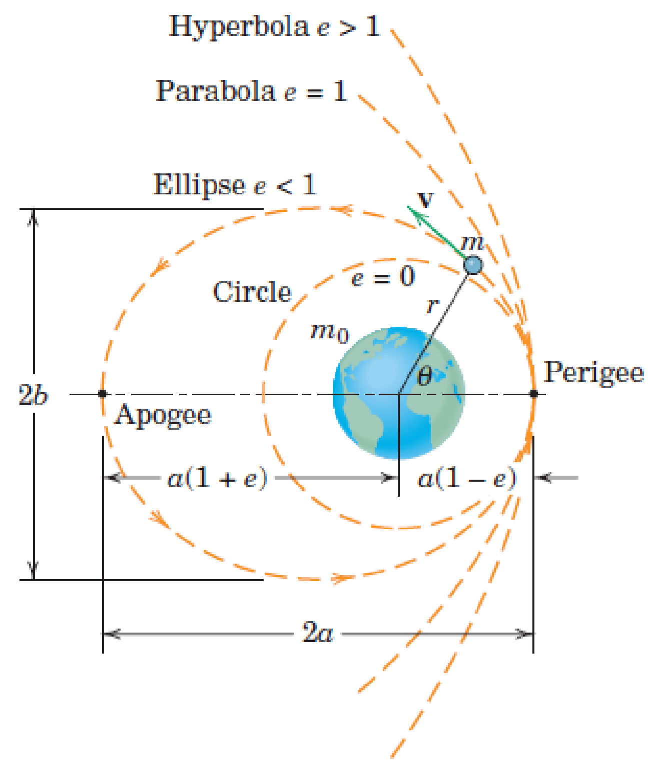
$$e = +\sqrt{1 + \frac{2Eh^2}{mg^2R^4}}$$

$e < 1$ (Ellipse), E is negative

$e = 1$ (Parabola), E is zero

$e > 1$ (Hyperbola), E is positive

These conclusions depend on the arbitrary selection of the datum condition for zero potential energy ($V=0$ when r tends to ∞)



Velocity of Moving Body: Central Force Motion

Conservative System

$$E = T + V = \frac{1}{2}m(v^2) - \frac{mgR^2}{r}$$

$$e = +\sqrt{1 + \frac{2Eh^2}{mg^2R^4}}$$

$$E = -\frac{gR^2m}{2a}$$

$$v^2 = 2gR^2\left(\frac{1}{r} - \frac{1}{2a}\right)$$

Elliptical Orbit

$$\boxed{egR^2 = h^2C}$$

$$\boxed{\frac{1}{C} = \frac{a(1-e^2)}{e}}$$

Velocity of Moving Body: Central Force Motion

Conservative System

$$v^2 = 2gR^2\left(\frac{1}{r} - \frac{1}{2a}\right)$$

$$v_{P(r_{min})} = R\sqrt{\frac{g}{a}}\sqrt{\frac{r_{max}}{r_{min}}}$$

$$v_{A(r_{max})} = R\sqrt{\frac{g}{a}}\sqrt{\frac{r_{min}}{r_{max}}}$$

Elliptical Orbit

$$r_{min} = a(1-e)$$

$$r_{max} = a(1+e)$$

$$b = a\sqrt{(1-e^2)}$$

Assumptions: Central Force Motion

1. The two bodies possess spherical mass symmetry so that they may be treated as if their masses were concentrated at their centers, that is, as if they were particles.

- ✓ Poor Assumption for: Artificial satellites in the very near vicinity of oblate planets.
- ✓ Good Assumption for: Bodies which are distant from the central attracting body, Most heavenly bodies

2. There are no forces present except the gravitational force which each mass exerts on the other.

- ✓ Poor Assumption: Aerodynamic drag on a low altitude earth satellite is a force which usually cannot be ignored in the orbital analysis

3. Mass m_o is fixed in space.

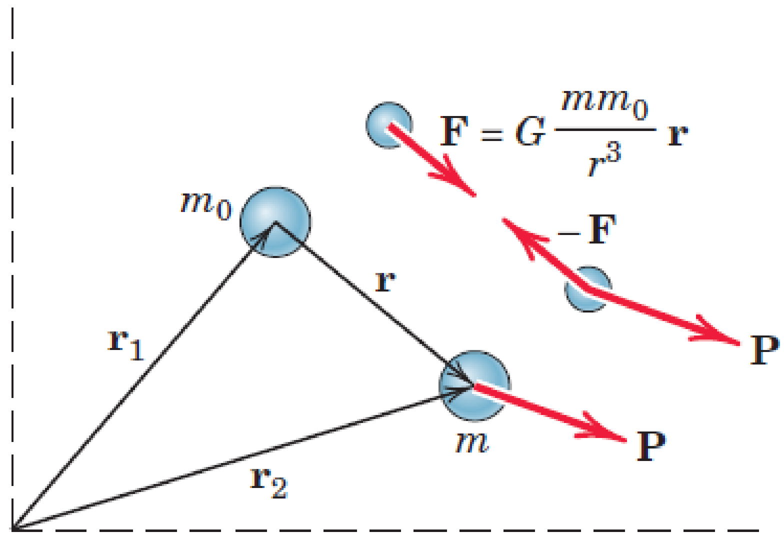
- ✓ Not so good assumption for earth–moon system. Mass of Moon= 1/81 times mass of Earth

Perturbed Two Body Problem

$$\frac{Gmm_0}{r^3}\mathbf{r} = m_o\ddot{\mathbf{r}}_1$$

$$-\frac{Gmm_0}{r^3}\mathbf{r} + \mathbf{P} = m\ddot{\mathbf{r}}_2$$

$$-\frac{Gm_0}{r^3}\mathbf{r} + \frac{\mathbf{P}}{m} = \ddot{\mathbf{r}}_2 \qquad \frac{Gm}{r^3}\mathbf{r} = \ddot{\mathbf{r}}_1$$



$$\ddot{\mathbf{r}}_2 - \ddot{\mathbf{r}}_1 = -\frac{Gm_0}{r^3}\mathbf{r} - \frac{Gm}{r^3}\mathbf{r} + \frac{\mathbf{P}}{m}$$

$$\ddot{\mathbf{r}} = -\frac{G(m_0+m)}{r^3}\mathbf{r} + \frac{\mathbf{P}}{m}$$

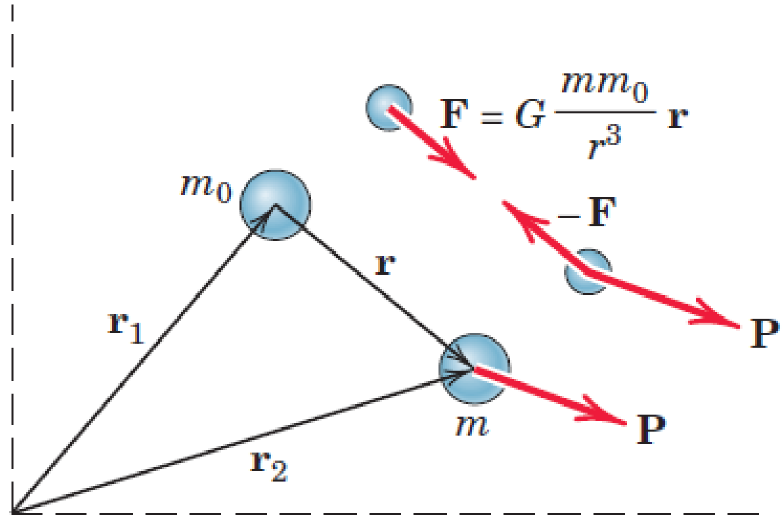
Solution of the equations yields the relative position vector **r** as a function of time

Restricted Two Body Problem

$$\ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3}\mathbf{r} + \frac{\mathbf{P}}{m}$$

$m_0 \gg m$ and P=0

$$\ddot{\mathbf{r}} + \frac{G(m_0)}{r^3}\mathbf{r} = 0$$



$$(\ddot{r} - r\dot{\theta}^2)\boldsymbol{e_r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\boldsymbol{e_\theta} + \frac{G(m_0)}{r^2}\boldsymbol{e_r} = 0$$

$$(\ddot{r} - r\dot{\theta}^2) + \frac{G(m_0)}{r^2} = 0 \qquad (r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0$$

Same expressions as derived assuming mass m_o fixed

Restricted Two Body Problem

$$\ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3}\mathbf{r} + \frac{\mathbf{P}}{m}$$

If P=0 $$\ddot{\mathbf{r}} = -\frac{G(m_0 + m)}{r^3}\mathbf{r}$$

$m_0 \gg m$ and P=0 $$\ddot{\mathbf{r}} + \frac{G(m_0)}{r^3}\mathbf{r} = 0$$

If we replace m_0 by $(m_0 + m)$ in the expressions derived with the assumption of m_0 fixed, then we obtain expressions which account for the motion of m_0

$$\boxed{\tau = \frac{2\pi a^{\frac{3}{2}}}{\sqrt{Gm_0}}}$$

Derived with the assumption of m_0 fixed

$$\boxed{\tau = \frac{2\pi a^{\frac{3}{2}}}{\sqrt{G(m + m_0)}}}$$

With m_0 NOT fixed

THANK YOU

Dynamics of Rigid Bodies

Dr. Amrita Puri
Assistant Professor
Department of Mechanical Engineering

Syllabus

- **Dynamics of Rigid bodies [12 Lectures]:** Translation/Rotation of rigid bodies, Chasles' theorem, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames. **[Dr. Amrita Puri]**

Rigid Body Kinematics

- **Particle Kinematics:** Development of the relationships governing the displacement, velocity, and acceleration of points as they moved along straight or curved paths
- **Rigid Body Kinematics:** One must also account for the rotational motion of the body. Thus, rigid-body kinematics involves both linear and angular displacements, velocities, and accelerations

Why study Rigid Body Kinematics?

- **Design based on motion and calculation of forces due to the motion:** we frequently need to generate, transmit, or control certain motions by the use of cams, gears, and linkages of various types. Here we must analyze the displacement, velocity, and acceleration of the motion to determine the design geometry of the mechanical parts. Furthermore, as a result of the motion generated, forces may be developed which must be accounted for in the design of the parts.
- **Calculation of motion based on action of forces:** Second, we must often determine the motion of a rigid body caused by the forces applied to it. Calculation of the motion of a rocket under the influence of its thrust and gravitational attraction is an example of such a problem.

Ideal Rigid Body and Assumption of Rigidity

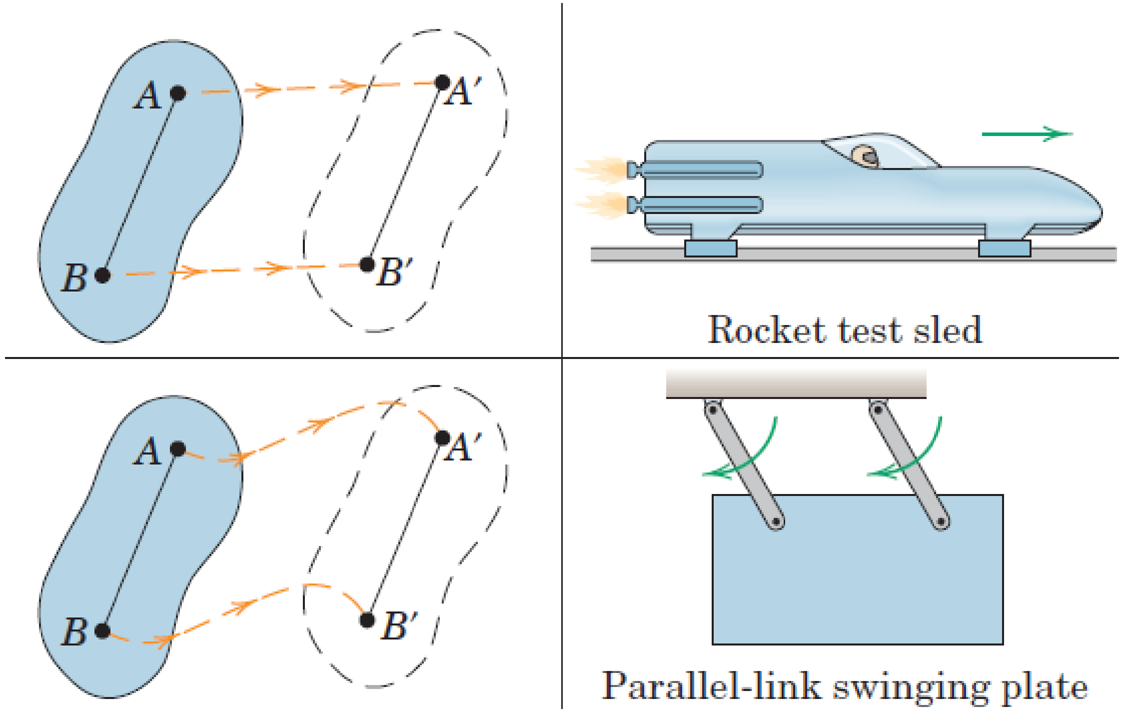
- A *rigid body* is defined as a system of particles for which the distances between the particles remain unchanged
- If each particle of such a body is located by a position vector from reference axes attached to and rotating with the body, there will be no change in any position vector as measured from these axes.
- If the movements associated with the changes in shape are very small compared with the movements of the body as a whole, then the assumption of rigidity is usually acceptable
- The displacements due to the flutter of an aircraft wing, for instance, do not affect the description of the flight path of the aircraft as a whole, and thus the rigid-body assumption is clearly acceptable
- the problem is one of describing, as a function of time, the internal wing stress due to wing flutter, then the relative motions of portions of the wing cannot be neglected, and the wing may not be considered a rigid body.

Plane Motion

- A rigid body executes plane motion when all parts of the body move in parallel planes. For convenience, we generally consider the *plane of motion* to be the plane which contains the mass center, and we treat the body as a thin slab whose motion is confined to the plane of the slab.

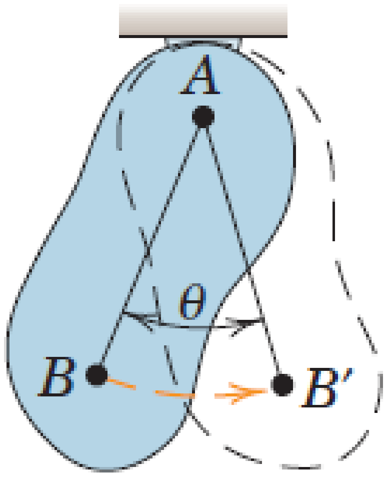
Translation in Plane Motion

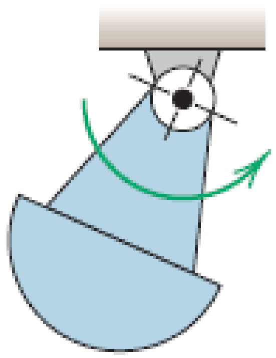
- ***Translation*** is defined as any motion in which every line in the body remains parallel to its original position at all times. In translation there is *no rotation of any line in the body*. In *rectilinear translation*, all points in the body move in parallel straight lines.
- In *curvilinear translation*, all points move on congruent curves.
- We note that in each of the two cases of translation, the motion of the body is completely specified by the motion of any point in the body, since all points have the same motion. Thus, our earlier study of the motion of a point (particle) in Chapter 2 enables us to describe completely the translation of a rigid body.



Rotation in Plane Motion

- *Rotation about a fixed axis is the angular motion about the axis. It follows that all particles in a rigid body move in circular paths about the axis of rotation, and all lines in the body which are perpendicular to the axis of rotation (including those which do not pass through the axis) rotate through the same angle in the same time. Again, our discussion in Chapter 2 on the circular motion of a point enables us to describe the motion of a rotating rigid body, which is treated in the next article.*

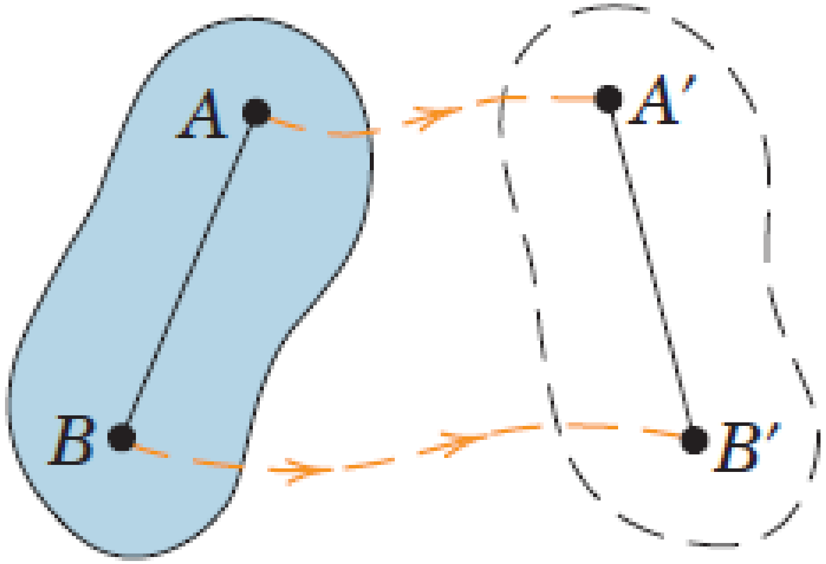


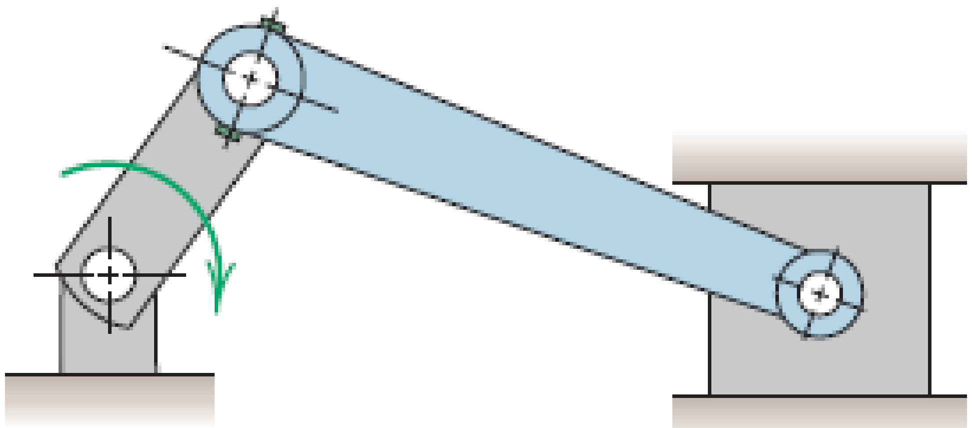


Compound pendulum

General Plane Motion

- ***General plane motion*** of a rigid body is a combination of translation and rotation. We will utilize the principles of relative motion covered in Art. 2/8 to describe general plane motion

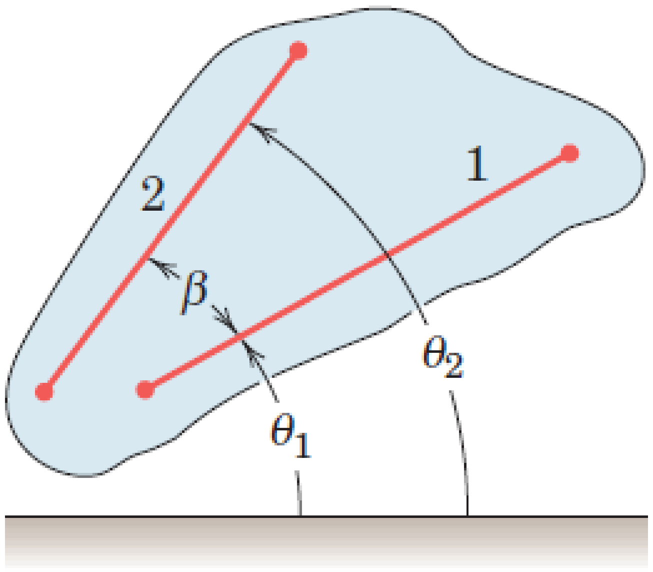




Connecting rod in a reciprocating engine

Rotation

- The rotation of a rigid body is described by its angular motion
- All lines on a rigid body in its plane of motion have the same angular displacement, the same angular velocity, and the same angular acceleration.
- The angular positions of any two lines 1 and 2 attached to the body are specified by θ_1 and θ_2 measured from any convenient fixed reference direction. Because the angle β is invariant, the relation $\beta = \theta_2 - \theta_1$ upon differentiation with respect to time gives $\dot{\theta}_2 = \dot{\theta}_1$ and $\ddot{\theta}_2 - \ddot{\theta}_1$, during a finite interval, $\Delta\theta_1 = \Delta\theta_2$.
- Angular motion of a line depends only on its angular position with respect to any arbitrary fixed reference and on the time derivatives of the displacement. Angular motion does not require the presence of a fixed axis, normal to the plane of motion, about which the line and the body rotate.



Angular Motion Relations

The angular velocity ω and angular acceleration α of a rigid body in plane rotation are, respectively, the first and second time derivatives of the angular position coordinate θ of any line in the plane of motion of the body.

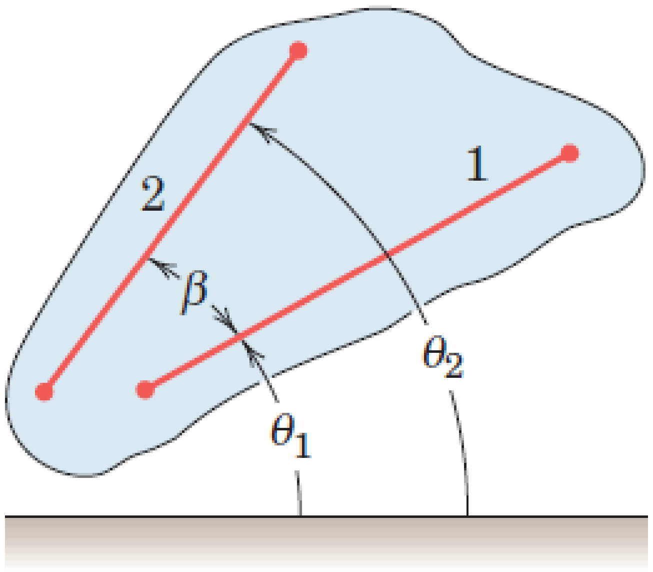
$$\omega = \frac{d\theta}{dt} = \dot{\theta}$$

$$\alpha = \frac{d\omega}{dt} = \dot{\omega} \qquad \alpha = \frac{d^2\theta}{dt^2} = \ddot{\theta}$$

$$\alpha d\theta = \frac{d\omega}{dt} d\theta = \omega d\omega \qquad \alpha d\theta = \frac{d^2\theta}{dt^2} d\theta = \ddot{\theta} d\theta$$

$$\alpha d\theta = \omega d\omega = \ddot{\theta} d\theta$$

$$\dot{\theta} d\dot{\theta} = \ddot{\theta} d\theta$$



Constant Angular Acceleration

For constant angular acceleration,

Integrating $\alpha = \dfrac{d\omega}{dt}$ $\Longrightarrow$ $\omega = \omega_0 + \alpha t$

Integrating $\alpha d\theta = \omega d\omega$ $\Longrightarrow$ $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$

Using $\omega = \omega_0 + \alpha t$, $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$ $\Longrightarrow$ $\theta = \theta_0 + \omega_0 t + \dfrac{1}{2}\alpha t^2$

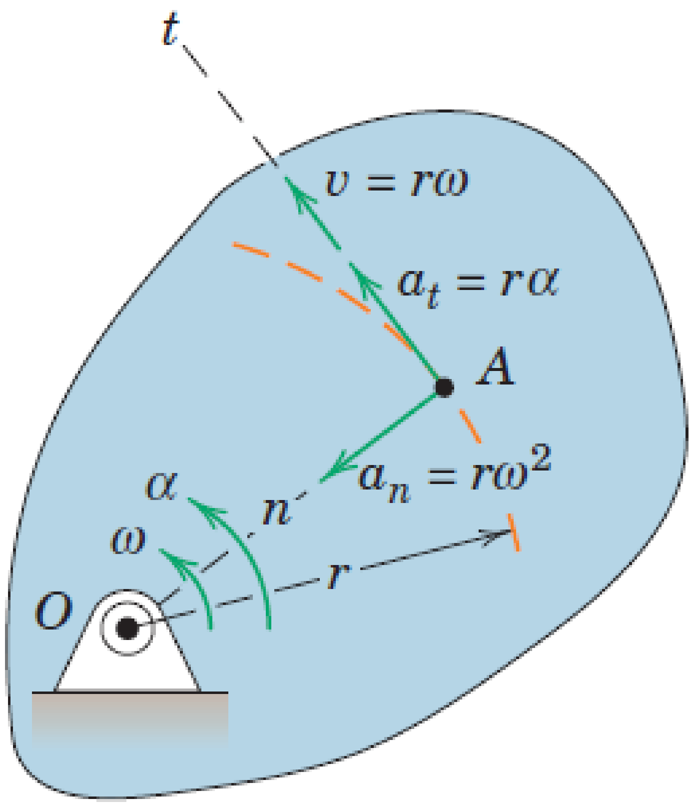
Rotation about a Fixed Axis

- When a rigid body rotates about a fixed axis, all points other than those on the axis move in concentric circles about the fixed axis. Thus, for the rigid body rotating about a fixed axis normal to the plane of the figure through *O*, any point such as *A* moves in a circle of radius *r*.
- Relationships between the linear motion of A and the angular motion of the line normal to its path, which is also the angular motion of the rigid body.

$$v = r\omega$$

$$a_n = r\omega^2 = \frac{v^2}{r} = v\omega$$

$$a_t = r\alpha$$



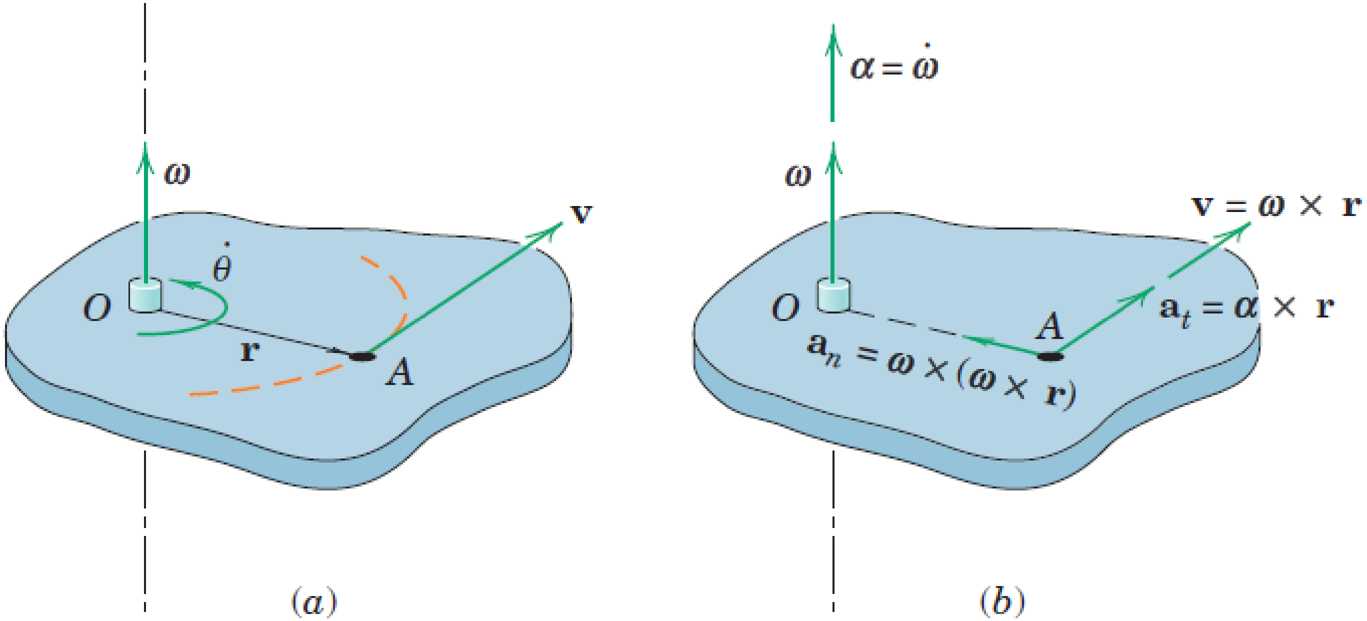
Vector Relations

The angular velocity of the rotating body may be expressed by the vector $\boldsymbol{\omega}$ normal to the plane of rotation and having a sense governed by the right-hand rule.

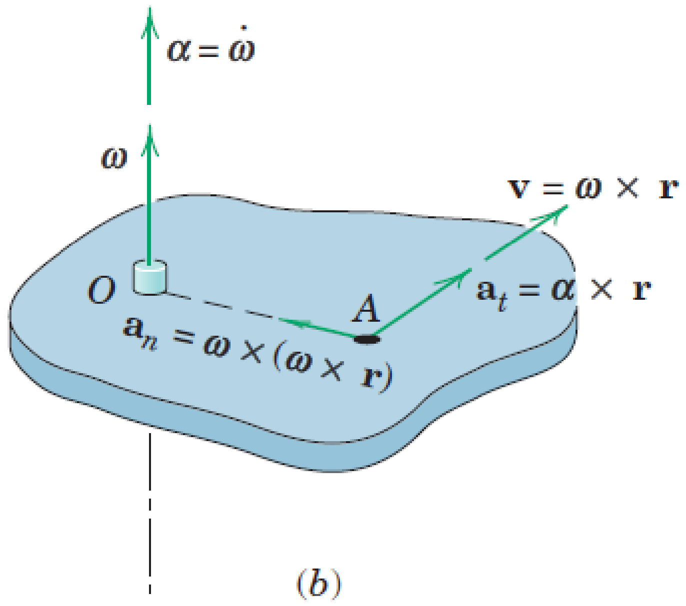
$$V = \dot{r} = \omega \times r$$

Order of the vectors to be crossed must be retained

$$r \times \omega = -v$$



Vector Relations



Acceleration of point A is obtained by differentiating the cross product expression for v

$$a = \dot{v} = \omega \times \dot{r} + \dot{\omega} \times r$$

$$a = \omega \times (\omega \times r) + \dot{\omega} \times r$$

$$a = \omega \times v + \alpha \times r$$

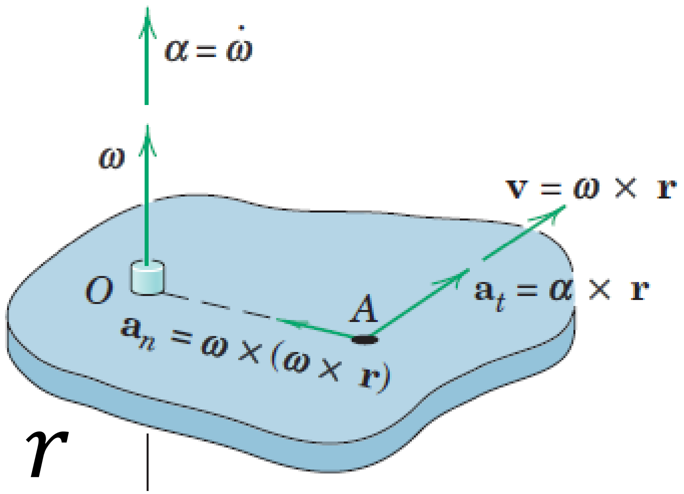
$\alpha = \dot{\omega}$ Angular acceleration of point A

$$v = \omega \times r$$

$$a_n = \omega \times (\omega \times r)$$

$$a_t = \alpha \times r$$

Vector Relations



Acceleration of point A is obtained by differentiating the cross product expression for v

$$a = \dot{v} = \omega \times \dot{r} + \dot{\omega} \times r$$

$$v = \omega \times r$$

$$a = \omega \times (\omega \times r) + \dot{\omega} \times r$$

$$a_n = \omega \times (\omega \times r)$$

$$a = \omega \times v + \alpha \times r$$

$$a_t = \alpha \times r$$

$\alpha = \dot{\omega}$ Angular acceleration of point A

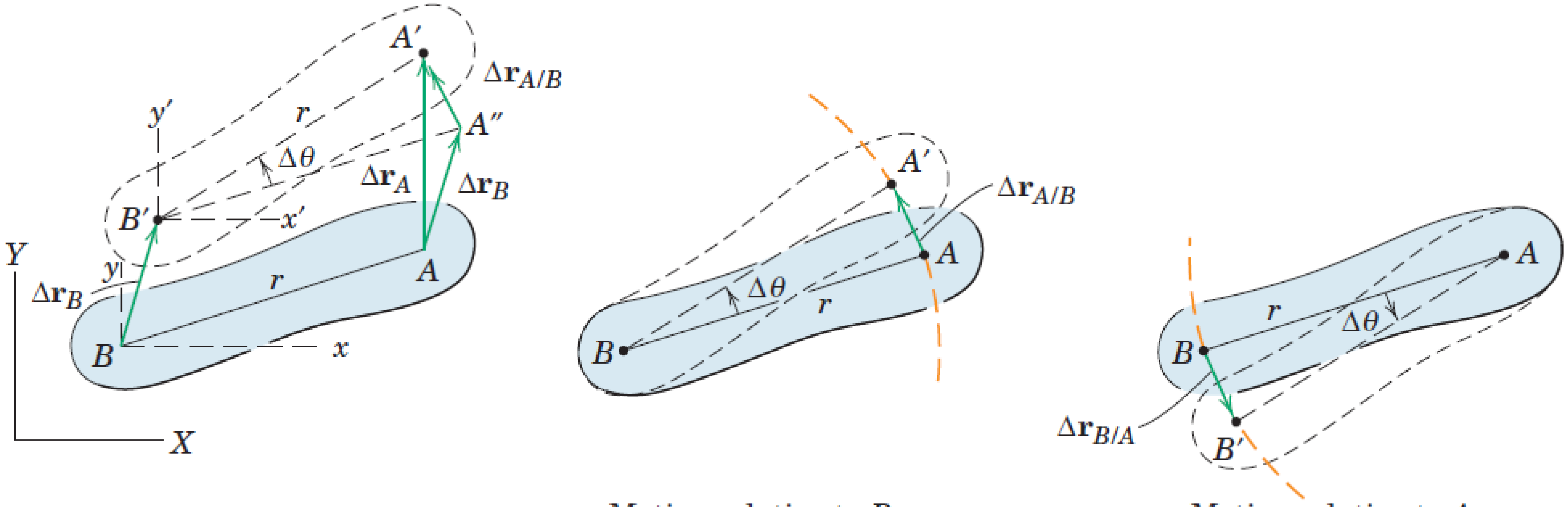
- For three-dimensional motion of a rigid body, the angular-velocity vector may change direction as well as magnitude, and in this case, the angular acceleration, α which is the time derivative of angular velocity, $\dot{\omega}$,will no longer be in the same direction as ω.

Absolute Motion

Acceleration of point A is obtained by differentiating the cross product expression for v

Relative Motion

- Two points on the same rigid body for our two particles
- The consequence of this choice is that the motion of one point as seen by an observer translating with the other point must be circular since the radial distance to the observed point from the reference point does not change.



(a)

Motion relative to B
(b)

Motion relative to A
(c)

Relative Motion

$$\Delta r_{B/A} = -\Delta r_{A/B}$$

$$\Delta r_A = \Delta r_B + \Delta r_{A/B}$$

$$V_A = V_B + V_{A/B}$$

$$v_{A/B} = \lim_{\Delta t \to 0} \left(\left| \Delta r_{A/B} \right| / \Delta t \right) = \lim_{\Delta t \to 0} \left(r \Delta \theta / \Delta t \right)$$

$$v_{A/B} = r\omega$$

Relative linear velocity is always perpendicular to the line joining the two points in question

$$v_{A/B} = \omega \times r$$

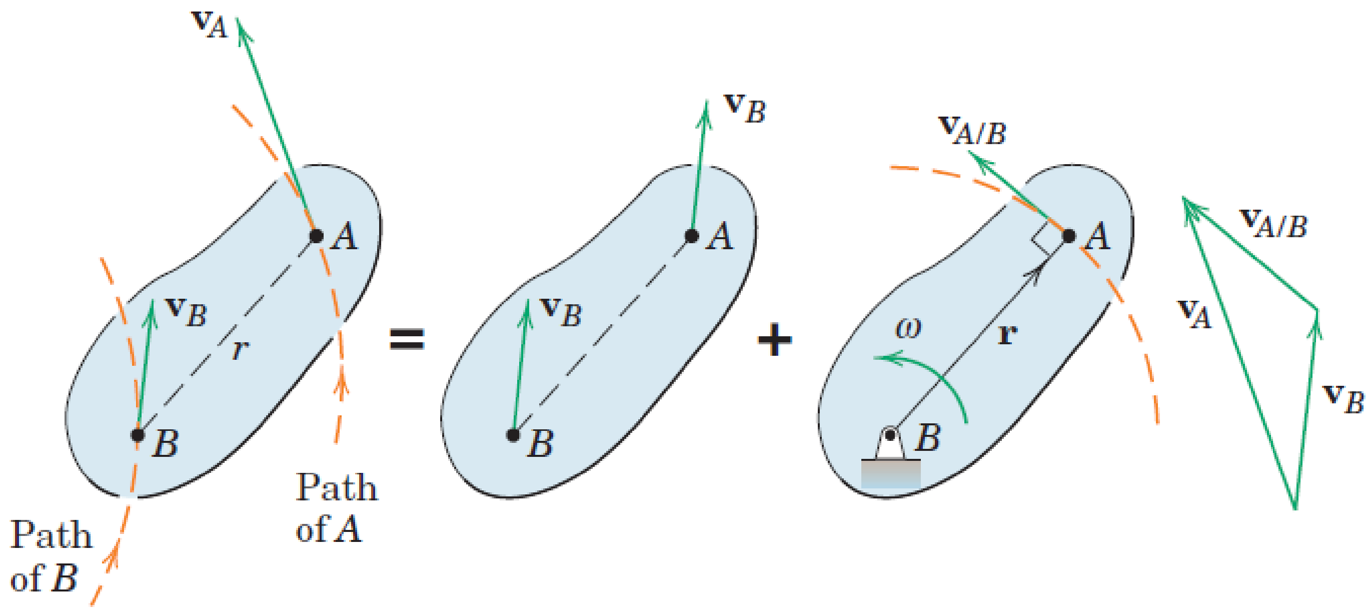
Relative Motion

$$V_A = V_B + V_{A/B}$$

$$v_{A/B} = r\omega$$

$$v_{A/B} = \omega \times r$$

- With B chosen as the reference point, the velocity of A is the vector sum of the translational portion $\boldsymbol{V_B}$, plus the rotational portion $\boldsymbol{V_{A/B}}$, which has the magnitude $v_{A/B} = r\omega$, where ω is the *absolute* angular velocity of AB.
- The fact that **the *relative linear velocity* is *always perpendicular*** to the line joining the two points in question is an important key to the solution of many problems



Instantaneous Center of Zero Velocity

Choosing a unique reference point which momentarily has zero velocity

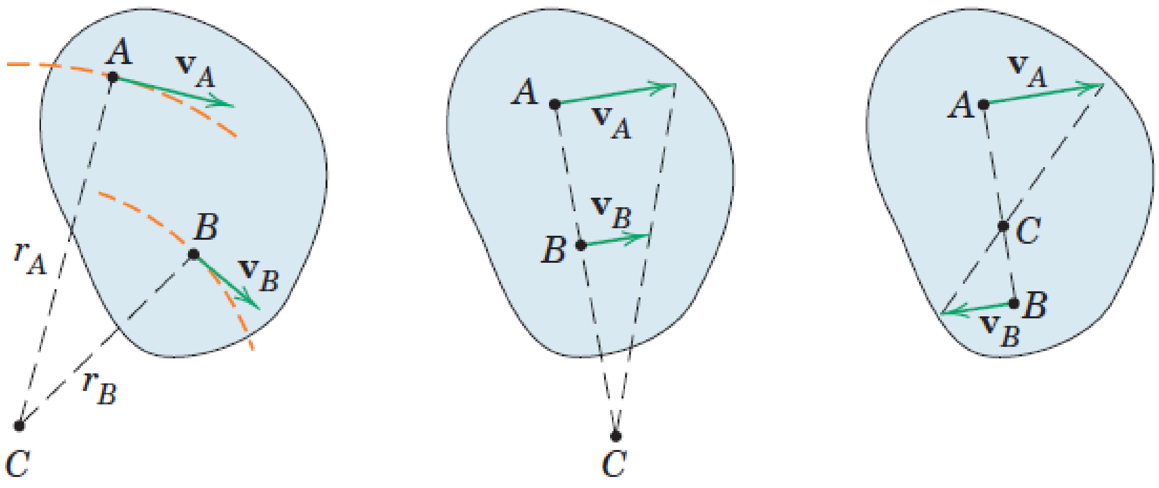
As far as velocities are concerned, the body may be considered to be in pure rotation about an axis, normal to the plane of motion, passing through this point.

This axis is called the instantaneous axis of zero velocity.

The intersection of this axis with the plane of motion is known as the instantaneous center of zero velocity.

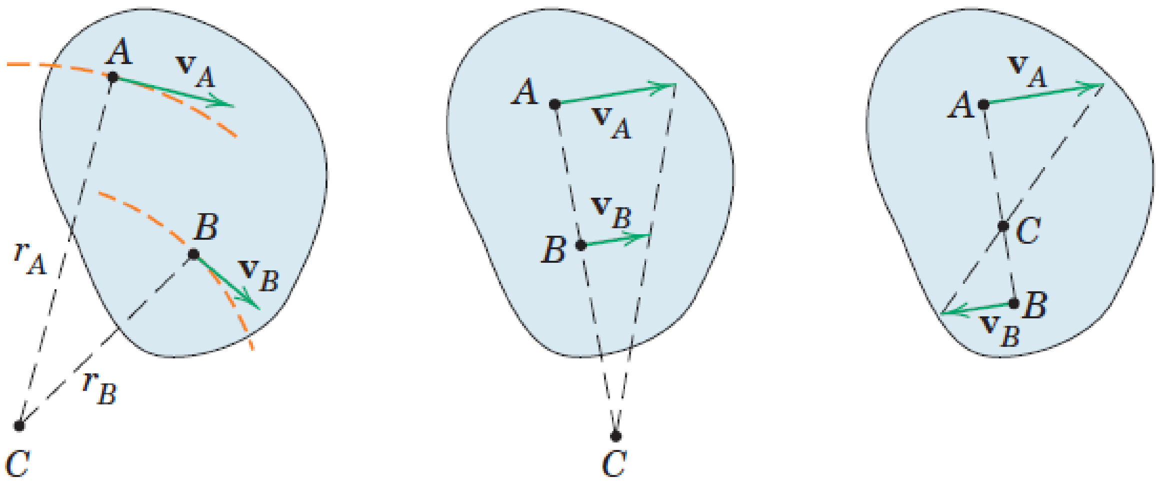
Instantaneous Center of Zero velocity

- Directions of the absolute velocities of any two points A and B on the body are known and are not parallel
- If there is a point about which *A* has absolute circular motion at the instant considered, this point must lie on the normal to v_A through *A*. Similar reasoning applies to *B*, and the intersection of the two perpendiculars fulfills the requirement for an absolute center of rotation *at the instant considered*.
- Point *C* is the instantaneous center of zero velocity and may lie on or off the body.
- If it lies off the body, it may be visualized as lying on an imaginary extension of the body.
- The instantaneous center need not be a fixed point in the body or a fixed point in the plane.



Instantaneous Center of Zero velocity

- If we also know the magnitude of the velocity of one of the points, say, v_A, we may easily obtain the angular velocity ω of the body and the linear velocity of every point in the body.
- Once the instantaneous center is located, the direction of the instantaneous velocity of every point in the body is readily found since it must be perpendicular to the radial line joining the point in question with C.

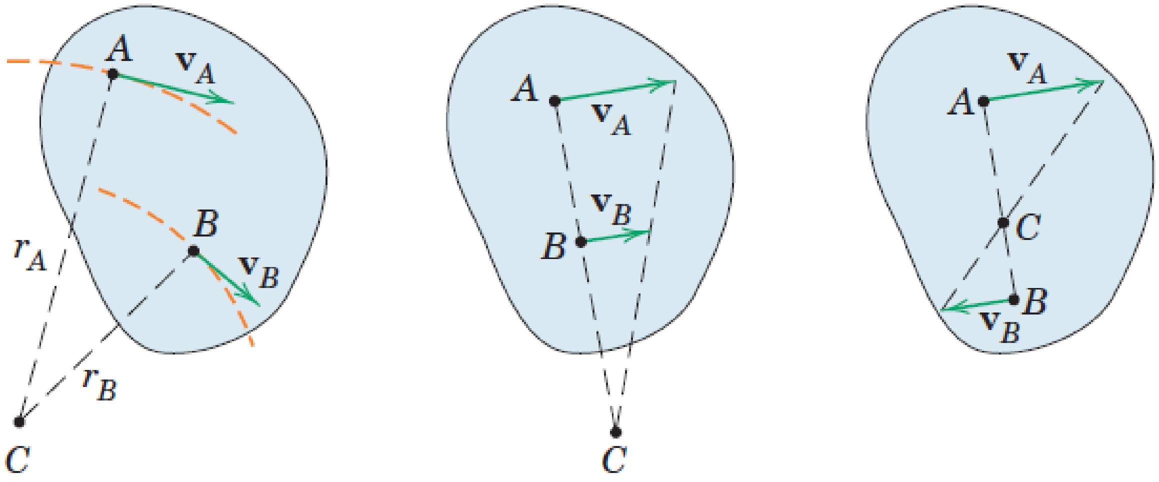


$$\omega = \frac{v_A}{r_A}$$

$$v_B = r_B\omega = (r_B/r_A)v_A$$

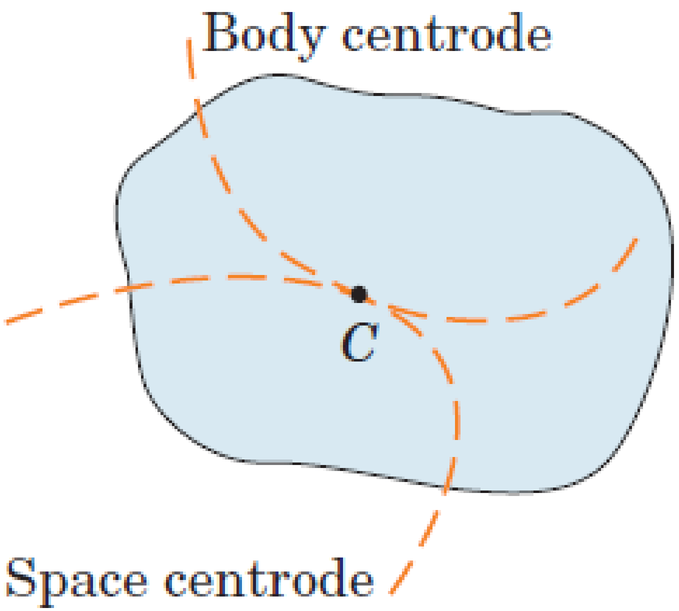
Instantaneous Center of Zero velocity

- If the **velocities of two points in a body having plane motion are parallel** and the line joining the points is perpendicular to the direction of the velocities, the instantaneous center is located by direct proportion.
- As the **parallel velocities become equal in magnitude**, the instantaneous center moves farther away from the body and approaches infinity in the limit as the body stops rotating and translates only.



Motion of Instantaneous Center of Zero velocity

- As the body changes its position, the instantaneous center C also changes its position both in space and on the body.
- The **locus of the instantaneous centers in space** is known as the **space centrode.**
- The **locus of the positions of the instantaneous centers on the body** is known as the **body centrode**.
- At the instant considered, the two curves are tangent at the position of point C.



Motion of Instantaneous Center of Zero velocity

- Although the instantaneous center of zero velocity is momentarily at rest, its acceleration generally is not zero. Thus, this point may not be used as an instantaneous center of zero acceleration in a manner analogous to its use for finding velocity.

- An instantaneous center of zero acceleration does exist for bodies in general plane motion.

Relative Acceleration

- The acceleration of point A equals the vector sum of the acceleration of point B and the acceleration which A appears to have to a nonrotating observer moving with B

$$v_A = v_B + v_{A/B}$$

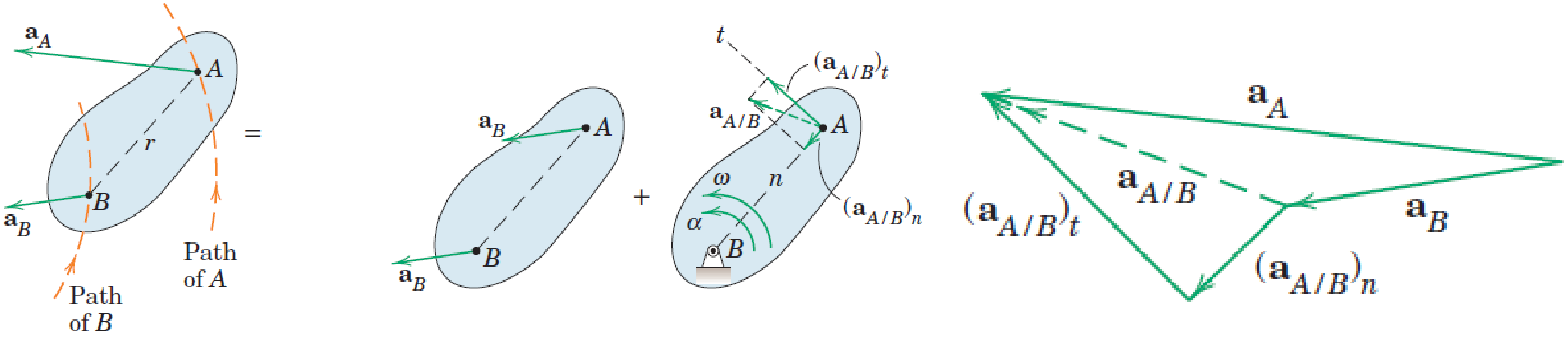
$$\dot{v}_A = \dot{v}_B + \dot{v}_{A/B}$$

$$a_A = a_B + a_{A/B}$$

Relative Acceleration due to Rotation

- If points A and B are located on the same rigid body and in the plane of motion, the distance r between them remains constant so that the observer moving with B perceives A to have circular motion about B
- Because the relative motion is circular, it follows that the relative-acceleration term will have both a normal component directed from A toward B due to the change of direction of $v_{A/B}$ and a tangential component perpendicular to AB due to the change in magnitude of $v_{A/B}$.

$$a_A = a_B + \left(a_{A/B}\right)_n + \left(a_{A/B}\right)_t$$



Relative Acceleration due to Rotation

$$a_A = a_B + \left(a_{A/B}\right)_n + \left(a_{A/B}\right)_t$$

$$\left(a_{A/B}\right)_n = V_{A/B}^2/r = r\omega^2$$

$$\left(\boldsymbol{a}_{A/B}\right)_n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$$

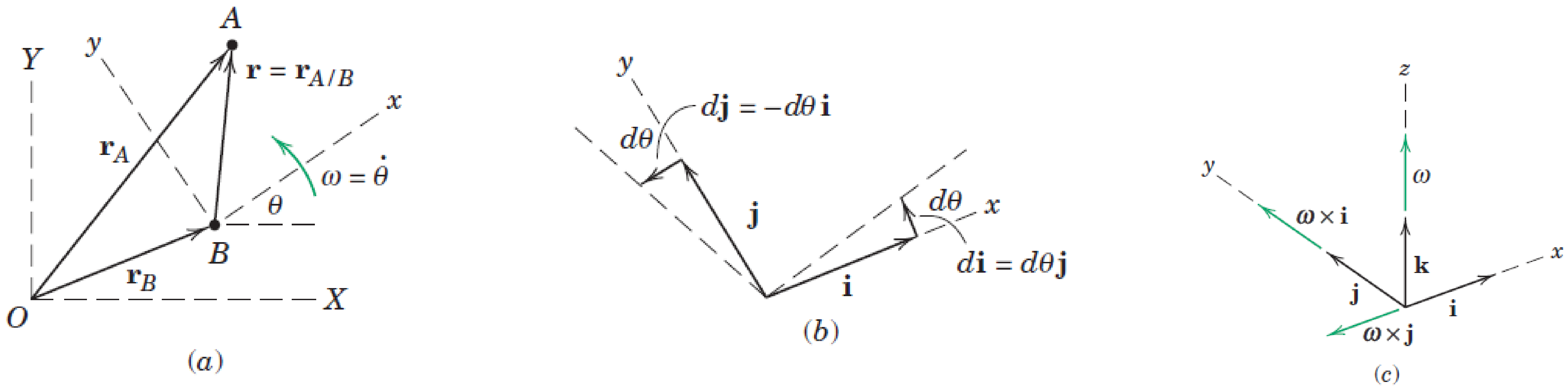
$$\left(a_{A/B}\right)_t = \dot{v}_{A/B} = r\alpha$$

$$\left(\boldsymbol{a}_{A/B}\right)_t = \boldsymbol{\alpha} \times \boldsymbol{r}$$

$\boldsymbol{\omega}$: Absolute angular velocity
$\boldsymbol{\alpha}$: Absolute angular acceleration

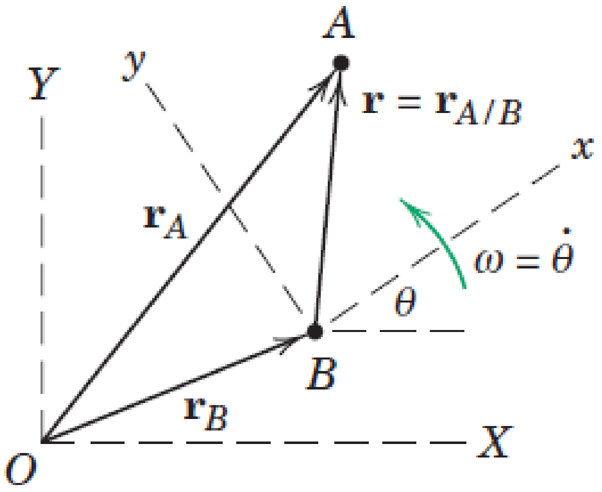
- *Relative* acceleration terms depend on the respective *absolute* angular velocity and *absolute* angular acceleration

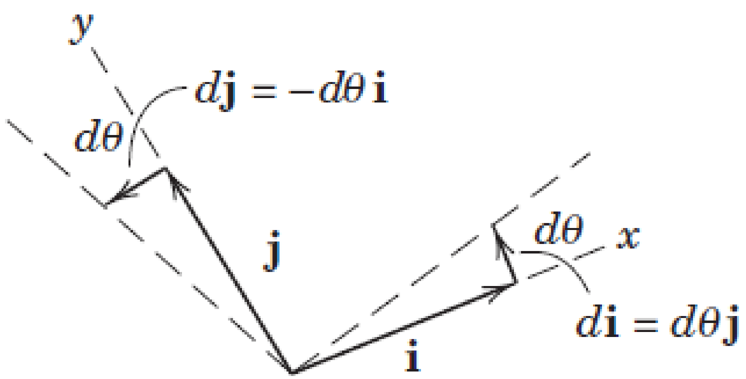
Motion Relative to Rotating Axes

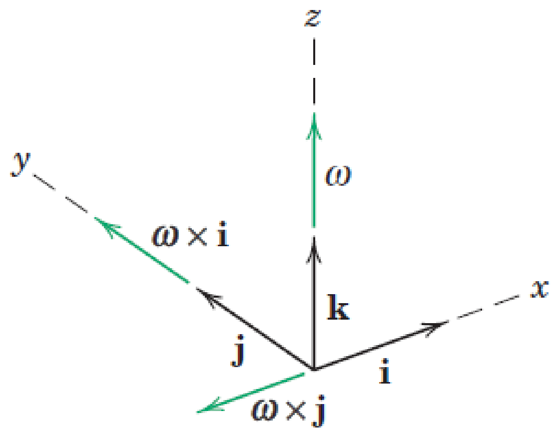


Use of rotating reference axes greatly facilitates the solution of many problems in kinematics where motion is generated within a system or observed from a system which itself is rotating. An example of such a motion is the movement of a fluid particle along the curved vane of a centrifugal pump, where the path relative to the vanes of the impeller becomes an important design consideration

Motion Relative to Rotating Axes

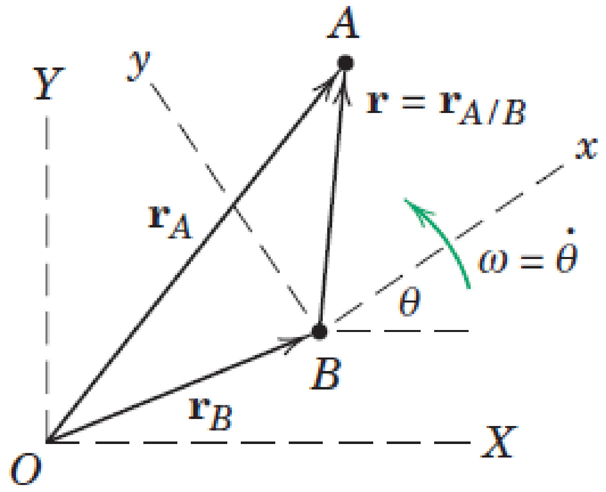


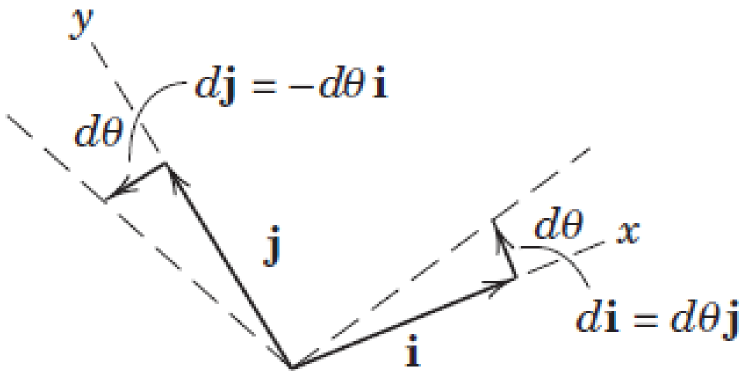


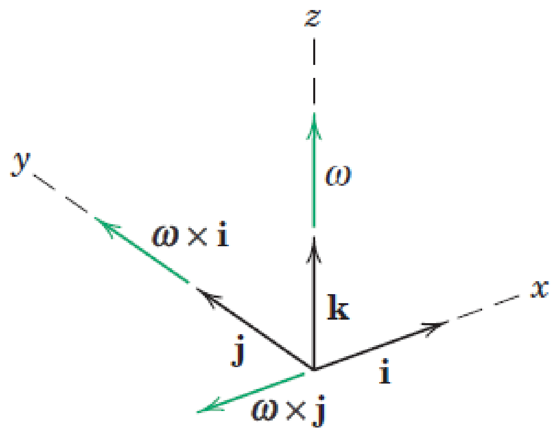


- Description of motion using rotating axes by considering the plane motion of two particles A and B in the fixed X-Y plane.
- A and B to be moving independently of one another
- Motion of A from a moving reference frame x-y which has its origin attached to B and which rotates with an angular velocity $\omega = \dot{\theta}$.
- In vector notation, $\boldsymbol{\omega} = \omega \boldsymbol{k} = \dot{\theta} \boldsymbol{k}$, where the vector is normal to the plane of motion where its positive sense is in the positive z-direction (out from the paper), as established by the right hand rule.

Motion Relative to Rotating Axes





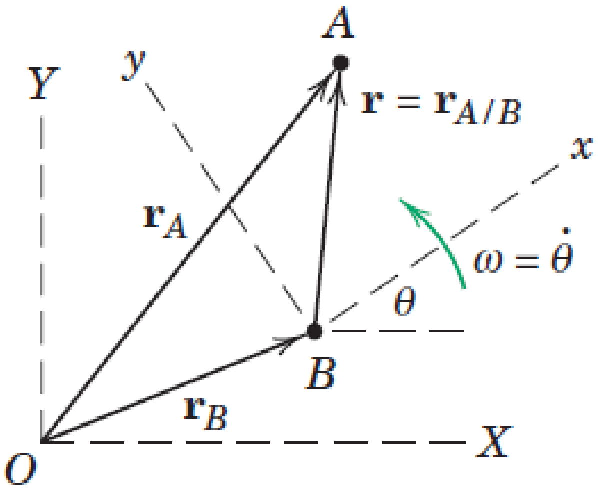


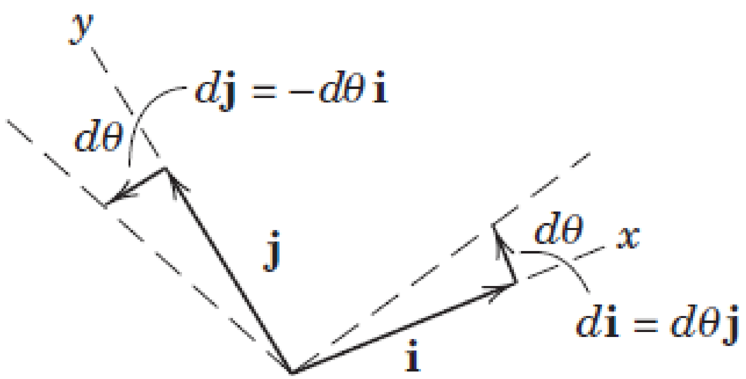
- The absolute position vector of A is given by

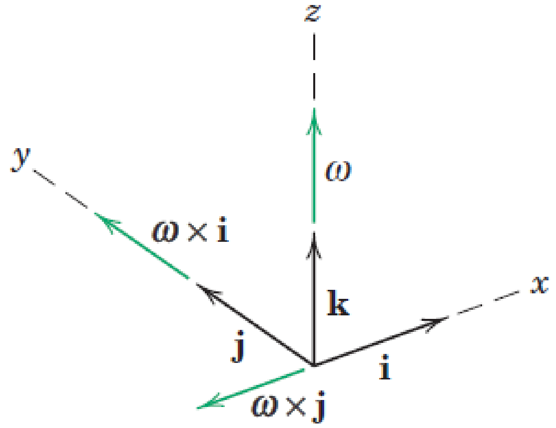
$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}$$

where i and j are unit vectors attached to the x-y frame and $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ stands for $r_{A/B}$, the position vector of A with respect to B.

Time Derivatives of Unit Vectors



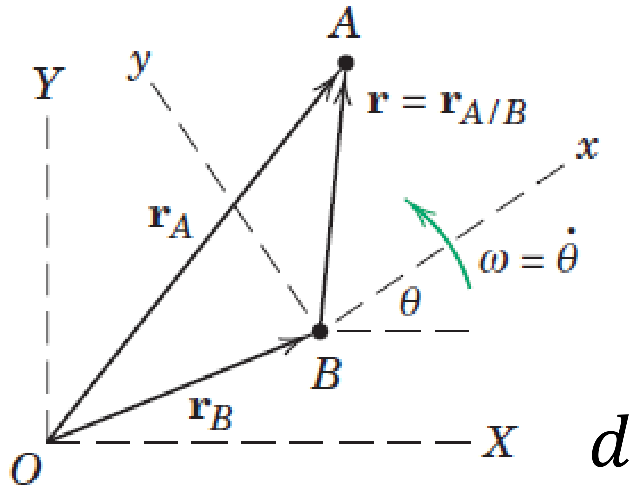


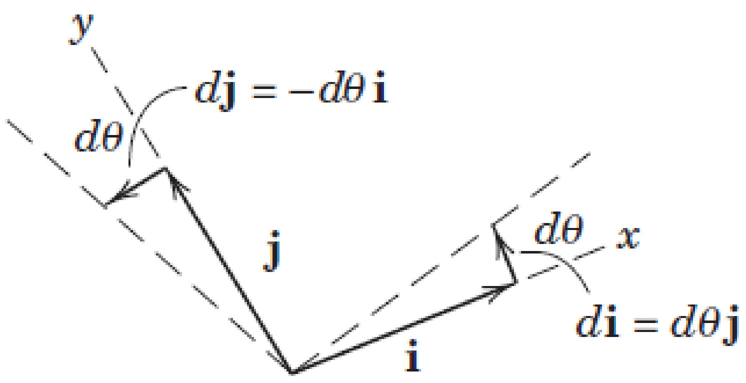


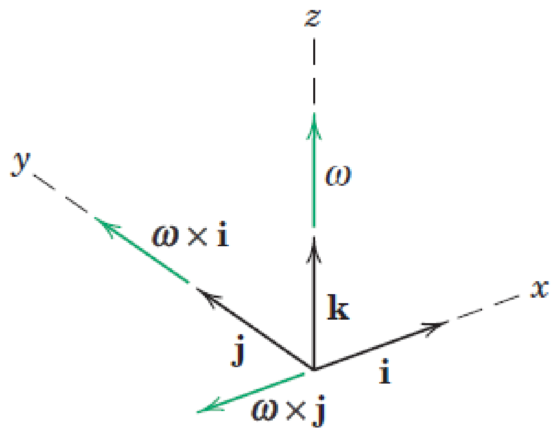
- To obtain the velocity and acceleration equations we must successively differentiate the position-vector equation with respect to time.
- In contrast to the case of translating axes, the unit vectors i and j are now rotating with the x-y axes and, therefore, have time derivatives which must be evaluated.
- The differential change in **i** is d**i**, and it has the direction of **j** and a magnitude equal to the angle d**j** times the length of the vector **i**, which is unity.

$$d\mathbf{i} = d\theta\ \mathbf{j} \qquad\qquad d\mathbf{j} = -\,d\theta\ \mathbf{i}$$

Time Derivatives of Unit Vectors







$$\frac{d\mathbf{i}}{dt} = \frac{d\theta}{dt}\,\mathbf{j}$$

$$\frac{d\mathbf{j}}{dt} = -\frac{d\theta}{dt}\,\mathbf{i}$$

$$\mathbf{i} = \omega\,\mathbf{j}$$

$$\mathbf{j} = -\,\omega\,\mathbf{i}$$

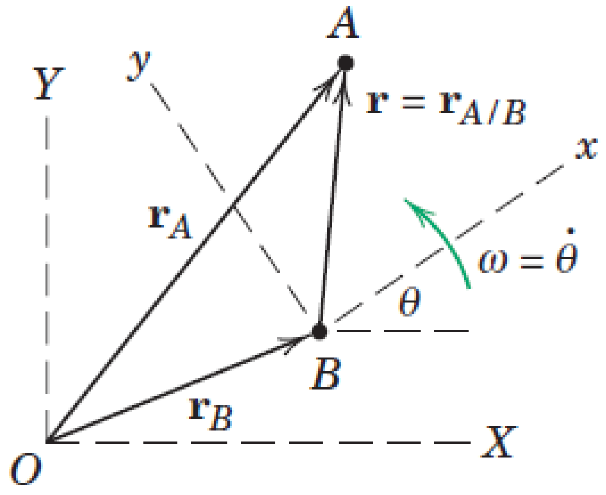
$$\mathbf{i} = \omega\,\mathbf{j} = \boldsymbol{\omega} \times \mathbf{i}$$

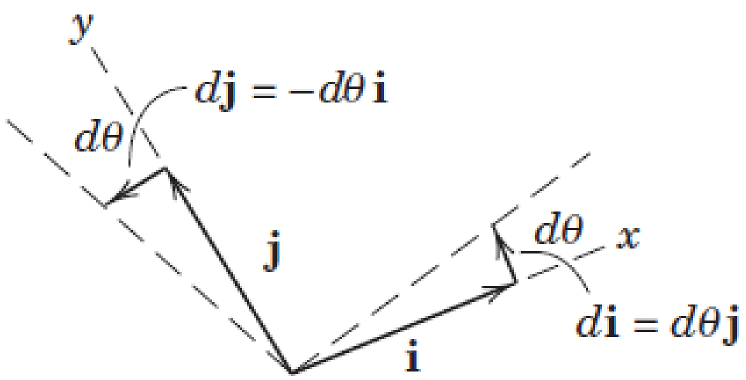
$$\mathbf{j} = -\omega\,\mathbf{i} = \boldsymbol{\omega} \times \mathbf{j}$$

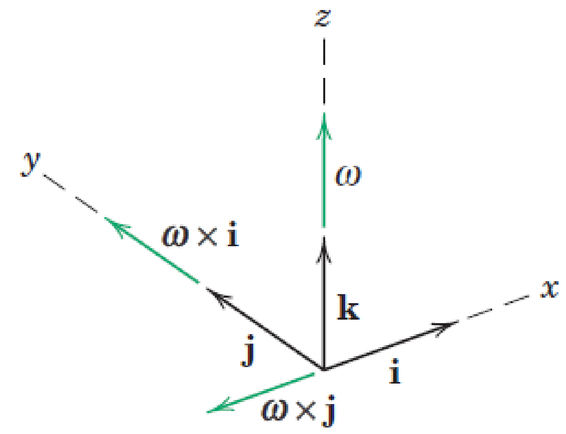
$$\mathbf{i} = \boldsymbol{\omega} \times \mathbf{i}$$

$$\mathbf{j} = \boldsymbol{\omega} \times \mathbf{j}$$

Relative Velocity







$$\mathbf{r}_A = \mathbf{r}_B + \mathrm{x}\mathbf{i} + \mathrm{y}\mathbf{j}$$

$$\dot{\mathbf{i}} = \boldsymbol{\omega} \times \mathbf{i}$$

$$\dot{\mathbf{j}} = \boldsymbol{\omega} \times \mathbf{j}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathrm{x}}\mathbf{i} + \dot{\mathrm{y}}\mathbf{j} + \mathrm{x}\dot{\mathbf{i}} + \mathrm{y}\dot{\mathbf{j}}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathrm{x}}\mathbf{i} + \dot{\mathrm{y}}\mathbf{j} + \mathrm{x}\boldsymbol{\omega} \times \mathbf{i} + \mathrm{y}\boldsymbol{\omega} \times \mathbf{j}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \boldsymbol{\omega} \times \mathrm{x}\mathbf{i} + \boldsymbol{\omega} \times \mathrm{y}\mathbf{j} + \mathbf{v}_{\mathbf{rel}}$$

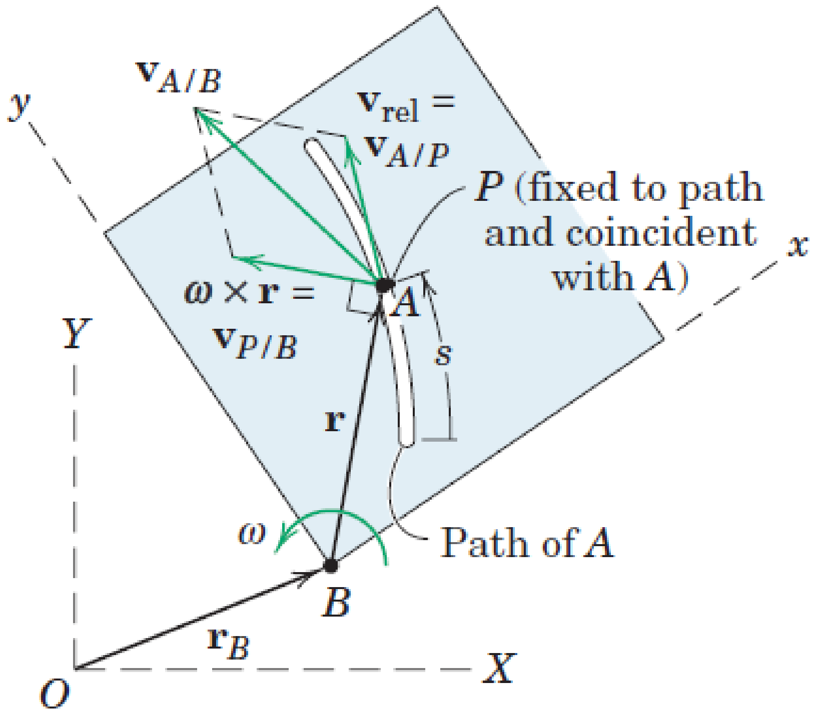
$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

$\boldsymbol{\omega} \times \mathbf{r}$ is the difference between the relative velocities as measured from nonrotating and rotating axes

Relative Velocity

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

- The motion of particle *A* relative to the rotating *x*-*y* plane is shown as taking place in a curved slot in a plate which represents the rotating *x*-*y* reference system.
- The velocity of *A* as measured relative to the plate, $\mathbf{v}_{\mathbf{rel}}$, would be tangent to the path fixed in the *x*-*y* plate and would have a magnitude $\dot{s}$, where s is measured along the path.
- This relative velocity may also be viewed as the velocity $\mathbf{v}_{A/P}$ relative to a point *P* attached to the plate and coincident with *A* at the instant under consideration.
- The term $\boldsymbol{\omega} \times \mathbf{r}$ has a magnitude and a direction normal to $\mathbf{r}$ and is the velocity of point *P* relative to *B* as seen from nonrotating axes attached to *B*.



Relative Velocity

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{P/B} + \mathbf{v}_{A/P}$$

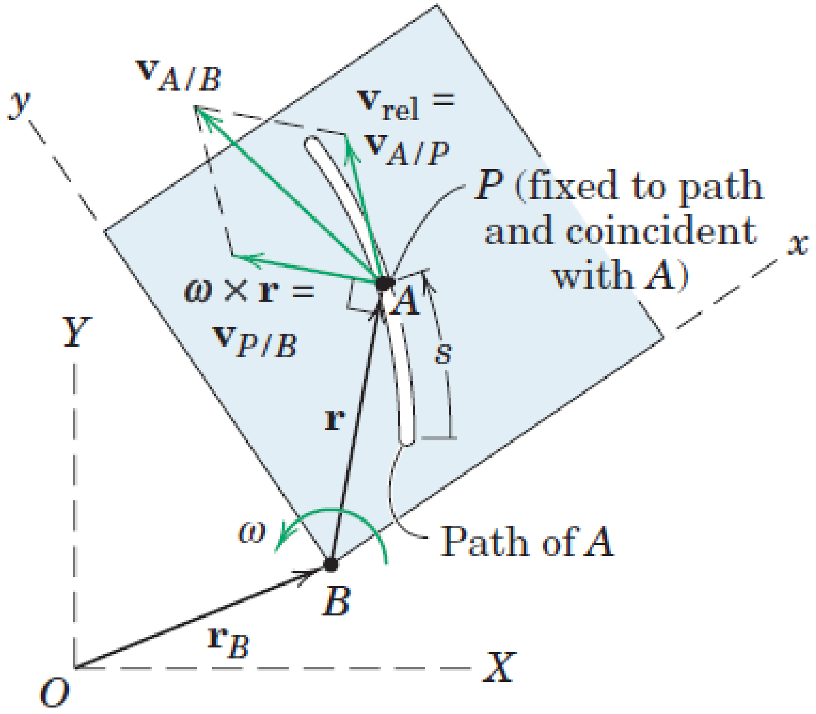
The term $\mathbf{v}_{P/B}$ is measured from a nonrotating position.

Term $\mathbf{v}_{A/P}$ is the same as $\mathbf{v}_{\mathbf{rel}}$ and is the velocity of *A* as measured in the *x*-*y* frame.

$$\mathbf{v}_A = \mathbf{v}_P + \mathbf{v}_{A/P}$$

$\mathbf{v}_P$ is the absolute velocity of *P* and represents the effect of the moving coordinate system, both translational and rotational

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$



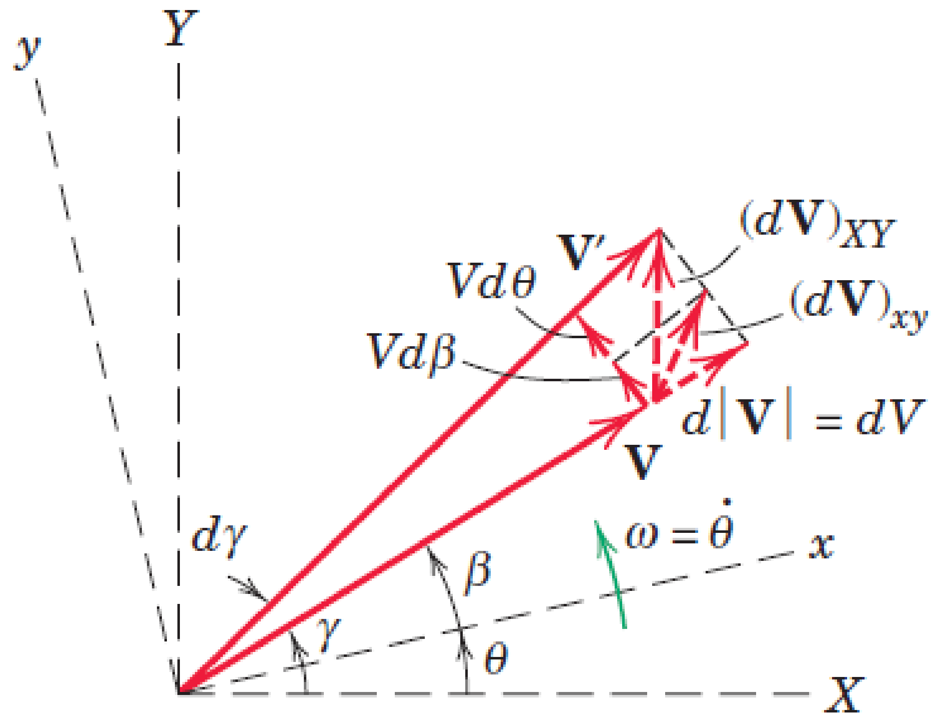
Transformation of a Time Derivative

$$\left(\frac{d\mathbf{V}}{dt}\right)_{XY} = \left(\dot{V}_x \mathrm{i} + \dot{V}_y \mathrm{j}\right) + \left(V_x \dot{\mathrm{i}} + V_y \dot{\mathrm{j}}\right)$$

- First two terms in the expression represent that part of the total derivative of **V** which is measured relative to the *x*-*y* reference system,
- Second two terms represent that part of the derivative due to the rotation of the reference system.

$$\left(\frac{d\mathbf{V}}{dt}\right)_{XY} = \left(\frac{d\mathbf{V}}{dt}\right)_{xy} + \boldsymbol{\omega} \times \mathbf{V}$$

$\boldsymbol{\omega} \times \mathbf{V}$ represents the difference between the time derivative of the vector as measured in a fixed reference system and its time derivative as measured in the rotating reference system



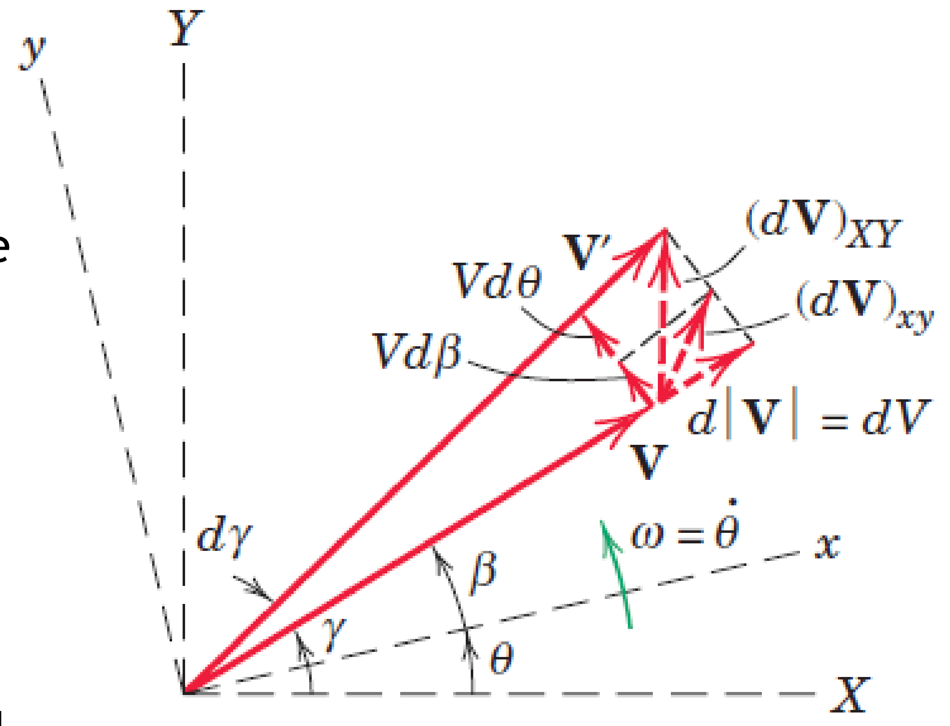
Transformation of a Time Derivative

- Figure shows the vector $\mathbf{V}$ at time t as observed both in the fixed axes X-Y and in the rotating axes x-y.

- During time dt, the vector swings to position $\mathbf{V}'$, and the observer in x-y measures the two components (a) dV due to its change in magnitude and (b) $Vd\beta$ due to its rotation $d\beta$ relative to x-y.

- To the rotating observer, then, the derivative $\frac{dV}{dt}_{\text{xy}}$ which the observer measures has the components $\frac{dV}{dt}$ and $V\frac{d\beta}{dt} = V\dot{\beta}$.

- The remaining part of the total time derivative not measured by the rotating observer has the magnitude $V\frac{d\theta}{dt}$ and, expressed as a vector, is $\boldsymbol{\omega} \times \mathbf{V}$.



$$\dot{\mathbf{V}}_{XY} = \dot{\mathbf{V}}_{xy} + \boldsymbol{\omega} \times \mathbf{V}$$

Relative Acceleration

$$\mathbf{v}_A = \mathbf{v}_B + \boldsymbol{\omega} \times \mathbf{r} + \mathbf{v}_{\mathbf{rel}}$$

$$\dot{\mathbf{v}}_A = \dot{\mathbf{v}}_B + \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times \dot{\mathbf{r}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \dot{\boldsymbol{r}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\dot{\boldsymbol{r}} = \frac{d(x\mathrm{i} + y\mathrm{j})}{dt} = x\dot{\mathbf{i}} + y\dot{\mathbf{j}} + \dot{x}\mathbf{i} + \dot{y}\mathbf{j} = \boldsymbol{\omega} \times \boldsymbol{r} + \mathbf{v}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r} + \mathbf{v}_{\mathbf{rel}}) + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

Relative Acceleration

$$\mathbf{v}_{\mathbf{rel}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$$

$$\dot{\mathbf{v}}_{\mathbf{rel}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} = \boldsymbol{a}_{\mathbf{rel}} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \dot{\mathbf{v}}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}} + \boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$$

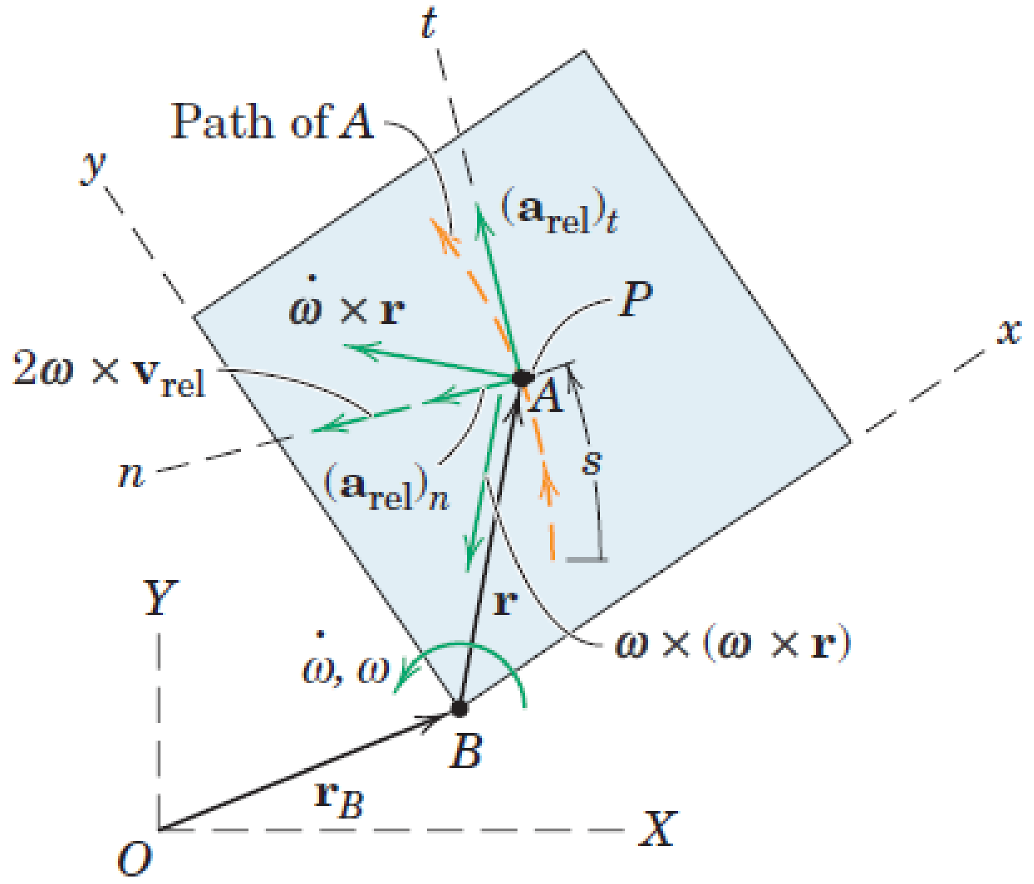
$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

Relative Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

Above equation is the general vector expression for the absolute acceleration of a particle *A* in terms of its acceleration $\boldsymbol{a}_{\mathbf{rel}}$ measured relative to a moving coordinate system which rotates with an angular velocity and an angular acceleration $\dot{\boldsymbol{\omega}}$.

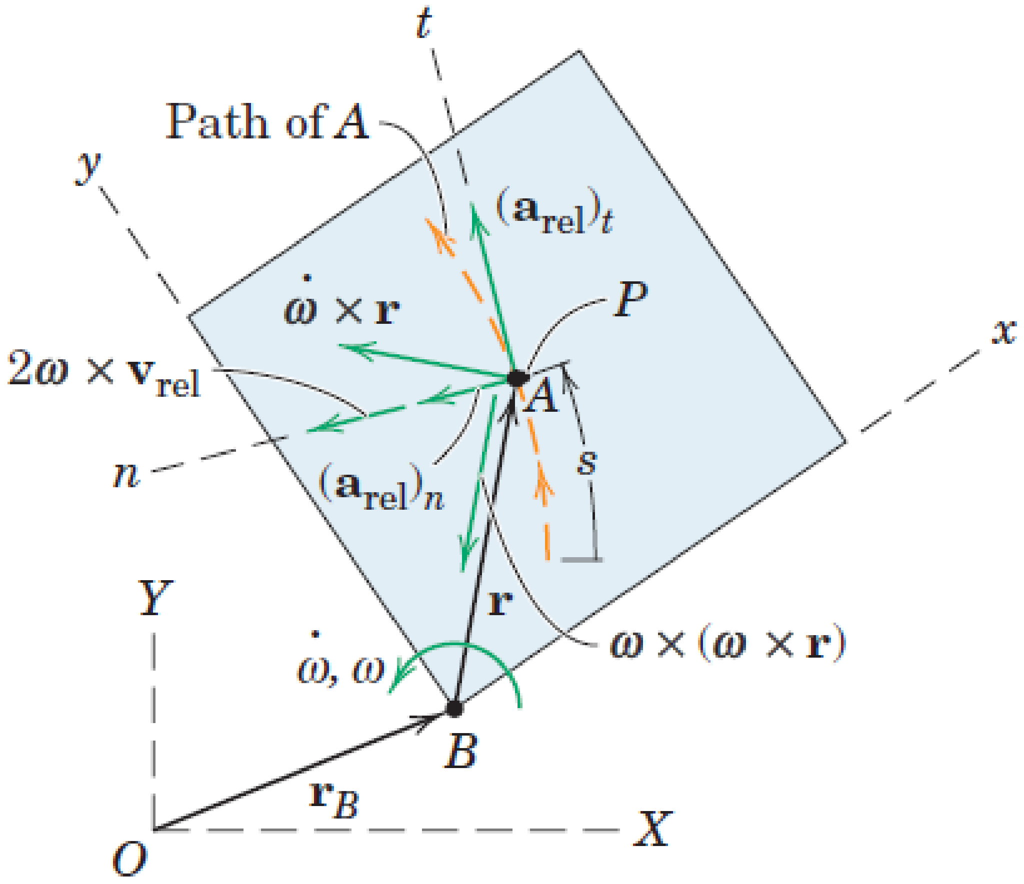
- $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ Tangential component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.
- $\boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$ normal component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.



Relative Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ Tangential component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.
- $\boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$ normal component of the acceleration $\boldsymbol{a}_P$ of the coincident point *P* in its circular motion with respect to *B*.
- This motion would be observed from a set of nonrotating axes moving with B.
- The magnitude of $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$ is $r\ddot{\theta}$ and its direction is tangent to the circle. The magnitude of $\boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$ is $\boldsymbol{\omega} \boldsymbol{r}^2$ and its direction is from P to B along the normal to the circle.



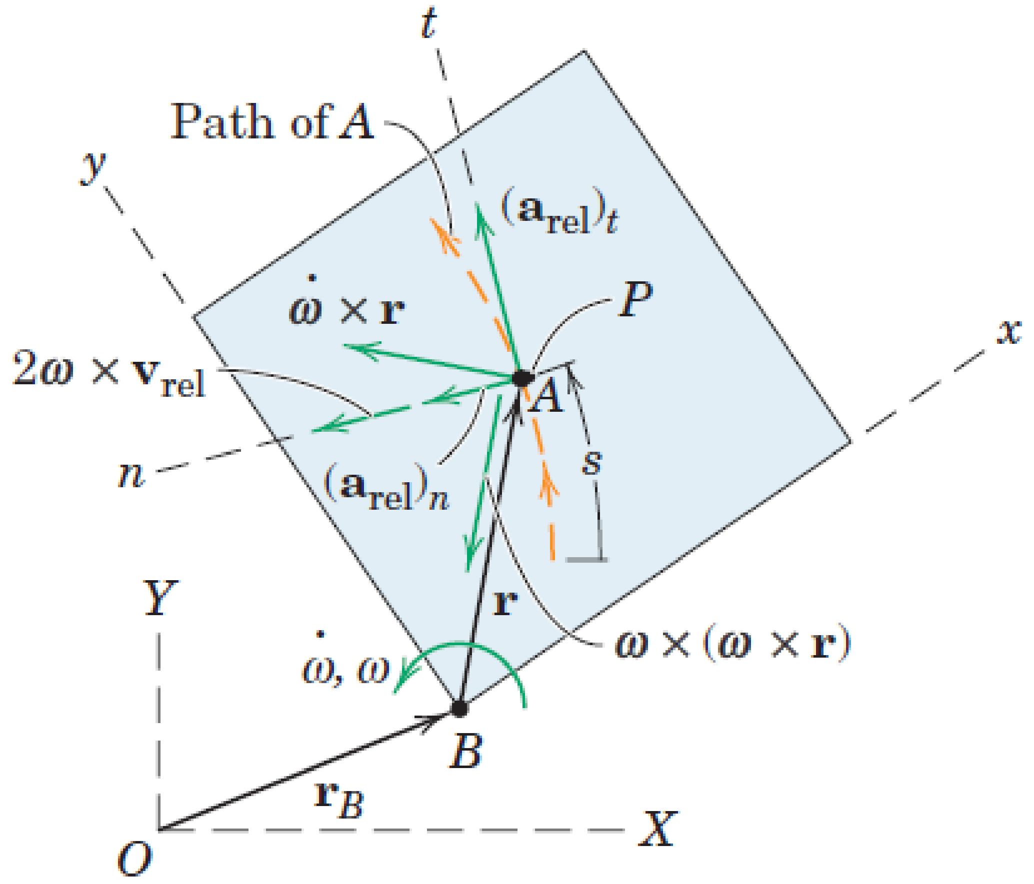
Relative Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v_{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- The acceleration of *A* relative to the plate along the path, $\boldsymbol{a}_{\mathbf{rel}}$, may be expressed in rectangular, normal and tangential, or polar coordinates in the rotating system. Frequently, *n*- and *t*-components are used.

- The tangential component has the magnitude $\boldsymbol{a}_{\mathbf{rel}_t} = \ddot{s}$, where s is the distance measured along the path to *A*.

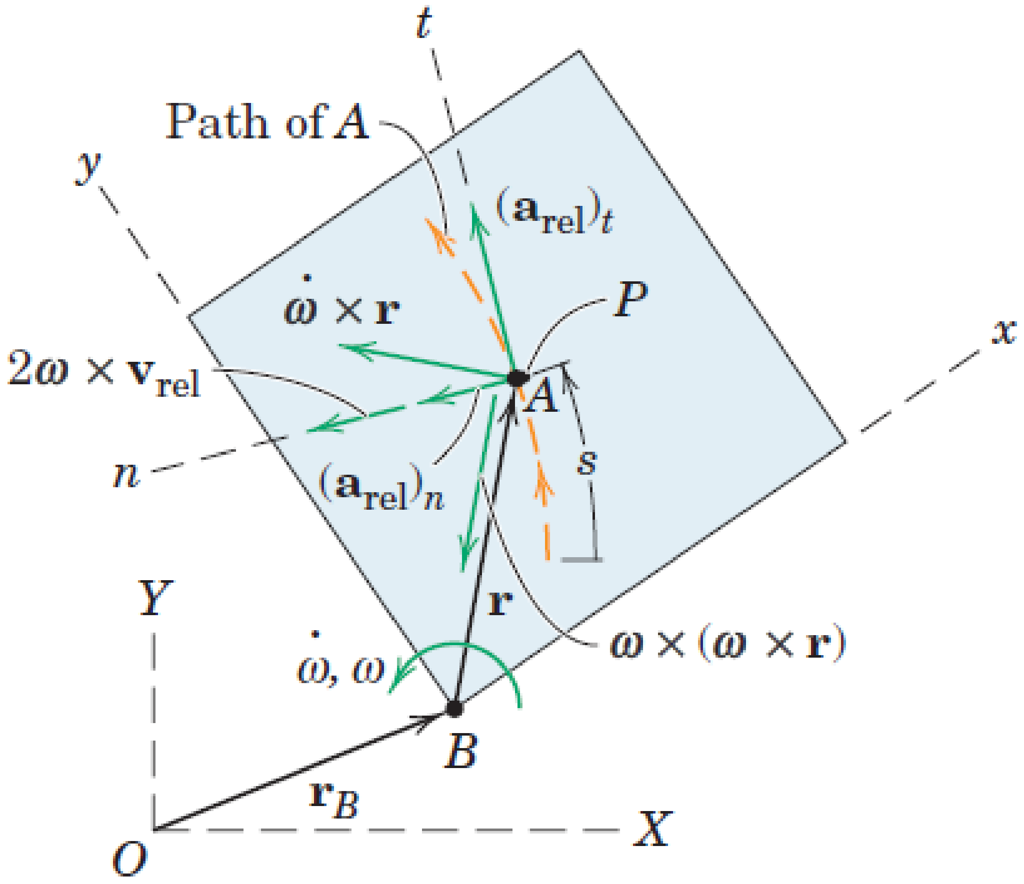
- The normal component has the magnitude $\boldsymbol{a}_{\mathbf{rel}_n} = \frac{v_{rel}^2}{\rho}$, where ρ is the radius of curvature of the path as measured in *x*-*y*. The sense of this vector is always toward the center of curvature.



Coriolis Acceleration

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- The term $2\boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}$ is called the ***Coriolis* acceleration**.
- Named after the French military engineer G. Coriolis (1792–1843), who was the first to call attention to this term.
- It represents the difference between the acceleration of *A* relative to *P* as measured from nonrotating axes and from rotating axes.
- The direction is always normal to the vector $\mathbf{v}_{\mathrm{rel}}$, and the sense is established by the right-hand rule for the cross product.

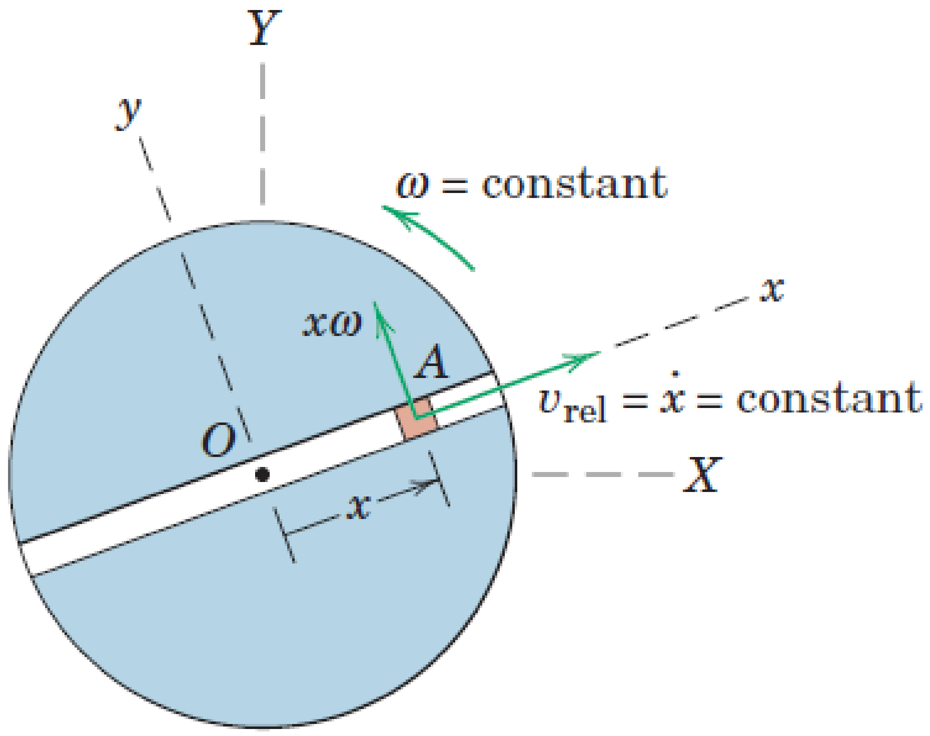


Coriolis Acceleration

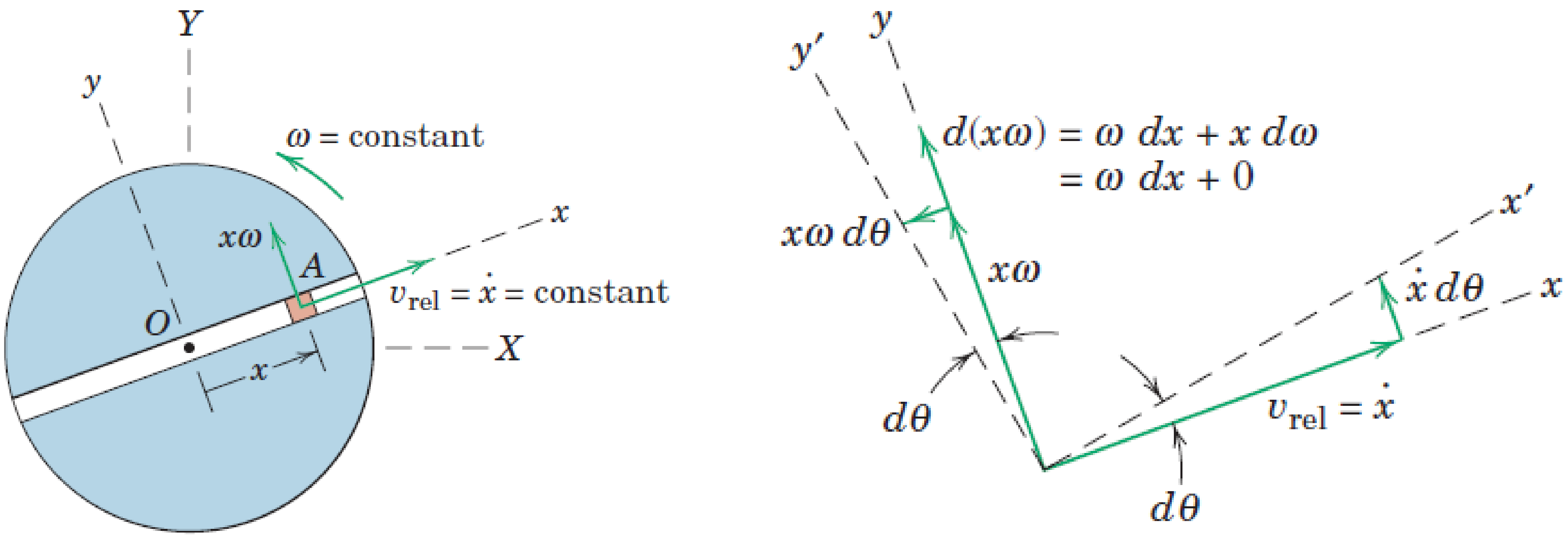
$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r}$$
$$+2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- The Coriolis acceleration $2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}}$ is difficult to visualize because it is composed of two separate physical effects. To help with this visualization, we will consider the simplest possible motion in which this term appears.

- We have a rotating disk with a radial slot in which a small particle *A* is confined to slide. Let the disk turn with a constant angular velocity $\boldsymbol{\omega}$ and let the particle move along the slot with a constant speed $v_{rel} = \dot{x}$ relative to the slot. The velocity of *A* has the two components (*a*) $\dot{x}$ due to motion along the slot and (*b*) $x\omega$ due to the rotation of the slot.

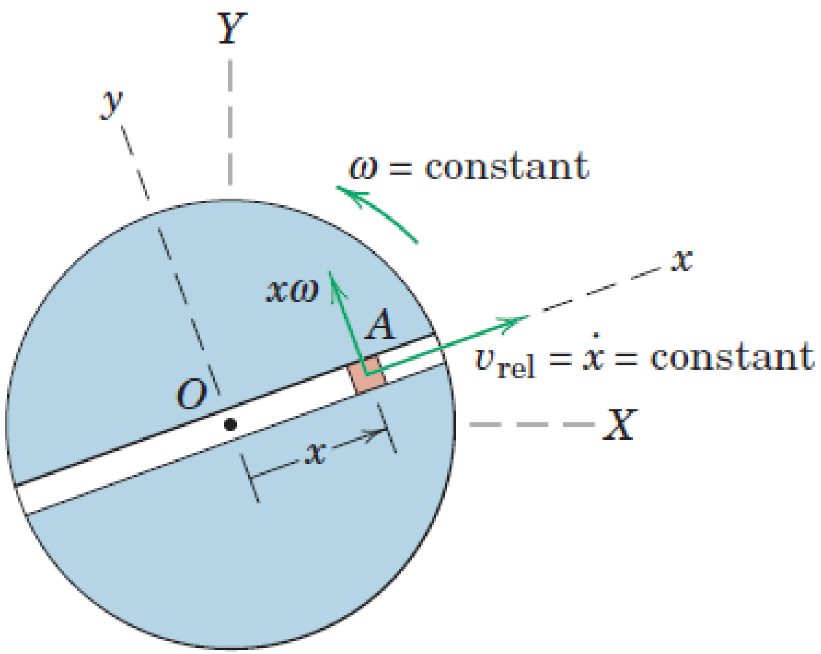


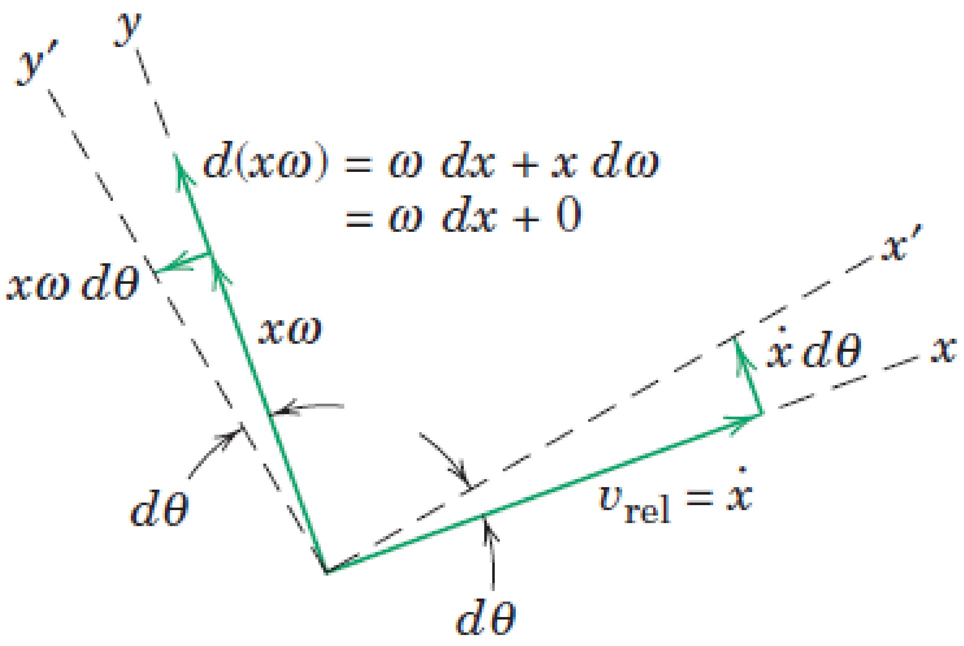
Coriolis Acceleration



- The changes in these two velocity components due to the rotation of the disk are shown in 2nd figure for the interval dt, during which the *x*-*y* axes rotate with the disk through the angle $d\theta$ to *x'*-*y'*.
- The velocity increment due to the change in direction of v_{rel} is $\dot{x}d\theta$ and that due to the change in magnitude of $x\omega$ is ωdx, both being in the y-direction normal to the slot.

Coriolis Acceleration



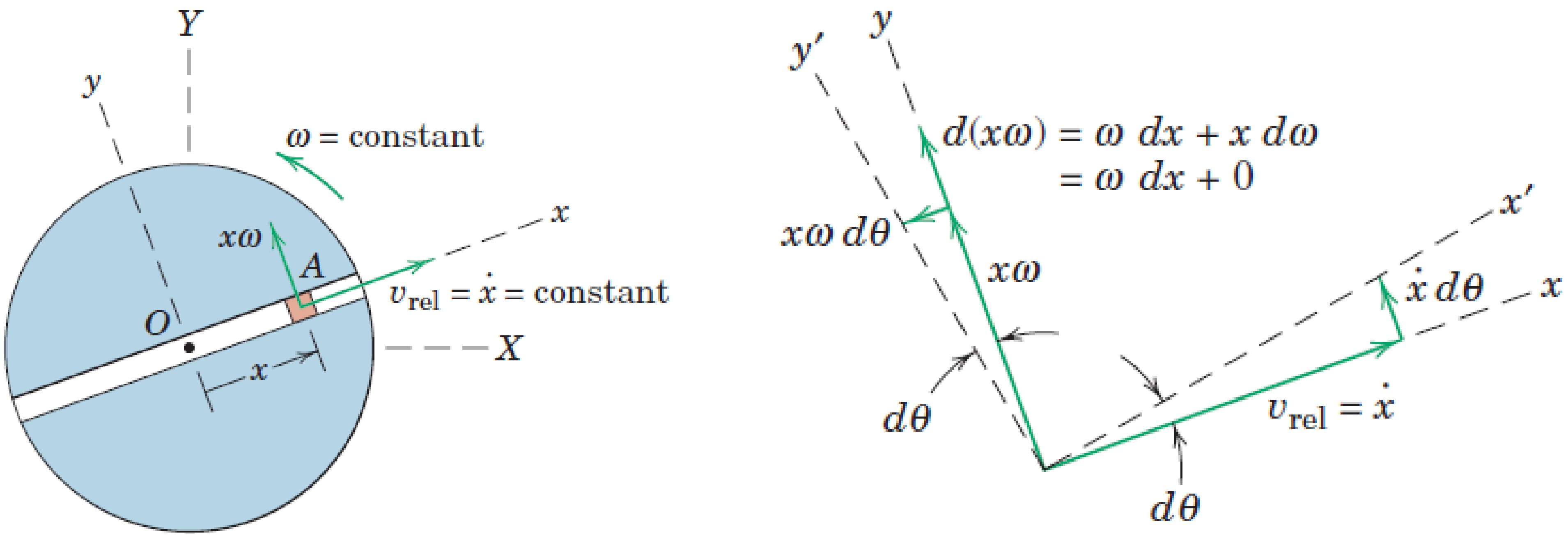


- The velocity increment due to the change in direction of v_{rel} is $\dot{x}d\theta$ and that due to the change in magnitude of $x\omega$ is ωdx, both being in the y-direction normal to the slot.

$$\frac{\dot{x}d\theta + \omega dx}{dt} = \dot{x}\dot{\theta} + \omega\dot{x} = \omega\dot{x} + \omega\dot{x} = 2\omega\dot{x}$$

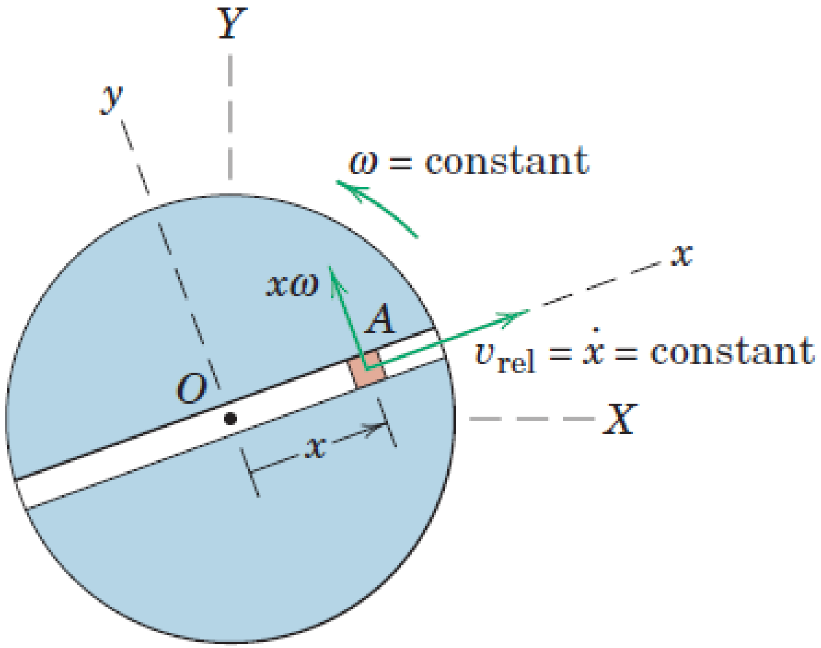
- For this case, $2\omega\dot{x}$ which is the magnitude of the Coriolis acceleration $\mathbf{2\boldsymbol{\omega}} \times \mathbf{v_{rel}}$

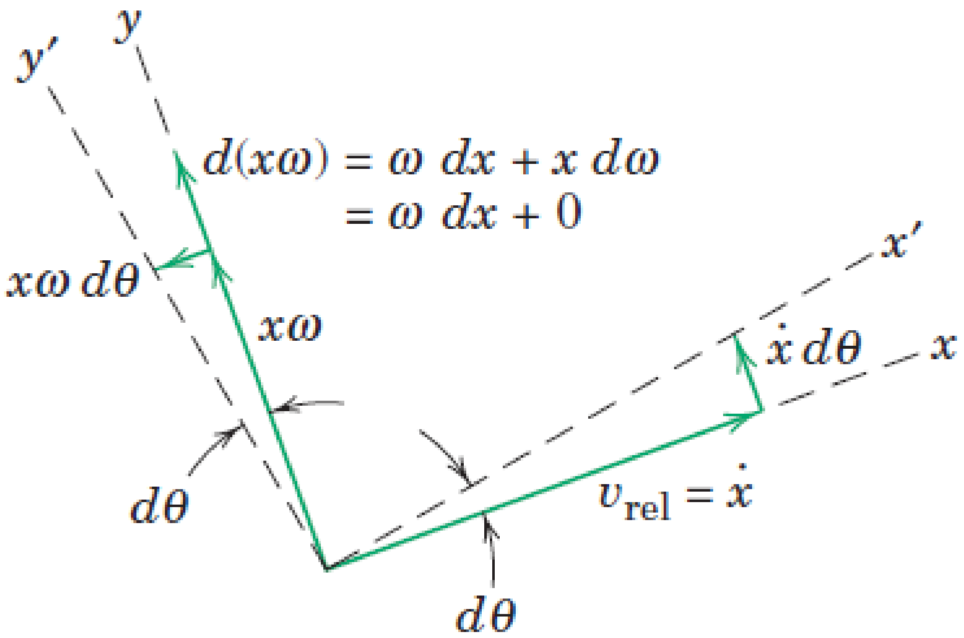
Coriolis Acceleration



➢ Dividing the remaining velocity increment $x\omega d\theta$ due to the change in direction of $x\omega$ by dt gives $x\omega\dot{\theta}$ or $x\omega^2$, which is the acceleration of a point *P* fixed to the slot and momentarily coincident with the particle *A*.

Coriolis Acceleration





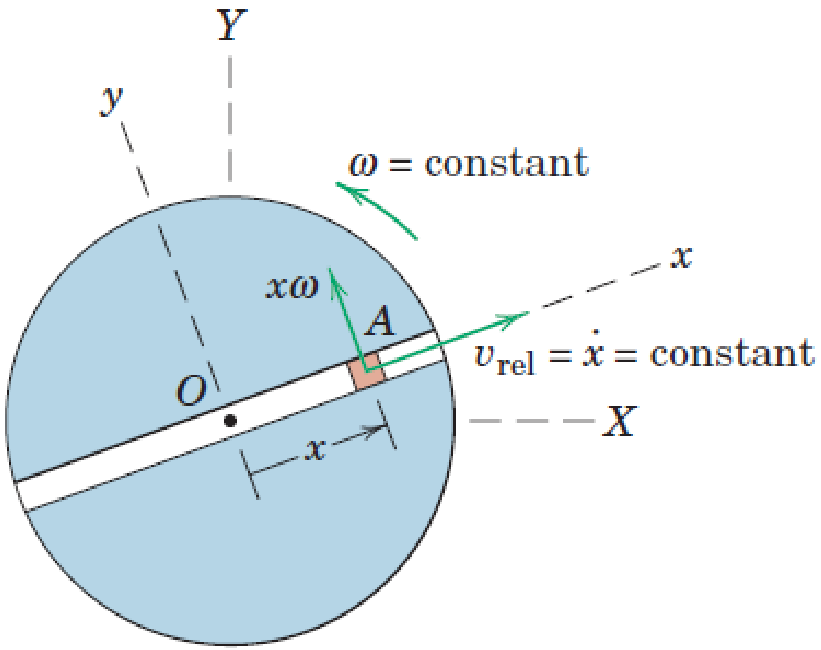
- With the origin B in that equation taken at the fixed center O, $\boldsymbol{a}_B = 0$.
- With constant angular velocity, $\dot{\boldsymbol{\omega}} \times \boldsymbol{r} = \mathbf{0}$
- With $\mathbf{v}_{\text{rel}}$ constant in magnitude and no curvature to the slot, $\boldsymbol{a}_{\text{rel}} = 0$

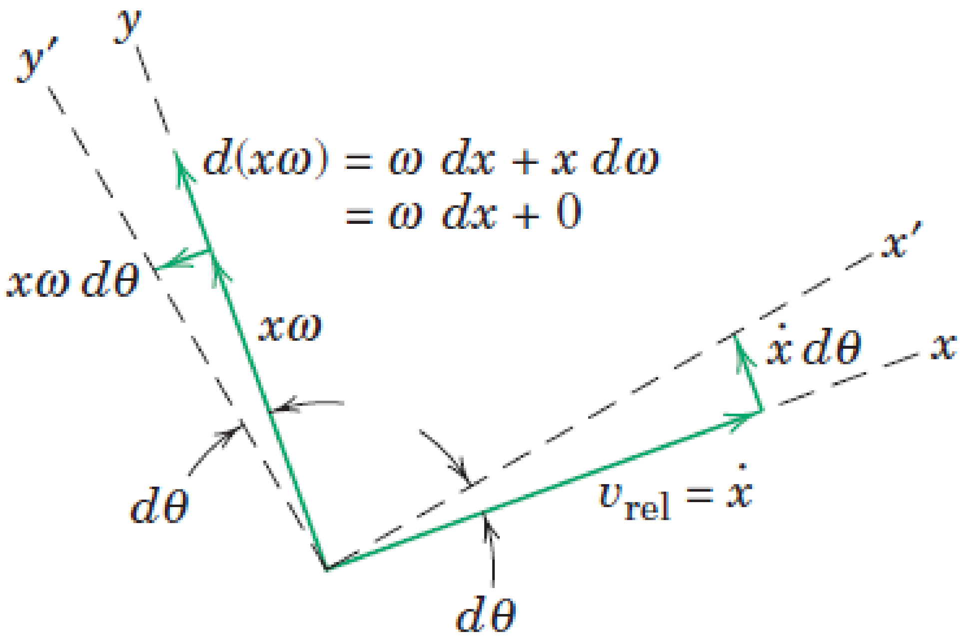
$$\boldsymbol{a}_A = \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}}$$

- Replacing $\boldsymbol{r}$ by $x\mathbf{i}$, $\boldsymbol{\omega}$ by $\omega\mathbf{k}$, and $\mathbf{v}_{\text{rel}}$ by $\dot{x}\mathbf{i}$ gives

$$a_A = -x\omega^2\mathrm{i} + 2\dot{x}\omega j$$

Coriolis Acceleration





- Replacing $\boldsymbol{r}$ by $x\mathbf{i}$, $\boldsymbol{\omega}$ by $\omega\mathbf{k}$, and $\mathbf{v}_{\mathbf{rel}}$ by $\dot{x}\mathbf{i}$ gives

$$a_A = -x\omega^2\mathrm{i} + 2\dot{x}\omega j$$

- If the slot in the disk had been curved, we would have had a normal component of acceleration relative to the slot so that $\boldsymbol{a}_{\mathbf{rel}}$ would not be zero.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

- From the third equation where $\boldsymbol{a}_B$, $\boldsymbol{a}_{\boldsymbol{P}/B}$ has been combined to give $\boldsymbol{a}_{\boldsymbol{P}}$, it is seen that the relative-acceleration term $\boldsymbol{a}_{A/P}$, unlike the corresponding relative-velocity term, is not equal to the relative acceleration $\boldsymbol{a}_{\mathbf{rel}}$ measured from the rotating *x*-*y* frame of reference.

- The Coriolis term $(2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}})$ is, therefore, the difference between the acceleration $\boldsymbol{a}_{A/P}$ of *A* relative to *P* as measured in a nonrotating system and the acceleration $\boldsymbol{a}_{\mathbf{rel}}$ of *A* relative to *P* as measured in a rotating system.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

- From the fourth equation, it is seen that the acceleration $\boldsymbol{a}_{A/B}$ of A with respect to B as measured in a nonrotating system is a combination of the last four terms in the first equation for the rotating system.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

- Writing the acceleration of *A* in terms of the acceleration of the coincident point *P*

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

- Because the acceleration of *P* is $\boldsymbol{a}_P = \boldsymbol{a}_B + \boldsymbol{a}_{\boldsymbol{P}/B}$
- When the equation is written in this form, point *P* may not be picked at random because it is the one point attached to the rotating reference frame coincident with *A* at the instant of analysis.

Rotating vs Non-Rotating Systems

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{r} + 2\boldsymbol{\omega} \times \mathbf{v}_{\mathbf{rel}} + \boldsymbol{a}_{\mathbf{rel}}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \qquad \boldsymbol{a}_{\boldsymbol{P}/B} \qquad + \qquad \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_P + \boldsymbol{a}_{A/P}$$

$$\boldsymbol{a}_A = \boldsymbol{a}_B + \boldsymbol{a}_{A/B}$$

$\mathbf{v}_B$: absolute velocity of the origin *B* of the rotating axes
$\boldsymbol{a}_B$: absolute acceleration of the origin *B* of the rotating axes
$\boldsymbol{r}$: position vector of the coincident point *P* measured from *B*
$\boldsymbol{\omega}$: angular velocity of the rotating axes
$\dot{\boldsymbol{\omega}}$: angular acceleration of the rotating axes
$\mathbf{v}_{\mathbf{rel}}$: velocity of *A* measured relative to the rotating axes
$\boldsymbol{a}_{\mathbf{rel}}$: acceleration of *A* measured relative to the rotating axes

DYNAMICS OF RIGID BODIES

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General equations of Motion

- The **kinetics** of rigid bodies treats the relationships between the external forces acting on a body and the corresponding translational and rotational motions of the body.
- **Plane Motion:** A body which can be approximated as a thin slab with its motion confined to the plane of the slab
- **Plane of Motion**: The plane of motion will contain the mass center, and all forces which act on the body will be projected onto the plane of motion
- A body which has appreciable dimensions normal to the plane of motion but is symmetrical about that plane of motion through the mass center may be treated as having plane motion.

General equations of Motion

- **Two force equations and one moment equation** or their equivalent are required to determine the state of rigid-body plane motion.
- Successful application of kinetics requires that you isolate the body or system to be analyzed
- For problems involving the instantaneous relationships among force, mass, and acceleration, the body or system should be explicitly defined by isolating it with its free-body diagram.
- When the principles of work and energy are employed, an active force diagram which shows only those external forces which do work on the system may be used in lieu of the free-body diagram.
- No solution of a problem should be attempted without first defining the complete external boundary of the body or system and identifying all external forces which act on it.

Mass Moment of Inertia

- **Mass moment of inertia of the body:** Property of the body which accounts for the radial distribution of its mass with respect to a particular axis of rotation normal to the plane of motion

General equations of Motion

Methods to obtain equations of motion

- Forces and moments to the instantaneous linear and angular accelerations
- Method of work and energy
- Method of impulse and momentum

Three types of motion:

- Translation
- fixed-axis rotation
- general plane motion

General equations of Motion

- Forces and moments to the instantaneous linear and angular accelerations
- Resultant $\sum F$ of the external forces acting on the body equals the mass m of the body times the acceleration of its mass center G.

$$\sum F = m\bar{a}$$

- Moment equation taken about the mass center, resultant moment about the mass center of the external forces on the body equals the time rate of change of the angular momentum of the body about the mass center.

$$\sum M_G = \dot{H}_G$$

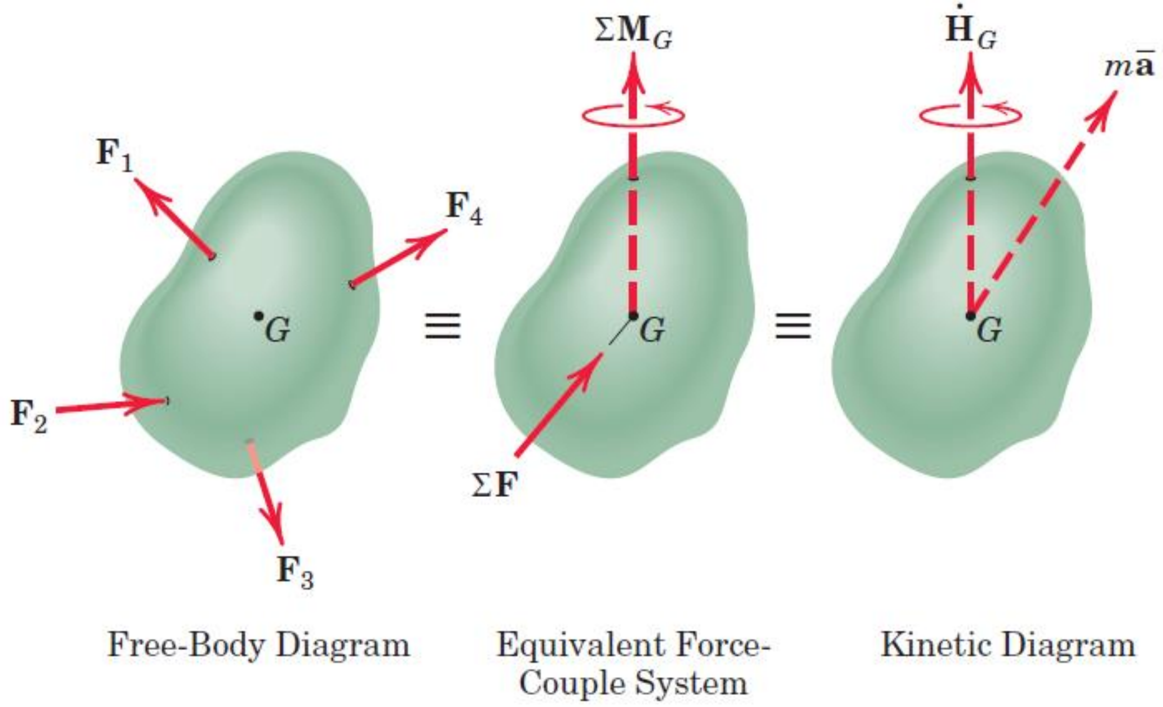
General equations of Motion

Free Body Diagram and Kinetic Diagram

- By replacing the external forces by their equivalent force-couple system in which the resultant force acts through the mass center, we may visualize the action of the forces and the corresponding dynamic response of the body with the aid of figure (

$$\sum F = m\bar{a}$$

$$\sum M_G = \dot{H}_G$$



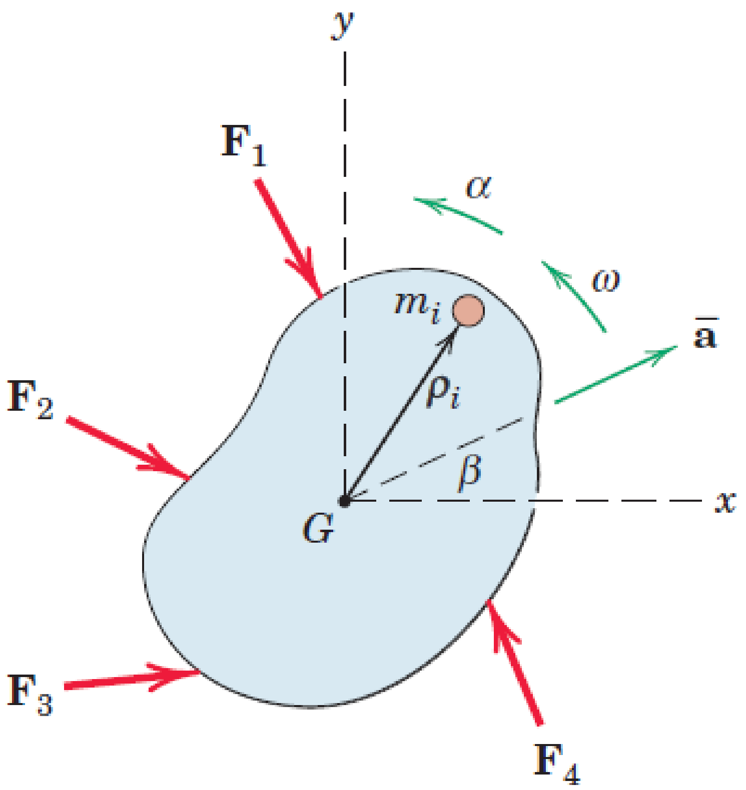
Plane Motion Equations

- Mass Center G
- Acceleration of mass center $\bar{\boldsymbol{a}}$
- Angular Velocity of body $\boldsymbol{\omega}\hat{\boldsymbol{k}}$
- Angular acceleration of body $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}\hat{\boldsymbol{k}}$

- Angular momentum about the mass center for the general system

$$\boldsymbol{H_G} = \sum \boldsymbol{\rho_i} \times \boldsymbol{m_i}\dot{\boldsymbol{\rho}}_{\boldsymbol{i}}$$

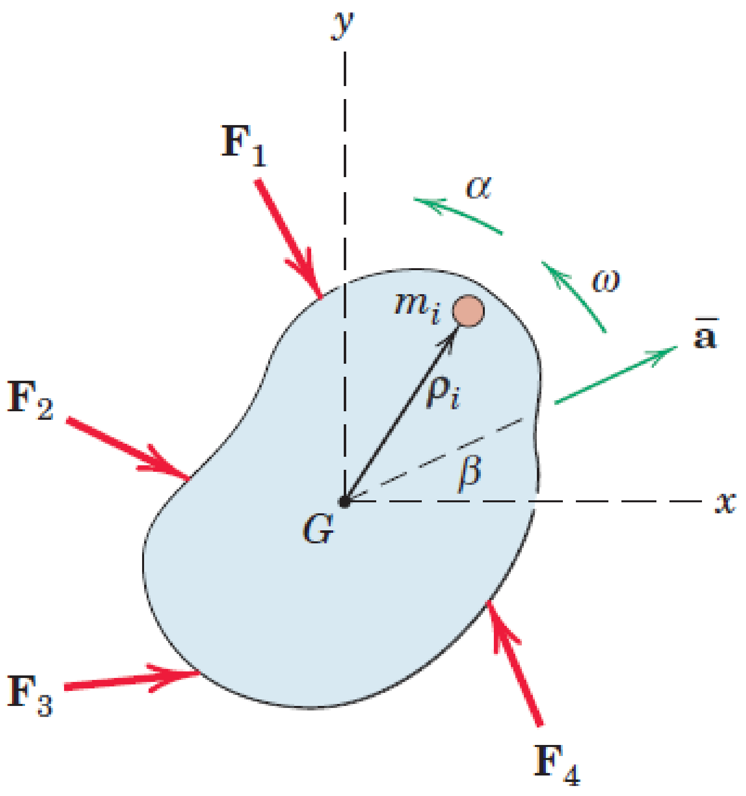
ρ_i is position vector of m_i relative to G



Plane Motion Equations

- Angular momentum about the mass center for the general system

$$H_G = \sum \rho_i \times m_i \dot{\rho}_i$$



ρ_i is position vector of m_i relative to G

- For rigid body, $\dot{\boldsymbol{\rho}}_i = \boldsymbol{\omega} \times \boldsymbol{\rho}_i$
- Direction of $\dot{\boldsymbol{\rho}}_i$ is perpendicular to $\boldsymbol{\rho}_i$ in x-y plane
- Direction of $\boldsymbol{H}_G$ is along $\boldsymbol{\omega}$
- Magnitude of $\boldsymbol{H}_G$ is $\sum \omega m_i \rho_i^2 = \omega \sum m_i \rho_i^2 = \omega \int \rho^2\, dm$
- **Mass moment of Inertia of the body about the z-axis through *G*,** $\bar{I} = \int \rho^2\, dm$

$$H_G = \omega \int \rho^2\, dm = \bar{I}\omega$$

Plane Motion Equations

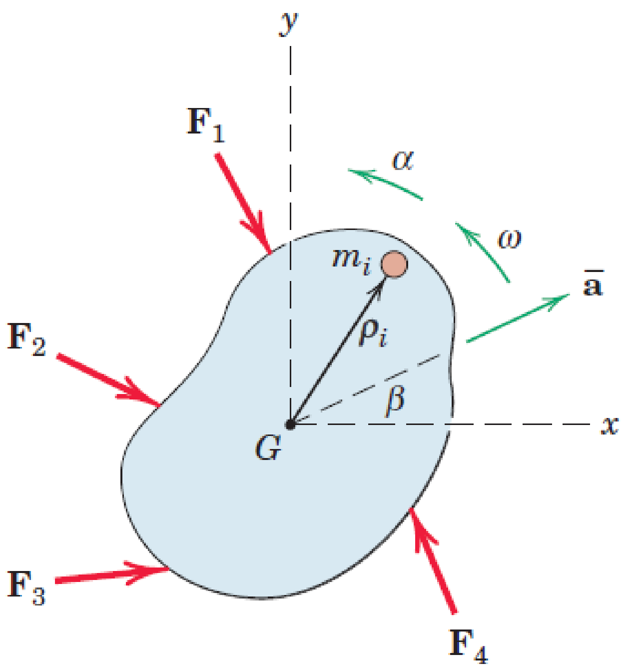
- Angular momentum of the rigid body about z-axis passing through *G*

$$H_G = \omega \int \rho^2 \, dm = \bar{I}\omega$$

- $\bar{I}$ is constant property of the body
- Measure of the rotational inertia, which is the resistance to change in rotational velocity due to the radial distribution of mass around the z-axis through *G*
- Moment Equation

$$M_G = \dot{H}_G = \bar{I}\dot{\omega} = \bar{I}\alpha$$

- $\dot{\omega} = \alpha$ is angular acceleration of the body



Plane Motion Equations

- Moment Equation

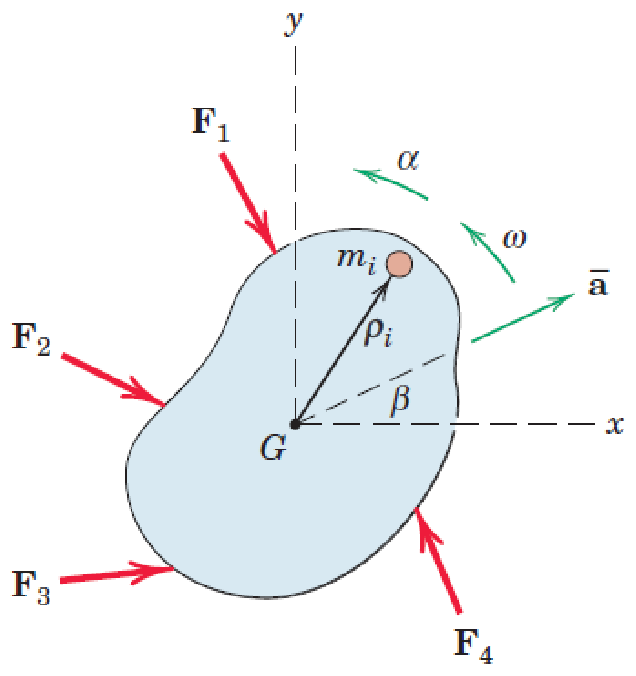
$$M_G = \bar{I}\alpha$$

- $\dot{\omega} = \alpha$ is angular acceleration of the body

- Force Equation

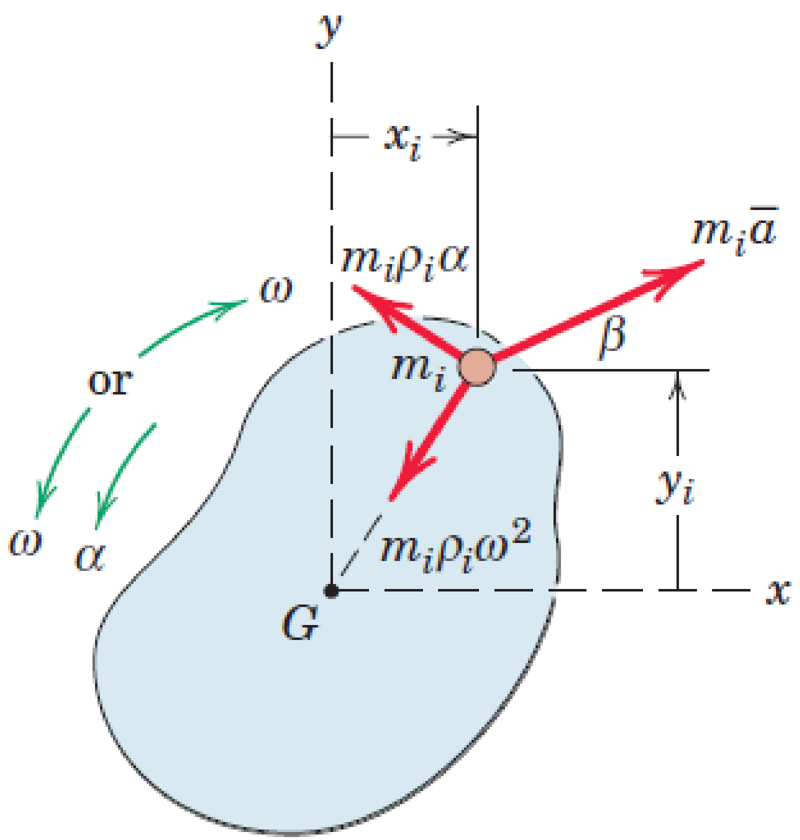
$$F = m\bar{a}$$

- $\bar{a}$ is acceleration of the mass center of body



Alternate derivations

- The acceleration of m_i equals the vector sum of $\bar{\boldsymbol{a}}$ and the relative terms $\rho_i\omega^2$ and $\rho_i\alpha$, where the mass center G is used as the reference point
- Resultant of all forces on m_i has the components $m_i\bar{\boldsymbol{a}}$, $m_i\rho_i\omega^2$ and $m_i\ \rho_i\alpha$ in the directions shown
- Sum of the moments of these force components about G in the sense of α
- $M_{G_i} = m_i\rho_i^2\alpha + m_i\bar{\boldsymbol{a}}\sin\beta\ x_i + m_i\bar{\boldsymbol{a}}\ \cos\beta\ \ y_i$
- $\sum M_G = \sum m_i\rho_i^2\alpha + \bar{\boldsymbol{a}}\sin\beta\ \sum m_i x_i + \bar{\boldsymbol{a}}\ \cos\beta\ \sum m_i y_i$



$$\sum m_i x_i = m\bar{x} = 0 \qquad \sum m_i y_i = m\bar{y} = 0$$

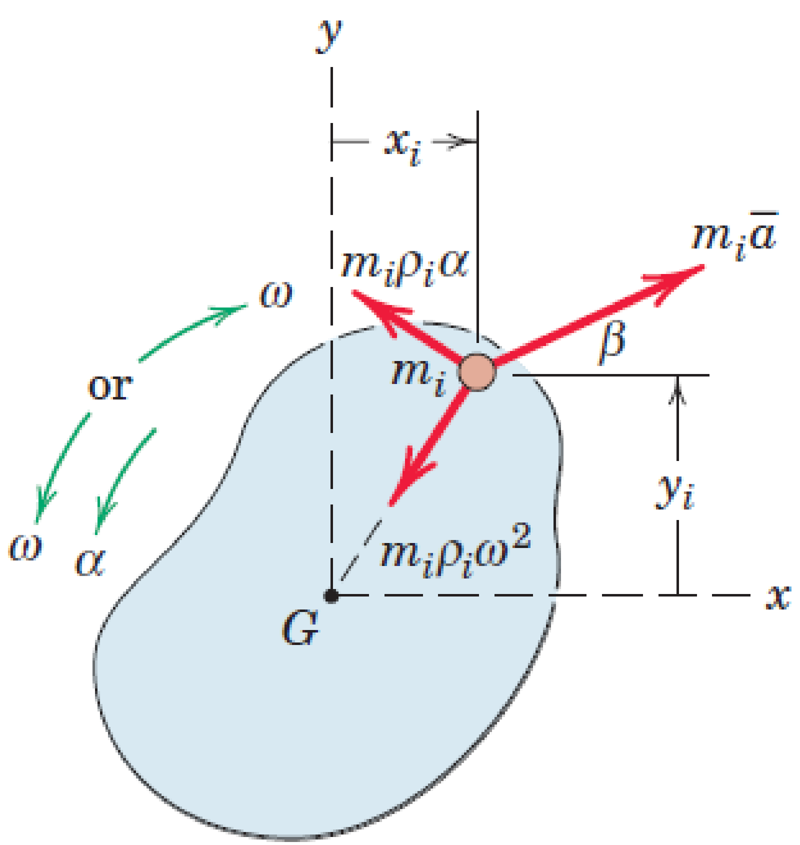
$$\sum M_G = \sum m_i\rho_i^2\alpha = \bar{I}\alpha$$

Alternate derivations

- The contribution to $\sum M_G$ of the forces internal to the body is **zero** since they occur in pairs of equal and opposite forces of action and reaction between interacting particles
- $\sum M_G$ represents the sum of moments about the mass center *G* of only the external forces acting on the body, as disclosed by the free-body diagram.

$$\sum M_G = \sum m_i \rho_i^2 \alpha = \bar{I}\alpha$$

- Force component $m_i \rho_i \omega^2$ has no moment about *G*
- The angular velocity ω has no influence on the moment equation about the mass center



Plane Motion Equations

- Moment Equation

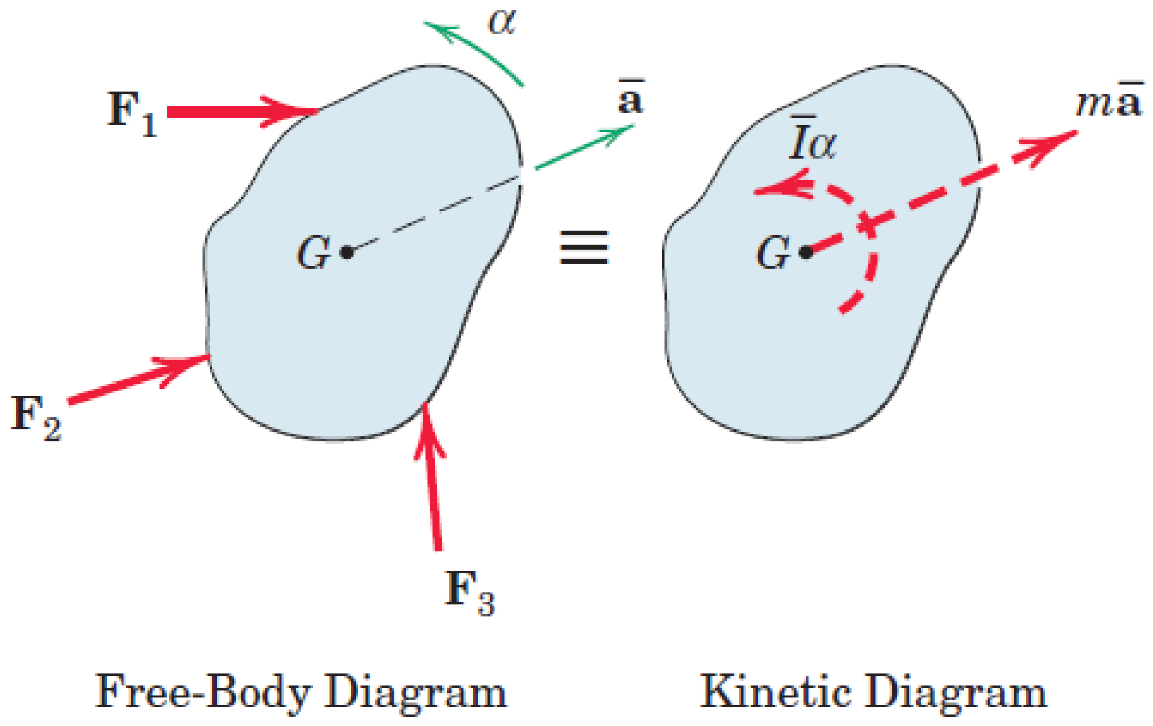
$$M_G = \bar{I}\alpha$$

- $\dot{\omega} = \alpha$ is angular acceleration of the body

- Force Equation

$$F = m\bar{a}$$

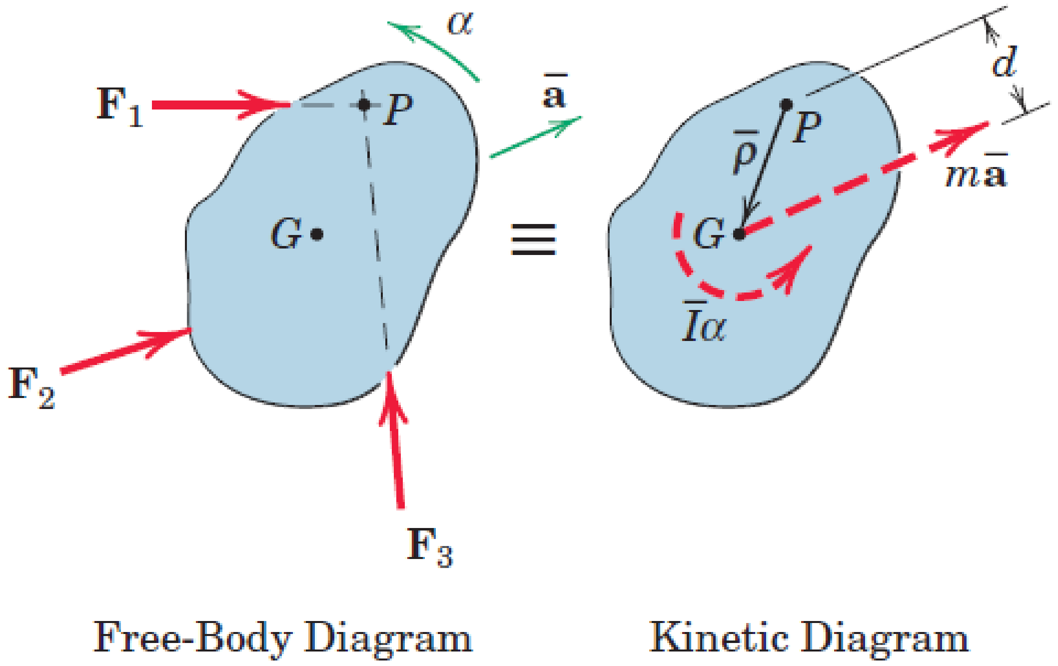
- $\bar{a}$ is acceleration of the mass center of body



Alternate derivations

- General equation for moments about an arbitrary point P

$$\sum M_p = \dot{\boldsymbol{H}}_{\boldsymbol{G}} + \overline{\boldsymbol{\rho}} \times m\overline{\boldsymbol{a}}$$



- $\overline{\boldsymbol{\rho}}$ is a vector from point P to point G
- $\overline{\boldsymbol{a}}$ is mass center acceleration

$$\dot{\boldsymbol{H}}_{\boldsymbol{G}} = \overline{\boldsymbol{I}}\boldsymbol{\alpha} \qquad \overline{\boldsymbol{\rho}} \times m\overline{\boldsymbol{a}} = m\bar{a}d$$

$$\sum M_p = \bar{I}\alpha + m\bar{a}d$$

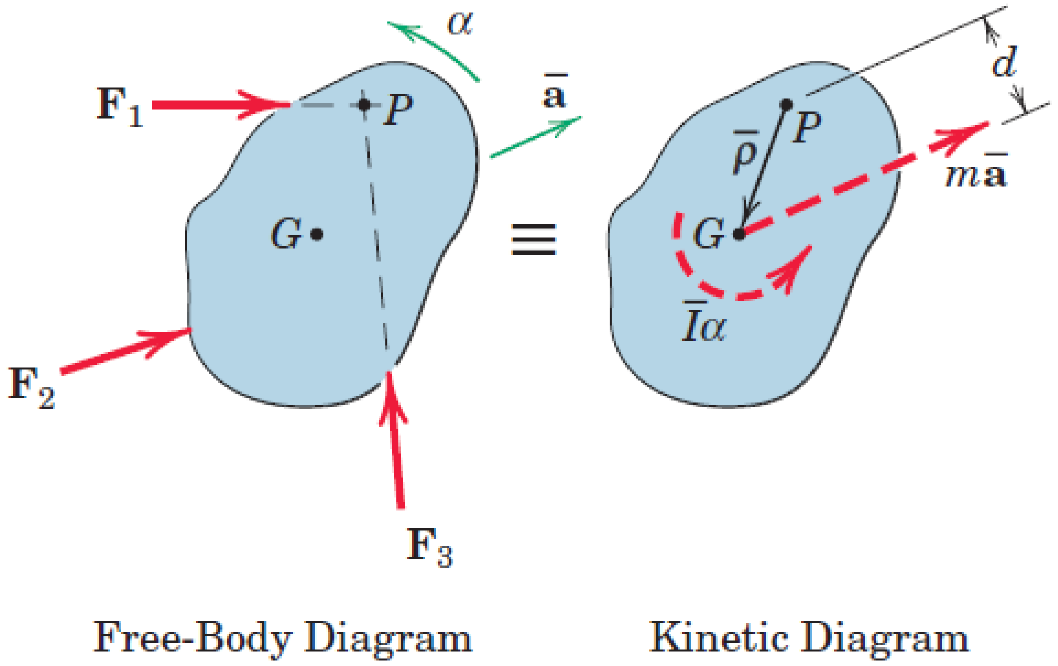
All three terms are positive in the counterclockwise sense for the example shown, and the choice of P eliminates reference to F_1 and F_3

Alternate derivations

- If we had wished to eliminate reference to F_2 and F_3, for example, by choosing their intersection as the reference point, then *P* would lie on the opposite side of the $m\bar{\boldsymbol{a}}$ vector, and the clockwise moment of $m\bar{\boldsymbol{a}}$ about P would be a negative term in the equation

$$\sum M_p = \bar{I}\alpha + m\bar{a}d$$

- It is merely an expression of the familiar principle of moments, where the sum of the moments about P equals the combined moment about P of their sum, expressed by the resultant couple $\sum M_G = \bar{I}\alpha$ and the resultant force $\sum F = m\bar{a}$



Alternate derivations

- Derived Earlier

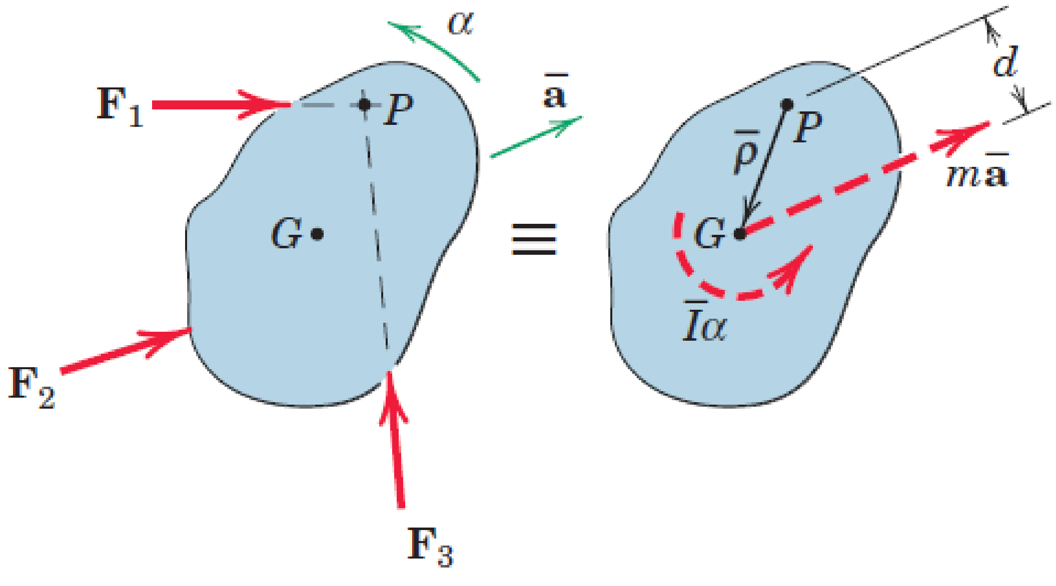
$$\sum \boldsymbol{M}_p = \left(\dot{\boldsymbol{H}}_p\right)_{rel} + \bar{\rho} \times m\boldsymbol{a}_{\boldsymbol{P}}$$

- It is merely an expression of the familiar

$$\left(\dot{\boldsymbol{H}}_p\right)_{rel} = I_P\alpha$$

Where I_P is the mass moment of inertia about an axis through P and α is the angular acceleration of the body

$$\sum \boldsymbol{M}_p = I_P\alpha + \bar{\rho} \times m\boldsymbol{a}_{\boldsymbol{P}}$$



Free-Body Diagram

Kinetic Diagram

Alternate derivations

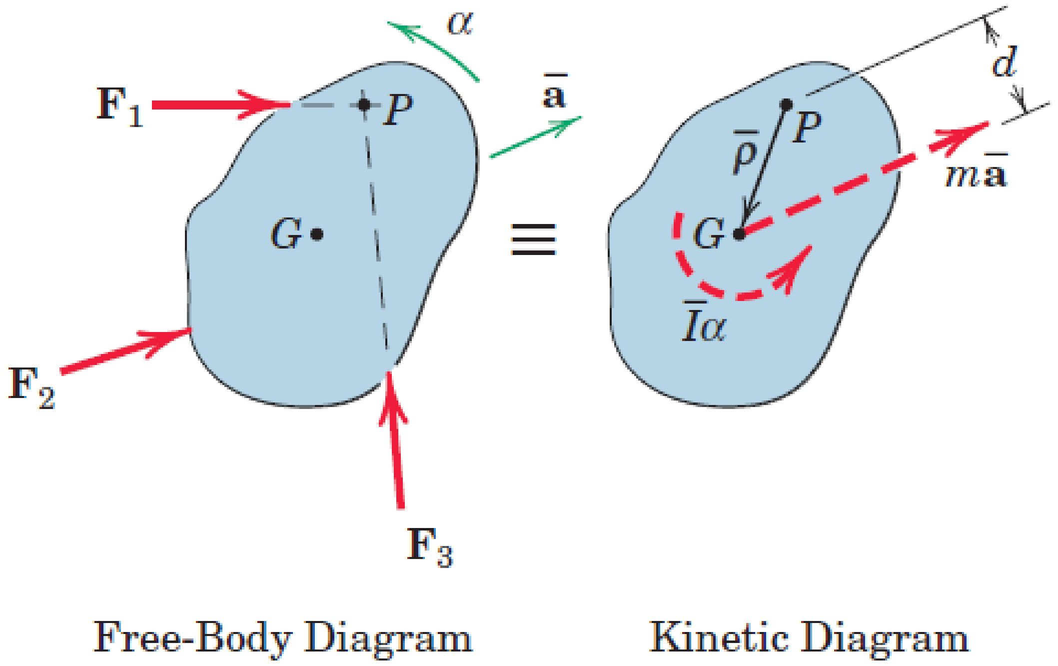
- If $\bar{\rho} = 0$

$$\sum \boldsymbol{M}_p = I_P \alpha + \bar{\rho} \times m\boldsymbol{a}_{\boldsymbol{P}}$$

$$\sum \boldsymbol{M}_G = I_G \alpha$$

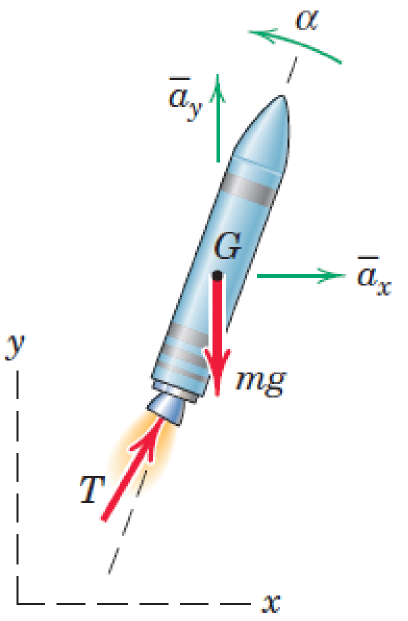
- If $\boldsymbol{a}_{\boldsymbol{P}} = 0$,

$$\sum \boldsymbol{M}_O = I_O \alpha$$

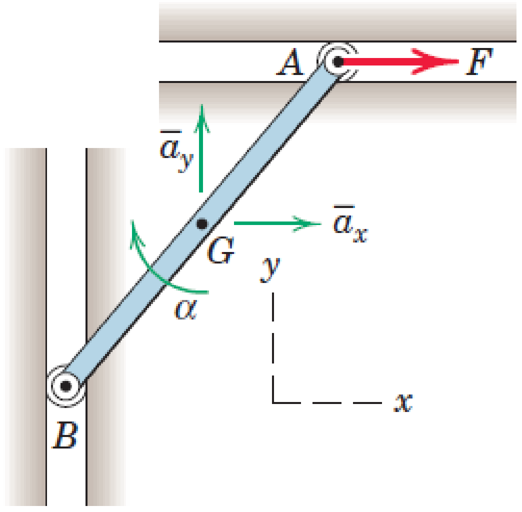


Unconstrained and Constrained Motion

- The rocket moving in a vertical plane is an example of unconstrained motion as there are no physical confinements to its motion
- The two components a_x and a_y of the mass-center acceleration and the angular acceleration α may be determined independently of one another.
- The bar undergoes a constrained motion, where the vertical and horizontal guides for the ends of the bar impose a kinematic relationship between the acceleration components of the mass center and the angular acceleration of the bar.
- Thus, it is necessary to determine this kinematic relationship from the principles established in Chapter 5 and to combine it with the force and moment equations of motion before a solution can be carried out.



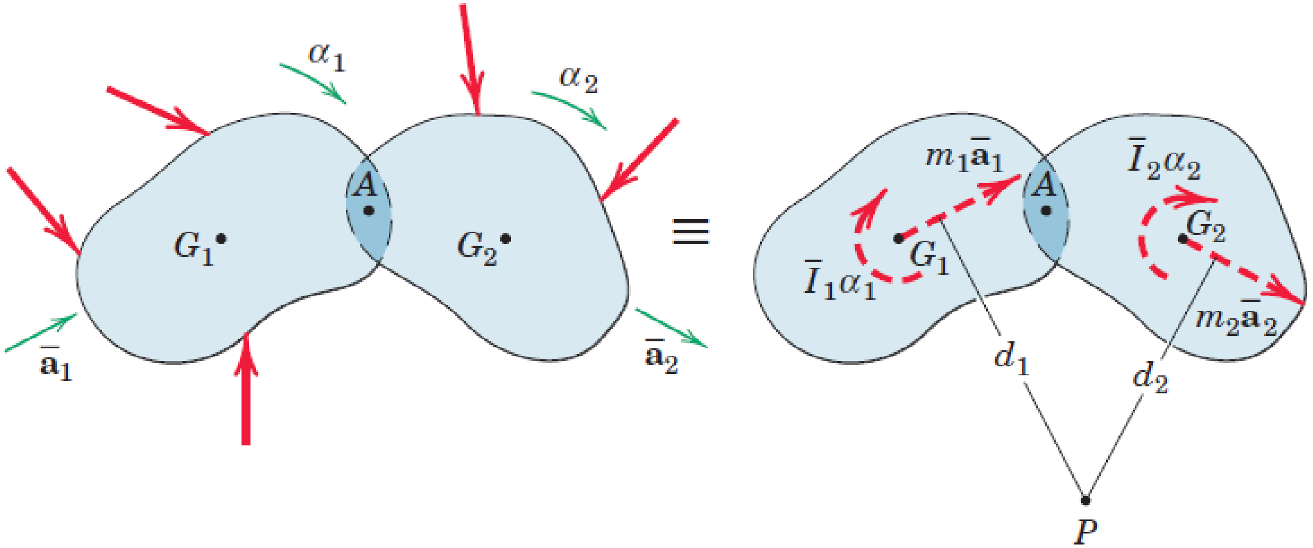
(a) Unconstrained Motion



(b) Constrained Motion

System of interconnected Bodies

- Figure illustrates two rigid bodies hinged at A and subjected to the external forces
- The forces in the connection at A are internal to the system and are not disclosed. The resultant of all external forces must equal the vector sum of the two resultants $m_1\bar{a}_1$and $m_2\bar{a}_2$, and the sum of the moments about some arbitrary point such as P of all external forces must equal the moment of the resultants, $\bar{I}_1\alpha_1 + \bar{I}_2\alpha_2 + m_1\bar{a}_1 d_1 + m_2\bar{a}_2 d_2$



Free-Body Diagram of System $\equiv$ Kinetic Diagram of System

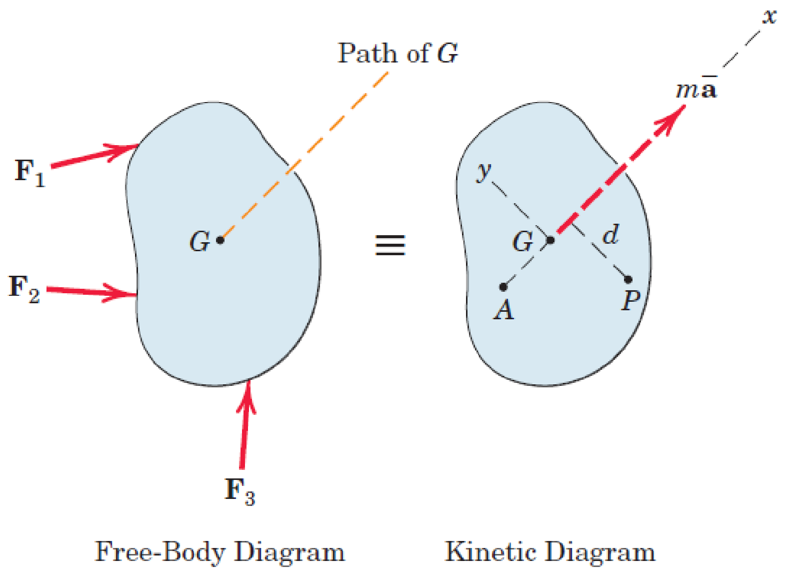
$$\sum \boldsymbol{F} = \sum \boldsymbol{m\bar{a}} \qquad \sum \boldsymbol{M_G} = \sum \boldsymbol{\bar{I}\alpha}$$

Summations on the right-hand side of the equations represent as many terms as there are separate bodies

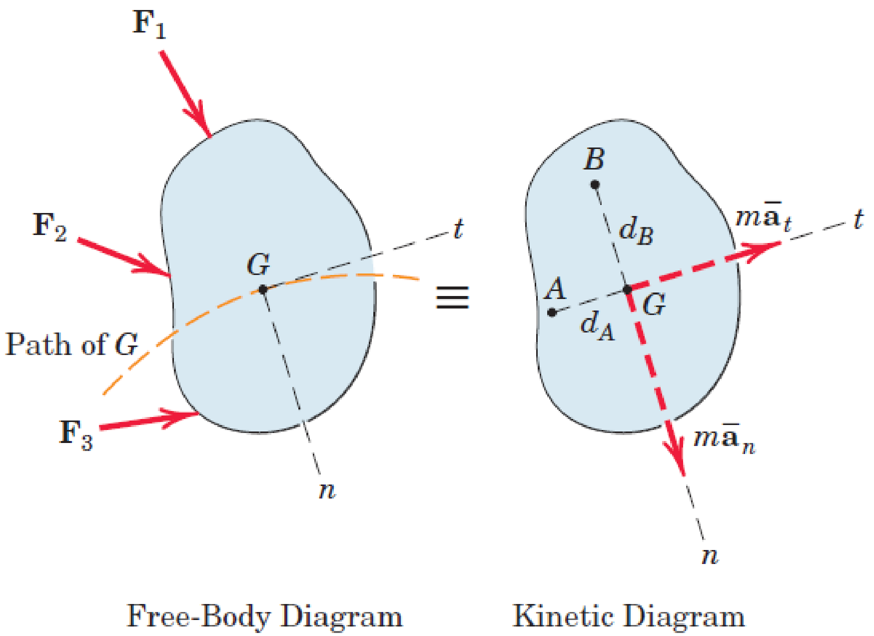
Translation

- Every line in a translating body remains parallel to its original position at all times. In rectilinear translation all points move in straight lines, whereas in curvilinear translation all points move on congruent curved paths.
- In either case, there is no angular motion of the translating body, so that both ω and α are zero.
- All reference to the moment of inertia is eliminated for a translating body

Translation



(a) Rectilinear Translation
($\alpha = 0,\ \omega = 0$)

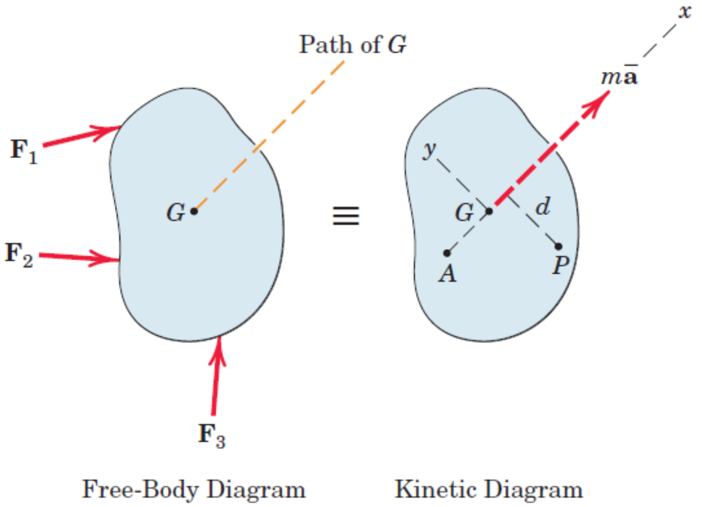


(b) Curvilinear Translation
($\alpha = 0,\ \omega = 0$)

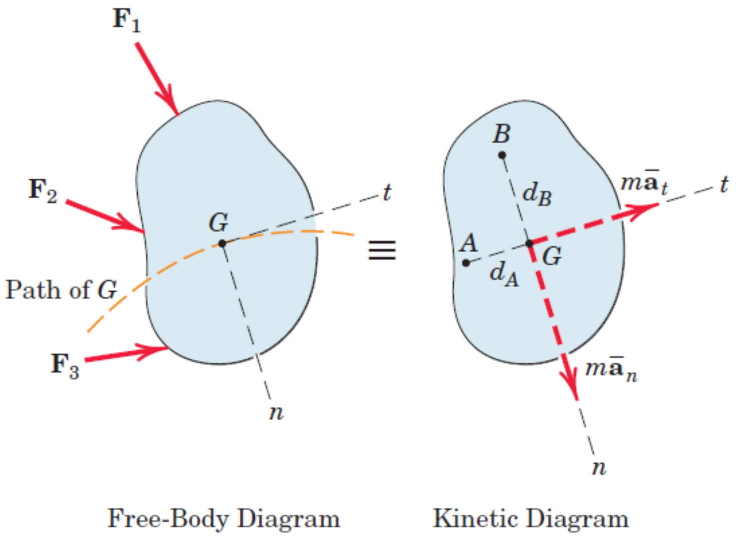
$$\sum F = m\bar{a}$$

$$\sum M_G = \bar{I}\alpha = 0$$

Translation



(a) Rectilinear Translation
$(\alpha = 0,\ \omega = 0)$



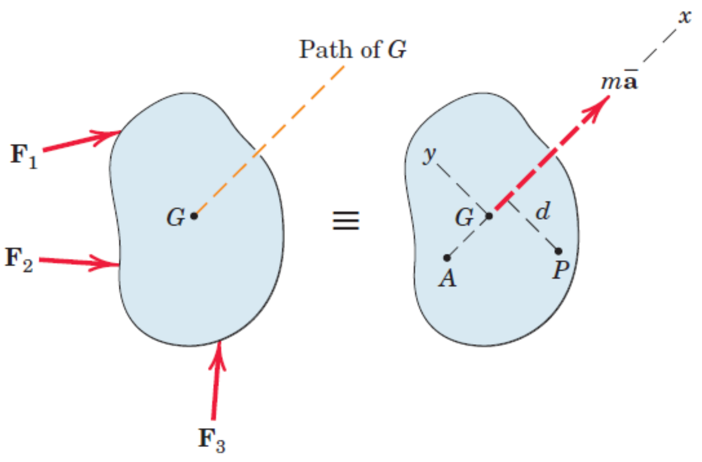
(b) Curvilinear Translation
$(\alpha = 0,\ \omega = 0)$

$$\sum F_x = m\bar{a}_x \qquad \sum F_y = m\bar{a}_y = 0$$

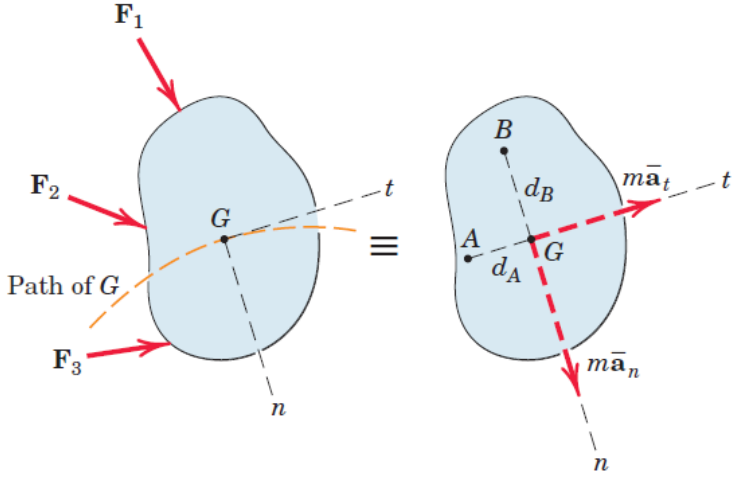
$$\sum F_n = m\bar{a}_n \qquad \sum F_t = m\bar{a}_t$$

$$\sum M_G = \bar{I}\alpha = 0$$

Translation



(*a*) Rectilinear Translation
($\alpha = 0,\ \omega = 0$)



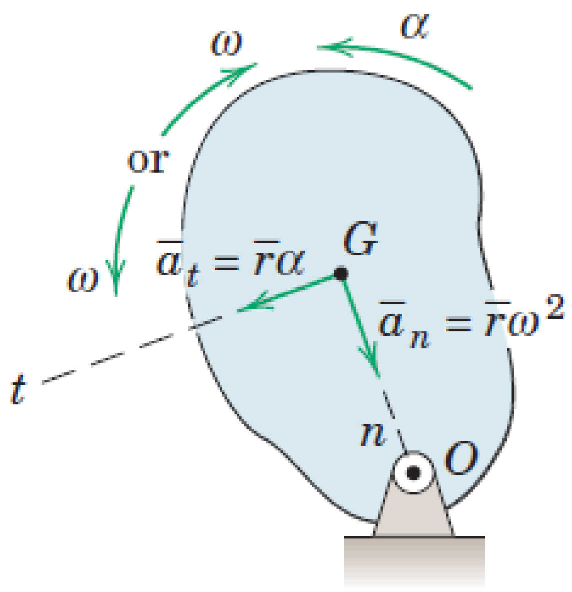
(*b*) Curvilinear Translation
($\alpha = 0,\ \omega = 0$)

$$\sum M_P = m\bar{a}d \qquad \sum M_A = 0$$

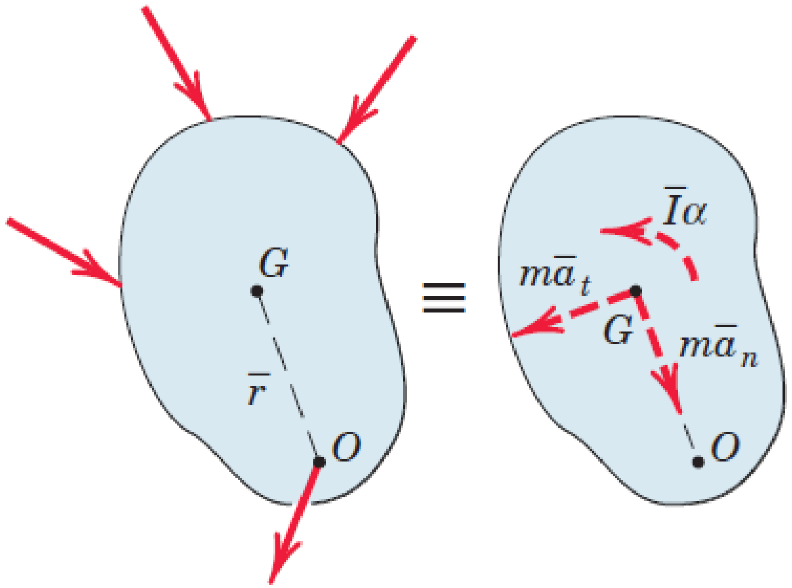
$$\sum M_A = m\bar{a}_n d_A \qquad \sum M_B = m\bar{a}_t d_B$$

Fixed Axis Rotation

- All points in the body describe circles about the rotation axis, and all lines of the body in the plane of motion have the same angular velocity ω and angular acceleration α



Fixed-Axis Rotation



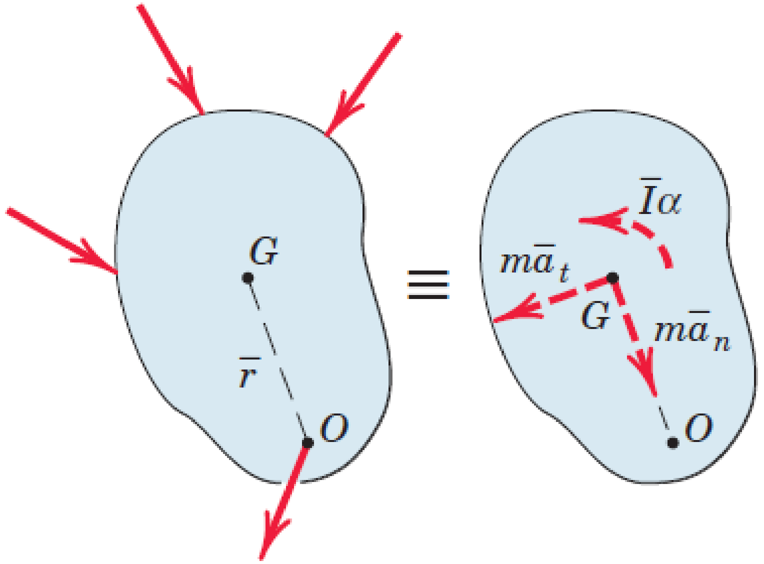
Free-Body Diagram Kinetic Diagram

Fixed Axis Rotation

$$\sum F_n = m\bar{a}_n = m\bar{r}\omega^2$$

$$\sum F_t = m\bar{a}_t = m\bar{r}\alpha$$

$$\sum M_G = \bar{I}\alpha$$



Free-Body Diagram Kinetic Diagram

- In applying the moment equation about G, one must account for the moment of the force applied to the body at O

Fixed Axis Rotation

$$\sum M_o = I_o\alpha$$

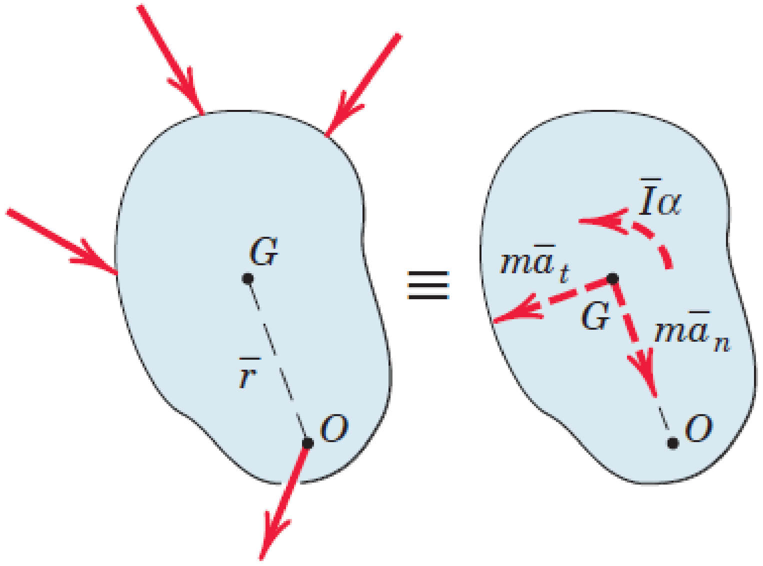
$$\sum M_o = \bar{I}\alpha + m\bar{a}_t\bar{r}$$

$$I_o = \bar{I} + m\bar{r}^2 \qquad I_o - m\bar{r}^2 = \bar{I}$$

$$\sum M_o = (I_o - m\bar{r}^2)\alpha + m\bar{a}_t\bar{r}$$

$$\sum M_o = I_o\alpha - m\bar{r}^2\alpha + m\bar{a}_t\bar{r} = I_o\alpha$$

- Moment equation about the rotation axis O



Free-Body Diagram Kinetic Diagram

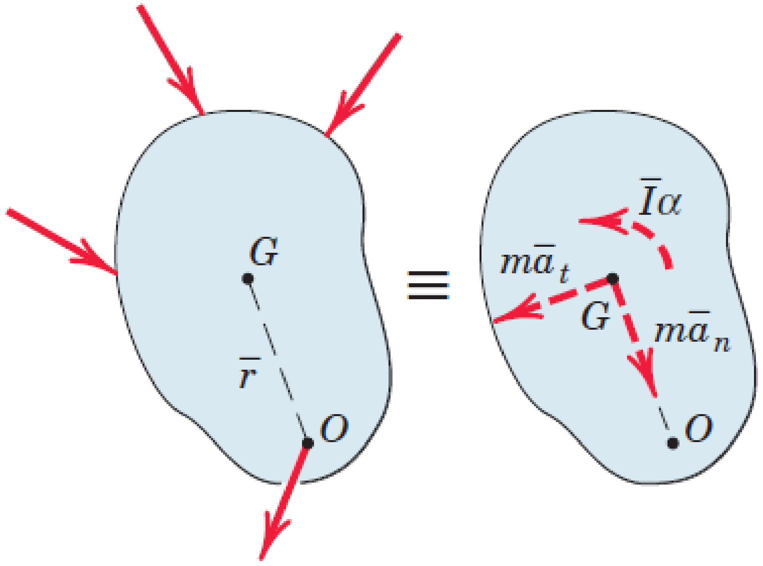
Fixed Axis Rotation

- Moment equation about the rotation axis O

For the case of rotation of a rigid body about a fixed axis through its mass center *G*,

$$\bar{a} = 0 \qquad \sum F = 0$$

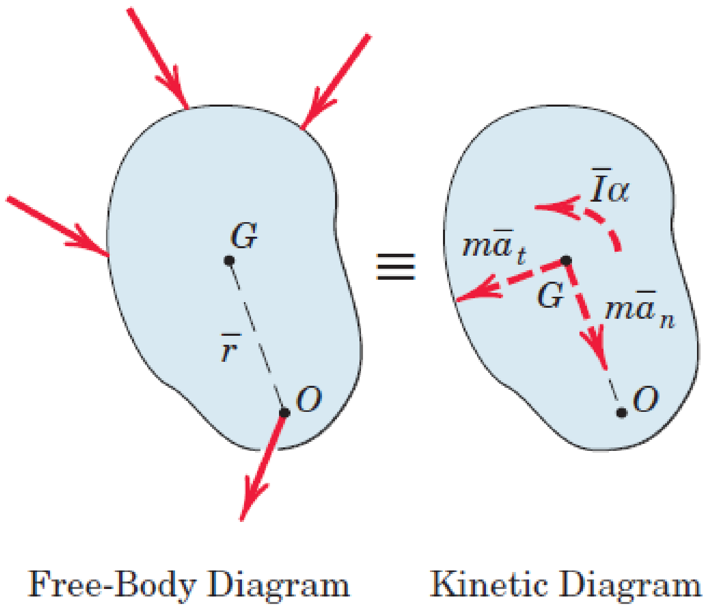
The resultant of the applied forces then is the couple $\bar{I}\alpha$



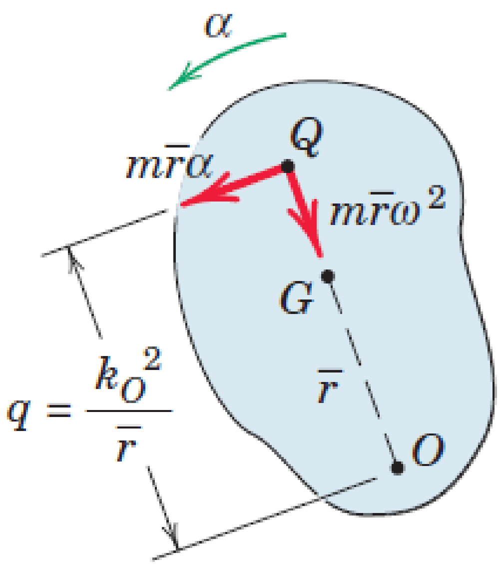
Free-Body Diagram Kinetic Diagram

Fixed Axis Rotation

- Shift $m\bar{r}\alpha$ to point Q
- $m\bar{r}\alpha q = \bar{I}\alpha + m\bar{r}\alpha\bar{r} = (I_o - m\bar{r}^2)\alpha + m\bar{r}\alpha\bar{r} = k_o^2 m\alpha$
- $m\bar{r}\alpha q = k_o^2 m\alpha$
- $q = \dfrac{k_o^2}{\bar{r}}$



- *Point Q is called the **center of percussion** and has the unique property that the resultant of all forces applied to the body must pass through it.*
- It follows that the sum of the moments of all forces about the center of percussion is always zero, $\sum M_Q = 0$

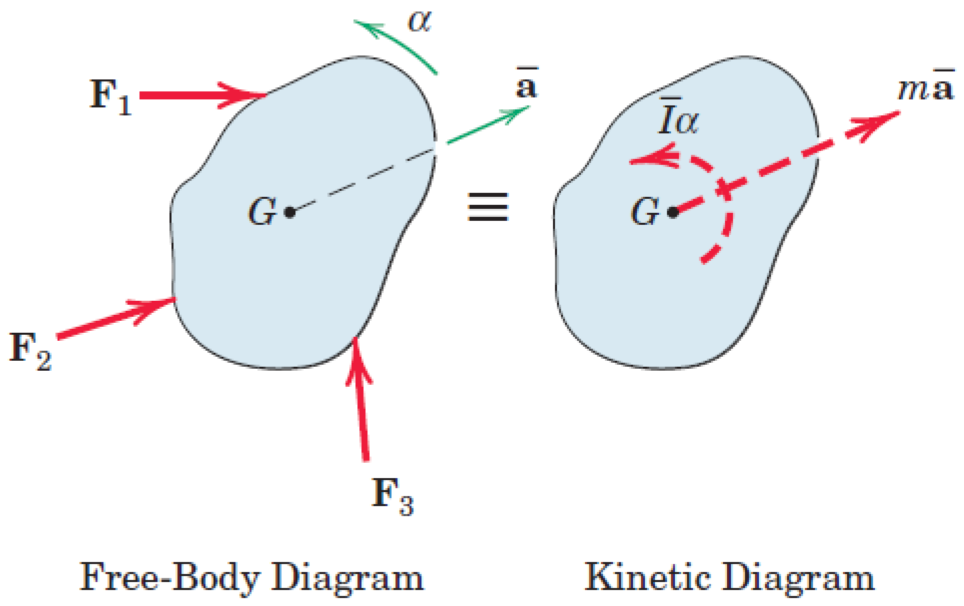


General Plane Motion

$$\sum F = m\bar{a}$$

$$\sum M_G = \bar{I}\alpha$$

- Direct application of these equations expresses the equivalence between the externally applied forces, as disclosed by the free-body diagram, and their force and moment resultants, as represented by the kinetic diagram



Work-Energy Relation: Work of Forces and Couples

Work done by a force F

$$U = \int \mathbf{F}.d\mathbf{r}$$

$$U = \int \mathbf{F}\ \cos\alpha\ \ ds$$

$d\mathbf{r}$is the infinitesimal vector displacement of the point of application

α is the angle between $\boldsymbol{F}$ and the direction of the displacement, $d\mathbf{r}$

ds is the magnitude of the vector displacement $d\mathbf{r}$

Work-Energy Relation: Work of Couples

Work done by a couple M which acts on a rigid body during its motion

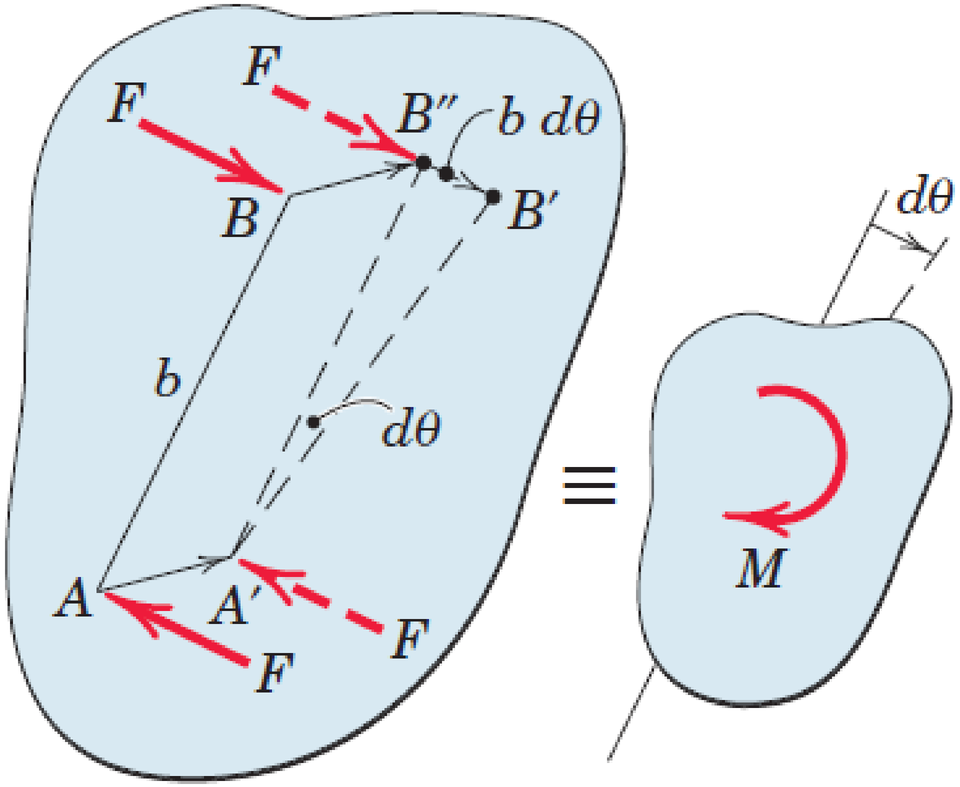
$$U = \int M \, d\theta$$

Body moves from AB to $A'B''$ to $A'B'$.

During translation from AB to $A'B''$, Work done by one force cancels work done by another force

Work is done during rotational motion only

$$dU = Fb \, d\theta = Md\theta$$

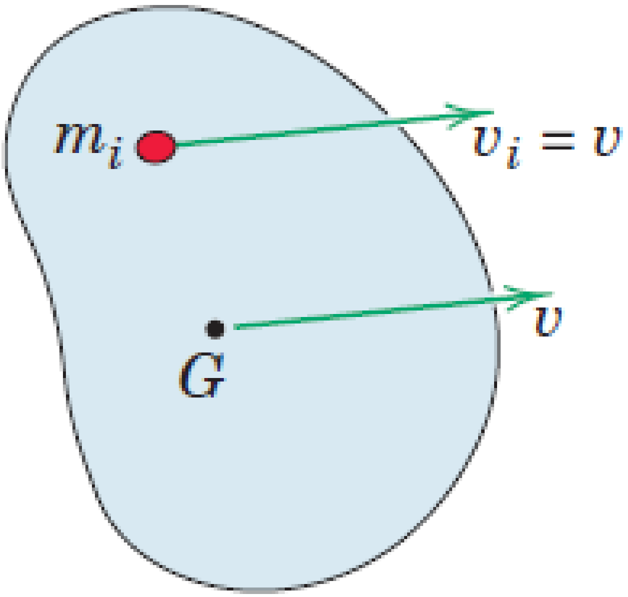


Kinetic Energy: Translation

Kinetic Energy of a particle of mass m_i

$$T_i = \frac{1}{2} m_i v_i^2$$

$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i v^2 = \frac{1}{2} v^2 \sum m_i = \frac{1}{2} m v^2$$



(a) Translation

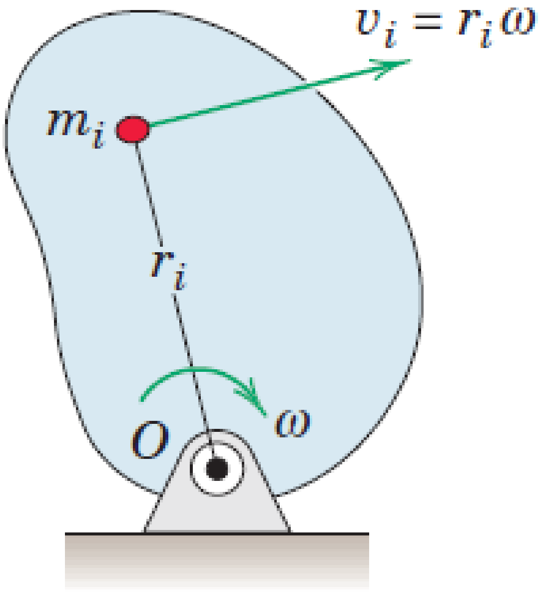
Kinetic Energy: Fixed Axis Rotation

Kinetic Energy of a particle of mass m_i

$$T_i = \frac{1}{2} m_i \omega^2 r_i^2$$

$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i \omega^2 r_i^2$$

$$= \frac{1}{2} \omega^2 \sum m_i r_i^2 = \frac{1}{2} I_o \omega^2$$



(b) Fixed-Axis Rotation

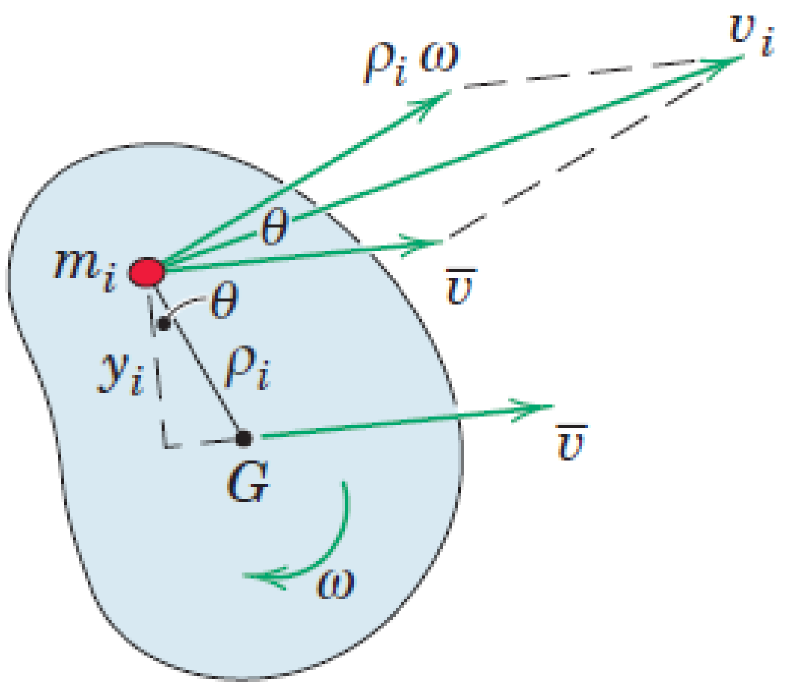
Kinetic Energy: General Plane Motion

Kinetic Energy of a particle of mass m_i

$$T_i = \frac{1}{2} m_i v_i^2$$

$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2$$

$$= \sum \frac{1}{2} m_i (\bar{v}^2 + (\rho_i \omega)^2 + 2\bar{v}\rho_i\omega \ \cos\theta)$$



(c) General Plane Motion

Kinetic Energy: General Plane Motion

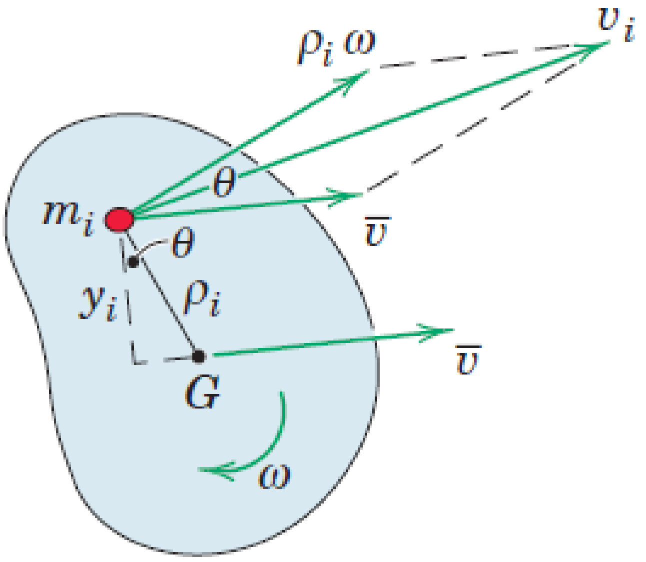
Kinetic Energy of a particle of mass m_i

$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i(\bar{v}^2 + (\rho_i\omega)^2 + 2\bar{v}\rho_i\omega\ \cos\theta)$$

$$T = \sum \frac{1}{2} m_i(\bar{v}^2 + (\rho_i\omega)^2 + 2\bar{v}\rho_i\omega\ \cos\theta)$$

$$T = \sum \frac{1}{2} m_i(\bar{v}^2 + (\rho_i\omega)^2) + \sum m_i\bar{v}\rho_i\omega\ \cos\theta$$

$$\sum m_i\bar{v}\rho_i\omega\ \cos\theta = \sum m_i\bar{v}\omega\rho_i \cos\theta = \bar{v}\omega \sum y_i\, m_i = 0$$



(c) General Plane Motion

Kinetic Energy: General Plane Motion

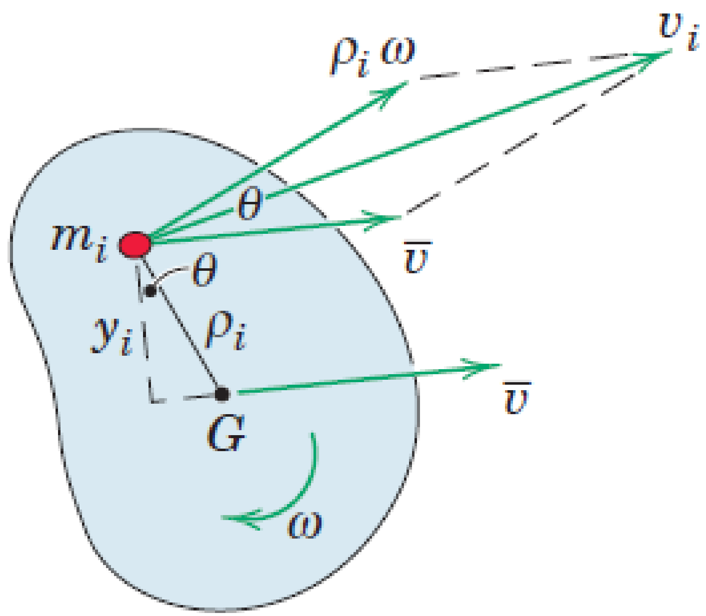
Kinetic Energy of a particle of mass m_i

$$T = \sum \frac{1}{2} m_i(\bar{v}^2 + (\rho_i\omega)^2)$$

$$T = \sum \frac{1}{2} m_i\bar{v}^2 + \sum \frac{1}{2} m_i(\rho_i\omega)^2$$

$$T = \frac{1}{2}\bar{v}^2 \sum m_i + \frac{1}{2}(\omega)^2 \sum m_i(\rho_i)^2$$

$$T = \frac{1}{2} m\bar{v}^2 + \frac{1}{2}\bar{I}(\omega)^2$$



(c) General Plane Motion

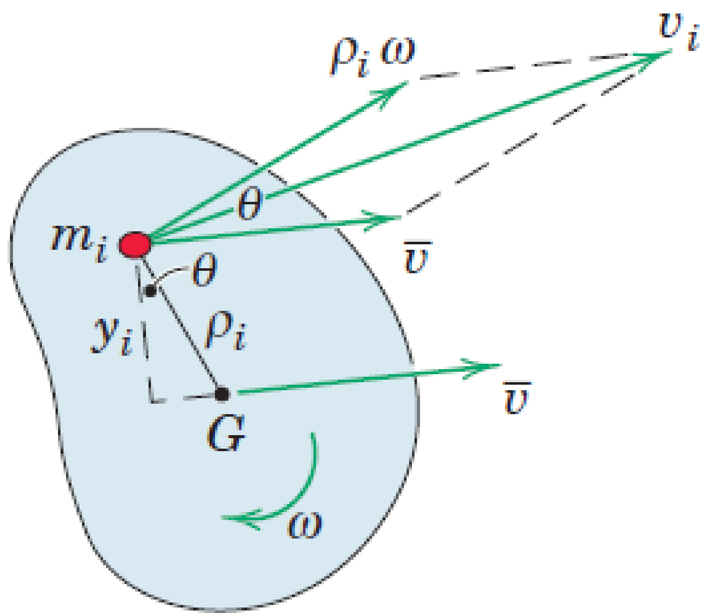
$\bar{I}$ is mass moment of Inertia about its mass center

Kinetic Energy: General Plane Motion

Kinetic Energy about instantaneous center of zero velocity

$$T = \frac{1}{2} I_C (\omega)^2$$

I_C is mass moment of Inertia about instantaneous center of zero velocity



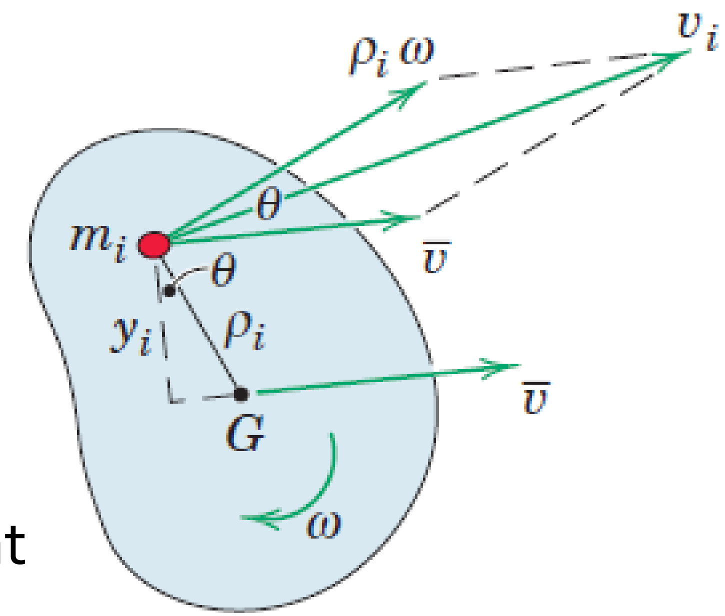
(c) General Plane Motion

Potential Energy and Work Energy Equation

$$T_2 = U_{1-2} + T_1$$

$$T_2 + V_2 = U'_{1-2} + T_1 + V_1$$

prime denotes the work done by all forces other than weight and spring forces



(c) General Plane Motion

Power

Power is the time rate at which work is performed

$$P = \frac{dU}{dt} = \frac{\mathbf{F}.d\boldsymbol{r}}{dt} = \mathbf{F}.\mathbf{v}$$

$d\boldsymbol{r}$: differential displacement

$\mathbf{v}$: velocity of point of application

$$P = \frac{dU}{dt} = \frac{\boldsymbol{M}.d\theta}{dt} = \boldsymbol{M}.\boldsymbol{\omega}$$

$d\theta$: differential angular displacement

$\boldsymbol{\omega}$: angular velocity

- If the senses of M and $\boldsymbol{\omega}$ are the same, the power is positive and energy is supplied to the body.
- Conversely, if M and $\boldsymbol{\omega}$ have opposite senses, the power is negative and energy is removed from the body.

Power

Power is the time rate at which work is performed

$$P = \frac{dU}{dt} = \frac{\mathbf{F}.\,d\boldsymbol{r}}{dt} = \mathbf{F}.\,\mathbf{v}$$

$d\boldsymbol{r}$: differential displacement

$\mathbf{v}$: velocity of point of application

$$P = \frac{dU}{dt} = \frac{\boldsymbol{M}.\,d\theta}{dt} = \boldsymbol{M}.\,\boldsymbol{\omega}$$

$d\theta$: differential angular displacement

$\boldsymbol{\omega}$: angular velocity

If the force **F** and the couple *M* act simultaneously, the total instantaneous power is

$$P = \mathbf{F}.\,\mathbf{v} + \boldsymbol{M}.\,\boldsymbol{\omega}$$

Power

Power by evaluating the rate at which the total mechanical energy of a rigid body or a system of rigid bodies is changing

$$dU' = dT + dV$$

dU' work of the active forces and couples applied to the body or to the system of bodies

dV accounts for work by gravitational forces and that of spring forces

$$\frac{dU'}{dt} = \frac{dT}{dt} + \frac{dV}{dt} = \dot{T} + \dot{V}$$

Power developed by the active forces and couples equals the rate of change of the total mechanical energy of the body or system of bodies

Power

$$\dot{T} = \frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\,\bar{\mathbf{v}}.\bar{\mathbf{v}} + \frac{1}{2}\bar{I}\,\omega^2\right)$$

$$\dot{T} = \frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\,\bar{\mathbf{v}}.\bar{\mathbf{v}} + \frac{1}{2}\bar{I}\,\omega^2\right)$$

$$\dot{T} = \frac{dT}{dt} = (m\,\bar{\mathbf{a}}.\bar{\mathbf{v}} + \bar{I}\,\omega\dot{\omega}) = (m\,\bar{\mathbf{a}}.\bar{\mathbf{v}} + \bar{I}\,\omega\,\alpha) = (\boldsymbol{R}.\bar{\mathbf{v}} + \bar{M}\,\omega)$$

R is the resultant of all forces acting on the body and $\bar{M}$ is the resultant moment about the mass center G of all forces.

Acceleration from Work Energy: Virtual Work

Work-Energy Equation

$$dU' = dT + dV$$

- The term dU' represents the total work done by all active nonpotential forces acting on the system under consideration during the infinitesimal displacement of the system
- The work of potential forces is included in the dV –term. If we use the subscript i to denote a representative body of the interconnected system, the differential change in kinetic energy T for the entire system becomes

$$dT = d\left(\sum \frac{1}{2} m_i \bar{v}_i^2 + \sum \frac{1}{2} \bar{I}_i \omega_i^2\right) = \left(\sum m_i \bar{v}_i \, d\bar{v}_i + \sum \bar{I}_i \, \omega_i \, d\omega_i\right)$$

$d\bar{v}_i$ and $d\omega_i$ are respective changes in velocities

Acceleration from Work Energy: Virtual Work

Work-Energy Equation

$$dU' = dT + dV$$

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$d\bar{v}_i$ and $d\omega_i$ are respective changes in velocities

$$m_i \bar{v}_i \, d\bar{v}_i = m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \, . \, d\bar{\mathbf{s}}_{\mathbf{i}}$$
$$\bar{I}_i \, \omega_i \, d\omega_i = \bar{I}_i \, \alpha_i \;\; d\theta_i$$

$d\bar{\mathbf{s}}_{\mathbf{i}}$ and $d\theta_i$ are infinitesimal linear displacement of center of mass and infinitesimal angular displacement of the body in the plane of motion

$$dT = \left(\sum m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \, . \, d\bar{\mathbf{s}}_{\mathbf{i}} + \sum \bar{I}_i \alpha_i \;\; d\theta_i\right)$$

Acceleration from Work Energy: Virtual Work

Work-Energy Equation

$$dU' = dT + dV$$

$$dT = \left(\sum m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \,.\, d\bar{\mathbf{s}}_{\mathbf{i}} + \sum \bar{I}_i \alpha_i \; d\theta_i\right)$$

$$dT = \left(\sum \boldsymbol{R}_i \,.\, d\bar{\mathbf{s}}_{\mathbf{i}} + \sum \boldsymbol{M}_{\boldsymbol{G}_i} \,.\, d\boldsymbol{\theta}_{\boldsymbol{i}}\right)$$

$\boldsymbol{R}_i$ and $\boldsymbol{M}_{\boldsymbol{G}_i}$ are resultant force and resultant couple acting on body

$$d\boldsymbol{\theta}_{\boldsymbol{i}} = d\theta_i \mathbf{k}$$

- Differential change in kinetic energy equals the differential work done on the system by the resultant forces and resultant couples acting on all the bodies of the system

Acceleration from Work Energy: Virtual Work

The term dV represents the differential change in the total gravitational potential energy V_g and the total elastic potential energy V_e

$$dV = dV_g + dV_e$$

$$dV = d\left(\sum m_i g h_i + \sum \frac{1}{2} k_j x_j^2\right) = \sum m_i g \, dh_i + \sum k_j x_j dx_j$$

- h_i represents the vertical distance of the center of mass of the representative body of mass m_i above any convenient datum plane
- x_j stands for the deformation, tensile or compressive, of a representative elastic member of the system (spring) whose stiffness is k_j

$$dU' = \sum m_i \bar{\mathbf{a}}_i \, . \, d\bar{\mathbf{s}}_\mathbf{i} + \sum \bar{I}_i \alpha_i \; d\theta_i + \sum m_i g \, dh_i + \sum k_j x_j dx_j$$

Acceleration from Work Energy: Virtual Work

$$dU' = \sum m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \, . \, d\bar{\mathbf{s}}_{\mathbf{i}} + \sum \bar{I}_i \alpha_i \; d\theta_i + \sum m_i g \; dh_i + \sum k_j x_j dx_j$$

- $m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \, . \, d\bar{\mathbf{s}}_{\mathbf{i}} \quad and \; \bar{I}_i \alpha_i \; d\theta_i$ will be positive if the accelerations are in the same direction as the respective displacements and negative if they are in the opposite direction

- Equation has the advantage of relating the accelerations to the active forces directly, which eliminates the need for dismembering the system and then eliminating the internal forces and reactive forces by simultaneous solution of the force-mass-acceleration equations for each member

Virtual Work

$$dU' = \sum m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \, . \, d\bar{\mathbf{s}}_{\mathbf{i}} + \sum \bar{I}_i \alpha_i \; d\theta_i + \sum m_i g \; dh_i + \sum k_j x_j dx_j$$

In above equation, the differential motions are differential changes in the real or actual displacements which occur

- For a mechanical system which assumes a steady-state configuration during constant acceleration, we often find it convenient to introduce the concept of virtual work.
- The concepts of virtual work and virtual displacement were introduced and used to establish equilibrium configurations for static systems of interconnected bodies

Virtual Work

$$\delta U' = \sum m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \,.\, \boldsymbol{\delta}\bar{\mathbf{s}}_{\mathbf{i}} + \sum \bar{I}_i \alpha_i \;\; \delta\theta_i + \sum m_i g \, \delta h_i + \sum k_j x_j \delta x_j$$

- A virtual displacement is any assumed and arbitrary displacement, linear or angular, away from the natural or actual position.
- For a system of connected bodies, the virtual displacements must be consistent with the constraints of the system.
- For example, when one end of a link is hinged about a fixed pivot, the virtual displacement of the other end must be normal to the line joining the two ends.
- Such requirements for displacements consistent with the constraints are purely kinematic and provide what are known as the ***equations of constraint***.

Virtual Work

$$\delta U' = \sum m_i \bar{\mathbf{a}}_{\boldsymbol{i}} \,.\, \boldsymbol{\delta}\bar{\mathbf{s}}_{\mathbf{i}} + \sum \bar{I}_i \alpha_i \;\; \delta\theta_i + \sum m_i g \; \delta h_i + \sum k_j x_j \delta x_j$$

- If a set of virtual displacements satisfying the equations of constraint and therefore consistent with the constraints is assumed for a mechanical system, the proper relationship between the coordinates which specify the configuration of the system will be determined by applying the work-energy relationship expressed in terms of virtual changes.
- It is customary to use the differential symbol d to refer to differential changes in the real displacements, whereas the symbol is used to signify virtual changes, that is, differential changes which are assumed rather than real.

Impulse-Momentum Equations

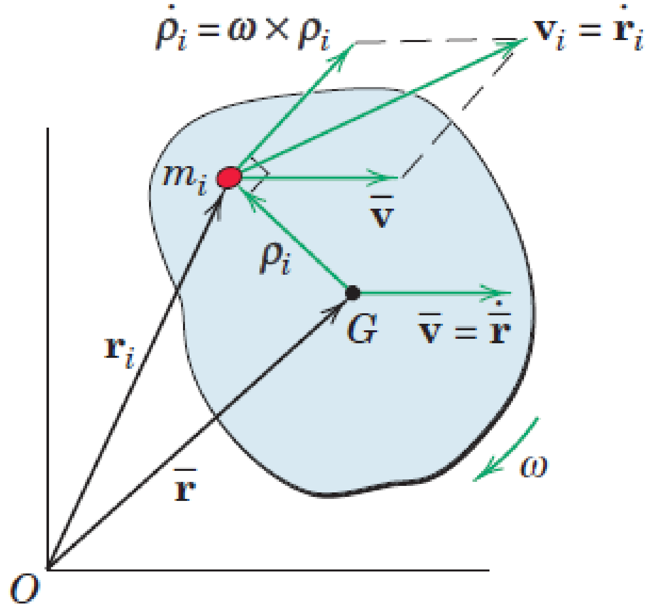
- Mass of particle m_i
- Momentum of particle $m_i v_i$
- Momentum of rigid body

$$G = \sum m_i v_i = \sum m_i \dot{r}_i = \frac{d \sum m_i r_i}{dt} = \frac{d\,(m\,\bar{r})}{dt} = m\bar{\mathbf{v}}$$

Where $\bar{\mathbf{v}}$ is velocity of mass center

- Linear momentum of any mass system, rigid or non-rigid,

$$G = m\bar{\mathbf{v}}$$



Impulse-Momentum Equations

- Mass of particle m_i
- Momentum of particle $m_i v_i$
- Momentum of rigid body

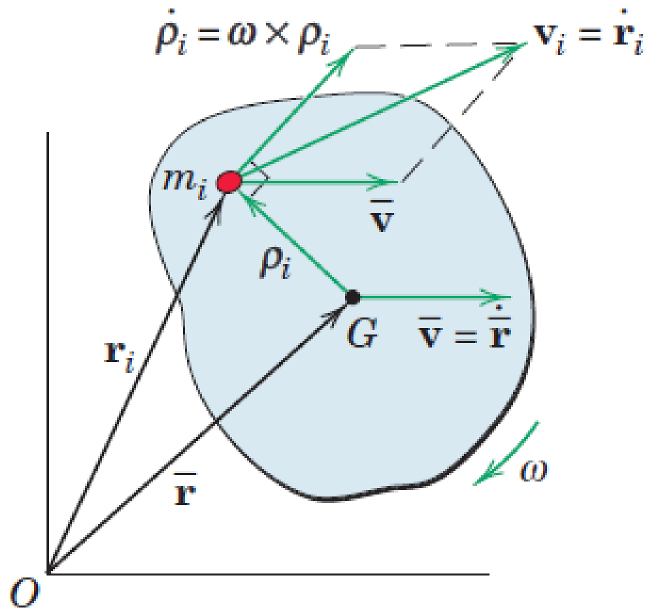
$$G = \sum m_i v_i = \sum m_i(\bar{\mathbf{v}} + \omega \times \rho_i) = \sum m_i \bar{\mathbf{v}} + \sum m_i \omega \times \rho_i$$

Where $\bar{\mathbf{v}}$ is velocity of mass center

$$G = \bar{\mathbf{v}} \sum m_i + \omega \times \sum m_i \rho_i = m\bar{\mathbf{v}} + 0$$

- Linear momentum of any mass system, rigid or non-rigid,

$$G = m\bar{\mathbf{v}}$$



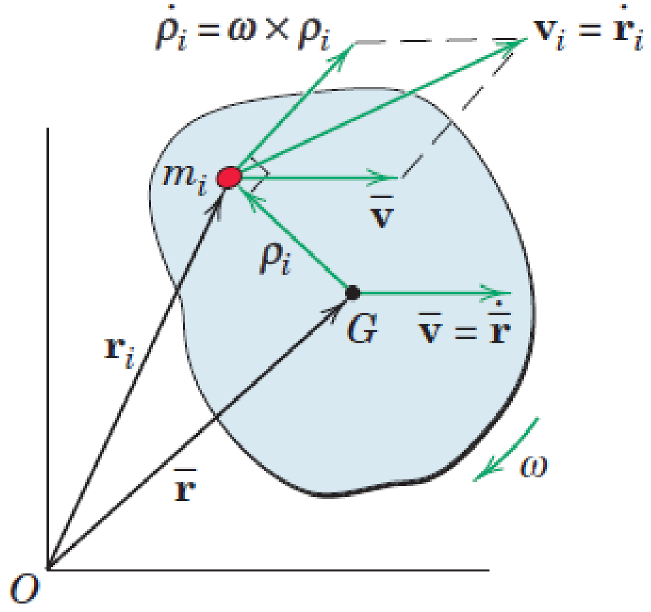
Impulse-Momentum Equations

$$\sum \mathbf{F} = \dot{\mathbf{G}}$$

Resultant force equals the time rate of change of momentum.

$$G_2 = \int_{t_1}^{t_2} F\ dt + G_1$$

Initial linear momentum plus the linear impulse acting on the body equals the final linear momentum.



Angular Momentum

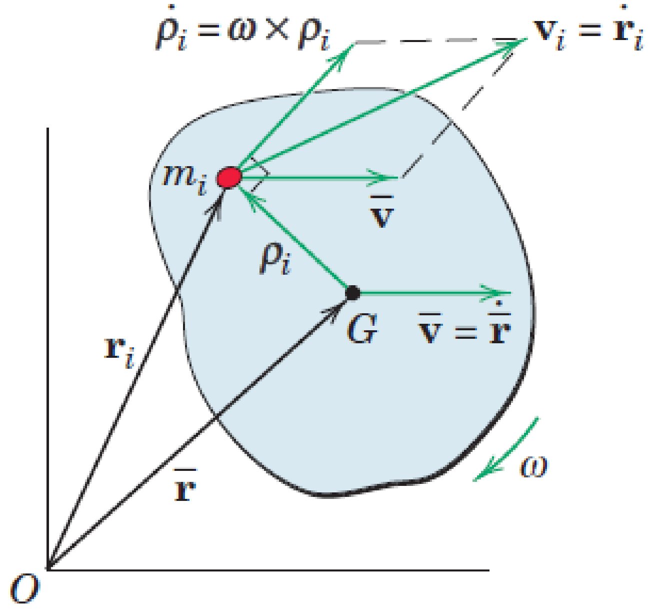
- Angular momentum is defined as the moment of linear momentum
- Angular momentum about the mass center of any prescribed system of mass

$$H_G = \sum \rho_i \times m_i v_i$$

Vector sum of the moments about G of the linear momenta of all particles.

$$H_G = \sum \rho_i \times m_i \dot{\rho}_i$$

$\dot{\rho}_i$ is velocity of mass m_i with respect to *G*.



Angular Momentum

Vector sum of the moments about G of the linear momenta of all particles.

$$H_G = \sum \rho_i \times m_i \dot{\rho}_i$$

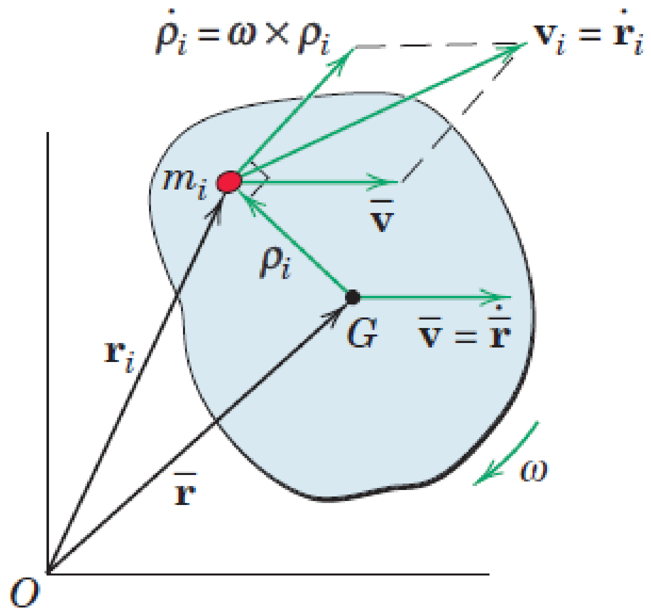
$\dot{\rho}_i$ is velocity of mass m_i with respect to *G.*

$$H_G = \sum \rho_i \times m_i \left(\omega \times \rho_i\right)$$

$$H_G = \sum \rho_i \times m_i \left(\omega \times \rho_i\right) = \sum m_i \rho_i \times \omega \times \rho_i = \sum \rho_i^2 m_i \omega \,\hat{\boldsymbol{k}}$$
$$= \bar{I}\boldsymbol{\omega}$$

$$\bar{I} = \sum \rho_i^2 m_i$$

Mass moment of inertia of the body about its mass center



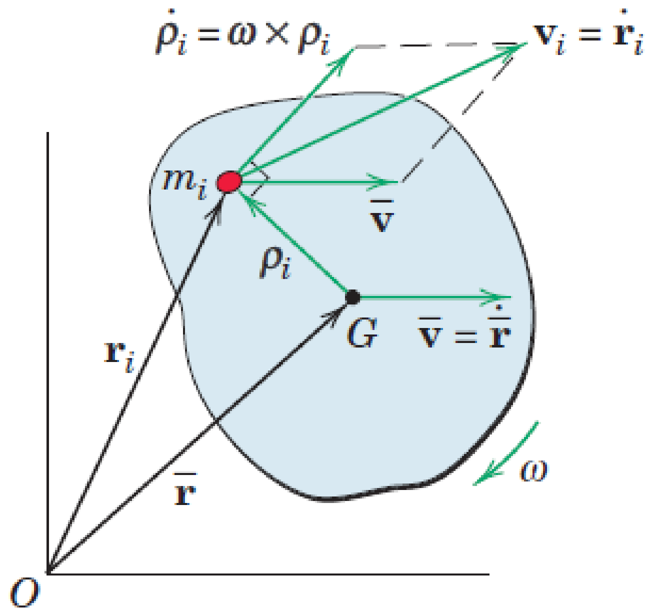
Angular Momentum

- Angular-momentum vector is always normal to the plane of motion
- Angular momentum about the mass center as the scalar

$$H_G = \bar{I}\omega$$

$$\sum M_G = \dot{H}_G$$

$$H_{G_2} = H_{G_1} + \int_{t_1}^{t_2} \sum M_G \; dt$$



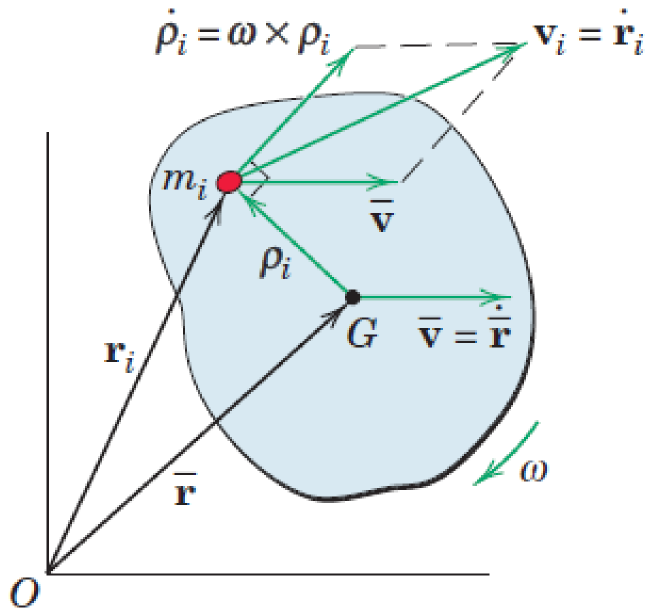
Angular Momentum

- Sum of the moments about the mass center of *all* forces acting on the body equals the time rate of change of angular momentum about the mass center.

$$\sum M_G = \dot{H}_G$$

- Initial angular momentum about the mass center *G* plus the external angular impulse about *G* equals the final angular momentum about *G*

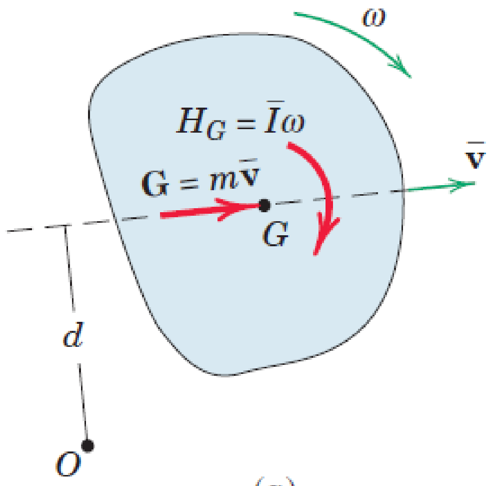
$$H_{G_2} = H_{G_1} + \int_{t_1}^{t_2} \sum M_G \; dt$$



Angular Momentum

- With the moments about G of the linear momenta of all particles accounted for by H_G it follows that we may represent the linear momentum $G = m\bar{\mathbf{v}}$ as a vector through the mass center G

- G and H_G have vector properties analogous to those of the resultant force and couple.



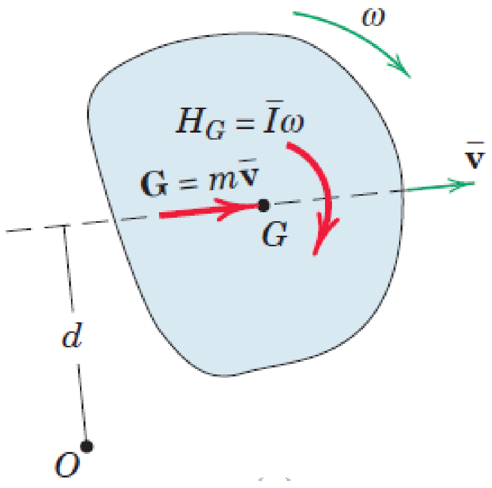
(a)

Angular Momentum

- Angular momentum H_o about any point O is

$$H_o = H_G + m\,\bar{\mathbf{v}}\,d$$

- This expression holds at any particular instant of time about O, which may be a fixed or moving point on or off the body



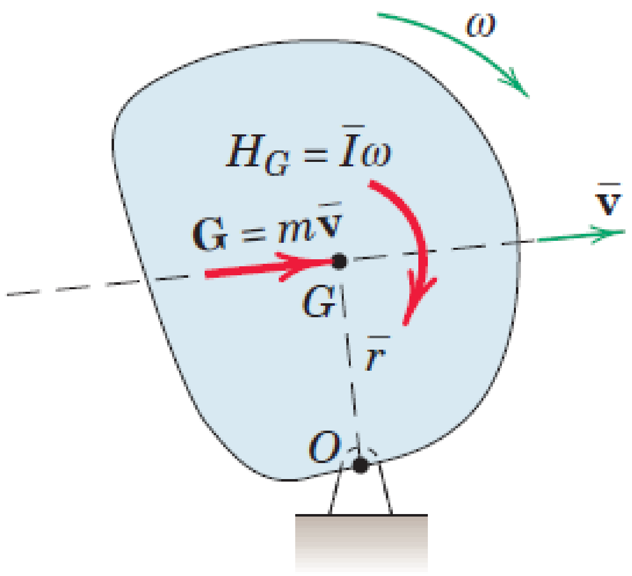
Angular Momentum

$$H_o = H_G + m\,\bar{\mathbf{v}}\,d$$

$$H_o = \bar{I}\omega + m\,\bar{r}\omega\,\bar{r}$$

$$H_o = \bar{I}\omega + m\,\bar{r}^2\omega = (\bar{I} + m\,\bar{r}^2)\omega$$

$$H_o = I_o\omega$$



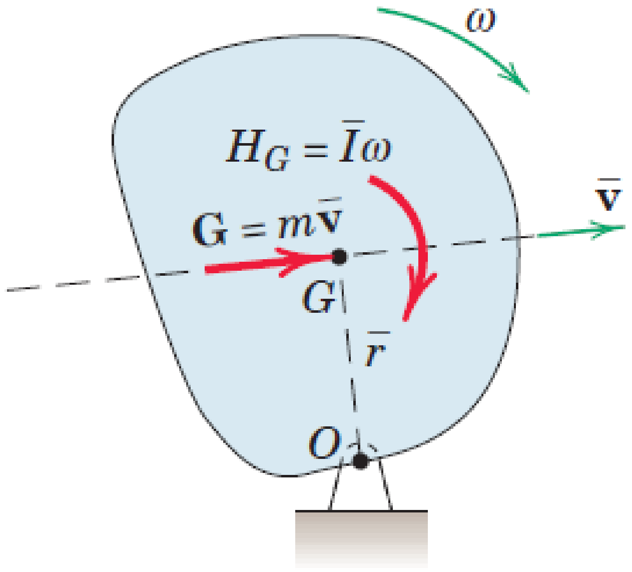
(b)

Angular Momentum

$$H_o = I_o\omega$$

$$\sum M_o = \dot{H}_o$$

$$H_{o_1} + \int_{t_1}^{t_2} M_o\, dt = H_{o_2}$$



(b)

- Do not add linear momentum and angular momentum for the same reason that force and moment cannot be added directly

Inter-Connected Rigid Bodies

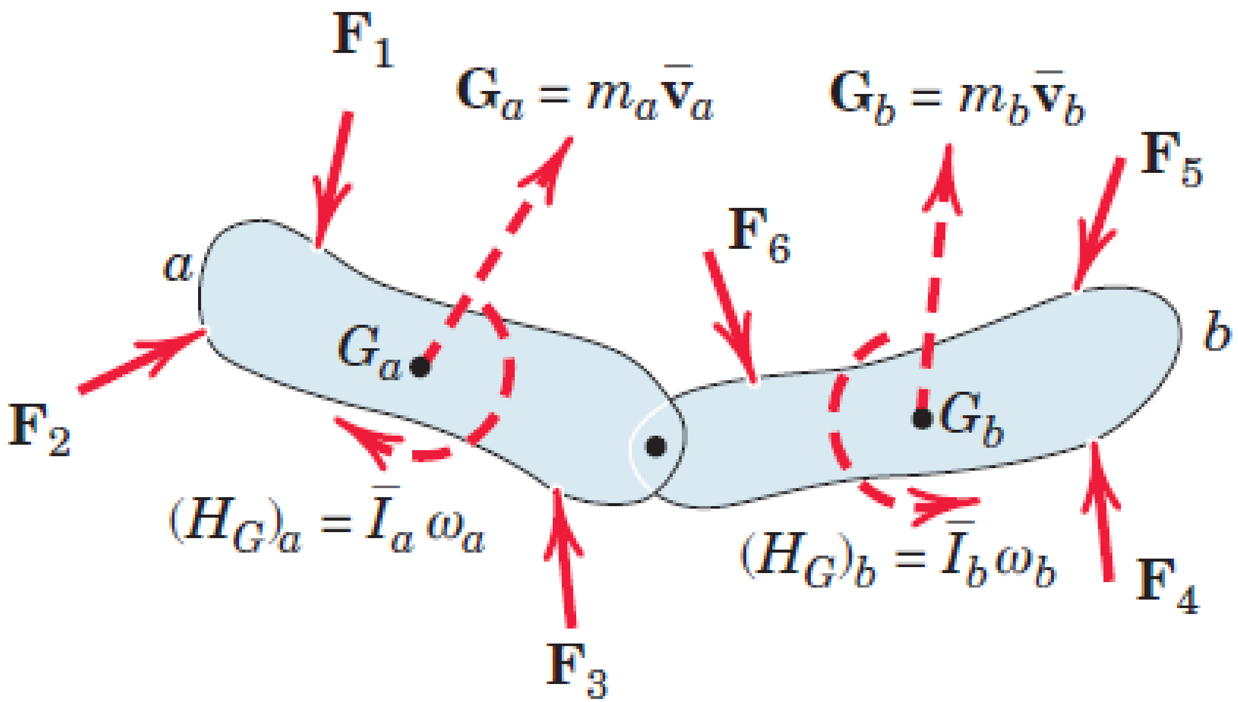
$$\sum F = \dot{G}$$

$$\sum M_o = \dot{H}_o$$

Where O is a fixed reference point

$$\sum F = \dot{G}_a + \dot{G}_b + \ \dots\dots.$$

$$\sum M_o = \dot{H}_{o\,a} + \dot{H}_{o\,b} + \cdots$$



$$H_{o_1} + \int_{t_1}^{t_2} M_o \, dt = H_{o_2}$$

O

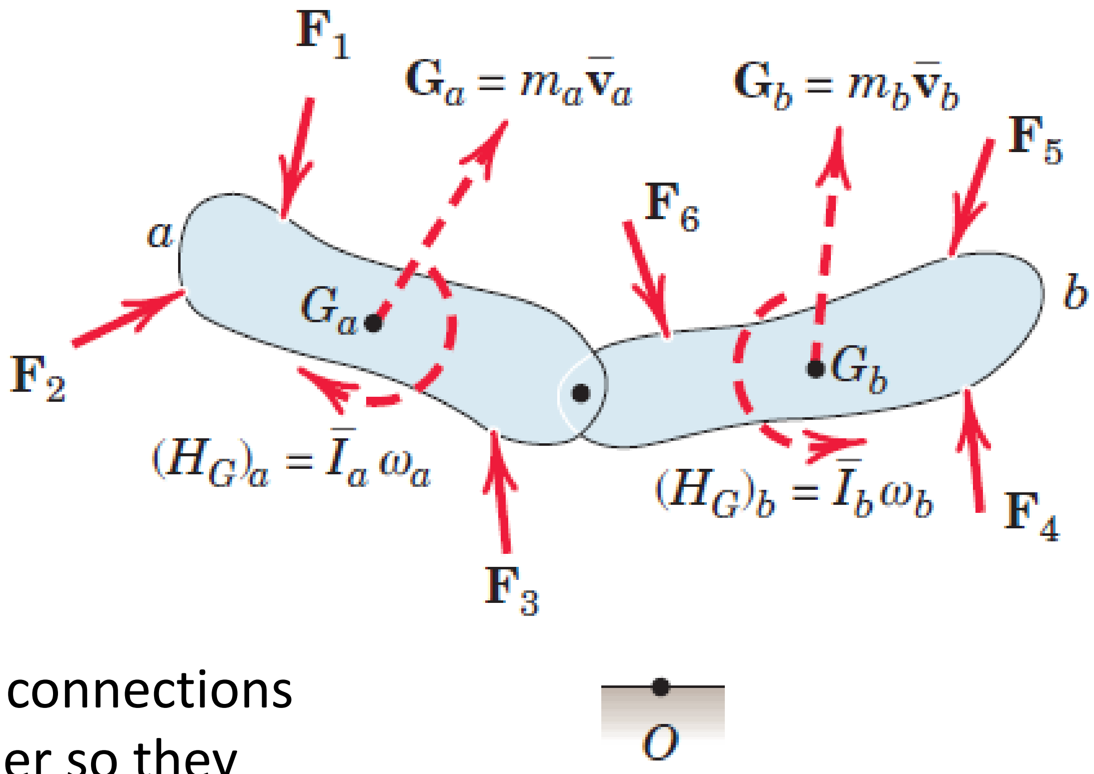
Inter-Connected Rigid Bodies

$$\sum F = \dot{G}_a + \dot{G}_b + \ \dots \dots .$$

$$\sum M_o = \dot{H}_{o\,a} + \dot{H}_{o\,b} + \cdots$$

$$\int_{t_1}^{t_2} \sum F \ \ dt = \Delta \, G_{system}$$

$$\int_{t_1}^{t_2} \sum M_o \ \ dt = \Delta \, \dot{H}_{o\,system}$$



- Equal and opposite actions and reactions in the connections are internal to the system and cancel one another so they are not involved in the force and moment summations.
- Point *O* is one fixed reference point for the entire system.

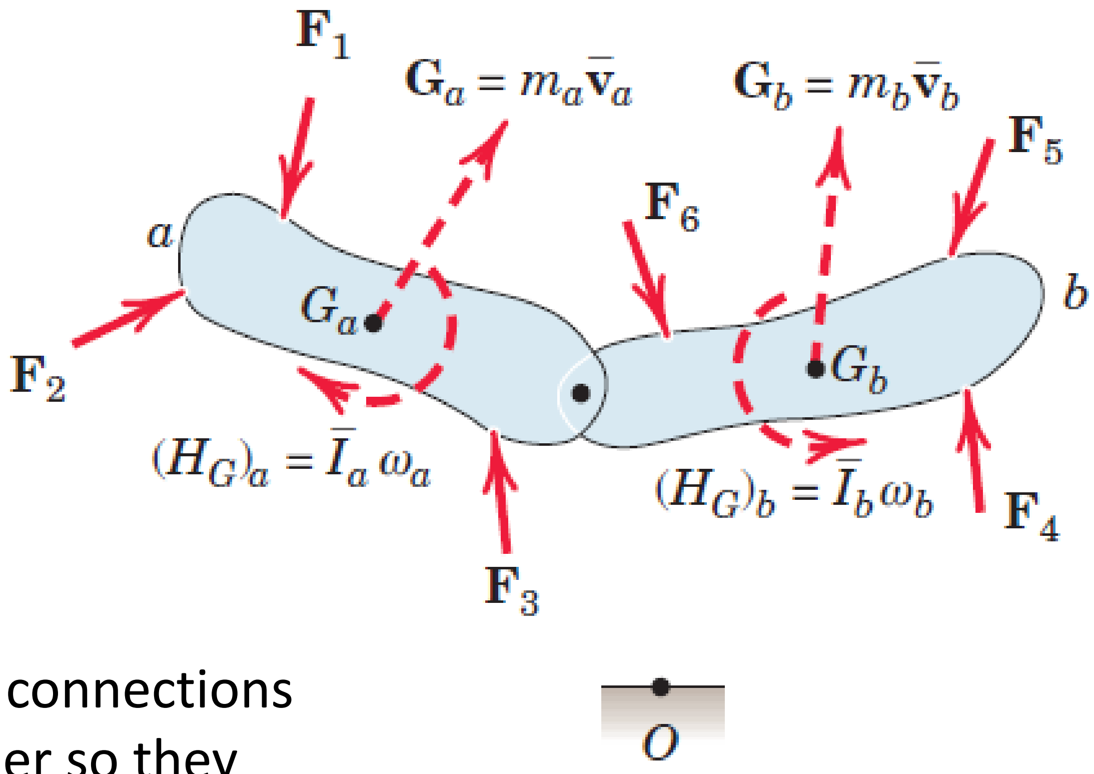
Conservation of Linear Momentum

$$\sum F = \dot{G}_a + \dot{G}_b + \ \ldots \ldots.$$

$$\sum M_o = \dot{H}_{o\,a} + \dot{H}_{o\,b} + \cdots$$

$$\int_{t_1}^{t_2} \sum F \quad dt = \Delta\, G_{system}$$

$$\int_{t_1}^{t_2} \sum M_o \quad dt = \Delta\, \dot{H}_{o\,system}$$



- Equal and opposite actions and reactions in the connections are internal to the system and cancel one another so they are not involved in the force and moment summations.
- Point *O* is one fixed reference point for the entire system.

Conservation of Angular Momentum

- If the resultant moment about a given fixed point O or about the mass center is zero during a particular interval of time for a single rigid body or for a system of interconnected rigid bodies,

$$H_{O_1} = H_{O_2}$$

$$H_{G_1} = H_{G_2}$$

- Angular momentum either about the fixed point or about the mass center undergoes no change in the absence of a corresponding resultant angular impulse

Conservation of Angular Momentum

- In the case of the interconnected system, there may be angular-momentum changes of individual components during the interval, but there will be no resultant angular momentum change for the system as a whole if there is no resultant angular impulse about the fixed point or the mass center

THANK YOU

THREE DIMENSIONAL DYNAMICS OF RIGID BODIES

Dr. Amrita Puri
Assistant Professor
Department of Mechanical Engineering
IIT Jodhpur

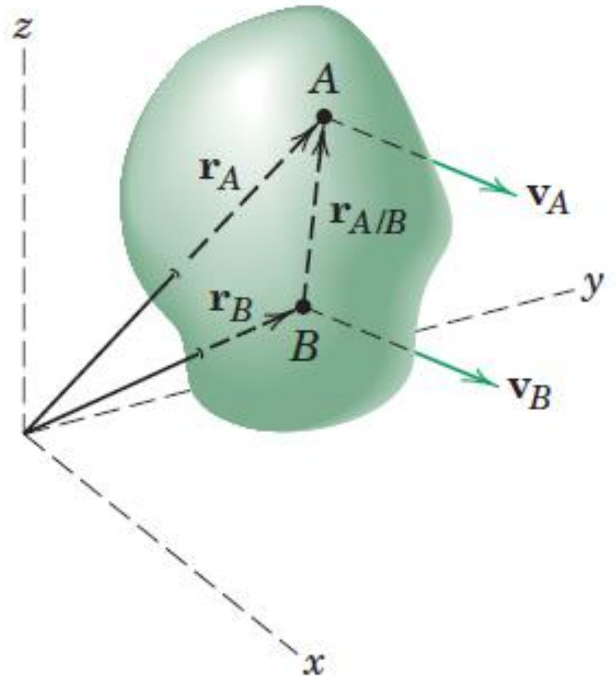
Translation

- Rectilinear Translation
- Curvilinear Translation

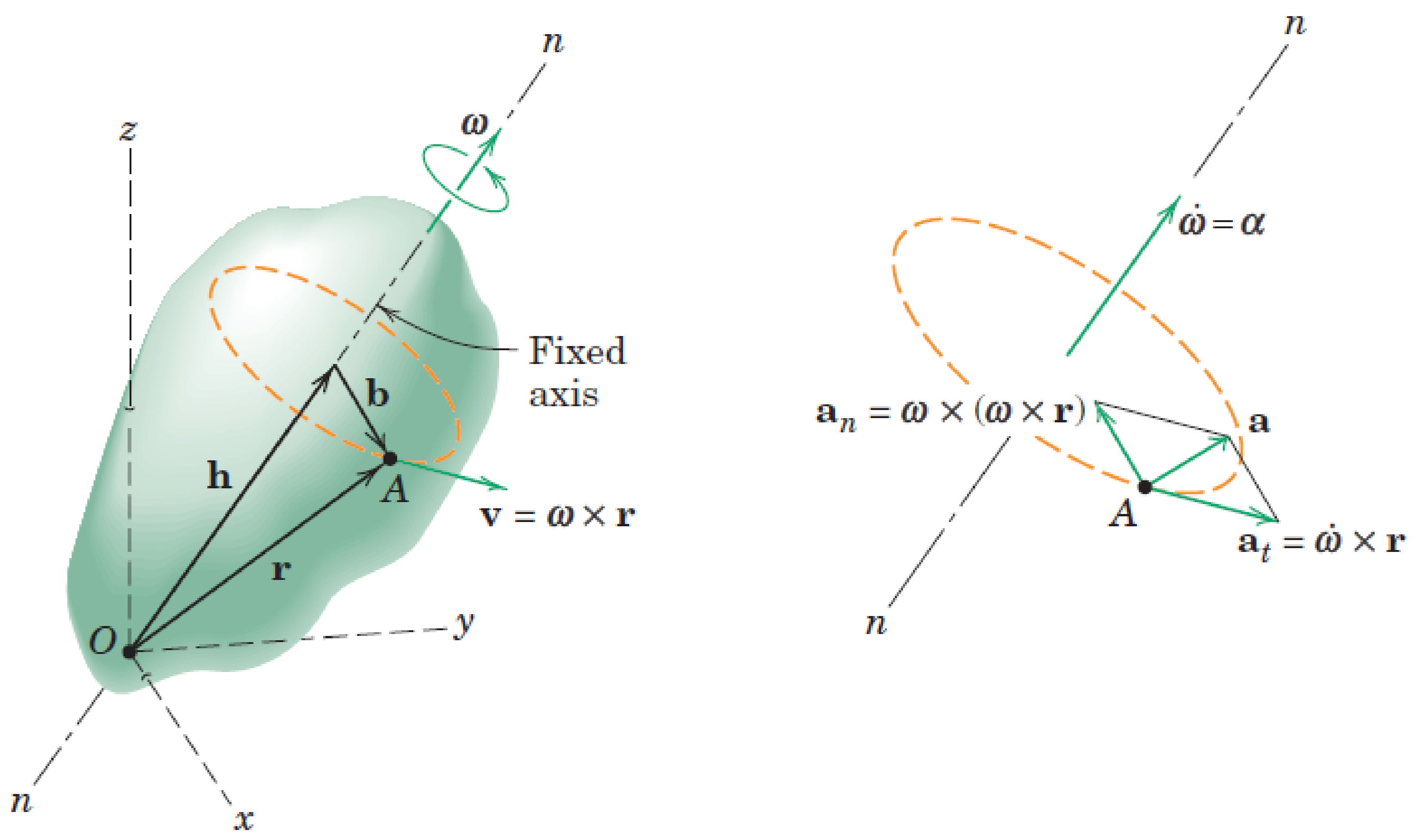
$$r_A = r_B + r_{A/B}$$

$$v_A = v_B$$

$$a_A = a_B$$



Fixed Axis Rotation



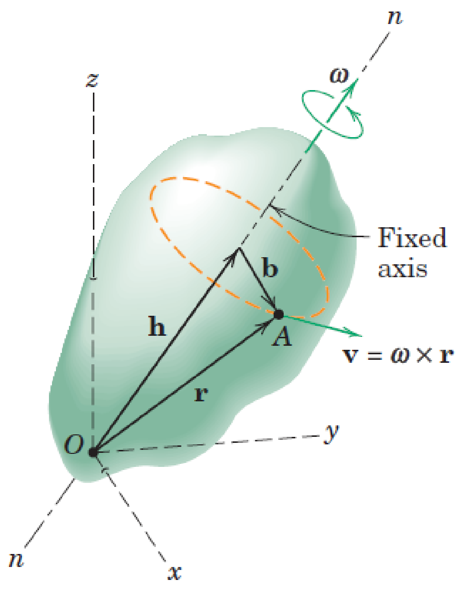
Fixed Axis Rotation

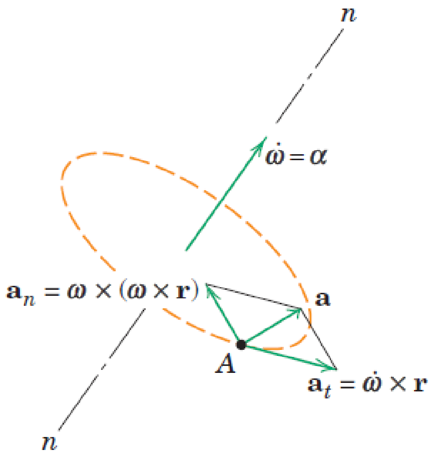
$$\mathrm{v} = \omega \times \mathrm{r}$$

$$a = \dot{\omega} \times r + \omega \times (\omega \times r)$$

$$\mathrm{r} = \mathrm{h} + \mathrm{b}$$

$$\mathrm{v} = \omega \times (\mathrm{h} + \mathrm{b}) = \omega \times \mathrm{b}$$





$$a_n = \omega \times (\omega \times r) = b\omega^2 \qquad a_t = \dot{\omega} \times r = b\alpha$$

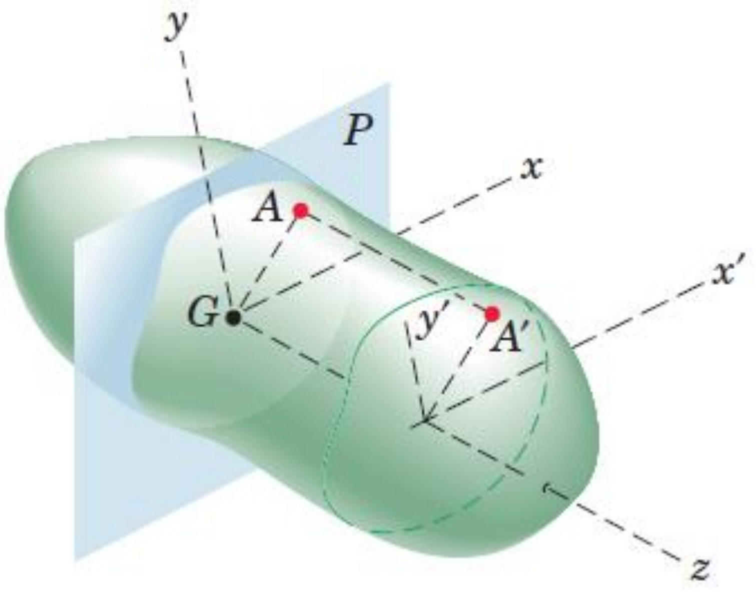
$$\mathrm{v}.\omega = 0 \qquad \mathrm{v}.\dot{\omega} = 0 \qquad \alpha.\omega = 0 \qquad \alpha.\dot{\omega} = 0$$

Parallel Plane Motion

Plane Motion: When all points in a rigid body move in planes which are parallel to a fixed plane *P*.

Plane of Motion: The reference plane is customarily taken through the mass center *G* and is called the *plane of motion*.

Because each point in the body, such as *A′*, has a motion identical with the motion of the corresponding point (*A*) in plane *P*, it follows that the kinematics of plane motion provides a complete description of the motion when applied to the reference plane.

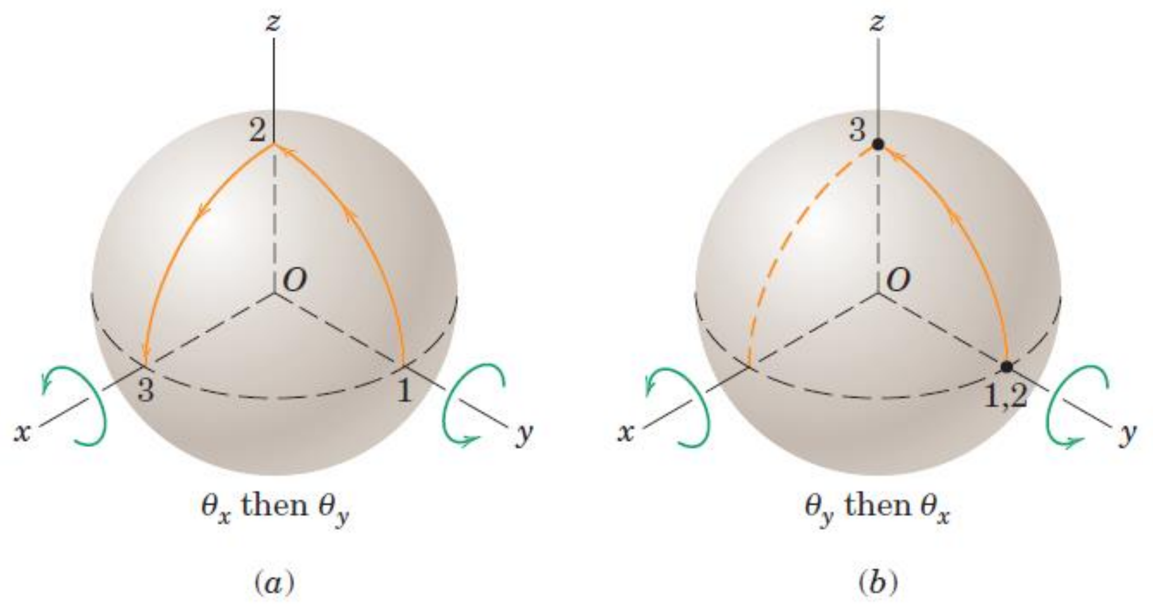


Rotation about a Fixed Point

Proper Vector: Vectors which obey the parallelogram law of addition

In left figure, two successive 90° rotations of the sphere about, first, the x-axis and, second, the y-axis result in the motion of a point which is initially on the y-axis in position 1, to positions 2 and 3, successively.

On the other hand, if the order of the rotations is reversed, the point undergoes no motion during the y-rotation but moves to point 3 during the 90° rotation about the x-axis. Thus, the two cases do not produce the same final position.



Finite rotations do not generally obey the parallelogram law of vector addition and are not commutative. Thus, **finite rotations may not be treated as proper vectors**.

Rotation about a Fixed Point

Infinitesimal rotations do obey the parallelogram law of vector addition.

Displacement of point A due to $\boldsymbol{d\theta_1}$ is $\boldsymbol{d\theta_1} \times \mathbf{r}$
Displacement of point A due to $\boldsymbol{d\theta_2}$ is $\boldsymbol{d\theta_2} \times \mathbf{r}$
Displacement of point A due to $\boldsymbol{d\theta_1}$ and $\boldsymbol{d\theta_2}$ is

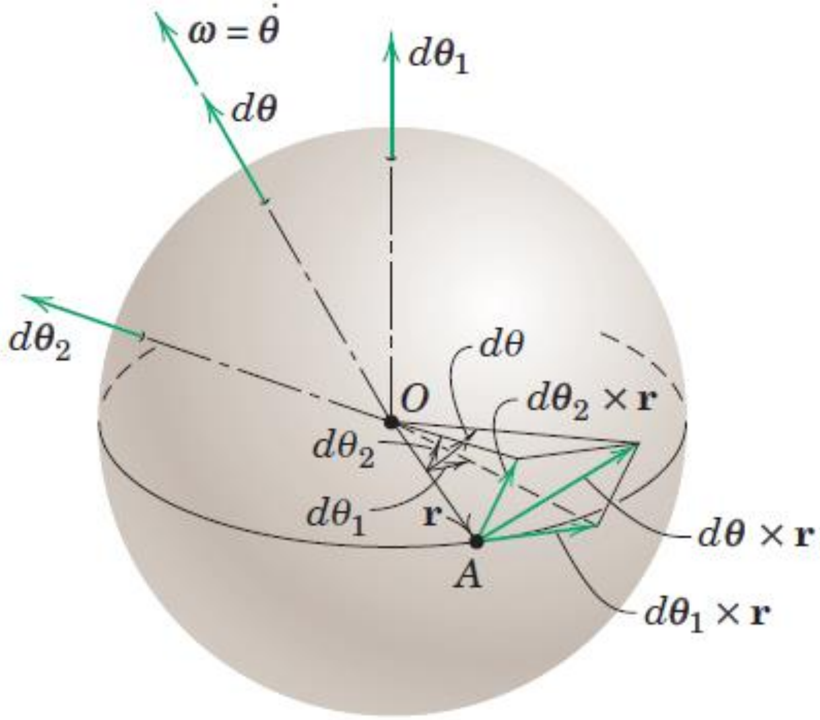
$$(\boldsymbol{d\theta_1} + \boldsymbol{d\theta_2}) \times \mathbf{r} = \boldsymbol{d\theta} \times \mathbf{r}$$

Two rotations are equivalent to one single rotation.

$$\omega_1 = \dot{\theta}_1 \qquad \omega_2 = \dot{\theta}_2$$

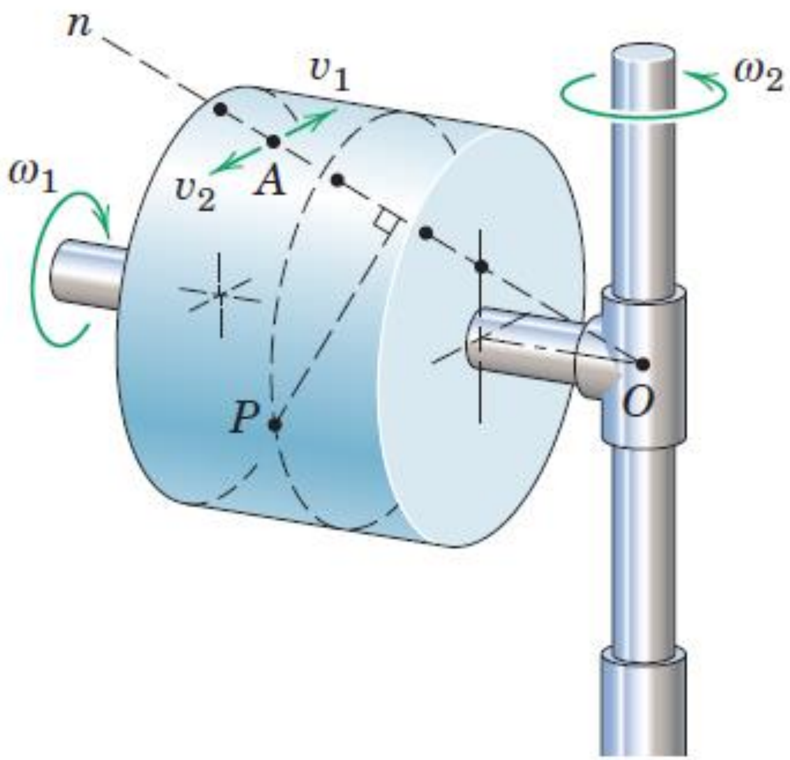
$$\omega = \omega_1 + \omega_2 = \dot{\theta}_1 + \dot{\theta}_2$$

We conclude, therefore, that **at any instant of time a body with one fixed point is rotating instantaneously about a particular axis passing through the fixed point.**



Instantaneous axes of Rotation

- *A solid cylindrical rotor made of clear plastic containing many black particles embedded in the plastic*
- *The rotor is spinning about its shaft axis at the steady rate ω_1 , and its shaft, in turn, is rotating about the fixed vertical axis at the steady rate ω_2 , with rotations in the directions indicated*
- *If the rotor is photographed at a certain instant during its motion, the resulting picture would show one line of black dots sharply defined, indicating that, momentarily, their velocity was zero*
- *This line of points with no velocity establishes the instantaneous position of the axis of rotation O-n.*
- *Any dot on this line, such as A, would have equal and opposite velocity components, v_1 due to ω_1 and v_2 due to ω_2.*

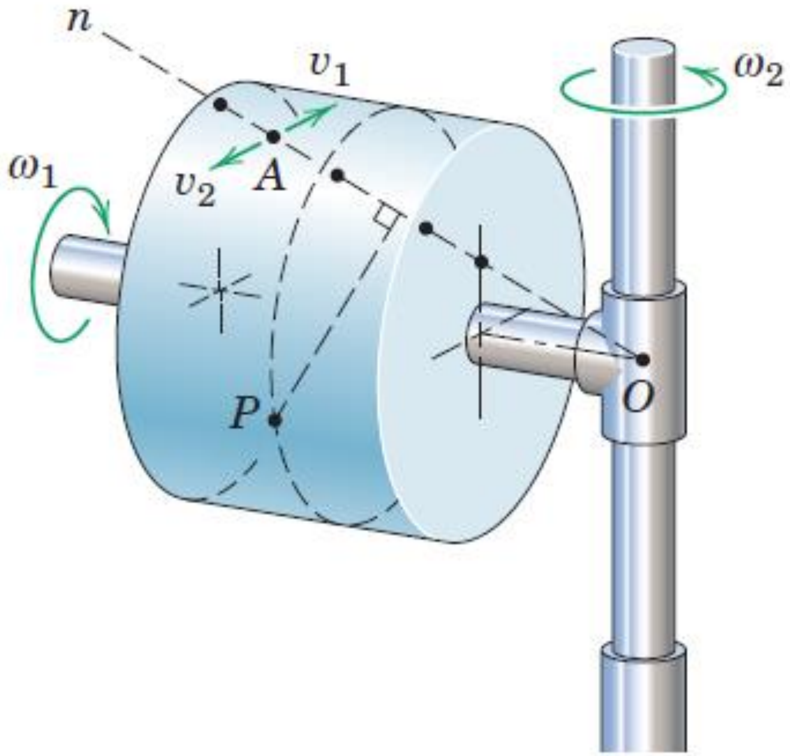


Instantaneous axes of Rotation

- *All other dots such as the one at P, would appear blurred, and their movements would show as short streaks in the form of small circular arcs in planes normal to the axis O-n. Thus, all particles of the body, except those on line O-n, are momentarily rotating in circular arcs about the instantaneous axis of rotation*

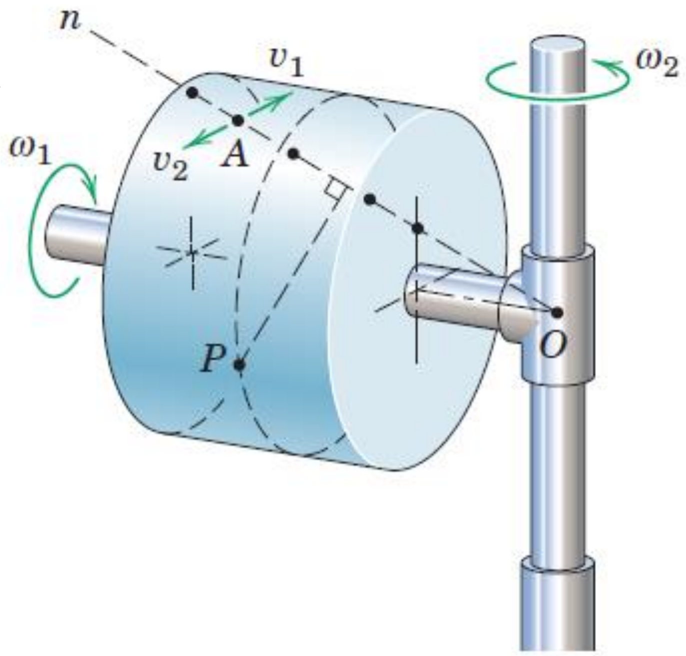
- *If a succession of photographs were taken, we would observe in each photograph that the rotation axis would be defined by a new series of sharply-defined dots and that the axis would change position both in space and relative to the body.*

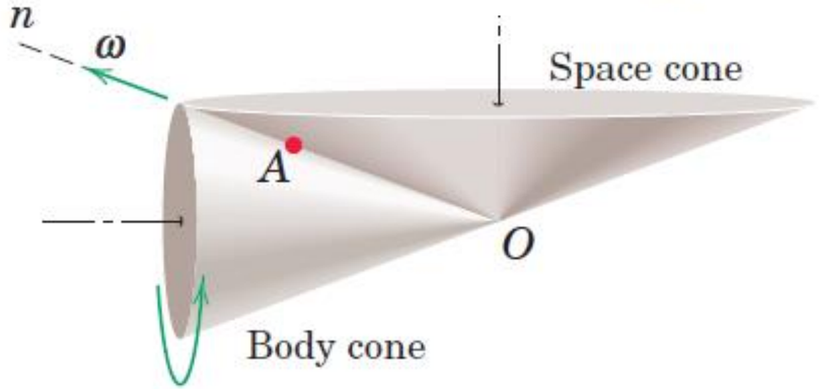
- *For rotation of a rigid body about a fixed point, then, it is seen that the rotation axis is, in general, not a line fixed in the body.*



Body and Space Cones

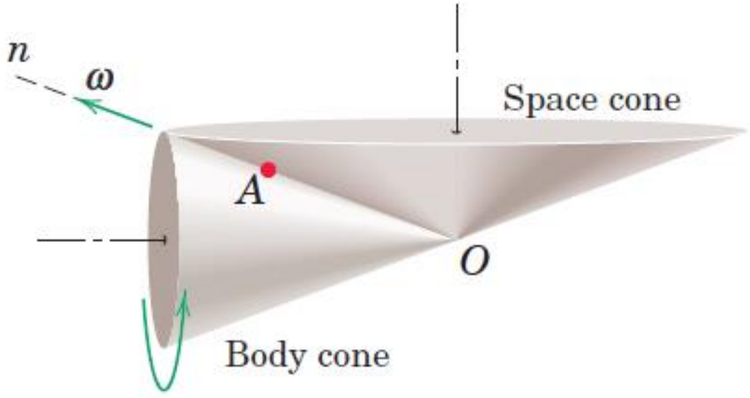
- Relative to the plastic cylinder, the instantaneous axis of rotation *O-A-n* generates a right-circular cone about the cylinder axis called the *body cone*.
- As the two rotations continue and the cylinder swings around the vertical axis, the instantaneous axis of rotation also generates a right-circular cone about the vertical axis called the *space cone*
- These cones are shown in figure for this particular example.

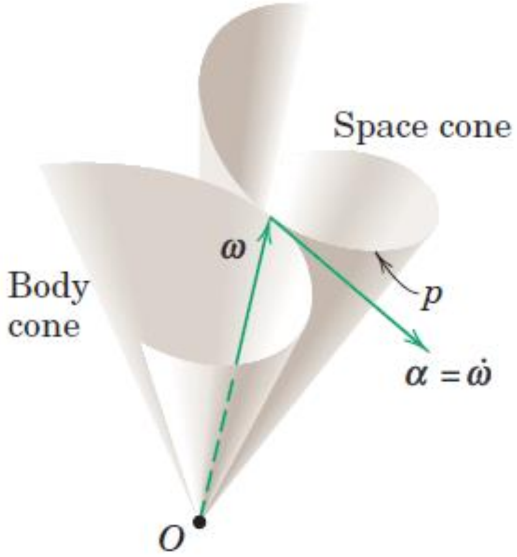




Body and Space Cones

- Body cone rolls on the space cone and that the angular velocity ω of the body is a vector which lies along the common element of the two cones
- For a more general case where the rotations are not steady, the space and body cones are not right-circular cones but the body cone still rolls on the space cone.



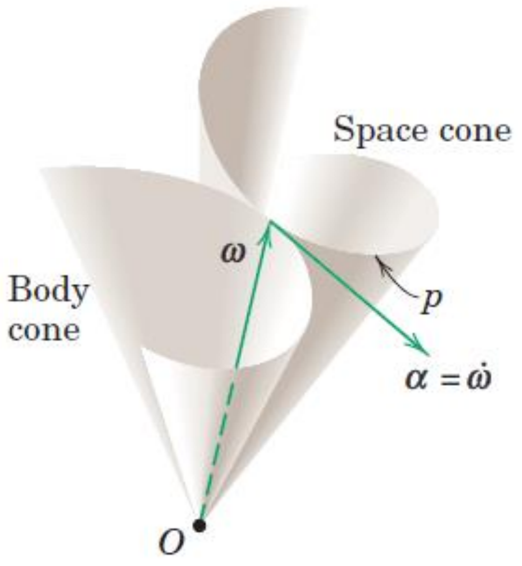


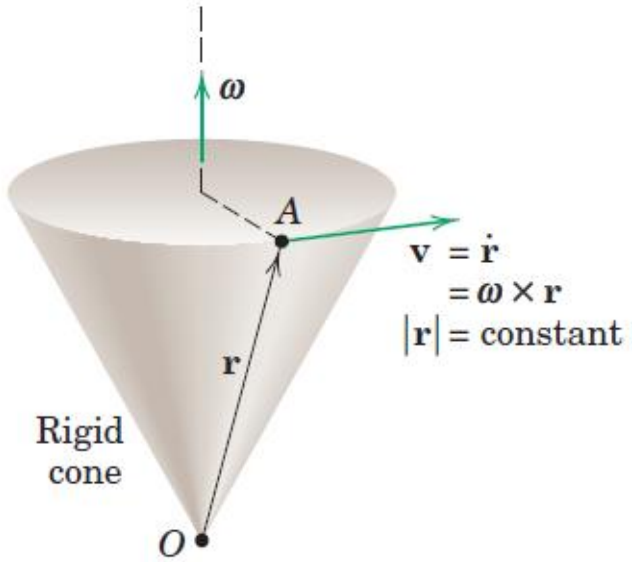
Angular Acceleration

- The angular acceleration $\boldsymbol{\alpha}$ of a rigid body in three-dimensional motion is the time derivative of its angular velocity, $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$

- In contrast to the case of rotation in a single plane where the scalar α measures only the change in magnitude of the angular velocity, in three-dimensional motion the vector $\boldsymbol{\alpha}$ reflects the change in direction of $\boldsymbol{\omega}$ as well as its change in magnitude.

- Thus in Figure where the tip of the angular velocity vector $\boldsymbol{\omega}$ follows the space curve p and changes in both magnitude and direction, the angular acceleration $\boldsymbol{\alpha}$ becomes a vector tangent to this curve in the direction of the change in $\boldsymbol{\omega}$.



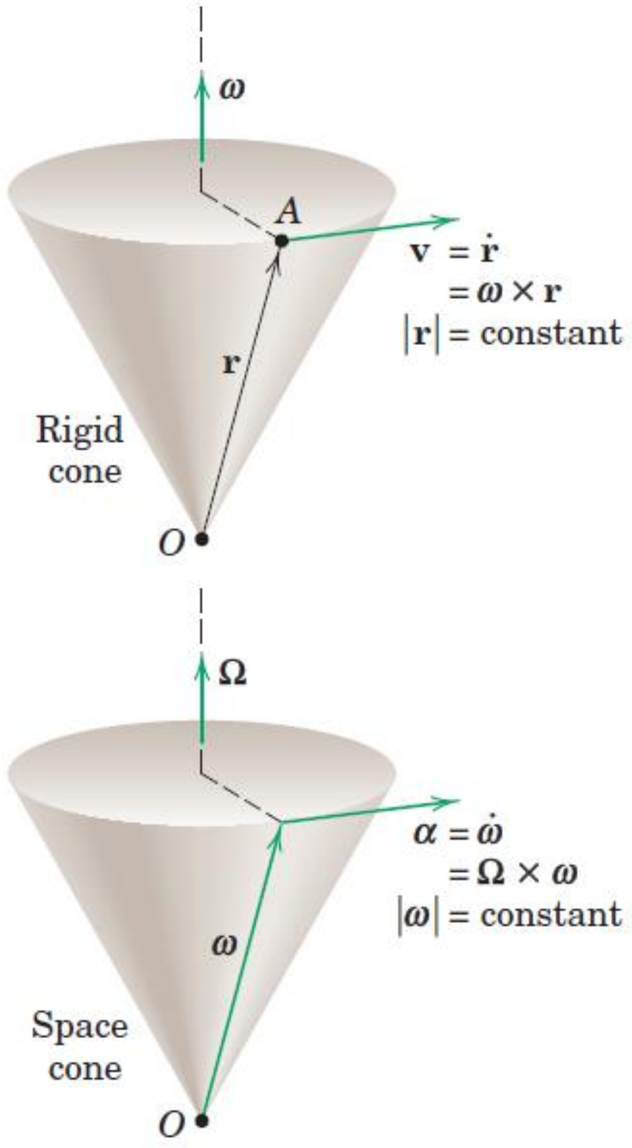


Angular Acceleration

- When the magnitude of $\boldsymbol{\omega}$ remains constant, the angular acceleration $\boldsymbol{\alpha}$ is normal to $\boldsymbol{\omega}$. For this case, if we let $\boldsymbol{\Omega}$ stand for the angular velocity with which the vector itself rotates (precesses) as it forms the space cone, the angular acceleration may be written

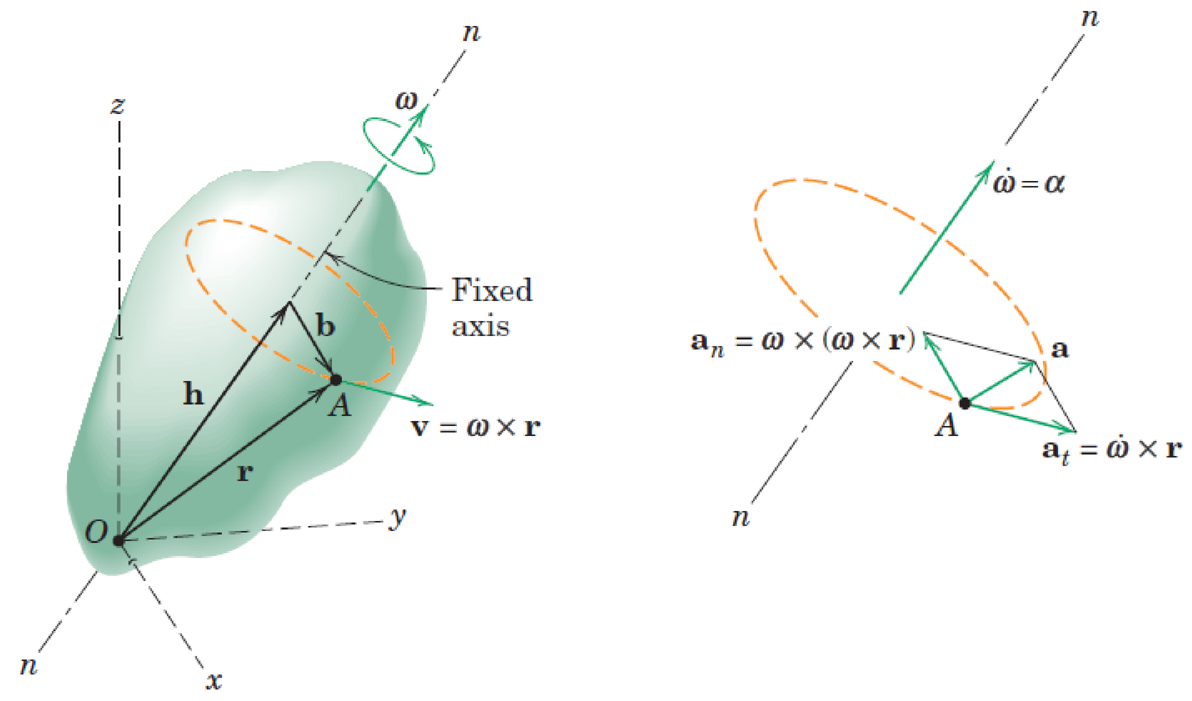
$$\boldsymbol{\alpha} = \boldsymbol{\Omega} \times \boldsymbol{\omega}$$

- The upper part of the figure relates the velocity of a point A on a rigid body to its position vector from O and the angular velocity of the body. The vectors $\boldsymbol{\alpha}$, $\boldsymbol{\omega}$ and $\boldsymbol{\Omega}$ in the lower figure bear exactly the same relationship to each other as do the vectors $\mathbf{v}$, $\mathbf{r}$ and $\boldsymbol{\omega}$ in the upper figure.



Angular Acceleration

Figure represent a rigid body rotating about a fixed point *O* with the instantaneous axis of rotation *n*-*n*, we see that the velocity $\mathbf{v}$ and acceleration $\boldsymbol{a}$ of any point A in the body are given by the same expressions as apply to the case in which the axis is fixed,



$$\mathrm{v} = \omega \times \mathrm{r}$$

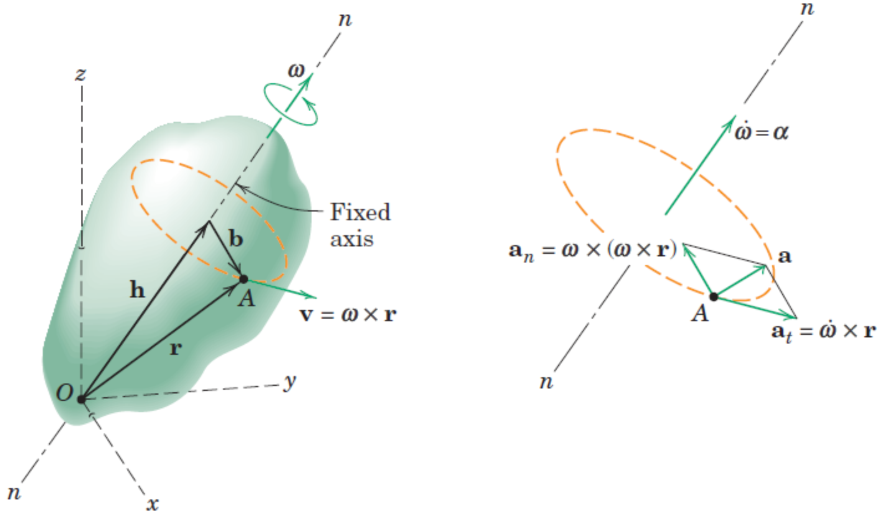
$$a = \dot{\omega} \times r + \omega \times (\omega \times r)$$

Angular Acceleration

- The one difference between the case of rotation about a fixed axis and rotation about a fixed point lies in the fact that for rotation about a fixed point, the angular acceleration $\boldsymbol{\alpha}$ will have a component normal to $\boldsymbol{\omega}$ due to the change in direction of $\boldsymbol{\omega}$ as well as a component in the direction of $\boldsymbol{\omega}$ to reflect any change in the magnitude of $\boldsymbol{\omega}$

- Although any point on the rotation axis n-n momentarily will have zero velocity, it will not have zero acceleration as long as it is changing its direction. On the other hand, for rotation about a fixed axis, $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$ has only the one component along the fixed axis to reflect the change in the magnitude of $\boldsymbol{\omega}$

- Furthermore, points which lie on the fixed rotation axis clearly have no velocity or acceleration.



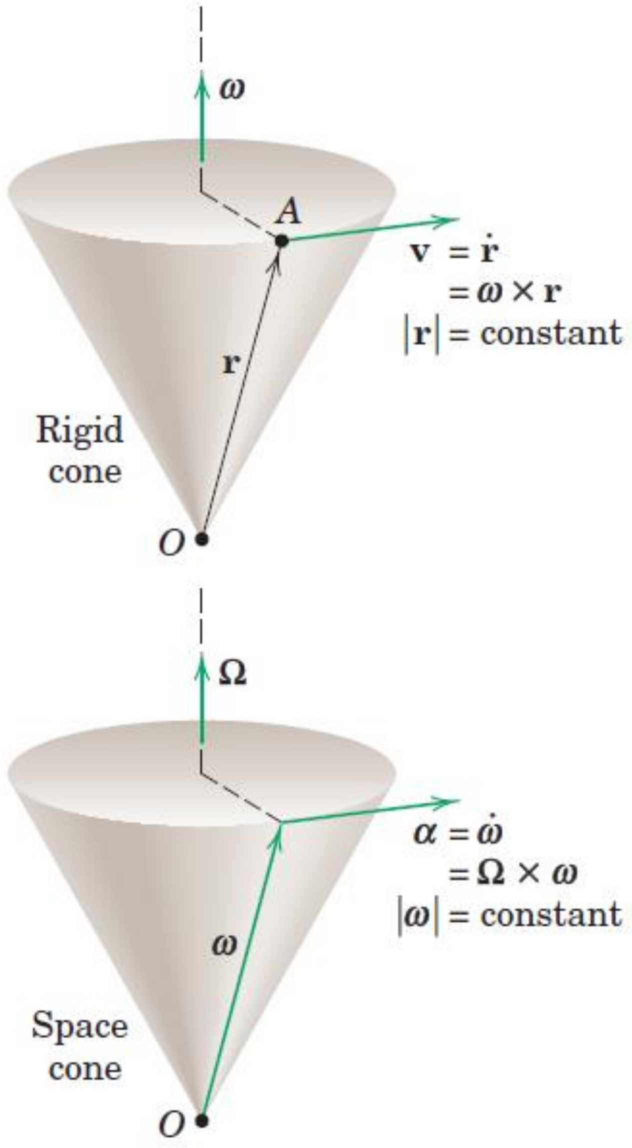
$$\mathrm{v} = \omega \times \mathrm{r}$$

$$a = \dot{\omega} \times r + \omega \times (\omega \times r)$$

Angular Acceleration

- Rotation is a function solely of angular change, so that the expressions for $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ do not depend on the fixity of the point around which rotation occurs

- Rotation may take place independently of the linear motion of the rotation point

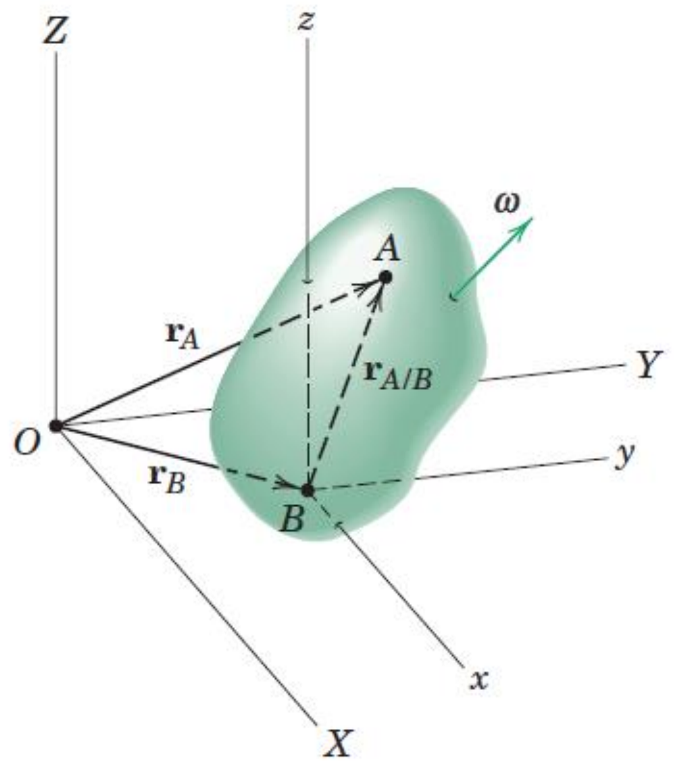


General Motion: Translating reference axes

- From an observer's position on *x-y-z*, the body appears to rotate about the point *B* and point *A* appears to lie on a spherical surface with *B* as the center.
- General motion as a translation of the body with the motion of *B* plus a rotation of the body about *B*.

$$\mathrm{v}_A = \mathrm{v}_B + \mathrm{v}_{A/B}$$

$$a_A = a_B + a_{A/B}$$

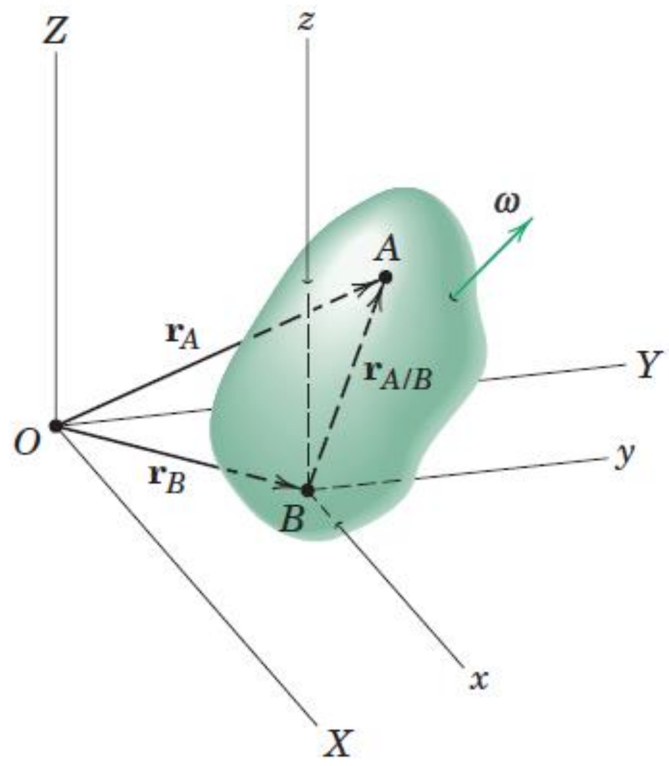


General Motion: Translating reference axes

- The relative-motion terms represent the effect of the rotation about B and are identical to the velocity and acceleration expressions discussed in the previous article for rotation of a rigid body about a fixed point.

$$\mathrm{v_A = v_B + \omega \times r_{A/B}}$$

$$a_A = a_B + \dot{\omega} \times r_{A/B} + \omega \times \left(\omega \times r_{A/B}\right)$$



- Point *A* is chosen as the reference point, the relative-motion equations

$$\mathrm{v_B = v_A + \omega \times r_{B/A}}$$

$$a_B = a_A + \dot{\omega} \times r_{B/A} + \omega \times \left(\omega \times r_{B/A}\right)$$

General Motion: Rotating reference axes

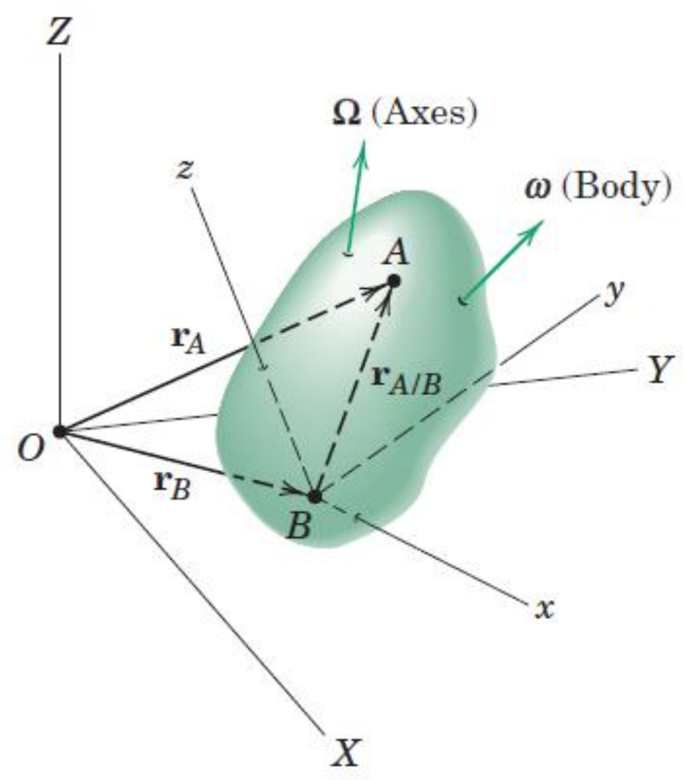
- Use of reference axes which rotate as well as translate

- To show reference axes whose origin is attached to the reference point B as before, but which rotate with an absolute angular velocity Ω which may be different from the absolute angular velocity ω of the body

$$\dot{\mathrm{i}} = \Omega \times \mathrm{i}$$

$$\dot{\mathrm{j}} = \Omega \times \mathrm{j}$$

$$\dot{\mathrm{k}} = \Omega \times \mathrm{k}$$



Time derivatives of the rotating unit vectors attached to *x-y-z*

General Motion: Rotating reference axes

- Expression for the velocity and acceleration of point A

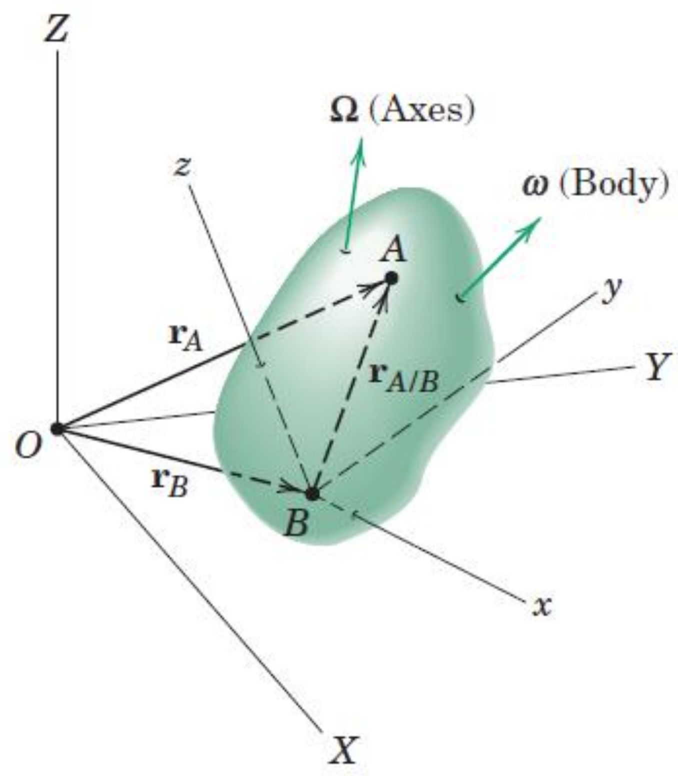
$$\mathbf{v}_A = \mathbf{v}_B + \Omega \times \mathbf{r}_{A/B} + \mathbf{v}_{rel}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\Omega} \times \mathbf{r}_{A/B} + \Omega \times (\Omega \times \mathbf{r}_{A/B}) + 2\Omega \times \mathbf{v}_{rel} + \mathbf{a}_{rel}$$

$$\mathbf{v}_{rel} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}$$

$$\mathbf{a}_{rel} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k}$$

Velocity and acceleration of point *A* measured relative to *x-y-z* by an observer attached to *x-y-z*



General Motion: Rotating reference axes

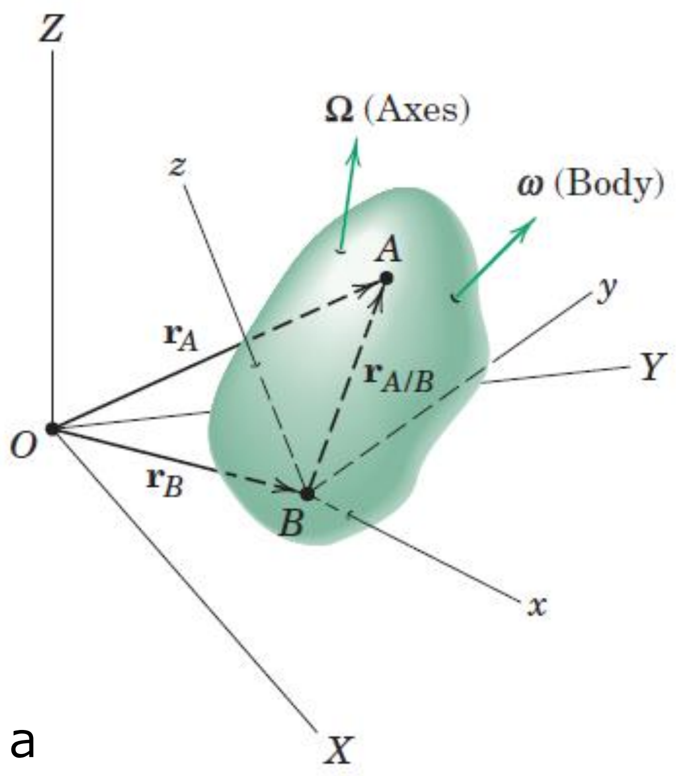
- Expression for the velocity and acceleration of point A

$$\mathrm{v_A = v_B + \Omega \times r_{A/B} + v_{rel}}$$

$$\mathrm{a_A = a_B + \dot{\Omega} \times r_{A/B} + \Omega \times (\Omega \times r_{A/B}) + 2\Omega \times v_{rel} + a_{rel}}$$

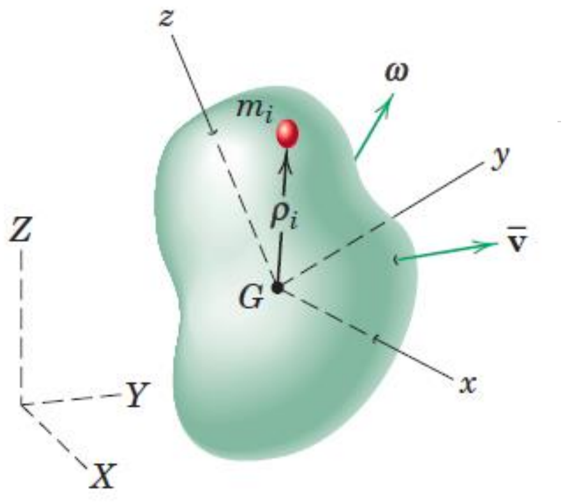
- Ω angular velocity of the axes and it may be different from the angular velocity ω of the body

- $\mathrm{r_{A/B}}$ remains constant in magnitude for points *A* and *B* fixed to a rigid body, but it will change direction with respect to *x*-*y*-*z* when the angular velocity of the axes is different from the angular velocity of the body



Angular Momentum

- Consider a rigid body moving with any general motion in space
- Axes x-y-z are attached to the body with origin at the mass center G
- The angular velocity of the body becomes the angular velocity of the x-y-z axes as observed from the fixed reference axes X-Y-Z.



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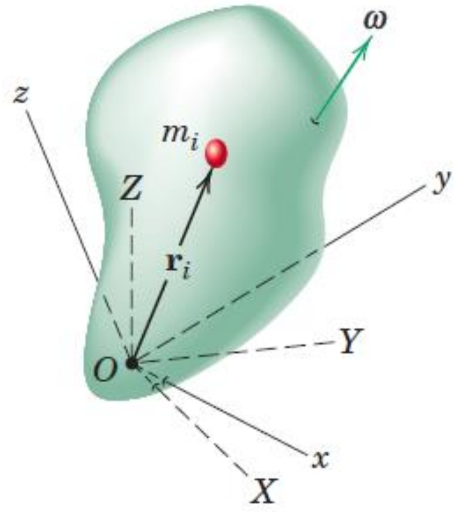
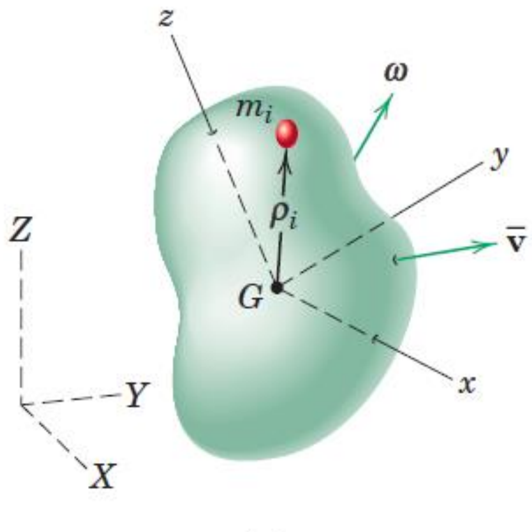


Angular Momentum

➢ Sum of the moments about G of the linear momenta of all elements of the body

$$H_G = \sum(\rho_i \times m_i v_i)$$

$\boldsymbol{v_i}$ is the absolute velocity of mass element

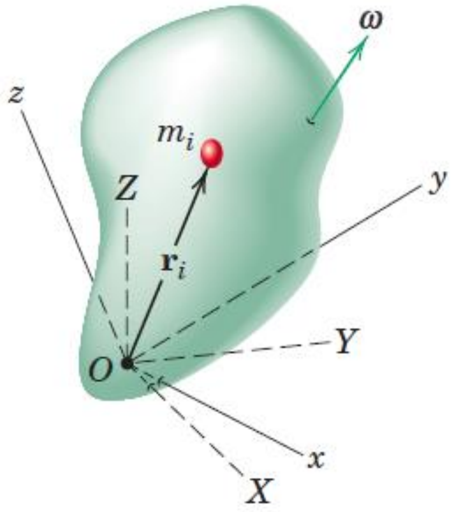


(a)

$$v_i = \bar{v} + \omega \times \rho_i$$

$$H_G = \sum(\rho_i \times m_i(\bar{v} + \omega \times \rho_i))$$

$$H_G = -\bar{v} \times \sum m_i \rho_i + \sum \rho_i \times m_i(\omega \times \rho_i)$$



Angular Momentum

$$H_G = \sum(\rho_i \times m_i v_i)$$

$$H_G = -\bar{v} \times \sum m_i \rho_i + \sum \rho_i \times m_i(\omega \times \rho_i)$$

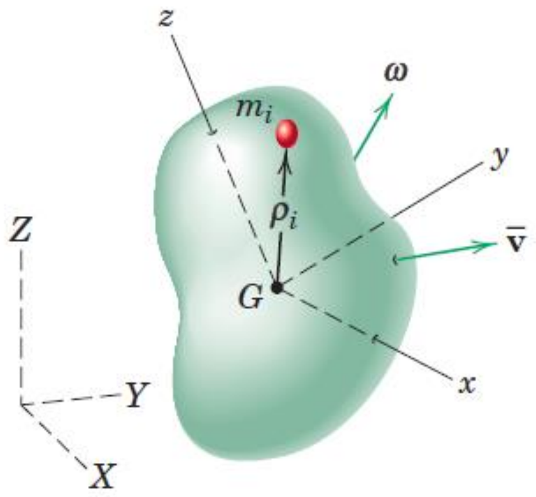
With the origin at the mass center *G*,

$$\sum m_i \rho_i = m\bar{\rho} = 0$$

dm for m_i and ρ for ρ_i

$$H_G = \int [\rho \times (\omega \times \rho)]\, \mathrm{d}m$$

$$H_0 = \int [r \times (\omega \times r)]\, \mathrm{d}m$$



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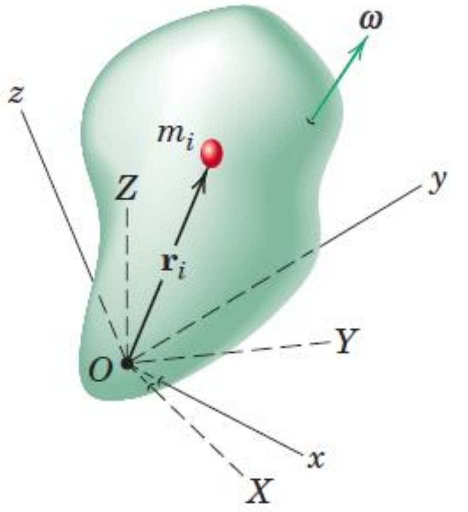


Moments and Product of Inertia

$$\mathbf{H_G} = \int [\rho \times (\omega \times \rho)]\, \mathrm{dm}$$

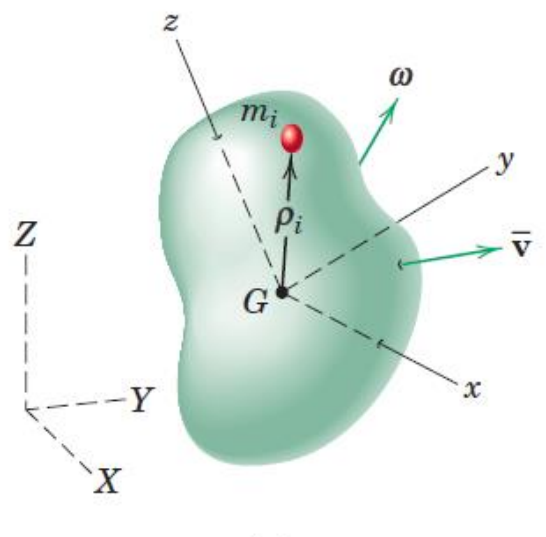
$$\mathbf{H_0} = \int [r \times (\omega \times r)]\, \mathrm{dm} = \int d\mathrm{H}$$

$$r = xi + yj + zk \qquad\qquad \omega = \omega_x i + \omega_y j + \omega_z k$$

$$d\mathbf{H} = \mathrm{i}\left[(y^2 + z^2)\omega_x - xy\omega_y - xz\omega_z\right]\mathrm{dm}$$

$$+j\left[-yx\omega_x + (z^2 + x^2)\omega_y - yz\omega_z\right]\mathrm{dm}$$

$$+k\left[-zx\omega_x - zy\omega_y + (x^2 + y^2)\omega_z\right]\mathrm{dm}$$



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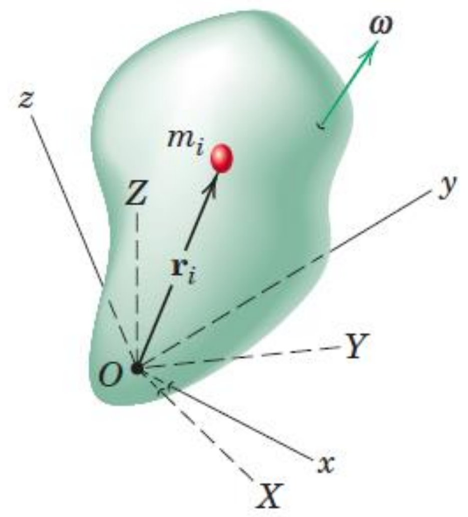
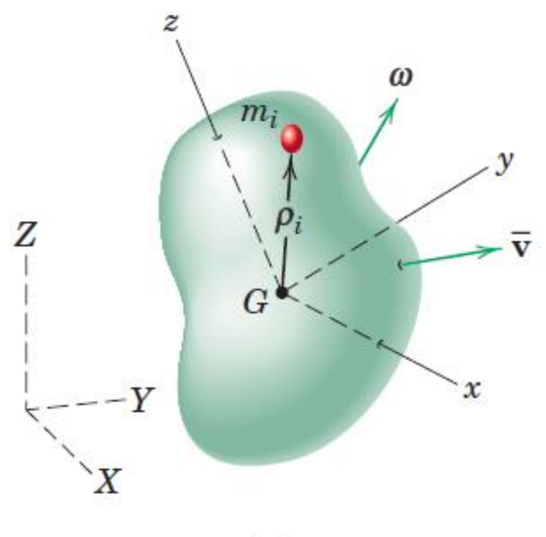


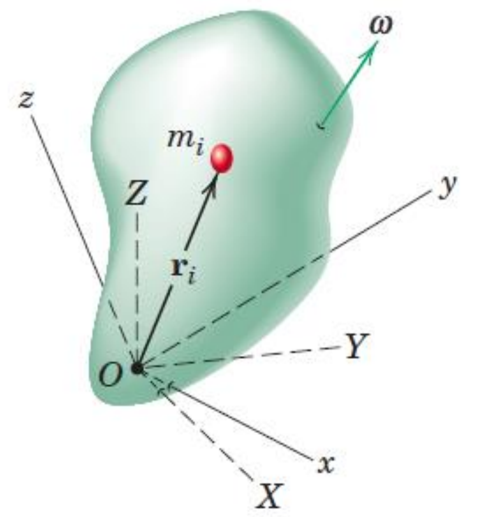
Moments and Product of Inertia

$$d\mathbf{H} = \mathrm{i}\left[(y^2+z^2)\omega_x - xy\omega_y - xz\omega_z\right]\mathrm{dm}$$
$$+j\left[-yx\omega_x + (z^2+x^2)\omega_y - yz\omega_z\right]\mathrm{dm}$$
$$+k\left[-zx\omega_x - zy\omega_y + (x^2+y^2)\omega_z\right]\mathrm{dm}$$



(a)

$$I_{xx} = \int (y^2+z^2)\,\mathrm{dm} \qquad I_{xy} = \int xy\,\mathrm{dm}$$
$$I_{yy} = \int (z^2+x^2)\,\mathrm{dm} \qquad I_{xz} = \int xz\,\mathrm{dm}$$
$$I_{zz} = \int (x^2+y^2)\,\mathrm{dm} \qquad I_{yz} = \int yz\,\mathrm{dm}$$



Moments and Product of Inertia

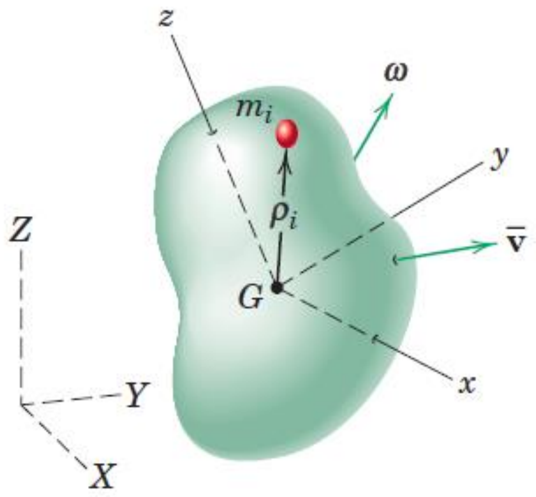
$$I_{xx} = \int (y^2 + z^2)\, \mathrm{d}m \qquad I_{xy} = \int xy\, \mathrm{d}m$$

$$I_{yy} = \int (z^2 + x^2)\, \mathrm{d}m \qquad I_{xz} = \int xz\, \mathrm{d}m$$

$$I_{zz} = \int (x^2 + y^2)\, \mathrm{d}m \qquad I_{yz} = \int yz\, \mathrm{d}m$$

Moments of inertia of the body about the respective axes,
$\boldsymbol{I_{xx}, I_{yy}, I_{zz}}$
Products of inertia with respect to the coordinate axes
$\boldsymbol{I_{xy}, I_{xz}, I_{yz}}$

These quantities describe the manner in which the mass of a rigid body is distributed with respect to the chosen axes.



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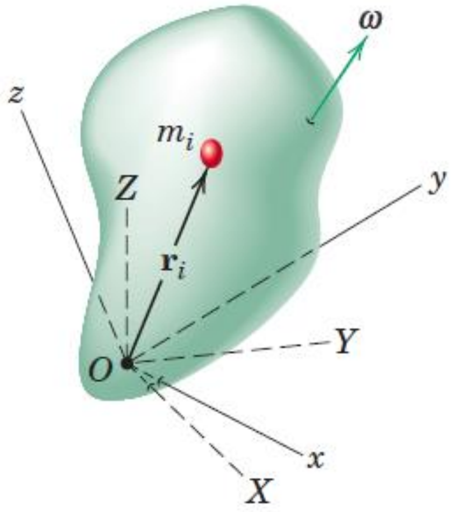


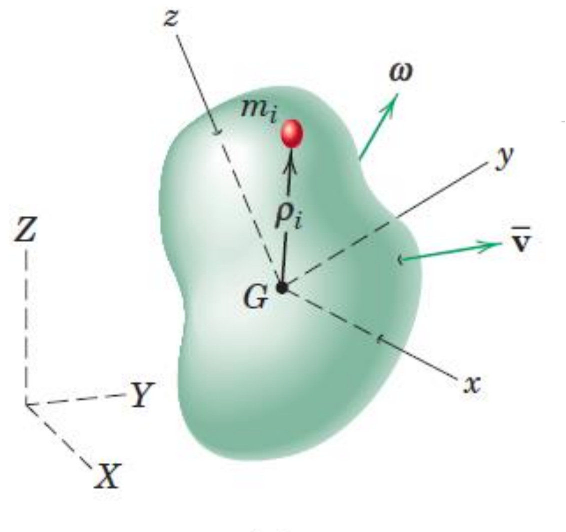
Moments and Product of Inertia

$$\mathbf{H} = \mathrm{i}\left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$
$$+j\left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$
$$+k\left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

$$I_{xx} = \int (y^2 + z^2)\,\mathrm{d}m \qquad I_{xy} = \int xy\,\mathrm{d}m$$

$$I_{yy} = \int (z^2 + x^2)\,\mathrm{d}m \qquad I_{xz} = \int xz\,\mathrm{d}m$$

$$I_{zz} = \int (x^2 + y^2)\,\mathrm{d}m \qquad I_{yz} = \int yz\,\mathrm{d}m$$



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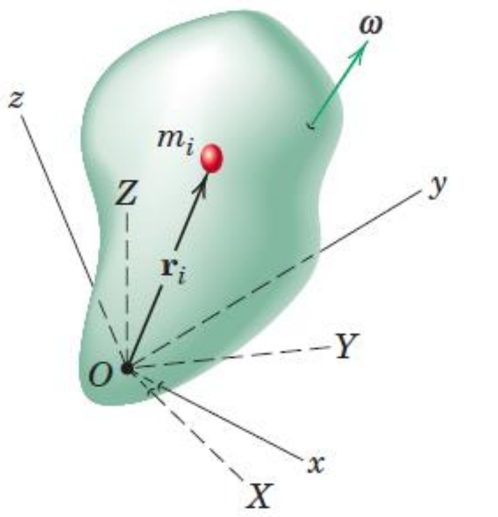


Moments and Product of Inertia

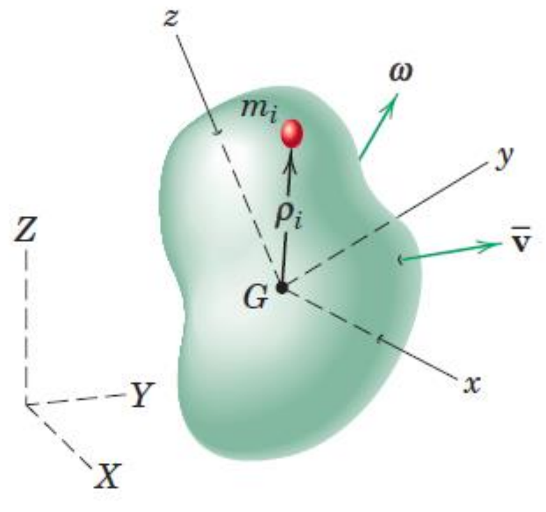
$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

- General expression for the angular momentum either about the mass center *G* or about a fixed point *O* for a rigid body rotating with an instantaneous angular velocity, ω

- In each of the two cases represented, the reference axes x-y-z are attached to the rigid body. This attachment makes the moment-of-inertia integrals and the product-of-inertia integrals invariant with time.



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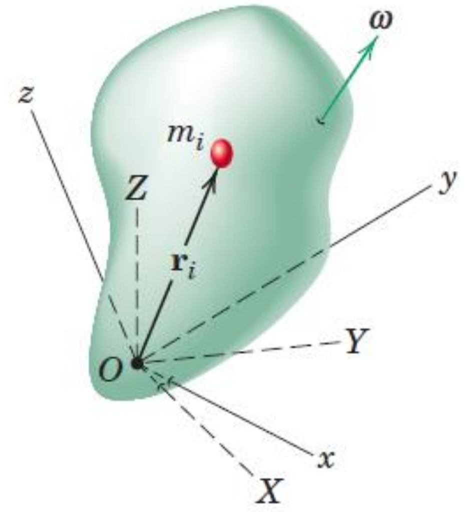


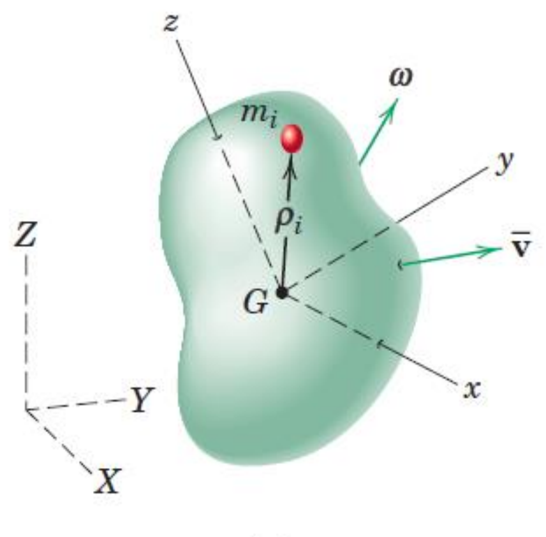
Moments and Product of Inertia

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + (I_{yy})\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

- If the x-y-z axes were to rotate with respect to an irregular body, then these inertia integrals would be functions of the time, which would introduce an undesirable complexity into the angular momentum relations.



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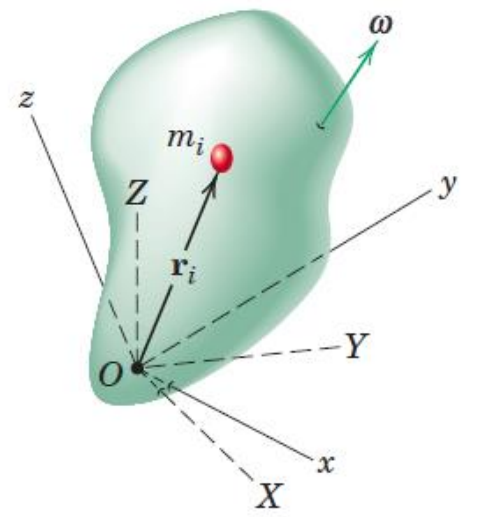
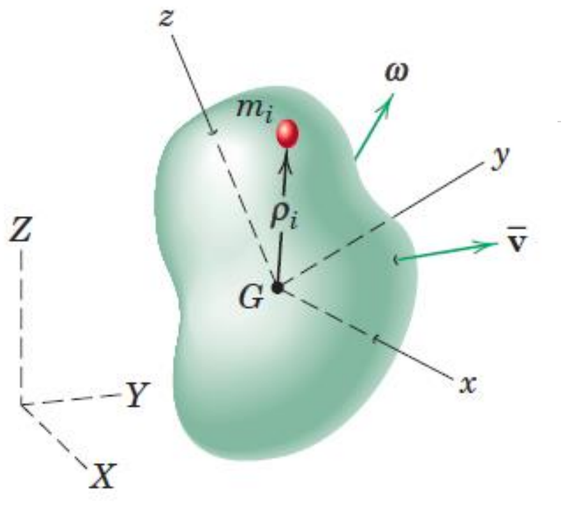


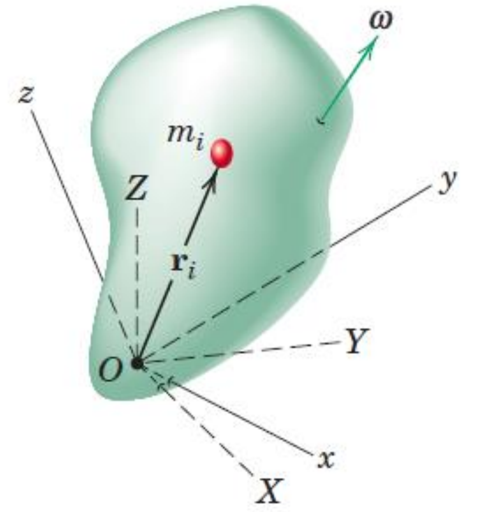
Inertia Matrix or Inertia tensor

$$\begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix}$$



(a)

- As we change the orientation of the axes relative to the body, the moments and products of inertia will also change in value. There is one unique orientation of axes x-y-z for a given origin for which the products of inertia vanish and the moments of inertia I_{xx}, I_{yy}, I_{zz} take on stationary values.



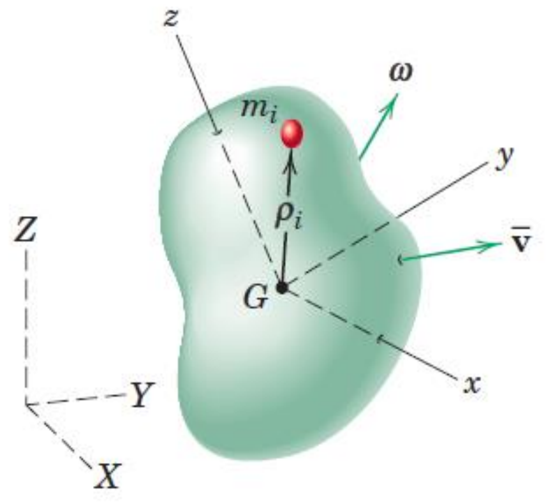
Principal moments of inertia

$$\begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

- The axes x-y-z for which the products of inertia vanish are called the **principal axes of inertia**

- I_{xx}, I_{yy}, and I_{zz} are called the **principal moments of inertia**

- The principal moments of inertia for a given origin represent the maximum, the minimum, and an intermediate value of the moments of inertia



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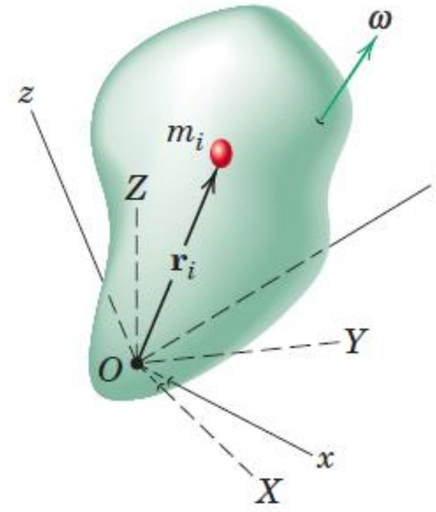
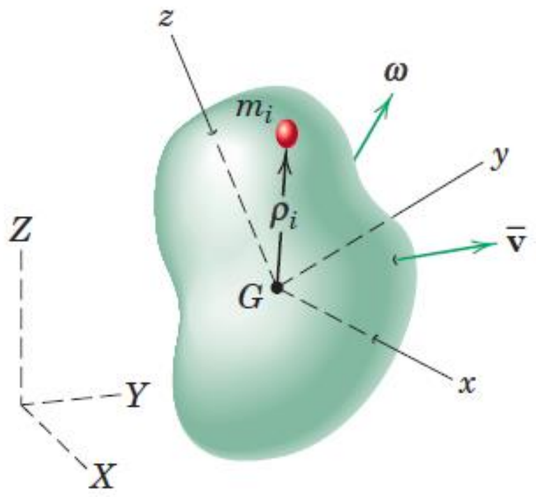


Angular Momentum

$$\begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

- The axes x-y-z for which the products of inertia vanish are called the **principal axes of inertia**
- I_{xx}, I_{yy}, and I_{zz} are called the **principal moments of inertia**

$$\mathbf{H} = I_{xx}\omega_x\mathbf{i} + I_{yy}\omega_y\mathbf{j} + I_{zz}\omega_z\mathbf{k}$$



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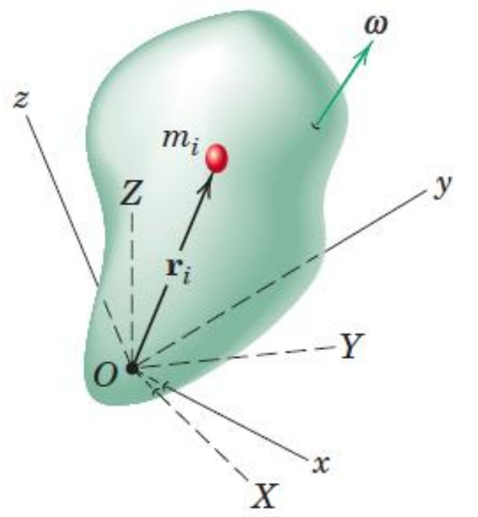
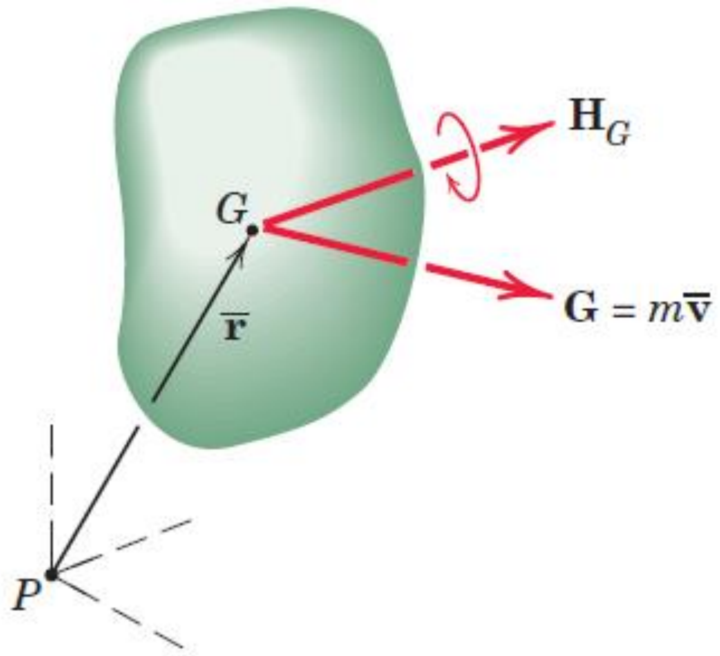


Transfer Principle for Angular Momentum

- The momentum properties of a rigid body may be represented by the resultant linear-momentum vector $\boldsymbol{G} = m\bar{\mathbf{v}}$ through the mass center and the resultant angular-momentum vector $\mathbf{H_G}$ about the mass center. Although $\mathbf{H_G}$ has the properties of a free vector, we represent it through G for convenience
- These vectors have properties analogous to those of a force and a couple. Thus, the angular momentum about any point P equals the free vector $\mathbf{H_G}$ plus the moment of the linear-momentum vector $\boldsymbol{G}$ about P
- Transfer theorem for angular momentum

$$H_P = H_G + \bar{r} \times G$$



Kinetic Energy

$$T = \frac{1}{2}m\bar{v}^2 + \sum \frac{1}{2}m_i|\dot{\rho}_i|^2$$

- $\bar{\mathbf{v}}$ is velocity of mass center G and ρ_i is position vector of mass element m_i with respect to mass center

- First term as the kinetic energy due to the translation of the system and the second term as the kinetic energy associated with the motion relative to the mass center

$$\frac{1}{2}m\bar{v}^2 = \frac{1}{2}m\dot{\bar{\boldsymbol{r}}}.\dot{\bar{\boldsymbol{r}}} = \frac{1}{2}\bar{\mathbf{v}}.\mathbf{G}$$

- $\mathbf{G}$ is linear momentum of the body

Kinetic Energy

$$T = \frac{1}{2} m \bar{v}^2 + \sum \frac{1}{2} m_i |\dot{\rho}_i|^2$$

$$\sum \frac{1}{2} m_i |\dot{\rho}_i|^2 = \sum \frac{1}{2} m_i (\omega \times \rho_i).(\omega \times \rho_i)$$

$$\mathbf{P} \times \mathbf{Q}.\mathbf{R} = \mathbf{P}.\mathbf{Q} \times \mathbf{R}$$

$$\sum \frac{1}{2} m_i (\omega \times \rho_i).(\omega \times \rho_i) = \sum \frac{1}{2} m_i \omega.\rho_i \times (\omega \times \rho_i) = \omega.\sum \frac{1}{2} m_i \rho_i \times (\omega \times \rho_i)$$

$$\omega.\sum \frac{1}{2} m_i \rho_i \times (\omega \times \rho_i) = \frac{1}{2}\omega.\int \rho \times (\omega \times \rho) dm = \frac{1}{2}\omega.\int \rho \times (\omega \times \rho) dm = \frac{1}{2}\boldsymbol{\omega}.\mathbf{H_G}$$

Kinetic Energy

General expression for the kinetic energy of a rigid body moving with mass-center velocity $\bar{\mathbf{v}}$ and angular velocity $\boldsymbol{\omega}$ is

$$T = \frac{1}{2}m\bar{v}^2 + \sum \frac{1}{2}m_i|\dot{\rho}_i|^2 = \frac{1}{2}\bar{\mathbf{v}}.\mathbf{G} + \frac{1}{2}\boldsymbol{\omega}.\mathbf{H_G}$$

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

Kinetic Energy

General expression for the kinetic energy of a rigid body moving with mass-center velocity $\bar{\mathbf{v}}$ and angular velocity $\boldsymbol{\omega}$ is

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$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\left(\bar{I}_{xx}\omega_x^2 + \bar{I}_{yy}\omega_y^2 + \bar{I}_{zz}\omega_z^2\right) - \left(\bar{I}_{xy}\omega_x\omega_y + \bar{I}_{xz}\omega_x\omega_z + \bar{I}_{yz}\omega_y\omega_z\right)$$

If the axes coincide with principal axes of inertia,

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\left(\bar{I}_{xx}\omega_x^2 + \bar{I}_{yy}\omega_y^2 + \bar{I}_{zz}\omega_z^2\right)$$

Kinetic Energy

When a rigid body is pivoted about a fixed point *O* or when there is a point *O* in the body which momentarily has zero velocity, the kinetic energy, *T*

$$T = \sum \frac{1}{2} m_i \dot{\boldsymbol{r}}_i . \dot{\boldsymbol{r}}_i$$

This expression reduces to

$$T = \frac{1}{2} \boldsymbol{\omega} . \mathbf{H}_{\mathrm{o}}$$

where $\mathbf{H}_{\mathrm{o}}$ is the angular momentum about *O.*

Momentum Equations

General linear- and angular-momentum equations for a system of constant mass

$$\sum F = \dot{G} \qquad \sum M = \dot{H}$$

In the derivation of the moment principle, the derivative of H was taken with respect to an absolute coordinate system.

When H is expressed in terms of components measured relative to a moving coordinate system *x*-*y*-*z* which has an angular velocity Ω, the moment relation becomes

$$\sum M = \left(\frac{\mathrm{d}H}{\mathrm{d}t}\right)_{xyz} + \Omega \times \mathrm{H}$$

Momentum Equations

When H is expressed in terms of components measured relative to a moving coordinate system x-y-z which has an angular velocity Ω, the moment relation becomes

$$\sum \boldsymbol{M} = \left(\frac{\mathrm{d}\boldsymbol{H}}{\mathrm{d}t}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

The terms in parentheses represent that part of $\dot{\boldsymbol{H}}$ due to the change in magnitude of the components of **H**, and the cross-product term represents that part due to the changes in direction of the components of **H**.

Momentum Equations

$$\sum \boldsymbol{M} = \left(\frac{\mathrm{d}\boldsymbol{H}}{\mathrm{d}t}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H_y}\Omega_\mathrm{z} + \mathrm{H_z}\Omega_\mathrm{y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H_z}\Omega_\mathrm{x} + \mathrm{H_x}\Omega_\mathrm{z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H_x}\Omega_\mathrm{y} + \mathrm{H_y}\Omega_\mathrm{x}\right)\mathrm{k}$$

Most general form of the moment equation about a fixed point *O* or about the mass center *G*
Ω's are the angular velocity components of rotation of the reference axes
In the *H* components of a rigid body, ω the angular velocity of the body is used

Momentum Equations

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H}_\mathrm{y}\Omega_\mathrm{z} + \mathrm{H}_\mathrm{z}\Omega_\mathrm{y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H}_\mathrm{z}\Omega_\mathrm{x} + \mathrm{H}_\mathrm{x}\Omega_\mathrm{z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H}_\mathrm{x}\Omega_\mathrm{y} + \mathrm{H}_\mathrm{y}\Omega_\mathrm{x}\right)\mathrm{k}$$

Most general form of the moment equation about a fixed point *O* or about the mass center *G*.
Ω's are the angular velocity components of rotation of the reference axes
In the *H* components of a rigid body, ω the angular velocity of the body is used

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

Momentum Equations

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H_y\Omega_z} + \mathrm{H_z\Omega_y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H_z\Omega_x} + \mathrm{H_x\Omega_z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H_x\Omega_y} + \mathrm{H_y\Omega_x}\right)\mathrm{k}$$

- Axes x-y-z attached to rigid body
- Under these conditions, when expressed in the x-y-z coordinates, the moments and products of inertia are invariant with time and $\Omega = \omega$

$$\sum M_x = \dot{H}_x - H_y\omega_z + H_z\omega_y$$

$$\sum M_y = \dot{H}_y - H_z\omega_x + H_x\omega_z \qquad \sum M_z = \dot{H}_z - H_x\omega_y + H_y\omega_x$$

- General moment equations for rigid-body motion with axes *attached to the body*
- They hold with respect to axes through a fixed point *O* or through the mass center *G*

Euler's Equations

$$\sum \mathrm{M} = (\dot{\mathrm{H}}_x - \mathrm{H}_y\Omega_z + \mathrm{H}_z\Omega_y)\mathrm{i} + (\dot{\mathrm{H}}_y - \mathrm{H}_z\Omega_x + \mathrm{H}_x\Omega_z)\mathrm{j} + (\dot{\mathrm{H}}_z - \mathrm{H}_x\Omega_y + \mathrm{H}_y\Omega_x)\mathrm{k}$$

For any origin fixed to a rigid body, there are three principal axes of inertia with respect to which the products of inertia vanish.
If the reference axes coincide with the principal axes of inertia with origin at the mass center *G* or at a point *O* fixed to the body and fixed in space, the factors I_{xy}, I_{yz} *and* I_{xz} will be zero. Then,

$$\sum M_x = I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z$$

$$\sum M_y = I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x$$

$$\sum M_z = I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y$$

- Euler's Equations
- Useful in the study of rigid-body motion

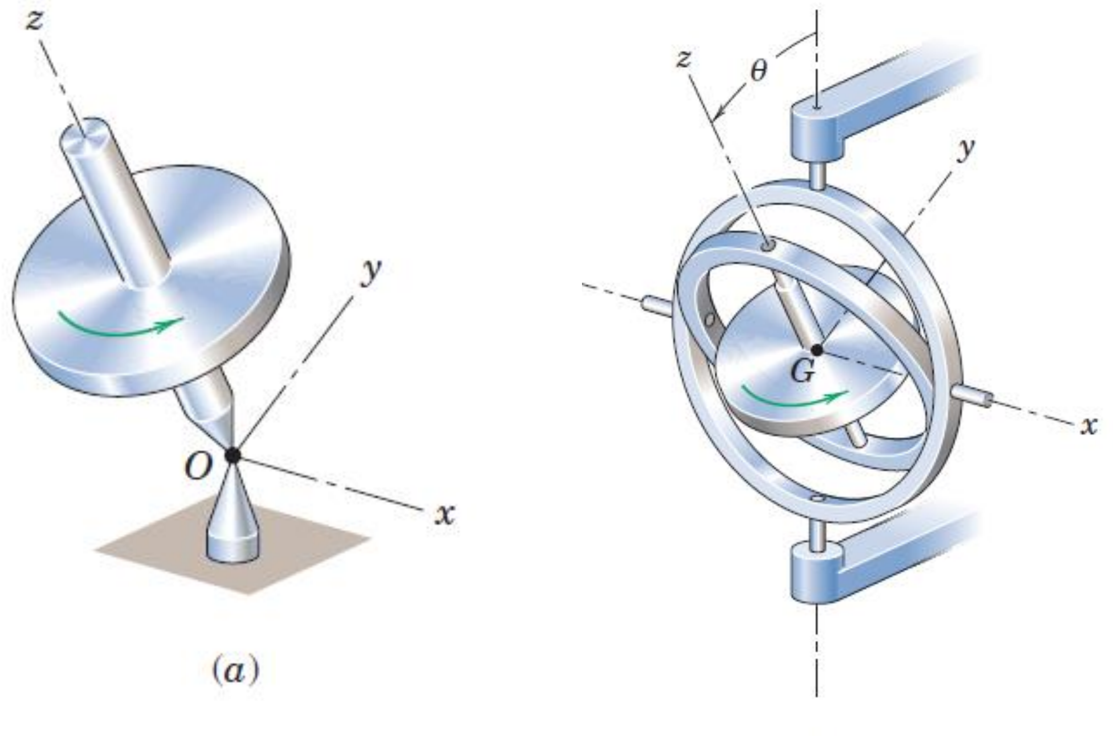
Gyroscopic Motion: Steady Precession

Gyroscopic motion: This motion occurs whenever the axis about which a body is spinning is itself rotating about another axis

Special Case: Gyroscopic motion when the **axis of a rotor spinning at constant speed turns** (precesses) about **another axis at a steady rate**

Gyroscopic Motion: Engineering Applications

- The gyroscope has important engineering applications. With a mounting in gimbal rings, the gyro is free from external moments, and its axis will retain a fixed direction in space regardless of the rotation of the structure to which it is attached. In this way, the gyro is used for inertial guidance systems and other directional control devices.

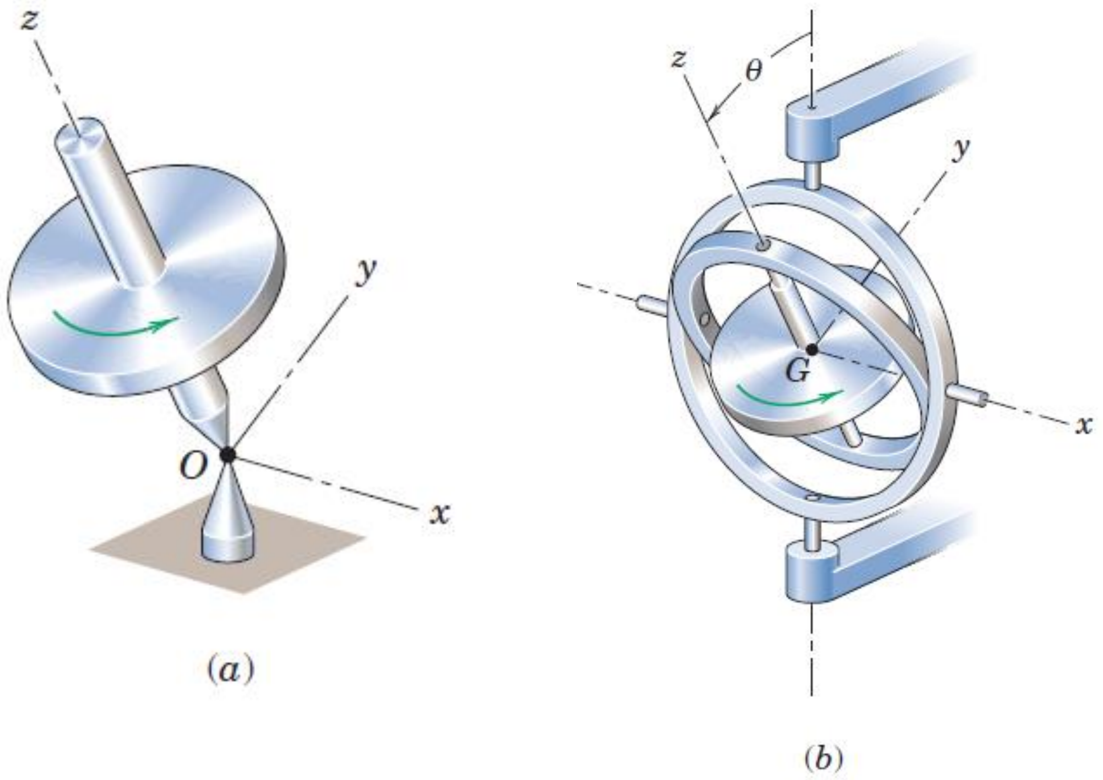


(a) (b)

Gyroscopic Motion: Engineering Applications

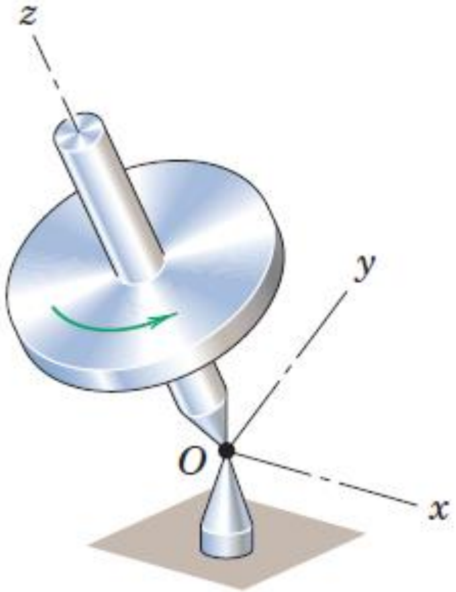
- With the addition of a pendulous mass to the inner gimbal ring, the earth's rotation causes the gyro to precess so that the spin axis will always point north, and this action forms the basis of the gyro compass

- The gyroscope has also found important use as a stabilizing device. The controlled precession of a large gyro mounted in a ship is used to produce a gyroscopic moment to counteract the rolling of a ship at sea

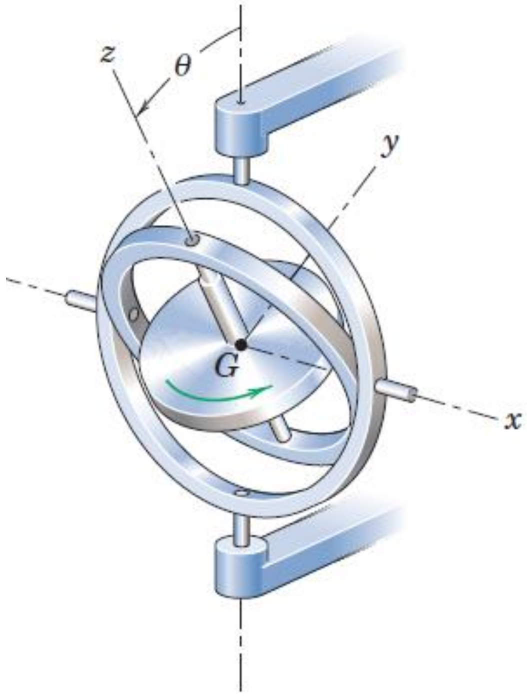


Gyroscopic Motion: Engineering Applications

- The gyroscopic effect is also an extremely important consideration in the design of bearings for the shafts of rotors which are subjected to forced precessions



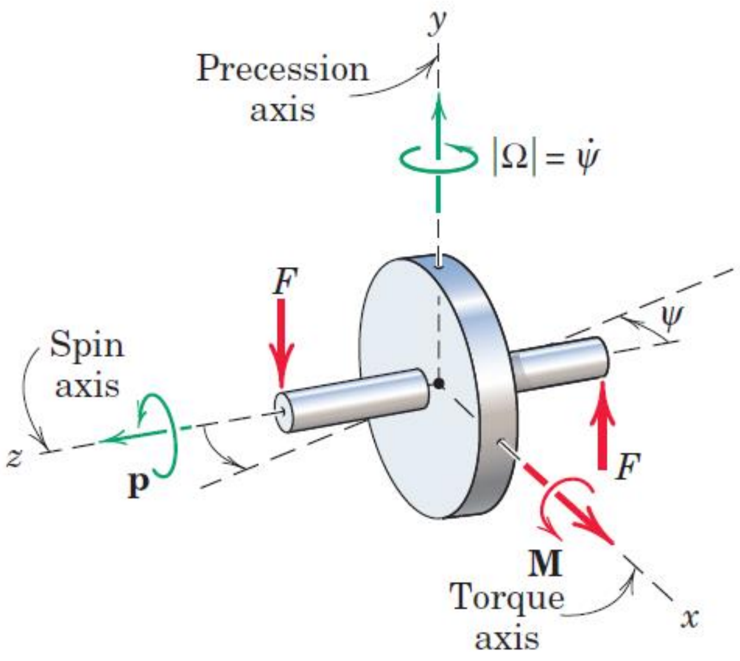
(a)



(b)

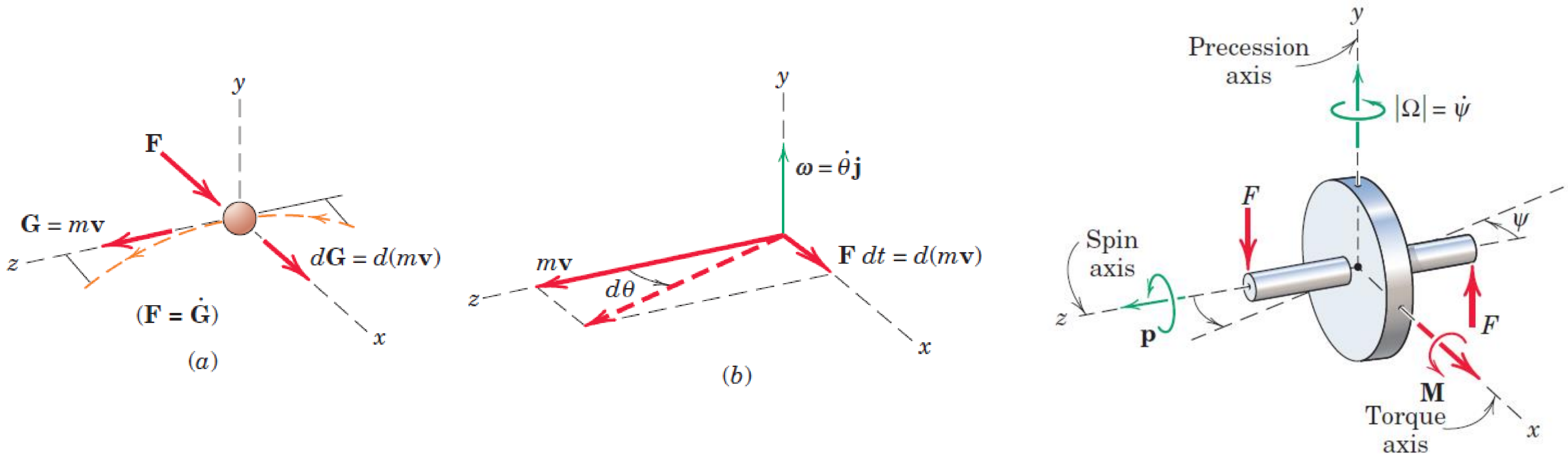
Gyroscopic Motion: Simplified Analysis

- Figure shows a symmetrical rotor spinning about the z-axis with a large angular velocity **p**, known as the **spin velocity**
- If we apply two forces **F** to the rotor axle to form a couple **M** whose vector is directed along the x-axis, we will find that the rotor shaft rotates in the x-z plane about the y-axis in the sense indicated, with a relatively slow angular velocity $\Omega = \dot{\psi}$ known as the **precession velocity**
- we identify the spin axis (p), the torque axis (M), and the precession axis (Ω), where the usual right-hand rule identifies the sense of the rotation vectors
- The rotor shaft does not turn about the x-axis in the sense of **M**, as it would if the rotor were not spinning



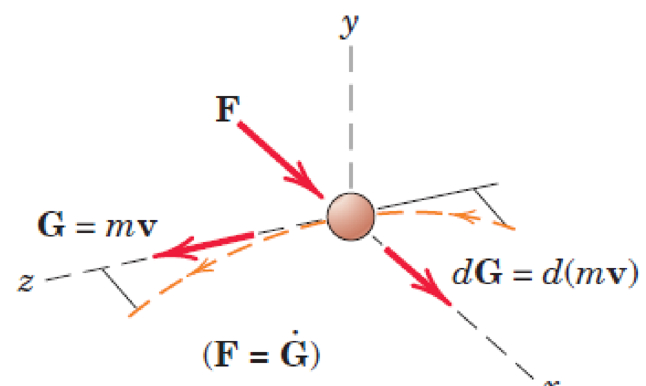
Gyroscopic Motion: Simplified Analysis

- To aid understanding of this phenomenon, a direct analogy may be made between the rotation vectors and the vectors which describe the curvilinear motion of a particle
- Figure shows a particle of mass m moving in the x-z plane with constant speed $|\mathbf{v}| = v$.

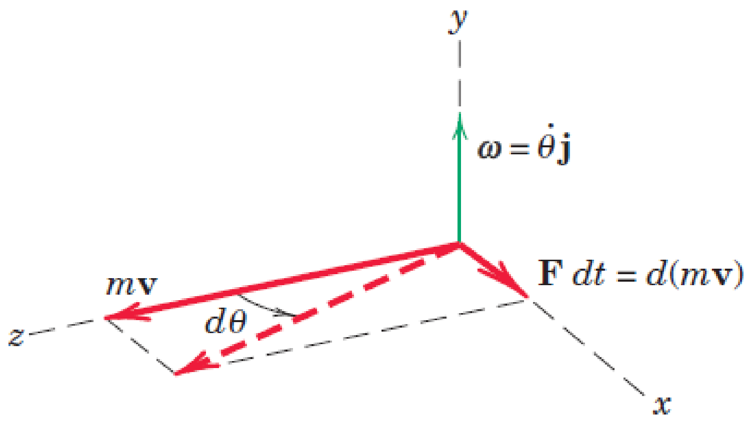


Gyroscopic Motion: Simplified Analysis

- The application of a force **F** normal to its linear momentum $\boldsymbol{G} = m\mathbf{v}$ causes a change $d\boldsymbol{G} = d(m\mathbf{v})$ in its momentum
- $d\boldsymbol{G}$, and thus $d\mathbf{v}$, is a vector in the direction of the normal force **F** according to Newton's second law $\boldsymbol{F} = \dot{\boldsymbol{G}}$ which may be written as $\boldsymbol{F}dt = d\boldsymbol{G}$
- In the limit, $\tan d\theta = d\theta = \mathbf{F}\,dt/m\mathbf{v}$ or $\mathbf{F} = m\mathbf{v}\dot{\boldsymbol{\theta}} = m\mathbf{v}\boldsymbol{\omega}$ In vector notation with $\boldsymbol{\omega} = \dot{\theta}\mathbf{j}$, the force becomes $\mathbf{F} = m\boldsymbol{\omega} \times \mathbf{v}$



(a)



(b)

Gyroscopic Motion: Simplified Analysis

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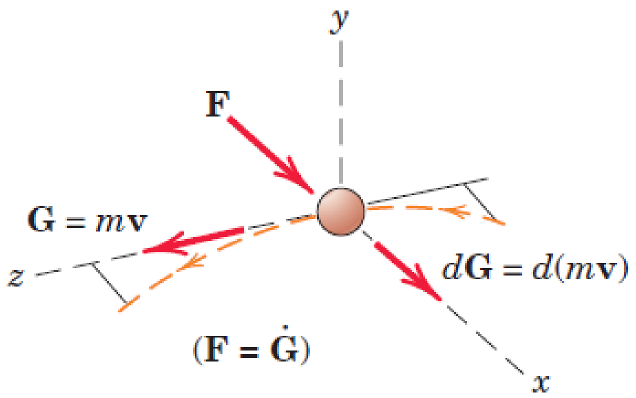
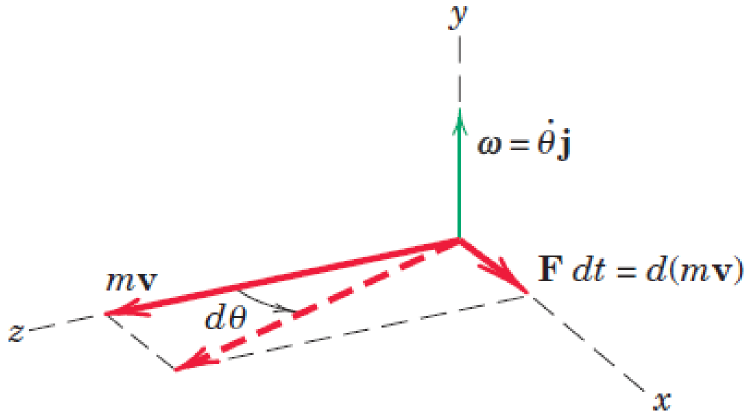


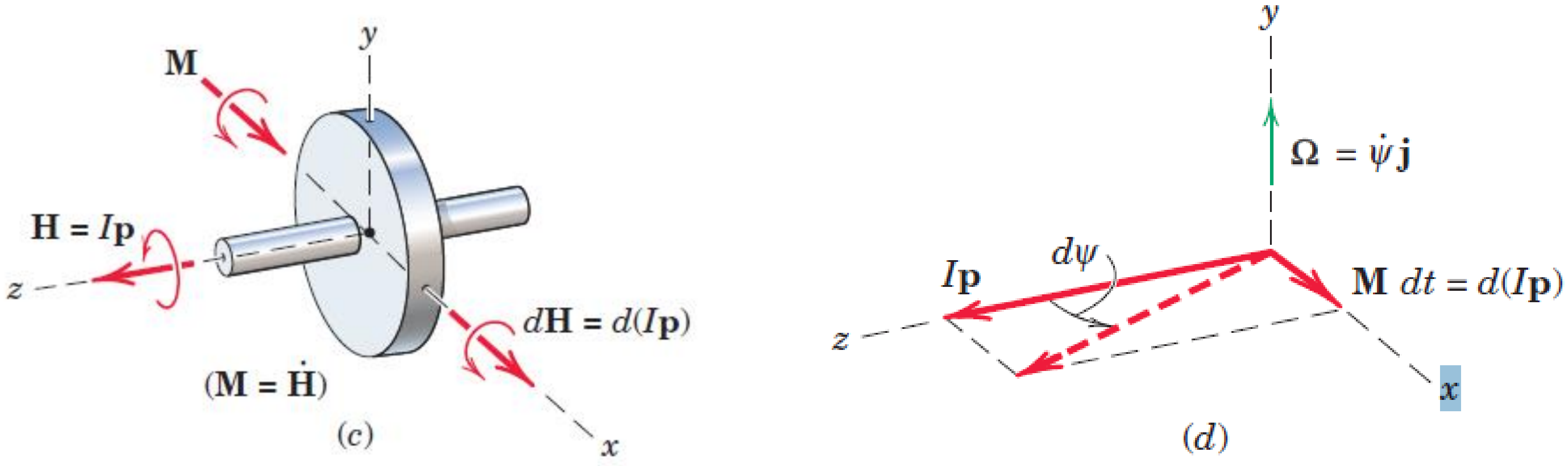
(a)



(b)

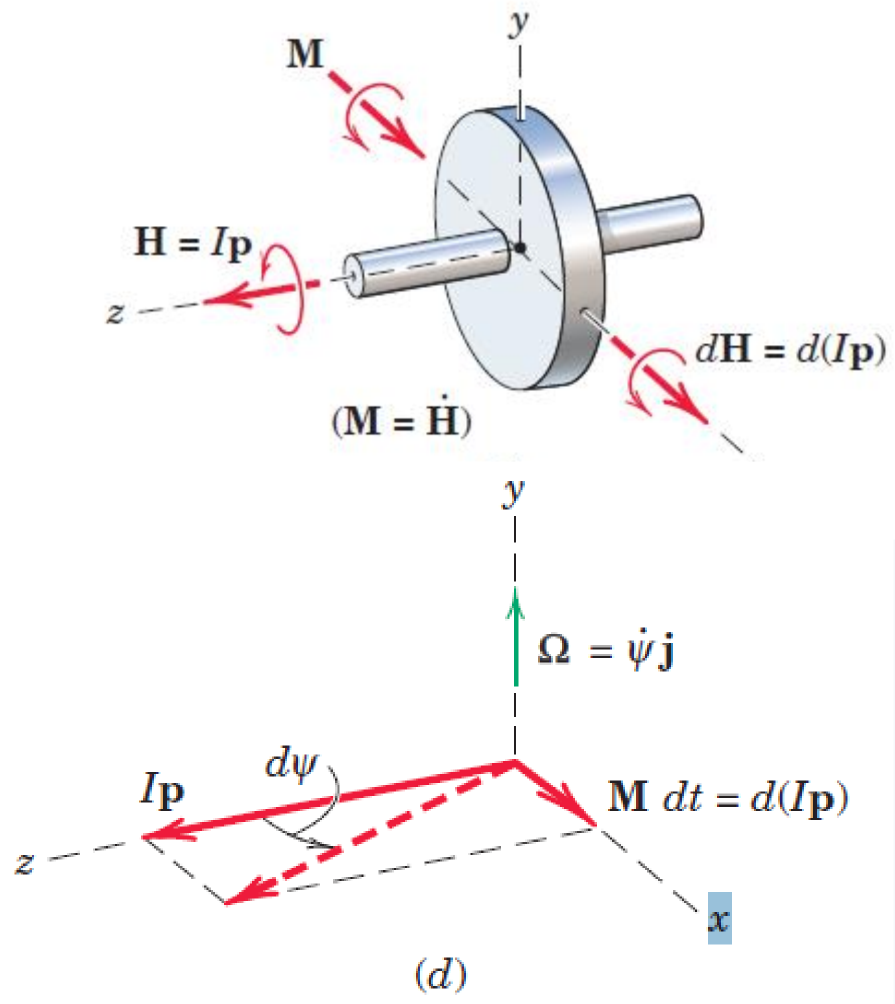
Gyroscopic Motion: Simplified Analysis

- Recall now the analogous equation $\mathbf{M} = \dot{\mathbf{H}}$ for any prescribed mass system, rigid or nonrigid, referred to its mass center or to a fixed point O
- Apply this relation to our symmetrical rotor, as shown in figure. For a high rate of spin $\mathbf{p}$ and a low precession rate $\boldsymbol{\Omega}$ about the y-axis, the angular momentum is represented by the vector $\boldsymbol{H} = I\boldsymbol{p}$, where $I = I_{zz}$ is the moment of inertia of the rotor about the spin axis



Gyroscopic Motion: Simplified Analysis

- Neglect the small component of angular momentum about the *y*-axis which accompanies the slow precession.
- The application of the couple **M** normal to **H** causes a change $d\mathbf{H}=d(I\mathbf{p})$ in the angular momentum
- $d\mathbf{H}$, and thus $d\mathbf{p}$, is a vector in the direction of the couple **M** and $\mathbf{M}\,dt=d\mathbf{H}$
- Just as the change in the linear-momentum vector of the particle is in the direction of the applied force, so is the change in the angular-momentum vector of the gyro in the direction of the couple
- Thus, we see that the vectors **M**, **H**, and $d\mathbf{H}$ are analogous to the vectors **F**, **G**, and $d\mathbf{G}$. With this insight, it is no longer strange to see the rotation vector undergo a change which is in the direction of **M**, thereby causing the axis of the rotor to precess about the *y*-axis.



Gyroscopic Motion: Simplified Analysis

- During time dt the angular-momentum vector Ip has swung through the angle $d\psi$ so that in the limit with $\tan d\psi = d\psi$

$$d\psi = \frac{\mathbf{M}\,dt}{I\mathbf{p}}$$

$$\mathbf{M} = I\mathbf{p}\frac{d\psi}{dt}$$

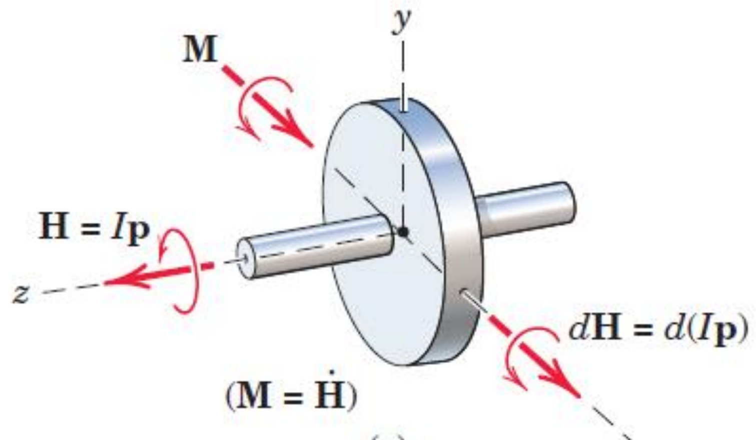
$$\Omega = \frac{d\psi}{dt}$$

$$\mathbf{M} = I\mathbf{p}\Omega$$

$\mathbf{M}$, $\boldsymbol{\Omega}$, and $\mathbf{p}$ as vectors are mutually perpendicular, and that their vector relationship may be represented by writing the equation in the cross-product form

$$\mathbf{M} = \boldsymbol{I\Omega} \times \mathbf{p}$$

Analogous to the foregoing relation $\mathbf{F} = \boldsymbol{m\omega} \times \mathbf{v}$ for the curvilinear motion of a particle



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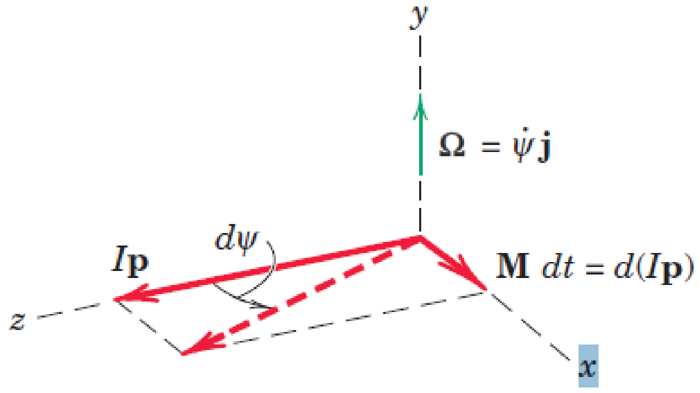


(d)

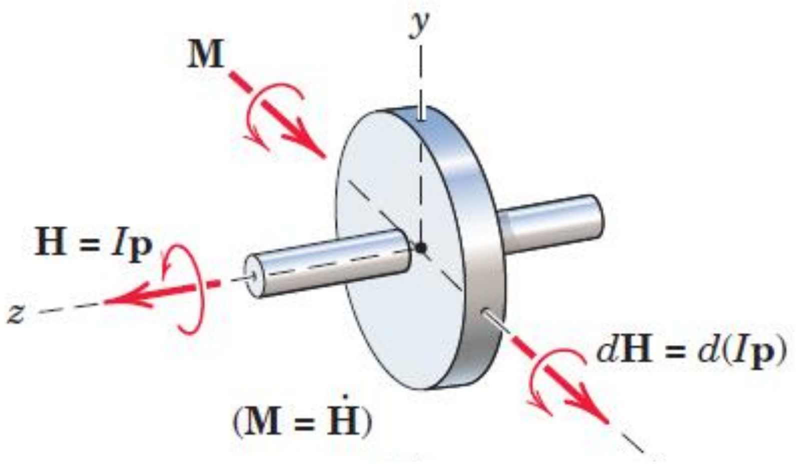
Gyroscopic Motion: Simplified Analysis

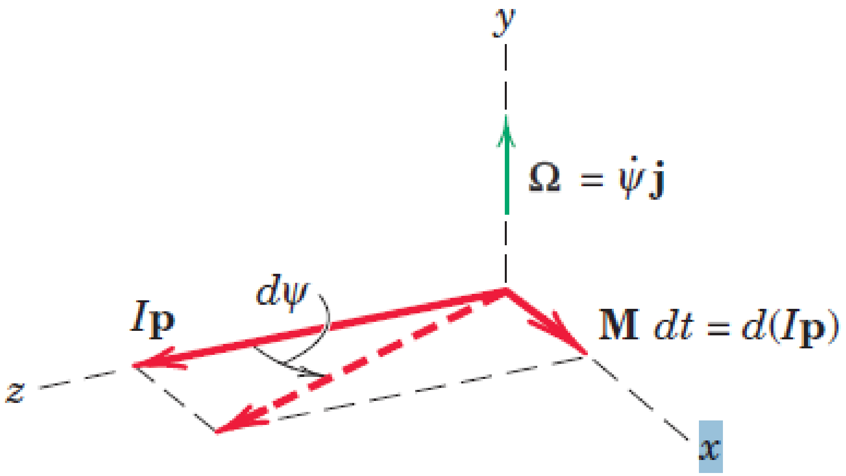
$$\mathbf{M} = I\mathbf{p}\Omega$$

M, $\boldsymbol{\Omega}$, and **p** as vectors are mutually perpendicular, and that their vector relationship may be represented by writing the equation in the cross-product form

$$\mathbf{M} = I\boldsymbol{\Omega} \times \mathbf{p}$$

- Equations apply to moments taken about the mass center or about a fixed point on the axis of rotation

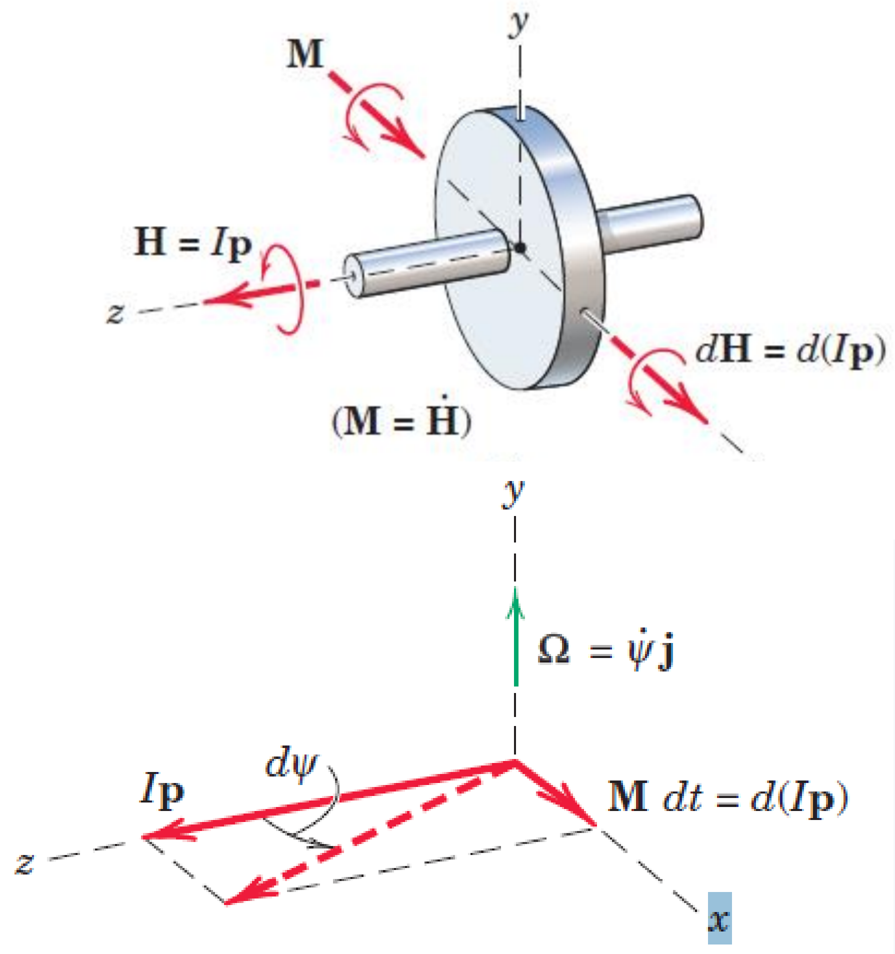




(d)

Gyroscopic Motion: Simplified Analysis

- The correct spatial relationship among the three vectors may be remembered from the fact that $d\mathbf{H}$, and thus $d\mathbf{p}$, is in the direction of **M**, which establishes the correct sense for the precession $\boldsymbol{\Omega}$. Therefore, the spin vector **p** always tends to rotate toward the torque vector **M**.



(d)

Gyroscopic Motion: Simplified Analysis

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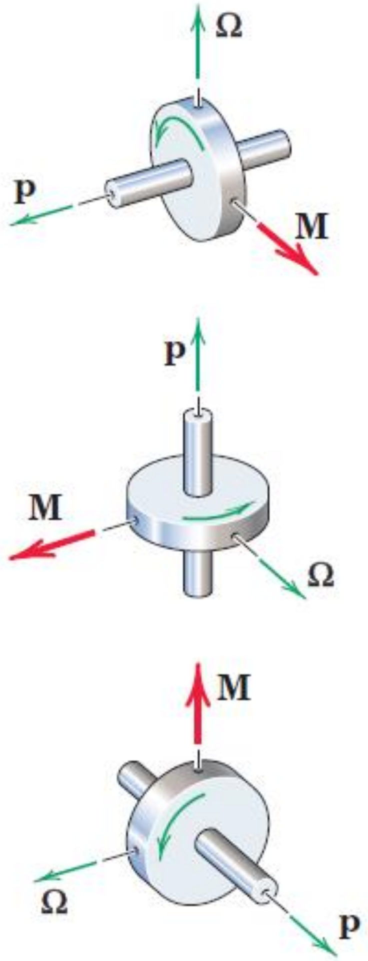
- Figure represents three orientations of the three vectors which are consistent with their correct order. Unless we establish this order correctly in a given problem, we are likely to arrive at a conclusion directly opposite to the correct one.

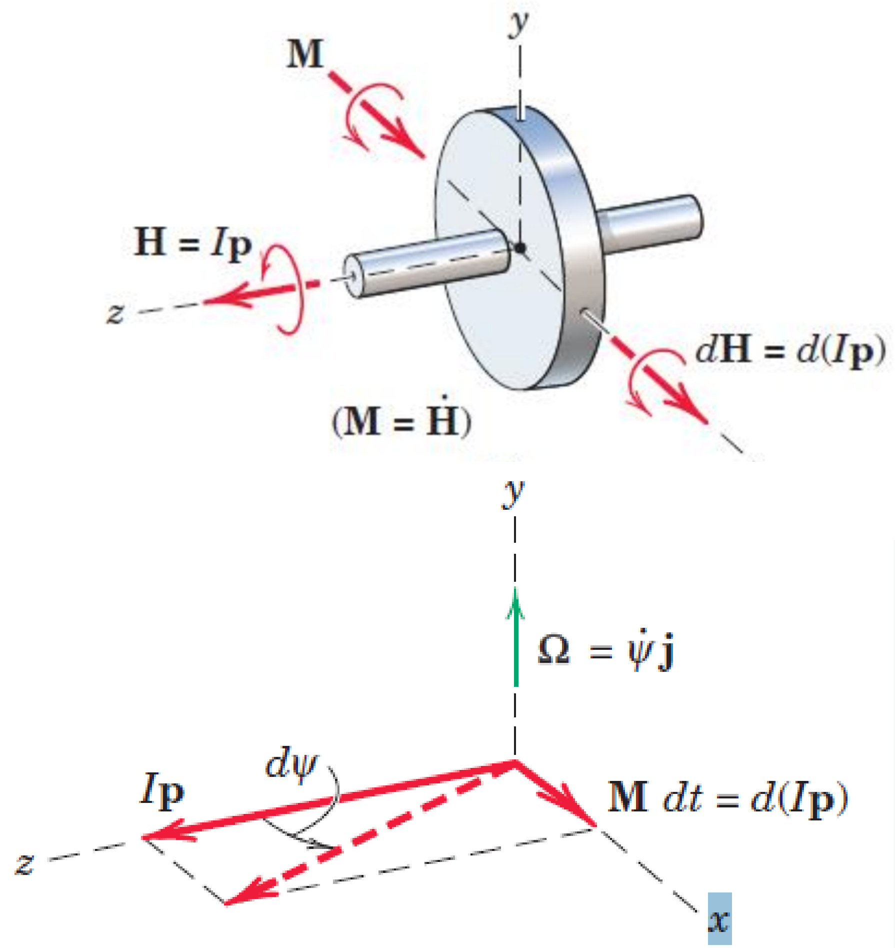


Gyroscopic Motion: Simplified Analysis

- $\boldsymbol{F} = m\boldsymbol{a}$ and $\boldsymbol{M} = I\boldsymbol{\alpha}$ equations of motion

The couple **M** represents the couple due to all forces acting on the rotor, as disclosed by a correct free-body diagram of the rotor

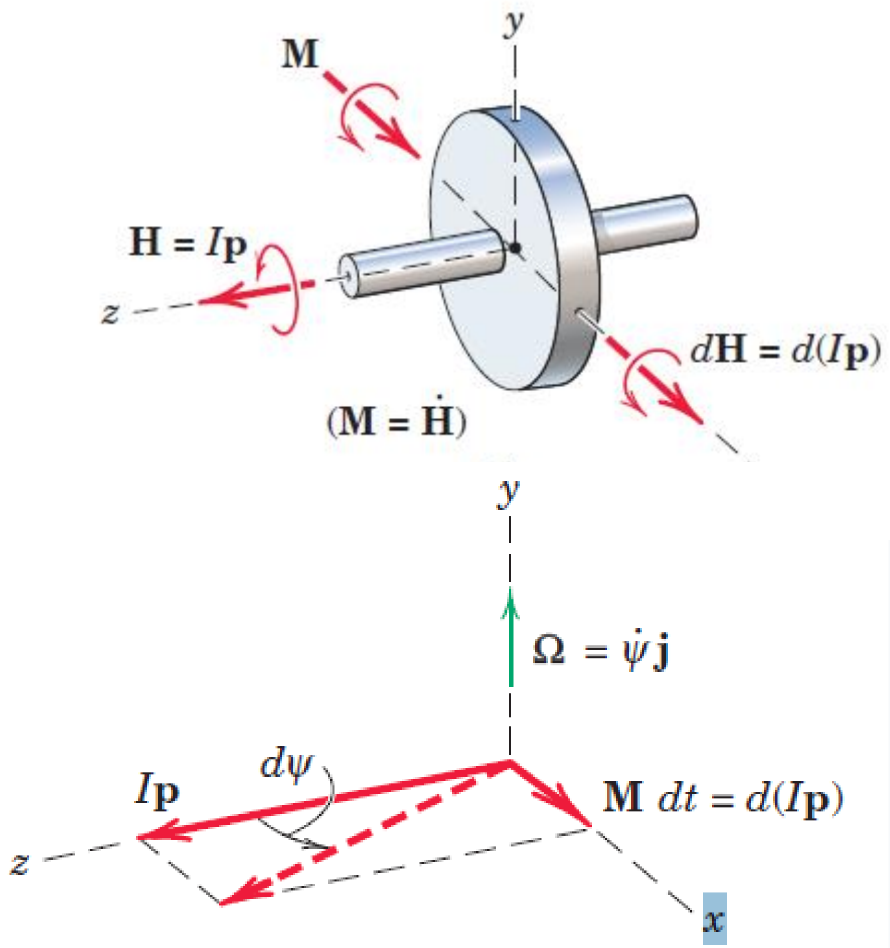
- Also note that, when a rotor is forced to precess, as occurs with the turbine in a ship which is executing a turn, the motion will generate a gyroscopic couple **M** which obeys $\mathbf{M} = I\boldsymbol{\Omega} \times \mathbf{p}$ in both magnitude and sense



(d)

Gyroscopic Motion: Simplified Analysis

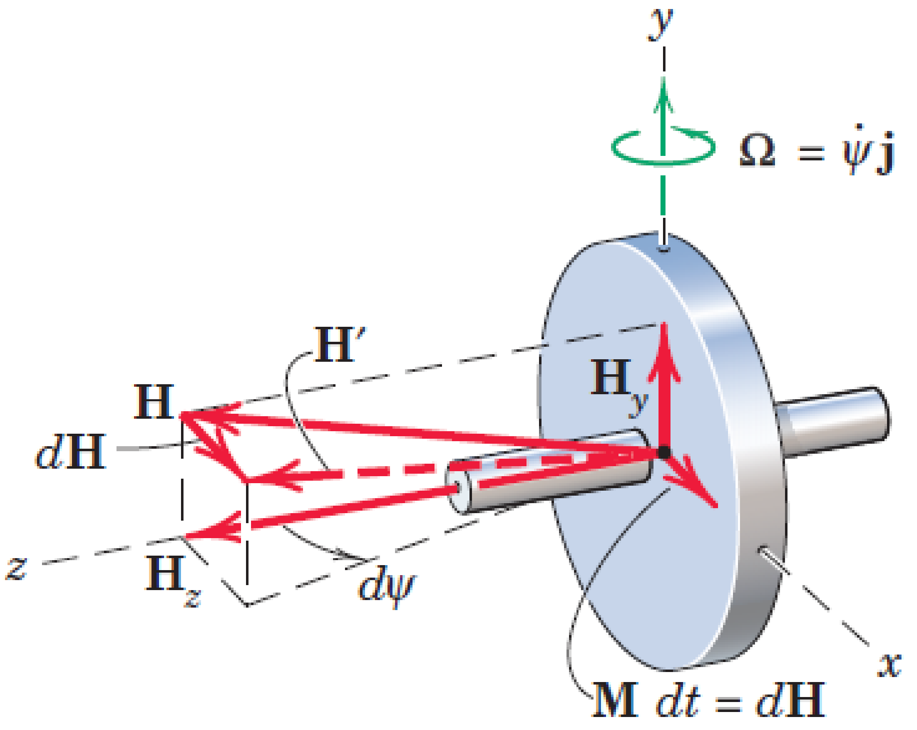
In the foregoing discussion of gyroscopic motion, it was assumed that the spin was large and the precession was small. Although we can see from $\mathbf{M} = I\mathbf{p}\Omega$ that for given values of I and M, the precession $\mathbf{\Omega}$ must be small if p is large, let us now examine the influence of $\mathbf{\Omega}$ on the momentum relations. Again, we restrict our attention to steady precession, where has $\mathbf{\Omega}$ a constant magnitude.



(d)

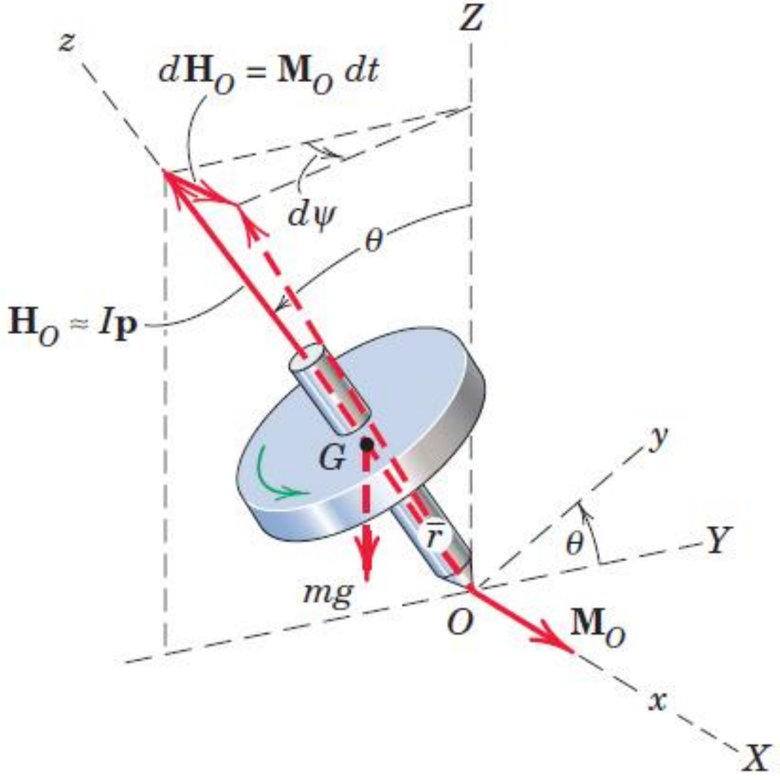
Gyroscopic Motion: Simplified Analysis

- Figure shows our same rotor again.
- Because it has a moment of inertia about the y-axis and an angular velocity of precession about this axis, there will be an additional component of angular momentum about the y-axis. Thus, we have the two components $H_z = Ip$ and $H_y = I_o\Omega$,where I_o stands for I_{yy} and I stands for I_{zz}.
- The total angular momentum is H as shown in figure. The change in H remains $d\mathbf{H} = \mathbf{M}\,dt$ as previously, and the precession during time dt is the angle $d\psi = \mathbf{M}\,dt/\mathbf{H}_z = \mathbf{M}\,dt/Ip$ as before.
- Thus, $\mathrm{M} = \mathrm{I}\Omega\mathrm{p}$ is still valid and for steady precession is an exact description of the motion as long as the spin axis is perpendicular to the axis around which precession occurs



Gyroscopic Motion: Simplified Analysis

- Consider now the steady precession of a symmetrical top spinning about its axis with a high angular velocity p and supported at its point O
- Here the spin axis makes an angle θ with the vertical z-axis around which precession occurs. Again, we will neglect the small angular-momentum component due to the precession and consider **H** equal to $I\mathbf{p}$, the angular momentum about the axis of the top associated with the spin only.
- The moment about O is due to the weight mg and is $mg\bar{r}\sin\theta$, where $\bar{r}$ is the distance from O to the mass center G.



Gyroscopic Motion: Simplified Analysis

- From the diagram, we see that the angular-momentum vector $\mathbf{H}_o$ has a change $d\mathbf{H}_o = \mathbf{M}_O dt$ in the direction of $\mathbf{M}_o$ during time dt and that θ is unchanged.
- The increment in precessional angle around the Z-axis is

$$d\psi = \frac{M_o dt}{Ip\ \sin\theta}$$

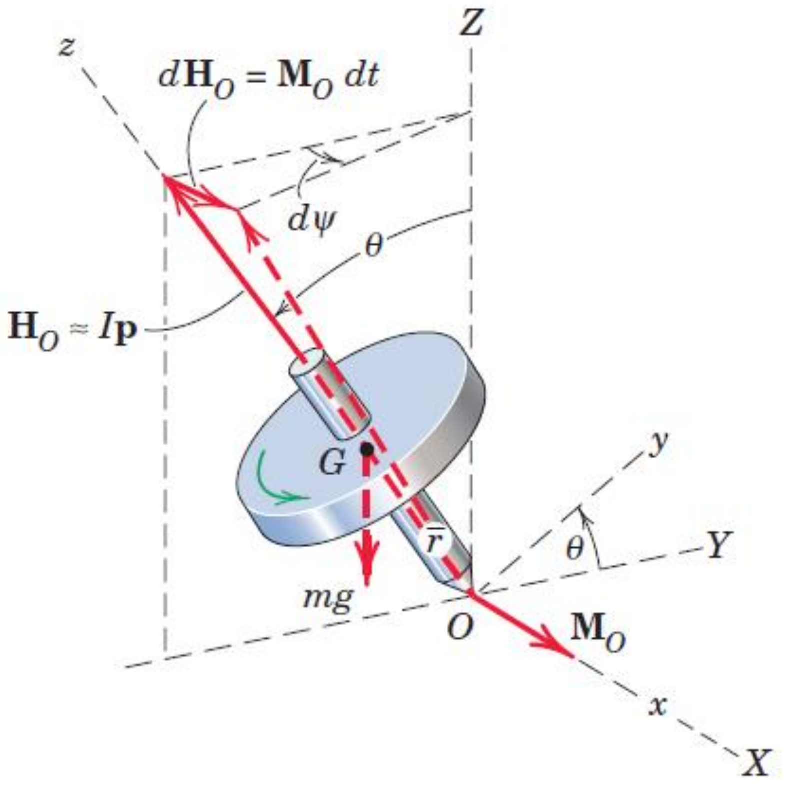
Substituting the values $M_o = mg\bar{r}\sin\theta$ and $\Omega = \frac{d\psi}{dt}$ gives

$$mg\bar{r} = Ip\Omega$$

which is independent of θ.

Introducing the radius of gyration so that $I = mk^2$ and solving for the precessional velocity give

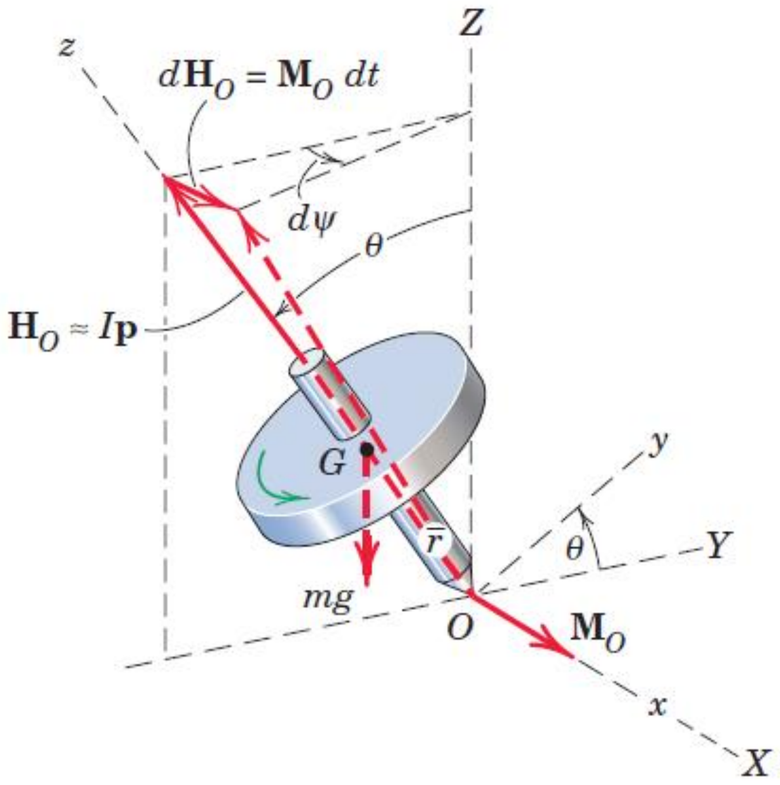
$$\Omega = \frac{g\bar{r}}{k^2 p}$$



Gyroscopic Motion: Simplified Analysis

- Unlike $\mathbf{M} = I\mathbf{p}\Omega$, which is an exact description for the rotor with precession confined to the x-z plane, $\Omega = \frac{g\bar{r}}{k^2 p}$ is an approximation based on the assumption that the angular momentum associated with Ω is negligible compared with that associated with p. We will see the amount of the error associated with this approximation when we reconsider steady-state precession later in this article.
- On the basis of our analysis, the top will have a steady precession at the constant angle θ only if it is set in motion with a value of Ω which satisfies $\Omega = \frac{g\bar{r}}{k^2 p}$.

  When these conditions are not met, the precession becomes unsteady, and θ may oscillate with an amplitude which increases as the spin velocity decreases. The corresponding rise and fall of the rotation axis is called **nutation**.



THANK YOU